



INTERNATIONAL ROADMAP FOR DEVICES AND SYSTEMS

INTERNATIONAL ROADMAP FOR DEVICES AND SYSTEMS

2024 UPDATE

BEYOND CMOS AND EMERGING RESEARCH MATERIALS

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ACKNOWLEDGMENTS

Sapan Agarwal
 Brad Aimone
 Hiro Akinaga
 Otitoaleke Akinola
 Mustafa Badaroglu
 Gennadi Bersuker
 Christian Binek
 Geoffrey Burr
 Leonid Butov
 Kerem Camsari
 Gert Cauwenberghs
 An Chen
 Winston Chern
 Supriyo Datta
 John Dallesasse
 Shamik Das
 Erik DeBenedictis
 Peter Dowben
 Tetsuo Endoh
 Ben Feinberg
 Thomas Ferreira de Lima
 Akira Fujiwara
 Elliot Fuller
 Michael Frank
 Paul Franzon
 Michael Fuhrer
 Satoshi Fujimura
 Mike Garner
 Saugata Ghose
 Chakku Goplan
 Bogdan Govoreanu
 Cat Graves
 Kohei Hamaya
 Masami Hane
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 D. Scott Holmes
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 Francesca Iacopi
 Gaku Ichihara
 Danielle Ilmeni
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 Jean Anne Incorvia
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 Koji Izunome

Satoshi Kamiyama
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 Asif Khan
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 Takehito Kozasa
 Suhas Kumar
 Ilya Krivorotov
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 Xiuling Li
 Xiang (Shaun) Li
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 Matthew Marinella
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 Koichi Matsumoto
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 Yoshiyuki Miyamoto
 Johannes Muller
 Azad Naeemi
 Mitchell A. Nahmias
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 Emre Neftci
 Kae Nemoto
 Kei Noda
 Mike Niemier
 Dmitri Nikonov
 Yutaka Ohno
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 Ferdinand Peper
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 R. Stanley Williams
 Justin Wong
 Dirk Wouters
 Patrick Xiao
 J. Joshua Yang
 Nobuyuki Yoshikawa
 Shusuke Yoshitake
 Victor Zhirnov

BEYOND CMOS

1. INTRODUCTION

1.1. SCOPE OF BEYOND-CMOS FOCUS TEAM

Dimensional and functional scaling¹ of CMOS is driving information processing² technology into a broadening spectrum of new applications. Scaling has enabled many of these applications through increased performance and complexity. As dimensional scaling of CMOS will eventually approach fundamental limits, new information processing devices and microarchitectures for both existing and new functions are being explored. This is driving interest in new devices for information processing and memory, new technologies for heterogeneous integration of multiple functions, and new paradigms for system architecture. This chapter, therefore, provides an IRDS perspective on emerging research device technologies and serves as a bridge between conventional CMOS and the realm of nanoelectronics beyond the end of CMOS scaling.

An overarching goal of this chapter is to survey, assess, and catalog viable emerging devices and novel architectures for their long-range potential and technological maturity and to identify the scientific/technological challenges gating their acceptance by the semiconductor industry as having acceptable risk for further development. This chapter also surveys beyond-CMOS devices for more than Moore (MtM) applications.

This goal is accomplished by addressing two technology-defining domains: 1) extending the functionality of the CMOS platform via heterogeneous integration of new technologies (“More Moore”), and 2) stimulating invention of new information processing paradigms (“Beyond CMOS”). The relationship between these domains is schematically illustrated in Figure BC1.1. Novel computing paradigms and application pulls (e.g., big data, Internet of Things (IoT), artificial intelligence, autonomous systems, exascale computing) introduce higher performance and efficiency requirements, which is increasingly difficult for the saturating More Moore technologies to fulfill. Beyond-CMOS technologies may provide the devices, processes, and architectures needed for the new era of computing.

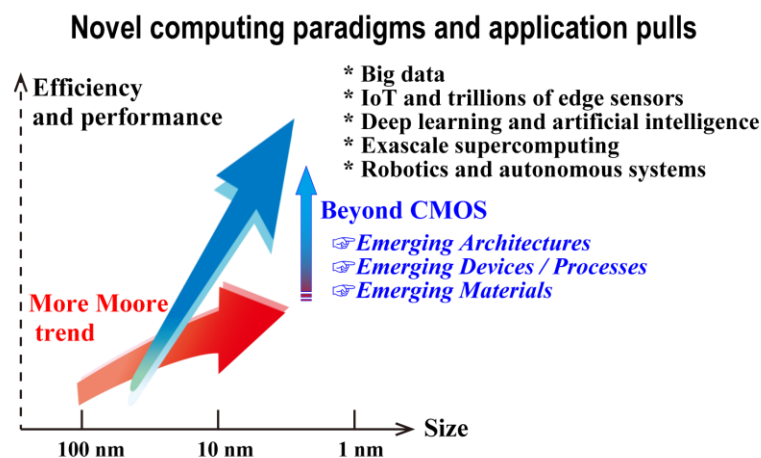


Figure BC1.1 Relationship of More Moore, Beyond CMOS, and Novel Computing Paradigms and Applications (Courtesy of Japan beyond-CMOS Group)

The chapter is intended to provide an objective, informative resource for the constituent nanoelectronics communities pursuing the following: 1) research, 2) tool development, 3) funding support, and 4) investment. These communities include universities, research institutes, industrial research laboratories, tool suppliers, research funding agencies, and the semiconductor industry. The potential and maturity of each emerging research device and architecture technology are reviewed and assessed to identify

¹ Functional Scaling: Suppose that a system has been realized to execute a specific function in a given, currently available, technology. We say that system has been functionally scaled if the system is realized in an alternate technology such that it performs the identical function as the original system and offers improvements in at least one of size, power, speed, or cost, and does not degrade in any of the other metrics.

² Information processing refers to the input, transmission, storage, manipulation or processing, and output of data. The scope of the BC Chapter is restricted to data or information manipulation, transmission, and storage.

the most important scientific and technological challenges that must be overcome for a candidate device or architecture to become a viable approach.

The chapter is divided into five sections: 1) emerging memory devices, 2) emerging logic and alternative information processing devices, 3) emerging device-architecture interaction, 4) beyond-CMOS devices for More-than-Moore applications, and 5) emerging materials integration (EMI). The former IRDS Emerging Research Materials (ERM) chapter is rolled into this chapter as a section named “emerging materials integration (EMI)”. Some detail is provided for each entry regarding operation principles, advantages, technical challenges, maturity, and current and projected performance. The chapter also discusses applications and architectural focus combining emerging research devices offering specialized, unique functions as heterogeneous core processors integrated with a CMOS platform technology. This represents the nearer term focus of the chapter, with the longer-term focus remaining on discovery of an alternate information processing technology beyond digital CMOS.

1.2. DIFFICULT CHALLENGES

1.2.1. INTRODUCTION

The semiconductor industry is facing some difficult challenges related to extending integrated circuit technology to new applications and to beyond the end of CMOS dimensional scaling. One class relates to propelling CMOS beyond its ultimate density and functionality by integrating a new high-speed, high-density, and low-power memory technology onto the CMOS platform. Another class is to extend CMOS scaling with alternative channel materials. The third class is information processing technologies substantially beyond those attainable by CMOS using an innovative combination of new devices, interconnect, and architectural approaches for extending CMOS and eventually inventing a new information processing platform technology. The fourth class is to extend ultimately scaled CMOS as a platform technology into new domains of functionalities and application, also known as “more than Moore”. The fifth class is to bridge the gap between novel devices and unconventional architectures and computing paradigms. These difficult challenges are summarized in Table BC1.1.

1.2.2. DEVICE TECHNOLOGIES

Difficult challenges gating development of beyond-CMOS devices include those related to memory technologies, information processing or logic devices, and heterogeneous integration of multi-functional components, a.k.a. More-than-Moore (MtM) or functional diversification.

One challenge is the need of a new memory technology that combines the best features of current memories in a fabrication technology compatible with CMOS process flow and that can be scaled beyond the present limits of SRAM and FLASH. This would provide a memory device fabrication technology required for both stand-alone and embedded memory applications. The ability of a chip to execute programs is limited by interaction between the processor and the memory, and scaling does not automatically solve this problem. The current evolutionary solution is to increase cache memory, thereby increasing the floor space that SRAM occupies on a chip. However, this trend eventually leads to a decrease of the net information throughput. Volatility of semiconductor memory requires external long-term storage media that tend to be slow to access (e.g., magnetic hard drives, optical CD, etc.). Therefore, development of electrically accessible non-volatile memory with high speed and high density would initiate a revolution in computer architecture. This development would provide a significant increase in information throughput beyond the traditional benefits of scaling.

Another challenge is to sustain scaling of CMOS logic technology. One approach to continuing performance gains as CMOS scaling matures is to replace the strained silicon MOSFET channel (and the source/drain) with an alternate material offering a higher potential quasi-ballistic-carrier velocity and higher mobility than strained silicon. Introduction of non-silicon materials into the channel and source/drain regions of an otherwise silicon MOSFET is fraught with difficult challenges. These challenges include fabrication of high-quality (i.e., defect free) channel and source/drain materials on non-lattice matched silicon, minimization of band-to-band tunneling in narrow bandgap channel materials, fabrication of high- κ gate dielectrics on the new channel materials, and elimination of Fermi level pinning in the channel/gate dielectric interface. Additional challenges are to sustain the required reduction in leakage currents and power dissipation in these ultimately scaled CMOS and to introduce new materials into MOSFET while simultaneously minimizing variations in critical dimensions and statistical fluctuations in the doping concentrations.

The industry is now addressing the increasing importance of a new trend of functional diversification, where added value to devices is provided by incorporating functionalities that do not necessarily scale according to “Moore's Law”. In this chapter, a “Beyond-CMOS devices for More-than-Moore Applications” section covers unconventional applications of beyond-CMOS technologies. The section currently covers emerging devices for hardware security.

Table BCI.1 *Beyond CMOS Difficult Challenges*

<i>Difficult Challenges</i>	<i>Summary of Issues and Opportunities</i>
Scale high-speed, dense, embeddable, volatile/non-volatile memory technologies to replace SRAM and FLASH in appropriate applications.	The scaling limits of SRAM and FLASH in two dimensional (2D) are driving the need for new memory technologies to replace SRAM and FLASH memories. Identify the most promising technical approach(es) to obtain electrically accessible, high-speed, high-density, low-power, (preferably) embeddable volatile and non-volatile memories. The desired material/device properties must be maintained through and after high temperature and corrosive chemical processing. Reliability issues should be identified and addressed early in the technology development.
Extend CMOS scaling	Develop new materials to replace silicon (or III-V, Ge) as alternate channel and source/drain to increase the saturation velocity and to further reduce V_{dd} and power dissipation in MOSFETs while minimizing leakage currents Develop means to control the variability of critical dimensions and statistical distributions (e.g., gate length, channel thickness, S/D doping concentrations, etc.) Accommodate the heterogeneous integration of dissimilar materials. The desired material/device properties must be maintained through and after high temperature and corrosive chemical processing. Reliability issues should be identified and addressed early in this development.
Continue functional scaling of information processing technology substantially beyond that attainable by ultimately scaled CMOS.	Invent and reduce to practice new information processing technologies with the potential to replace CMOS as the performance driver. Ensure that new information processing technologies have compatible memory technologies and interconnect solutions. New information processing technologies must be compatible with system architectures that can fully utilize new devices. Non-binary data representations or non-Boolean logic may be required to employ new devices for information processing, which will drive the need for new system architectures. Bridge the gap that exists between materials behaviors and device functions. Accommodate the heterogeneous integration of dissimilar materials. Reliability issues should be identified and addressed early in the technology development.
Extend ultimately scaled CMOS as a platform technology into new domains of functionalities and applications (“more than Moore, MtM”).	Discover and reduce to practice new device technologies and primitive-level architectures to provide optimized special-purpose functional accelerator functions heterogeneously integrable with CMOS. Provide added value by incorporating functionalities that do not necessarily scale according to “Moore’s Law”. Heterogeneous integration of digital <i>and</i> non-digital functionalities into compact systems that will be the key driver for a wide variety of applications, such as communication, automotive, environmental control, healthcare, security, and entertainment.
Bridge the gap between emerging devices and novel architectures and computing paradigms.	Identify suitable opportunities in unconventional architectures and computing paradigms that can utilize unique characteristics of emerging devices. Identify emerging devices that can implement computing functions and architectures more efficiently than CMOS and Boolean logic.

A longer-term challenge is invention and reduction to practice of a manufacturable information processing technology addressing “beyond CMOS” applications. For example, emerging research devices might be used to realize special purpose processor cores that could be integrated with multiple CMOS CPU cores to obtain performance advantages. These new special purpose cores may provide a particular system function much more efficiently than a digital CMOS block, or they may offer a uniquely new function not available in a CMOS-based approach. Solutions to this challenge beyond the end of CMOS scaling may also lead to new opportunities for such an emerging research device technology to eventually replace the CMOS gate as a new information processing primitive element. A new information processing technology must also be compatible with a system architecture that can fully utilize the new device. A non-binary data representation and non-Boolean logic may be required to employ a new device for information processing. These requirements will drive the need for new system architectures. The requirements and opportunities correlating emerging devices and architectures are discussed in the “Emerging Device-Architecture Interaction” section.

1.2.3. MATERIALS TECHNOLOGIES

The most difficult challenge for Beyond CMOS is to deliver materials with controlled properties that will enable operation of emerging research devices in high density at the nanometer scale. To improve control of material properties for high-density

devices, research on materials synthesis must be integrated with work on new and improved metrology and modeling. These important objectives are addressed in the “Emerging Materials Integration” section.

1.3. NANO-INFORMATION PROCESSING TAXONOMY

Information processing systems to accomplish a specific function, in general, require several different interactive layers of technology. One comprehensive top-down list of these layers begins with the required application or system function, leading to system architecture, micro- or nano-architecture, circuits, devices, and materials. A different bottom-up representation of this hierarchy begins with the lowest physical layer represented by a computational state variable and ends with the highest layer represented by the architecture. In this representation focused on generic information processing at the device/circuit level, a fundamental unit of information (e.g., a bit) is represented by a computational state variable, for example, the position of a bead in the ancient abacus calculator or the charge (or voltage) state of a node capacitance in CMOS logic. The electronic charge as a binary computational state variable serves as the foundation for the von Neumann computational system architecture. A device provides the physical means of representing and manipulating a computational state variable among its two or more allowed discrete states. Eventually, device concepts may transition from simple binary switches to devices with more complex information processing functionality, perhaps with multiple fan-in and fan-out. The device is a physical structure resulting from the assemblage of a variety of materials possessing certain desired properties obtained through exercising a set of fabrication processes. An important layer, therefore, encompasses the various materials and processes necessary to fabricate the required device structure, which is a focus of the “Beyond CMOS (BC)” chapter. The data representation is how the computational state variable is encoded by the assemblage of devices to process the bits or data. Two of the most common examples of data representation are binary digital and continuous or analog signal. This layer is within the scope of the BC chapter. The architecture layer encompasses three subclasses of this taxonomy: 1) nano-architecture or the physical arrangement or assemblage of devices to form higher level functional primitives to represent and execute a computational model, 2) the computational model that describes the algorithm by which information is processed using the primitives, e.g., logic, arithmetic, memory, cellular nonlinear network (CNN), and 3) the system-level architecture that describes the conceptual structure and functional behavior of the system exercising the computational model.

2. EMERGING MEMORY DEVICES

The emerging research memory technologies tabulated in this section are a representative sample of published research efforts (circa 2020 – 2022) describing alternative approaches to established memory technologies.³ The scope of this section also includes a new section on the Properties of Memory Devices for In-Memory Computing, as well as updated subsections addressing the Select Device required for a crossbar memory application and updated section on Storage Class Memory.

Figure BC2.1 is a taxonomy of the prototypical and emerging memory technologies. An overarching theme is the need to monolithically integrate each of these memory options onto a CMOS technology platform in a seamless manner. Fabrication technologies are sought that are modifications of or additions to established CMOS platform technologies. A goal is to provide the end user with a device that behaves similarly to the familiar silicon memory chip.

This memory portion of this section is organized around a set of eight technology entries shown in the column headers of Table BC2.1. These entries were selected using a systematic survey of the literature to determine areas of greatest worldwide research activity. Each technology entry listed has several sub-categories of devices that are grouped together to simplify the discussion. Key parameters associated with the technologies are listed in the table. For each parameter, two values for performance are given: 1) theoretically predicted performance values based on calculations and early experimental demonstrations, 2) up-to-date experimental values of these performance parameters reported in the cited technical references.

The tables have been extensively footnoted, and details may be found in the indicated references. The text associated with the table gives a brief summary of the operating principles of each device and significant scientific and technological issues, not captured in the table, but which must be resolved to demonstrate feasibility.

The purpose of many memory systems is to store massive amounts of data, and therefore memory capacity (or memory density) is one of the most important system parameters. In a typical memory system, the memory cells are connected to form a two-dimensional array, and it is essential to consider the performance of memory cells in the context of this array architecture. A

³ Including a particular approach in this section does not in any way constitute advocacy or endorsement. Conversely, not including a particular concept in this section does not in any way constitute rejection of that approach. This listing does point out that existing research efforts are exploring a variety of basic memory mechanisms.

memory cell in such an array can be viewed as being composed of two fundamental components: the ‘storage node’ and the ‘select device’, the latter of which allows a given memory cell in an array to be addressed for read or write. Both components impact scaling limits for memory. For several emerging resistance-based memories, the storage node can, in principle, be scaled down below 10 nm,¹ and the memory density will be limited by the select device. Planar transistors (e.g., FET or BJT) are typically used as select devices. In a two-dimensional layout using in-plane select FETs the cell layout area is $A_{\text{cell}}=(6-8)F^2$. In order to reach the highest possible 2-D memory density of $4F^2$, a vertical select transistor can be used.

Table BC2.1 *Emerging Research Memory Devices—Demonstrated and Projected Parameters*

2.1. MEMORY TAXONOMY

Figure BC2.1 provides a simple visual method of categorizing memory technologies. At the highest level, memory technologies are separated by the ability to retain data without power. Non-volatile memory offers essential use advantages, and the degree to which non-volatility exists is measured in terms of the length of time that data can be expected to be retained. Volatile memories also have a characteristic retention time that can vary from milliseconds to (for practical purposes) the length of time that power remains on. Non-volatile memory technologies are further categorized by their maturity. Flash memory is considered the baseline non-volatile memory because it is highly mature, well optimized, and has a significant commercial presence. Flash memory is the benchmark against which prototypical and emerging non-volatile memory technologies are measured. Prototypical memory technologies are at a point of maturity where they are commercially available (generally for niche applications), and have a large scientific, technological, and systems knowledge base available in the literature. The focus of this section is Emerging Memory Technologies. These are the least mature memory technologies in Figure BC2.1, but they have been shown to offer significant potential benefits if various scientific and technological hurdles can be overcome. This section provides an overview of these emerging technologies, their potential benefits, and the key research challenges that will allow them to become viable commercial technologies.

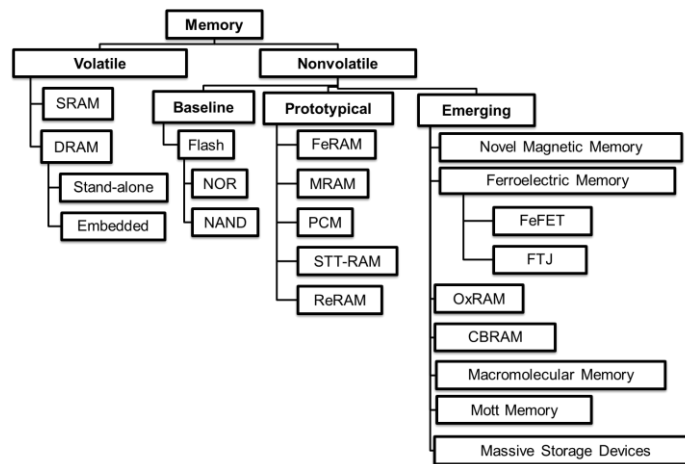


Figure BC2.1 *Taxonomy of Emerging Memory Devices*

2.2. EMERGING MEMORY DEVICES

2.2.1. NOVEL MAGNETIC MEMORIES

This section is divided into three categories of devices based on different mechanisms, i.e., spin-transfer torque, spin-orbital torque, and voltage-controlled magnetic anisotropy.

2.2.1.1. SPIN-TRANSFER TORQUE

Spin-transfer torque (STT) MRAM has entered the commercial production stage for both embedded and standalone Flash-like applications. Various foundries announce the readiness of embedded MRAM (TSMC, GlobalFoundries, Samsung) for production, and standalone MRAM products with various density (Mb-Gb) are also available on the market for IoT and data center applications (Avalanche Tech, Everspin). For embedded applications, STT-MRAM possesses non-volatility, high-endurance, scalability, low power, and fewer masks than the embedded Flash.² It also provides great area savings and lower leakage compared with SRAM. A STT-MRAM cell consists of a magnetic tunnel junction (MTJ) with two ferromagnetic layers, i.e., a

free layer (FL) and a reference layer (RL) sandwiching a tunnel barrier. When a current enters the RL, the electrons' spin is polarized according to that of the RL. After tunneling through to the FL, the spin momentum of these spin-polarized electrons is transferred to the FL, thus writing the FL orientation to that of the RL. The opposite operation can be achieved by flipping the current polarity. Compared with its in-plane counterpart, perpendicular STT-MRAM has higher density, lower switching current, and is easier to control in large scale manufacturing.³

Since Flash-like STT-MRAM is entering mass production at all major foundries, this topic is covered in the More Moore Chapter of IRDS. For cache-level applications, research has shown 3 ns, 63 μA switching of 16-nm perpendicular STT-MRAM with endurance above 10^{12} cycles and WER below 10^{-6} .⁴ Reliable 10 ns, 0.12 pJ switching of 50 μA with sub-ppm error rates is also recently demonstrated in 30 nm perpendicular STT-MTJs satisfying the last level cache (LLC) application requirements.⁵ A system-level benchmarking study also indicates the use of 34 nm perpendicular STT-MRAM to replace SRAM as a LLC for high performance computing (HPC) at 5 nm node provides significant read and write energy gains at about 43% of the total macro area, achieving a nominal access latency <2.5 ns and <7.1 ns for read and write respectively.⁶ (see Table BC2.1)

For SRAM-like STT-MRAM development, a major challenge is to reduce the write power consumption at sub-10 ns write speed. The large write power at sub-ns speed using STT requires a large access transistor, reducing density as well as reduced endurance due to tunnel barrier damage from higher writing current. Multiple factors contribute to this high switching current at sub-10 ns regime including limited spin torque efficiency, incubation delay, and intrinsic magnetization precession frequency on the order of GHz. A great amount of work has been devoted to increasing the switching speed, such as decreasing the FL saturation magnetization, lowering the FL damping factor,⁷ and increasing the STT effect via double-RL design⁸. However, because at most 100% of the current can be polarized, there is an upper bound of the write efficiency using the STT effect. In contrast, the use of writing mechanisms of fundamentally different physics, such as spin-orbit torque (SOT) and voltage-controlled magnetic anisotropy (VCMA) effect, may help propel the next-generation of MRAM technologies for SRAM-like cache applications. Secondly, as the critical dimension of STT-MRAM scales down to less than 20 nm, the requirement of thermal stability calls for higher interfacial perpendicular magnetic anisotropy (PMA). Double-MgO barrier MTJs have shown 2x improvement in MTJ with diameters above 10 nm,⁹ whereas the use of shape-anisotropy induced PMA from elongated ferromagnetic pillars may further the scaling of STT-MRAM below 10 nm diameter.¹⁰ Lastly, STT-MRAM with higher density to replace DRAM is still an open area for research. Stacking of STT-MRAM dies with logic or memory dies using through silicon vias,^{11,12} and high density 3D integration of STT-MRAM using selector-MTJ crossbar architecture¹³ are two high potential approaches.

2.2.1.2. SPIN-ORBIT TORQUE

Spin-orbit torque (SOT)-driven magnetization switching recently emerges as an alternative write mechanism beyond STT for SRAM-like cache-level applications. Though at rather early stage of research, sub-ns SOT writing has been demonstrated at current density of 20-40 MA/cm^2 ,^{14,15} compared with 3-10 ns switching of STT-MRAM at a current density of 7 MA/cm^2 .^{5,16} (see Table BC2.1). A SOT-MRAM cell consists of a magnetic tunnel junction (MTJ) with its free layer (FL) sitting on top of a strip of material with large spin-orbit coupling (SOC), such as heavy metal^{17,18}. When current flows through this long strip of SOT material, spin-polarized current emerges and diffuses into the adjacent FL. Like the STT case, the spin-polarized current exerts a damping-like spin torque on the ferromagnetic layer, thus switching the FL orientation. As the write path is separated from the read path, a much larger read voltage can increase read speed. The major advantage of SOT over STT is that unlike STT where the filtered spin-polarized current is smaller than the charge current, the SOT efficiency (spin-polarized current over charge current) can be larger than one in the SOT case.¹⁹ From a physics perspective, multiple mechanisms have been found to contribute to this large damping-like SOT, including spin Hall effect (SHE)¹⁷, Rashba-Edelstein effect¹⁸, and spin-momentum locking from topological protected electronic states¹⁹. Most experimental work has discovered a damping-like SOT in the in-plane transverse direction with respect to the current flow direction. However, there also exists a field-like SOT in many of the experimental works, which acts on the free layer like a static magnetic field with a fixed orientation.

There are three types of SOT-MRAM configurations, i.e., in-plane MTJ with easy axis oriented along the current direction (type X), in-plane MTJ with easy axis oriented orthogonal to the current direction (type Y), and perpendicular MTJ (type Z). Note that only type Y can achieve field-free switching, while both type X and Z require the breaking of symmetry for deterministic switching. Experimentally, 0.5-ns switching with a current density of 40 MA/cm^2 has been demonstrated in a 100 x 400 nm^2 type X MTJ, which will lead to 51- μA , 0.3-V, and 8-fJ write performance when scaled down to a channel width of 50 nm.¹⁴ Another work shows 0.5-ns switching with a current density of 18 MA/cm^2 in 30 x 190 nm^2 type Y MTJ with a write error rate (WER) of 10^{-6} .¹⁵ For perpendicular MTJ (type Z), research has shown 0.5-ns and 220-fJ write operation with a current density of 180 MA/cm^2 of a 60 nm perpendicular MTJ with endurance up to 10^{11} and WER down to 10^{-5} . Field-free switching is realized in this work by using an elongated biasing ferromagnet deposited on top of the SOT-MTJ.²⁰

Lowering the write energy while maintaining sub-ns switching speed is the main challenge to push SOT-MRAM into cache-level applications. A multitude of novel SOT materials beyond traditional heavy metal are being intensively investigated for higher

SOT efficiency. Several new research directions include heavy metal alloys²¹, topological insulator and semimetal^{22,23}, antiferromagnets^{24,25}, and complex oxides^{26,27}. The need of high SOT efficiency is especially critical for type Z as it inherently shows a larger switching current than type X and Y²⁸. Second, SOT-MRAM suffers from a large cell size due to the three-terminal configuration needed to perform separate write and read functions.²⁹ A two-terminal perpendicular SOT-MRAM has been demonstrated by increasing the density of the current flowing in-plane in the SOT underlayer while suppressing that of the current flowing perpendicular in the MTJ³⁰. Meanwhile, the scheme of multiple SOT-MTJs sharing one single SOT write line can partially alleviate the density disadvantage of SOT-MRAM.³¹ Third, a tradeoff exists between writing speed and field-free switching. For type Y, the FL aligning to the transverse direction does not experience any SOT. Therefore, an initial perturbation of the FL away from the easy axis is required for fast switching. A large field-like SOT or Oersted field due to SOT current can provide this perturbation.³² For type X and Z, a maximal SOT exists as the FL is aligned perpendicular to the spin current direction, thus enabling high-speed switching. There have been several approaches to break the symmetry for realizing field-free type X and Z switching, such as lateral structural and shape-induced asymmetry^{28,33,34}, use of interlayer exchange coupling³⁵, exchange bias²⁴, and dipolar external magnetic field³⁶. Meanwhile, new studies show field-free switching of type Z MTJ using a damping-like SOT with perpendicular polarization arising from crystalline materials with broken in-plane symmetry³⁷ and interfaces with SOC^{38,39}. Last, the scalability of all three SOT types remains an open question. Type X and Y scaling are challenging due to variations in MTJ shape, while type X and Z scaling face obstacles in implementing field-free switching at scaled nodes.

2.2.1.3. VOLTAGE-CONTROLLED MAGNETIC ANISOTROPY

Contrary to current-driven writing mechanisms such as STT and SOT, voltage-assisted writing using electron-mediated voltage-controlled magnetic anisotropy (VCMA) effect⁴⁰ enables lower write energy ($\sim$ fJ) and smaller cell size due to reduced current, and therefore joule heating and select transistor size (see Table BC2.1). A VCMA-MTJ is almost the same as an STT-MTJ, except that the tunnel barrier MgO thickness is increased to suppress the tunneling current and enhance the capacitive characteristics of the tunnel barrier.⁴¹ When a voltage is applied across the VCMA-MTJ, charge accumulation or depletion takes place at the FL/barrier interface, leading to a change of electron occupancies among different Fe 3d orbitals. Because the interfacial perpendicular magnetic anisotropy (PMA) originates from the Fe 3d and O 2p orbitals hybridization, this change of electron occupancy results in the modulation of PMA and thus the energy barrier between the two FL stable states.^{40,42,43} The VCMA effect is, therefore, a useful handle to reduce the energy barrier during the write operation, while the energy barrier is restored for retention purposes after writing by simply removing the VCMA bias.

There are two main types of VCMA-assisted magnetization switching schemes. First, removal of the entire energy barrier by the VCMA effect facilitates a precessional motion of the FL along an in-plane bias field direction (built-in or applied). By precise timing of the VCMA pulse width, the FL can switch from one state to the other in half the precession period.⁴⁴ Research has shown switching energy of 6 fJ/bit, switching speed of 0.5 ns, write voltage of 1.96 V, current density of 0.3 MA/cm² with a WER of 10^{-5} using perpendicular MTJs with a VCMA coefficient of 30 fJ/V-m.⁴⁵ Another recent work further shows 0.15 ns precessional switching of 120 nm perpendicular VCMA-MTJ at a write voltage of 3.06 V, a current density of 0.3 MA/cm² with a WER of $<10^{-6}$.⁴⁶ Second, the VCMA effect can be utilized to reduce the write energy in in-plane and perpendicular SOT-MTJs further.^{31,47} Research has demonstrated VCMA-assisted (VCMA bias of 1 V) SOT writing of 30 x 80 nm² to 50 x 120 nm² in-plane MTJ using 2-ns pulse with a current density of 12 MA/cm² with a high endurance of 10^{13} write cycles.⁴⁸ Another work shows 5-ns 62 μ A SOT current writing (VCMA bias of 1.2 V) of 30 x 80 nm² in-plane MTJ with WER $<10^{-8}$ and endurance over 10^{12} cycles, the VCMA coefficient in this device is about 100 fJ/V-m.⁴⁹

The major roadblock of VCMA in either precessional switching or assisting SOT switching is the rather small VCMA coefficient of around 100 fJ/V-m, as defined by the interfacial PMA change under given electric field applied at the MgO barrier.⁵⁰ Though $\sim$ fJ-level write performance has already been demonstrated, further scaling of MTJs requires higher VCMA coefficient (>300 fJ/V-m) for advanced nodes cache or storage applications.⁵¹ New materials research using Cr and Ir-based crystalline MTJs have shown a high VCMA coefficient of up to 1000 fJ/V-m.^{52,53} Meanwhile, detailed chemical and structural characterizations of VCMA-MTJs recently reveal that metal-oxides at the FL/MgO interface lead to large VCMA effect.⁵⁴ Another challenge facing VCMA is the longer read time, because the thicker MTJ tunnel barrier leads to a much larger MTJ resistance. One way to resolve this is using a large read voltage (V_{DD}) which has reverse polarity compared with the write voltage to increase read speed and reduce read disturbance.⁵⁵ In terms of the precessional switching scheme, another significant challenge is the non-deterministic nature of the writing process, which results in large WER and narrow write pulse window. The use of pulse shape engineering and reverse biasing can partially help^{56,57}, whereas combining VCMA with deterministic writing mechanisms such as type Y SOT⁴⁷ and STT⁵⁸ may solve this challenge.

2.2.2. OXIDE-BASED RESISTIVE MEMORY (OxRAM)

The redox-based nanoionic memory operation is based on a *change in resistance* of a MIM structure caused by ion (cation or anion) migration combined with redox processes involving the electrode material or the insulator material, or both.^{59,60,61} Three classes of electrically induced phenomena have been identified that involve chemical effects, i.e., effects that relate to redox

processes in the MIM cell. In these three ReRAM classes, there is a competition between thermal and electrochemical driving forces involved in the switching mechanism. Two major types of ReRAM exist: i) those based on metal oxide (OxRAM), which involve oxygen ions/vacancies motion, and ii) conducting bridge-based RAM (CBRAM), which involves metal cation motion. This section covers the three categories of OxRAM, while conducting bridge-based RAM (CBRAM) is covered in the following section. Beyond CMOS has sub-categorized oxide ReRAM (OxRAM) based on the electrical switching type (bipolar versus unipolar) and whether a conductive filament is formed in the device. Most of the literature fits into the three categories: bipolar filamentary, unipolar filamentary, and nonfilamentary.⁶²

In most cases, the conduction is of a filamentary nature, and hence a one-time formation process is required before the bipolar switching can be started. If this process can be controlled, memories based on this switching process can be scaled to very small feature sizes. The switching speed is limited by the ion transport. If the active distance over which the anions or cations move is small (in the <10 nm regime) the switching time can be below few nanoseconds, down to sub-nanoseconds range.^{63,64} Many of the finer details of the ReRAM switching mechanisms are still under investigation. Developing an understanding of the physical mechanisms governing switching of the redox memory is a key challenge for this technology. Nevertheless, experimental demonstrations of scalability,⁶⁵ retention,⁶⁶ and endurance⁶⁷ are encouraging.

2.2.2.1. BIPOLAR-FILAMENTARY OXRAM

Bipolar filamentary OxRAM is the most common form of oxide-based ReRAM. At any given defect density, the number of current paths through the dielectric, in the virgin or fresh state, is proportional to the device area, and consequently the total current is area dependent. In addition, the current magnitude tends to fluctuate from device to device due to randomness of the initial distribution of vacancies/ions. However, cell area dependency is eliminated when the current is dominated by a single conductive path, called conductive filament (CF). The CF provides an ultimate scaling advantage since it is only limited to the active filament size, which potentially may be as small as a few nm.

A one-time forming process is required for most types of OxRAM devices to create a conduction filament across the dielectric layer linking the electrodes. A stable preferential conduction path is known to form through oxide films subjected to electrical stress: under the applied voltage, a current abruptly increases at some point in time indicating the occurrence of a dielectric breakdown (BD) resulting in the formation of a CF. During the forming process, electrons injected from the cathode electrode may lead to their trapping at defect sites in the dielectric material inducing chemical bonds breakage and the generation of anion vacancies (Oxygen or Nitrogen).^{68,69}

Post-forming switching events between high and low conductive states, which are operated at significantly smaller voltages, are believed to modify the filament conductivity by rupturing/recovering a section of the filament (primarily in the vicinity of the metal electrode) or changing the filament cross-section. The specific mechanisms in filament-type switching depend on the materials (dielectric and metal electrodes) employed in the fabrication of the memory cell and may include more than one type of a conduction mode. The operation of these devices involves redox reactions of the dielectrics sandwiched between two electrodes.^{70,71,72} The dielectrics are mostly comprised of one or a few layers of insulating materials⁷³ (e.g., oxide AlO_x , HfO_x , TaO_x , TiO_x , WO_x , ZrO_x , oxynitrides AlO_xN_y , or nitrides including AlN_x and CuN_x). TaO_x and HfO_x are the leading candidates among the aforementioned dielectrics, due to their superior performance (e.g., endurance) and CMOS compatibility.

Since the demonstration of a single crosspoint HfO_x device with a 10 nm dimension in 2011⁷⁴, scaling to a smaller size has been achieved by employing a sidewall electrode in a $1 \times 3 \text{ nm}^2$ cross-sectional HfO_x -based OxRAM device with reasonable performance in terms of both endurance and retention.⁷⁵ Up to 10^{12} cycles has been demonstrated with $\text{Zr}:\text{SiO}_x$ sandwiched by graphene oxide layers.⁷⁶ Some of the filament-based metal-oxide RRAMs implemented with metal electrodes and a variety of fab-friendly transition-metal-oxides (i.e., HfO_2 , ZrO_2 , TiO_2 , etc.) and nitride devices demonstrated sub-nanosecond,^{77,78} switching with high (up to 10^{12} cycles) endurance⁷⁹ and retention of more than 10 years. Extrapolated retention at 85°C by stressing TaO_x in the temperature range from 300°C to 360°C is estimated to be years with an activation energy of 1.6 eV.⁸⁰ Reliable switching operations have been demonstrated at 340°C with devices based on 2D layered heterostructures (e.g., graphene/ MoS_2 - xO_x /graphene).⁸¹

Unconventional electrodes such as graphene have been paired with HfO_x dielectrics to yield a low power consumption, a write/erase energy of 230 fJ per bit for a single programming transition.⁸² $\text{Pt/BMO}((\text{Bi}, \text{Mn})\text{O}_x)/\text{Pt}$ structured OxRAM device was used to demonstrate an even lower write/erase energy per transition, of the order of 3.8 pJ/bit for read and 20 pJ/bit for write operation.⁸³

Large scale integration of OxRAM switching based on 1T1R schemes has been carried out by Toshiba, Panasonic and IMEC. In 2013, Toshiba announced the 32 Gb RRAM chip integrated with 24 nm CMOS.⁸⁴ In 2014, Panasonic and IMEC demonstrated the encapsulated cell structure with an $\text{Ir/Ta}_2\text{O}_5/\text{TaO}_x/\text{TaN}$ stack on a 2-Mbit chip at the 40 nm node. In addition, passive integration of 1S1R scheme has been reported by Crossbar on a 4-Mbit chip, but the material stack of the OxRAM switch has

not been revealed.⁸⁵ Ultra-fast (down to 100 ps), compliance-free, low power (< pJ) switching was demonstrated with 1R devices using TiN/HfO₂/TiN stack.⁸⁶

A number of technical challenges hampering the commercialization of OxRAM still remain despite the significant advancements made in the field. One of the main challenges is the fact that the switching currents for devices based on the currently most mature materials (e.g., HfO_x and TaO_x) are still too high (above tens of μ A) for large arrays. Apart from that, the filament formation and rupture processes are stochastic in nature, which leads to variation in switching parameters like the voltage and resistance distribution of the switching. This is especially detrimental to certain applications such as multilevel cell memory.

2.2.2.2. BIPOLAR NON-FILAMENTARY OXRAM

The *Bipolar Non-Filamentary* OxRAM is a non-volatile bipolar resistive switching device composed of one or more oxide layers. One layer is a conductive metal oxide (CMO), which is usually a perovskite such as PrCaMnO₃ or Nb:SrTiO₃.⁸⁷ In contrast to *Unipolar* and *Bipolar Filamentary* OxRAM devices – typically based on binary oxides such as TiO_x, NiO_x, HfO_x, TaO_x or combinations thereof—the resistance change effect of the *Bipolar Non-Filamentary* OxRAM is *uniform*. Depending on the materials choice and structure the current is conducted across the entire electrode area, or at least across the majority of this area. A forming step to create a conductive filament is *not* needed. Non-volatile memory functionality is achieved by the field-driven redistribution of oxygen vacancies close to the contact resulting in a change of the electronic transport properties of the interface (e.g., by modifying the Schottky barrier height). Oxygen can be exchanged between layers due to the exponential increase in ion mobility at high fields. Low current densities, uniform conduction, and bipolar switching imply that substantial self-heating is not involved. Typical R_{OFF} to R_{ON} ratios are on the order of 10.

One class of the Bipolar Non-Filamentary OxRAM includes a deposited ion conductive tunnel layer (Tunnel ReRAM), e.g., ZrO₂. Here, a redistribution of oxygen vacancies causes a change of the electronic transport properties of the tunnel barrier. Low current densities and area scaling of device currents enable ultra-high-density memory applications. Set, reset, and read currents scale with device area. In addition, set, reset, and write currents are controlled by the tunnel oxide and hence, can be adjusted by changing the tunnel barrier thickness. Both set and reset IV characteristics are highly nonlinear enabling true 1R cross-point architectures *without* the need for an additional selector device for asymmetric arrays up to 512×4096 bit. No external circuitry is needed for current control during set operation. A continuous transition between on and off states allows straightforward multi-level programming without the need for precise current control.

The typical thickness of the CMO is greater than 5 nm and the tunnel barrier is typically 2–3 nm. If a tunnel barrier is present, the adjacent electrode needs to be an inert metal such as Pt to prevent oxidation during operation. For the case of PCMO cell, low deposition temperatures of less than 425°C of all layers enables back end integration schemes.

Currently, the technology is in the research and development stage. Depending on material system and structure cycling endurance over 10,000 cycles and up to a billion cycles as well as data retention from days to months at 70°C has been achieved on single devices.^{88,89,90} Within the Bipolar Non-Filamentary OxRAM device family the Tunnel OxRAM is probably the most developed technology. Single device functionality is demonstrated down to 30 nm. Set, reset, and read currents scale with area and tunnel oxide thickness facilitating sub μ A switching currents with read currents in the order of a few nA to a few 100 nA. BEOL integration schemes and CMOS/OxRAM functionality are verified for 200 nm devices on 200 mm CMOS wafers. True cross-point array (1R) functionality utilizing the self-selecting non-linear device IV characteristics and transistor-less array operation is demonstrated on fully decoded 4kb true cross-point arrays (1R) build on top of CMOS base wafers. SLC and MLC operations are demonstrated within 4kb arrays.

Major challenges to be resolved towards the commercialization of Bipolar Non-filamentary OxRAM are, in order of priority, a) improvement of data retention, b) the integration of conductive metal oxide layer (perovskites) via ALD or the replacement of CMO by more process-friendly materials, and c) the replacement of Pt electrodes by a non-reactive, more process-friendly electrode material.

The most important issue is the improvement of retention and the “voltage-time dilemma.” This dilemma hypothesizes physical reasons as to why it is difficult in a particular device and material system to simultaneously obtain a long retention, with short low read voltages, and fast switching at moderate write voltages.⁹¹ Even though the exact mechanism is still under investigation there is a common agreement that oxygen vacancies are moved by the external electric field resulting in different resistance states of the memory cell. Vacancy drift at room temperature is possible due to a field dependent mobility, which increases exponentially with field at fields of 1 MV/cm and larger. However, current models based on a field-dependent mobility underestimate the experimentally observed ratio between set/reset times and data retention indicating that the mechanism is only partly understood. More theoretical work is needed to understand the kinetics of programming and retention mechanisms. Once understood, materials need to be chosen to maximize the ratio between set/reset and retention times. The goal is to set/reset devices at low temperatures and meet retention requirement of 10 years at 70°C, 85°C, and 125°C, depending on the application. A multi-layer

ReRAM structure ($\text{HfO}_2/\text{A}_2\text{O}_3$) was shown to improve retention by suppressing tail bit failure due to decreased oxygen ion diffusivity.⁹²

Memory cells using conductive perovskite material as an electrode have proven to show excellent device-to-device and wafer-to-wafer reproducibility with yields close to 100%. One of the reasons might be that perovskites display high oxygen vacancy mobilities and tolerate large variations in the oxygen content while maintaining its crystal structure. From an integration perspective, ALD is the method of choice for advanced technology nodes and future 3D integration schemes. Key issues are the control of the metal ratio (perovskites are ternary or quaternary oxides), the control of the oxygen stoichiometry in the cell, oxygen loss in the presence of reducing atmospheres like H_2 , as well as high temperatures required for crystallization. Eventually a migration to binary oxides with comparable properties might be required to resolve the integration challenges.

Platinum or other noble electrodes display superior device performance over fab-friendly electrodes like TiN. On the one hand it was observed that the oxidation resistance of TiN is not sufficient to prevent oxidation and the formation of TiO_2 during operation. On the other hand, inert electrodes such as Pt or Pt-like metals are difficult to integrate. New oxidation-resistant electrodes and Pt alternatives are required to reduce integration challenges and enable 3D integration schemes.

2.2.2.3. UNIPOLAR FILAMENTARY OXRAM

Note that unipolar filamentary OxRAM has been removed from the memory tracking tables, due to lack of research over the period covered by this Beyond CMOS chapter. However, this text section has been maintained to provide background on earlier unipolar OxRAM work, due to the close relationship and key differences with bipolar OxRAM.

Unipolar OxRAM is another resistive switching device, also referred to in the literature as thermochemical memory (TCM)⁵⁹ due to its primary switching mechanism. The device structure consists of a top electrode metal / insulator / bottom electrode metal (MIM) structure. Typical insulator materials are metal-oxides such as NiO_x , HfO_x , etc., and common metal electrodes include TiN, Pt, Ni, and W. In general, the device can be asymmetric (i.e., top electrode material differs from bottom electrode material), but unlike other types of ReRAM, asymmetry is not required.

The first reported resistive switching in these MIM structures after 2000 was unipolar in nature (see reference⁹³ for the first integrated device work that put metal oxide ReRAM in the spotlight). Unipolar is defined as switching where the same polarity of voltage needs to be applied for changing the resistance from high to low (SET) or from low to high (RESET). Note that in the general case, polarity is still important (e.g., repeatable SET/RESET switching only occurs for one polarity of voltage with respect to one of the electrodes⁹⁴). Only in symmetric structures (e.g., Pt/ HfO_2 /Pt), nonpolar behavior can be obtained, where SET and RESET are occurring irrespective of voltage polarity.⁹⁵

The switching process is generally understood as being filamentary, where conduction is caused by a filamentary arrangement of defects (e.g., oxygen vacancies) throughout the thickness of the insulator film. As with other filamentary OxRAM devices, an initial high voltage “electroforming” step is required to form the conduction filament, while subsequent RESET/SET switching is thought to occur through local breaking/restoration of this conduction path.

The unipolar character of the switching indicates that drift (of charged defects) in an electric field plays a less important role (than it does in bipolar switching resistive memory), but that thermal effects probably dominate.^{96,97} On the other hand, polarity effects indicate anodic oxidation (e.g., at Ni or Pt electrodes) is responsible for RESET.⁹⁴ These findings suggest a thermochemical “fuse” model for describing this unipolar switching. It has been shown for different MIM structures that both unipolar and bipolar switching mechanisms can be induced, depending on the operation conditions.^{98,99,100,101} An interesting work reporting on the Scaling Effect on Unipolar and Bipolar Resistive Switching of Metal Oxides was published.¹⁰²

A unipolar switching device is seen as advantageous for making scaled memory arrays, as it only requires a selector device as simple as a diode that can be stacked vertically with the memory device in a dense crossbar array.⁹³ In addition, the use of a single program voltage polarity greatly simplifies the circuitry.

On the other hand, as has been exemplified in mixed mode (unipolar/bipolar) operation of memory cells, there are important trade-offs between the unipolar and bipolar switching modes. On the positive side, unipolar switching mode typically shows a higher ON/OFF resistance ratio. On the negative side, unipolar switching is typically obtained at higher switching power (higher currents) than the bipolar mode, and also endurance is much more limited. As a result, major research and development work on resistive memories has shifted towards bipolar switching mechanisms. Yet, some interesting recent development work has been reported.^{103,104,105,106,107,108} One paper¹⁰³ shows an endurance of over 10^6 cycles with a resistive window of over 5 orders of magnitude (and a reset current ~ 1 mA). Others^{104,105,106} demonstrate how unipolar RRAM elements can be integrated in a very simple way in an existing CMOS process (known as Contact ReRAM technology). This may provide a very inexpensive embedded ReRAM technology. Recently, integration unipolar ReRAM with a 28 nm CMOS process was reported.¹⁰⁶ The key attributes were a small cell size ($0.03 \mu\text{m}^2$), switching voltage of less than 3V, RESET current of less than 60 μA , endurance >

10^6 cycles, and short SET and RESET times of 500 ns and 100 μ s, respectively. One paper¹⁰⁷ shows 4 Mb array data using this same Contact-RRAM technology, fabricated using a 65 nm CMOS process. To accommodate the low logic VDD process, an on-chip charge pump was applied. Set and reset voltages are less than 2 V. Another paper¹⁰⁸ reports on a novel approach using thermal assisted switching to lower the switching current.

As stated above, large OFF to ON resistance ratio is an attribute of unipolar switching. The low resistance window and large intrinsic variability of bipolar switching OxRAM may require complex and time-consuming switching operation schemes (e.g., the so-called verify scheme). Further study of the stability and control of the large resistance window (at low current levels), are required to determine if unipolar OxRAM variability can be improved, potentially even allowing for multi-level cell operation.

Major challenges to be resolved are the high switching current that seems inherent to the unipolar operation mode. Reset currents less than 100 μ A are achieved but need further reduction to less than 10 μ A.^{104,105,106} Recently, a possible solution incorporating thermally assisted switching has been presented.¹⁰⁸

2.2.3. CONDUCTING BRIDGE MEMORY

Conductive Bridge RAM (CBRAM), also referred to as Programmable Metallization Cell (PMC), and electrochemical metallization cells, solid-electrolyte switch and atom(ic) switch, is a device that utilizes electrochemical control of nano-scale quantities of metal in thin dielectric films or solid electrolytes to perform the resistive switching operation.^{109, 110} The basic CBRAM cell is a metal-ion conductor-metal (MIM) system consisting of an electrode made of an electrochemically active material such as Ag, Cu or Ni, an electrochemically inert electrode such as W, Ta, Ru, or Pt, and a thin film of solid electrolyte sandwiched between both electrodes.¹¹¹ Large, non-volatile resistance changes are caused by the oxidation and reduction of the metal ions by the application of low bias voltages. Key attributes are low voltage, low current, rapid write and erase, good retention and endurance, and the ability for the storage cells to be physically scaled to a few tens of nm. The material class for the dielectric film or the solid electrolyte is comprised of oxides, higher chalcogenides (including glasses), semiconductors, as well as organic compounds including polymers.¹¹²

CBRAM is a strong emerging memory candidate primarily due to scalability (~ 10 nm),¹¹³ ultra-low energy operations due to fast read, write and erase times, and low voltage requirements.¹¹⁴ Maturity of the CBRAM technology development can be assessed by the fact that many companies are either shipping products based on CBRAM or are in advanced stages of commercialization. Recent publications show CBRAM technology application in various markets including SSDs,¹¹⁵ embedded NVM,¹¹⁶ and serial interface non-volatile memory replacement.¹¹⁷ In 2014, a 16 GB CBRAM array based on a CuTe CBRAM cell was demonstrated.^{118, 119} Such efforts are critical to identify core technology challenges¹²⁰ and fundamental materials and mechanisms.¹²¹ An improved thermal stability of the conducting bridge in CBRAM gives a new opportunity for automotive applications.¹²²

Novel applications such as reconfigurable switch¹²³ and synaptic elements in Neuromorphic systems¹²⁴ based on CBRAM are also gaining prominence and are expected to expand the application base for this technology. An atom-switch-based field-programmable gate array (FPGA) is also released using Cu conduction bridge in a new polymer solid electrolyte.¹²⁵ Low-power and rad-hard operation of the FPGA using the atom switch has been demonstrated in orbit.¹²⁶ For further advancing the FPGA performance, a via-switch FPGA is proposed¹²⁷, in which the via-switch is a stacked device consisting of two atom switches with one selector and integrated in BEOL. By replacing the select transistor to the selector, all the switch cell is placed in BEOL, resulting in closing the performance gap between FPGAs and ASICs.¹²⁸

As with other filamentary ReRAM technologies, CBRAM is challenged by bit level variability,¹²⁰ the random nature of reliability failure such as retention or endurance, and random telegraph noise potentially contributing to read disturbs.¹²⁹ Such issues require large populations of bits to be studied, which suggests collaboration between universities and industry may be beneficial. Focus on fundamental understanding and simultaneously addressing some mitigation path such as error correction schemes, redundancy and algorithm development would enable closing the technology gap.

Engineering hurdles include the availability and integration of new materials used in CBRAM at advanced process nodes especially when there could be issues with compatibility of thermal budgets and process tooling. The availability of integrated array level information suggests that some of these challenges are being resolved in the recent years.^{117,123} Active participation from semiconductor equipment vendors and material suppliers could assist in overcoming manufacturing hurdles rapidly.

2.2.4. MACROMOLECULAR (POLYMER) MEMORY

Macromolecular memory is a category of memory that focuses on structures incorporating a layer of polymer - the polymer perhaps containing nano-particles, small molecules and nanoparticles - that is sandwiched between two metal electrodes. This structure allows two different stable electrical states controlled through an external electrical voltage. These two stable electrical states, which are often called ON and OFF states (or 0 and 1), exhibit resistive, ferroelectric or capacitive natures according to

the physical properties of the sandwich. The first fully-organic memory devices, based on nano-composite (a blend of poly-vinyl-phenol (PVP) and Bucky-ball (C60)) was presented in Materials Research Society in 2005.¹³⁰ Around the same time, memory devices using gold nano-particles and 8-hydroxyquinoline, dispersed in a polystyrene matrix, were also demonstrated.¹³¹ Since then, the interest to use an admixture of nano-particles, small molecules and polymers in the manufacture of electronic memory devices, is on the rise. Non-volatile memory effects with a non-destructive read have been reported for a surprisingly large variety of polymeric/organic materials and blends of polymers with nanoparticles and molecules. Unlike the other four categories, this category is based on the material used in the switching layer(s) of the cell, but the mechanism is not specified. Both bipolar and unipolar (all pulses of the same polarity) switching have been demonstrated. Macromolecular ReRAM may have a mechanism placing it in one of the four main ReRAM categories listed above. However, the other mechanisms behind the electrical bistability, such as capacitive and ferroelectric, have also been reported.

Depending on the structure of the polymer, a variety of mechanisms can be operative. For polymers supporting transport of inorganic ions, formation of metallic filaments is reported. In semiconducting polymers supporting ion transport, dynamic doping due to migration of inorganic ions occurs. Ferroelectric polymers in blends with semiconducting material give rise to a memory effect-based modification of charge injection barriers by the ferroelectric polarization. However, for many polymeric materials, the origin of the resistive switching is not well understood. To date no specific design criteria for the polymer are known, although clear correlations between memory effect and electronic properties of the polymer have been demonstrated.

Stability of the memory states at high temperatures (85 °C, 2×10^4 s) has been demonstrated.^{132,133} Programming at very low power (70 nW) has been realized.¹³³ Assuming a 15 ns switching time for the same system, one might achieve a write energy of 6×10^{-15} J/bit. Furthermore, low programming voltages have been realized: +1.4 and -1.3 V for the two states with good retention time ($>10^4$ s).¹³⁴ Downscaling of polymer resistive memory cell to the 100 nm length scale has been reported.¹³⁵ At this length scale, integration of memory cells into an 8×8 array could be shown. Polymer memory cells on a flexible substrate have been shown.¹³⁶ For amorphous carbon, downscaling to nanometer sized cells has been published (1×10^3 nm²).¹³⁷ Using carbon nanotubes as macromolecular electrodes and aluminum oxide as interlayer, isolated, non-volatile, rewriteable memory cells with an active area of essentially 36 nm² have been achieved, requiring a switching power less than 100 nW, with estimated switching energies below 10 fJ per bit.¹³⁸ With regards to the mechanism of operation, extensive work on the class of polyimide polymers has shown clear correlations between electronic structure of the polymer and memory effects, although a comprehensive picture for the operation has not yet emerged. A number of studies have indicated an active role of the interface between macromolecular material and (native) oxide layers in the operation of the memory involving charge trapping.^{139,140} The recent and past studies show resistive,¹⁴¹ capacitive (charge storage, based on electric dipole formation)^{142,143} and ferroelectric behavior¹⁴⁴ of such devices. Thus, there is a need to open up a further discussion on the right pathway to realize such memory.

In macromolecular memory, a large variety of operation mechanisms can be operative. A key research question concerns distinguishing different mechanisms and evaluating the potential and possibilities of each mechanism. A second subsequent step would be to identify model systems for each mechanism. Having such a model system then provides a possibility to benchmark the operation of the macromolecular materials. These research steps would be crucial for establishing and securing the collaboration of the chemical industry. For design, synthesis and development of the next generation macromolecular materials for memory applications, clear guidelines on the required structural and electronic properties of the macromolecular material are needed. For instance, memory effect originating for metallization and formation of metallic filaments requires macromolecular materials that support transport of ions and have appropriate internal free volume for ion conduction. Here, the field could benefit from interaction with the field of polymer batteries. Ferroelectric polymers have been shown to give rise to resistive memory¹⁴⁵ and could benefit enormously from development of new macromolecular polymeric materials with combined ferroelectric switching and semiconducting structural units. Finally, a number of macromolecular memories involve oxide layers. Here, mutually beneficial interaction with the (research) community on metal oxide ReRAM switching could spring, because at the macromolecular / oxide interface trap states can be engineered by tuning the electron levels of the macromolecular material.

In a nutshell, this area certainly needs an attention from theoretical physicists, materials scientists, chemists and device engineers. There are a number of issues that need to be addressed before we can embark on extending these devices to the real world. Such issues involve understanding of the electrical bistability mechanism in nano-composite (there are a number of contradicting theories), maintaining the difference between low and high conduction states for a longer period time by ensuring the stability of the high and low states, selecting environmentally friendly materials required for fabrication of nano-composite/polymer materials, and developing a cost-effective methodology for the fabrication of devices.

It is not possible to replace silicon-based memory devices with polymers in the foreseeable future. However, there are a number of other applications where “cheap” electronic memory devices can play a vital role. For example, nano-composite based memory devices can be directly printed on medicine bottles/packages and the information about the patient and schedule of taking medicine can be stored on the printed device.

2.2.5. FERROELECTRIC MEMORY

Coding digital memory states by the electrically alterable polarization direction of ferroelectrics has been successfully implemented and commercialized in capacitor-based Ferroelectric Random Access Memory (covered in Table BC2.1). However, in this technology the identification of the memory state requires a destructive read operation and largely depends on the total polarization charge on a ferroelectric capacitor, which in terms of lateral dimensions is expected to shrink with every new technology node. In contrast to that, alternative device concepts, such as the ferroelectric field effect transistor (FeFET) and the ferroelectric tunnel junction (FTJ), allow for a non-destructive detection of the memory state and promise improved scalability of the memory cell. The current status of and key challenges for these emerging ferroelectric memories will be assessed within this section.

2.2.5.1. FERROELECTRIC FET

The FeFET is best described as a conventional MISFET that contains a ferroelectric oxide in addition to or instead of the commonly utilized SiO_x , SiON or HfO_2 insulators. The former case requires the direct and preferably epitaxial contact of the ferroelectric to the semiconductor channel (metal-ferroelectric-semiconductor-FET, MFSFET), whereas the latter and commonly applied case maintains a buffer layer between the channel material and the ferroelectric (metal-ferroelectric-insulator-semiconductor-FET, MFISFET). When additionally introducing a floating gate in-between the buffer layer and the ferroelectric, a metal-ferroelectric-metal-insulator-semiconductor structure (MFMISFET) may be obtained that shares its equivalent circuit representation with the MFISFET approach. By applying a sufficiently high voltage pulse to the gate of the FeFET (i.e., voltage drop across the ferroelectric layer larger than its coercive voltage V_c), the polarization direction of the ferroelectric can be set to either assist in the inversion of the channel or to enhance its accumulation state. This results in a polarization dependent shift of the threshold voltage V_T , which allows for a non-destructive read operation and a 1T memory operation comparable to that of FLASH devices.

In order to assess the material and device requirements for a reliable and scalable FeFET technology the following two intrinsic relations in a ferroelectric gate stack need to be considered. First the aforementioned V_T -shift (memory window) in FeFET devices is primarily determined by the V_c of the implemented ferroelectric rather than its remanent polarization P_r .¹⁴⁶ This results in a scaling versus memory window trade-off as V_c is proportional to the coercive field E_c and thickness d_{FE} of the ferroelectric. The technical challenges of the perovskite-based FeFETs are not solely because of the poor CMOS compatibility but also their low E_c (SBT: 10-100 kV/cm, PZT: ~50 kV/cm, summarized in¹⁴⁷) that requires large d_{FE} to achieve a reasonable memory window. A solution to this scaling retardation is provided by the high coercive field (1-2 MV/cm) and thickness-scalable FE- HfO_2 .¹⁴⁸ This CMOS-compatible material innovation enabled the demonstration of a FeFET technology scaled to the 28 nm node utilizing a conventional HKMG technology.¹⁴⁹ The close resemblance of the HKMG transistor and the FE- HfO_2 -based memory transistor proves especially useful for the realization of an embedded memory solution with greatly reduced mask counts as compared to embedded FLASH.

The second noteworthy and important characteristic of the FeFET is related to the electric field distribution across the gate stack, particularly across the non-ferroelectric insulator or so-called the interlayer. The polarization charge of the order of $10 \mu\text{C}/\text{cm}^2$ induces large electric field ($> 30 \text{ MV}/\text{cm}$) in the low-k interlayer, which in turn induces an enormous amount of charge trapping (trap density $> 10^{14} \text{ cm}^{-2}$) at the interface.¹⁵⁰ The repeated trapping and detrapping of this large amount of charges when the polarization is switched its direction cause the degradation of interface characteristics and reduce the memory window. This degradation is the main challenge that limits the cycling endurance of FeFET. Moreover, during the data retention period with no external biasing, the non-zero polarization charge causes the non-zero electric field across the interlayer and the semiconductor substrate, which then leads to a buildup of a permanent electric field inside the ferroelectric layer through the capacitive voltage divider. The so-called depolarization field building up in the ferroelectric is opposed to the polarization direction of the ferroelectric and to the electric field induced in the insulator¹⁵¹. This endurance- and retention-critical distribution of the electric field within the gate stack may be optimized by choosing the insulator capacitance as high as possible and the ferroelectric capacitance as low as possible. In the perovskite-based FeFET this is achieved by utilizing high-k buffer layers and is additionally fostered by the unavoidably large physical thickness of the perovskite ferroelectrics.^{4, 152} In the case of the aggressively scaled FE- HfO_2 -based FeFET, the small thickness of the ferroelectric is compensated by the comparably low permittivity of HfO_2 , the possibility to use ultra-thin interfacial layers, and by the depolarization resilience of the high E_c .^{147, 153} This leads to the situation that despite the markedly different stack dimensions and materials used, the electrically obtained characteristics are quite similar. Fast switching speed ($\leq 100 \text{ ns}$), switching voltages in the range of 4-6 V, and 10-year data retention and endurance in the range of 10^{12} switching cycles have been demonstrated for FE- HfO_2 -^{148, 149, 154, 155} as well as for perovskite-based FeFETs.^{146, 156, 157} In the case of cycling endurance, however, the interface degradation due to enormous charge trapping limits endurance of 10^5 cycles

observed in FE-HfO₂-based FeFETs with ultra-thin interfacial layer enabling excellent data retention.¹⁵⁸ Nevertheless, in an alternative approach utilizing a thicker insulator and sub-loop operation it was demonstrated that at the cost of retention a cycling endurance $>10^{12}$ may still be obtained.¹⁵⁴ In the current stage of development this endurance versus retention trade-off may be tailored, spanning the application range from embedded NOR-FLASH replacement with high retention requirements to low refresh rate 1T DRAM requiring high cycling endurance.

Entirely overcoming this endurance versus retention trade-off will require an improved stack design that may include a tailored polarization hysteresis (low P_r and high P_r/P_s ratio)¹⁴⁶, a reduced trap density at the interfaces,¹⁵⁸ an optimized capacitive voltage divider by area scaling in the MF-MISFET approach,¹⁵⁹ the realization of a MFSFET device by implementing recent breakthroughs in the epitaxial growth of FE-HfO₂,¹⁶⁰ or the introduction of oxide-semiconductor channel materials that are robust against trap-induced interface degradation.¹⁶¹ Despite promising results obtained for perovskite-based FeFET devices implemented into 64Kb NAND-Arrays at a feature size of 5 μm ¹⁵⁷, little is known about the variability and array characteristics of FeFET devices scaled to technology nodes approaching the grain or domain size of the implemented ferroelectrics. Initial investigations on phase and grain distribution in doped HfO₂ based ferroelectric thin films and the effects of such granularity on device level characteristics of scaled FeFETs (such as on the statistical nature of switching) have recently been reported in Refs.^{162,163,164}. Recently, 64 kb and 32 Mb FeFET arrays were demonstrated in the 28 nm¹⁶⁵ and the 22 nm FD-SOI CMOS platform,¹⁶⁶ respectively—in each case, a clear low and high V_T separation at the array level was demonstrated. Nevertheless, in order to fully judge the variability of ferroelectric phase stability at the nanoscale and to guide material optimization and fundamental understanding of the phenomenon, larger array statistics in the kB to Mb range and high-resolution PFM data will be required. Besides, recent demonstration of non-volatile memory operation based on antiferroelectricity—a phenomenon closely related to ferroelectricity—in work-function engineered ZrO₂ thin film capacitors may allude to new way of addressing and potentially solving some of these challenges in FeFETs.¹⁶⁷

2.2.5.2. FERROELECTRIC TUNNEL JUNCTION

The ferroelectric tunnel junction, a ferroelectric ultra-thin film commonly sandwiched by asymmetric electrodes and/or interfaces, exhibits ferroelectric polarization induced resistive switching by a non-volatile modulation of barrier height. With the tunneling current depending exponentially on the barrier height, the ferroelectric dipole orientation either codes for a high or a low resistance state in the FTJ, which can be read out non-destructively. The resulting tunneling electroresistance (TER) effect of FTJs, the ratio between HRS and LRS, is usually in the range of 10 to 100 (¹⁶⁸ and references therein). However, giant TER of $>10^4$ has most recently been reported in a super-tetragonal BiFeO₃ based FTJ by Yamada et al.¹⁶⁹ A similarly high TER was demonstrated by Wen and co-workers¹⁷⁰ for a BaTiO₃ tunnel barrier by replacing one metal electrode of the FTJ with a semiconducting electrode. With this new junction design, the modulation of tunneling current does not only rely on barrier height, but due to a variable space charge region in the semiconductor, also on a barrier width modification. With these most recent findings, two strategies to achieve giant TER have been identified: either use a ferroelectric barrier with a large polarization such as BiFeO₃ or use a semiconductor as electrode material to modulate the barrier width by field-induced carrier depletion.

The MFM-based structure of FTJs may be able to enable a retention time (>10 years) and very high cycling endurance ($>10^{14}$) properties of conventional FRAM. Nonetheless, in order to have a significant tunneling current, ferroelectric films in FTJs usually have a thickness ranging from several unit cells to ~ 5 nm, which is much thinner than in commercialized 1T-1C FRAM (>50 nm). Due to larger interface contributions and increased leakage currents at reduced thickness, experimental data of these material systems might strongly deviate from their thick film behavior and need to be assessed separately.¹⁷¹ However, even though only limited data are available up to this point, promising single cell characteristics have already been demonstrated, such as the most recent demonstration of 4×10^6 endurance cycles and extrapolated data retention of 10 years at room temperature for a BiFeO₃-based FTJ.¹⁷² In the context of retention, it should be noted that despite improved TER, the newly proposed MFS-FTJ structure will give rise to a depolarization field, which will most likely degrade memory retention in a similar manner as described for the FeFET in Section 2.2.5.1. The highly energy efficient electric field switching, common to all ferroelectric memories, enables fast (10 ns¹⁷³) and low voltage (1.4 V¹⁷²) switching in FTJ devices and results in a minimal power consumption during write operation (1.4 fJ/bit, calculation based on the device characteristics given¹⁷⁴). Due to the availability of non-destructive read-out and the further reduced ferroelectric thickness in FTJ devices as compared to conventional FRAM, improved voltage scaling and total energy consumption may be expected from this technology.

Similar to most other two-terminal resistor-based memories with insufficient self-rectification, the elimination of sneak currents in large crossbar arrays is most efficiently suppressed utilizing 1T-1R or 1D-1R cell architecture. In terms of scaling, this two-element memory cell, as well as the scalability of the selector device itself, has to be considered.¹⁷⁵ Simply based on the lateral dimensions of the FTJ element (assuming unlimited scalability of the selector), scaling below 50 nm²,¹⁷⁴ based on PFM data,¹⁷⁶ most likely appears possible. However, with further scaling a simultaneous enhancement of the LRS current density is required to maintain readability in massively parallel memory architectures. A recent breakthrough of 1.4×10^5 A/cm² current density at

300 nm feature size has been achieved by Bruno et al.¹⁷⁷ utilizing low resistivity nickelate electrodes. Based on these results maintaining 10 μ A read current for feature sizes <100 nm appears possible. New FTJ concepts are also emerging; for example, engineered domain walls within the ferroelectric layer in an FTJ structure can lead to exotic quantum phenomenon such as resonant electron tunneling and quantum oscillations in the electrical conductance albeit at low temperatures.¹⁷⁸

FTJ based memories are currently at a very early development stage, and most of the research activity is focused on perovskite-based ferroelectrics. Further investigations reaching beyond single device characterization will be needed to fully judge the scalability of FTJ as well as its MLC capability suggested in Ref. ¹⁷⁹. So far, no conclusions can be drawn on retention and statistical distribution of the polarization induced resistance states in large arrays. However, when considering the collective phenomenon of ferroelectricity with multiple dipoles contributing to a resistance change as opposed to filament-type resistive switching, advanced scalability may be expected. First results have shown that the FTJ is very similar to ReRAM in terms of electrical behavior and memory design, albeit distinct physical mechanisms. It should be noted that current prototypes could actually have both FTJ and ReRAM traits, as resistive switching is common among oxides including ferroelectric perovskites (¹⁸⁰ and references therein). For future development, the ferroelectric film in an ideal FTJ should be as thin as possible to allow scalability (while maintaining sufficient read current) and much less defective than that in ReRAM (e.g., with fewer oxygen vacancies), so that the mechanism of ferroelectric switching can dominate electrical behavior with little influence from mechanisms related to conducting filaments. The manufacturability of the rather complex electrode-perovskite ferroelectric-system of the FTJ concept will largely rely on the availability of high throughput and CMOS-compatible epitaxial growth techniques for large substrates or alternatively on the unrestricted feasibility demonstration of a polycrystalline FTJ. In this context it is worth noting that the CMOS-compatibility and advanced thickness scalability of ferroelectrics based on HfO₂ and its doped variant¹⁵⁵ as well as recent breakthroughs in its epitaxial growth¹⁶⁰ might yield great potential for the manufacturability of competitive FTJs. Experimental demonstrations of FTJs based on doped variant of HfO₂ were recently reported in Refs. ^{181,182,183}.

2.2.6. MASSIVE STORAGE DEVICES

Device scaling has become a matter of strategic importance for modern and future information storage technologies, which motivates an exploration of unconventional materials with competitive performance attributes. By 2040 the conservative estimate the worldwide amount of stored data is 10²⁴ bits, and the high estimate is $\sim 10^{29}$ bits¹⁸⁴ (these estimates are based on research by Hilbert and Lopez¹⁸⁵). In nature, much of the data about the structure and operation of a living cell is stored in the molecule of deoxyribonucleic acid (DNA) and using nucleic acids molecules, such as DNA, for memory storage has been proposed. DNA has an information storage density that is several orders of magnitude higher than any other known storage technology: 1 kg of DNA stores 10²⁴ bits, for which $>10^9$ kg of silicon Flash memory would be needed.¹⁸⁴ Thus, a few tens of kilograms of DNA could meet all of the world's storage needs for centuries to come.

A number of recent studies have shown that DNA can support scalable, random-access and error-free information storage.^{186,187,188} A state-of-the-art operating system developed at the University of Washington with an industry partner is a DNA-based archival storage framework that supports random access from a DNA key-value store.¹⁸⁹ The DNA-stored files are compatible with mainstream digital format, and large-scale DNA storage up to 200 MB has been demonstrated.¹⁹⁰ There are still many unknowns regarding both DNA operations in cell and with regard to the potential of DNA technology for massive storage applications. DNA volumetric memory density far exceeds (10^3 – $10^7\times$) projected ultimate electronic memory densities. Also, in the living cell, the memory read/write operations occur at high speed (<100 μ s/bit) and require very low energy of $\sim 10^{-17}$ J/bit or 10^{-11} W/GB.¹⁹¹ DNA can store information stably at room temperature for hundreds of years with zero power requirements, thus making it an excellent candidate for large-scale archival storage.¹⁹¹ Also, DNA is an extremely abundant and totally recyclable material. Recently, a method for efficient encoding of information—including a full computer operating system—into DNA was presented, which approaches the theoretical maximum for information stored per nucleotide.¹⁹² One of the goals for research efforts is to demonstrate miniaturized, on-chip integrated DNA storage. New methods for DNA synthesis and sequencing are key components for these developments.

Two major categories of technical challenges remain:

- Physical Media: Improving scale, speed, cost of synthesis and sequencing technologies.
- Operating System: Creating scalable indexing, random access and search capabilities.

The key technological and scientific challenges are in improving performances beyond the life sciences industry. In the life science industry applications require perfect synthesis and perfect sequencing, while scale, throughput and cost are secondary considerations. For data storage, high read and write error rates can be tolerated, and information encoding schemes can be used. In this application, scale and throughput and cost are primary considerations. Current DNA storage workflows can take several days to write and then read data, due to reliance on life sciences technologies that were not designed for use in the same system. The demonstrated DNA write-read cycle is too slow and costly to support exascale archival data storage. Solving this problem

will require: 1) Substantial reductions in the cost of DNA synthesis and sequencing, and 2) Deployment of these technologies in a fully automated end-to-end workflow.

2.2.7. MOTT MEMORY

Mott memory is a metal/insulator/metal capacitor structure consisting of a correlated electron insulator (or Mott insulator). Correlated electron insulators often show the electronic phase transition accompanied by a drastic change in their resistivity under external stimuli such as temperature, magnetic field, electric field, and light. Mott memory exploits this electronic phase transition (called Mott metal-to-insulator transition or Mott transition¹⁹³) induced by an electric field. A mechanism of the Mott memory has been theoretically proposed in terms of the interfacial Mott transition induced by the carrier accumulation at a Schottky-like interface between a metal electrode and a correlated electron insulator.¹⁹⁴ The theory also predicted that the resistive switching due to the interfacial Mott transition has a non-volatile-memory functionality, because the Mott transition is a first-order phase transition due to its nature.¹⁹³ In addition, Mott memory based on the Mott transition involving a large number of carriers (more than 10^{22} cm^{-3}) has in principle an advantage in device scaling, because there are a sufficient number of carries for the Mott transition even in a nanoscale device. In an ideal Mott transition, the electrons localized due to the strong electron-electron correlation come to be itinerant, via the stimuli, such as application of an electric field, and so forth. It needs no dopants, and the mechanism withstands the miniaturization of the (silicon) devices.

The Mott transition induced by an electric field or carrier injection has been experimentally demonstrated in a correlated electron material of $\text{Pr}_{1-x}\text{Ca}_x\text{MnO}_3$.¹⁹⁵ After this demonstration, two-terminal devices such as switches and memories have been intensively studied using such correlated electron oxides as $\text{Pr}_{1-x}\text{Ca}_x\text{MnO}_3$,^{196,197} VO_2 ,^{198,199,200} SmNiO_3 ,²⁰¹ NiO ,^{202,203} Ca_2RuO_4 ,²⁰⁴ and NbO_2 ,^{205,206} and using Mott-insulator chalcogenides of AM_4X_8 ($\text{A}=\text{Ga, Ge; M}=\text{V, Nb, Ta; X}=\text{S, Se}$).^{207,208,209,210} In addition to these inorganic materials, reversible and non-volatile resistive switching based on the electronic phase change between charge-crystalline state and quenched charge glass has recently been demonstrated in the organic correlated materials of $\theta\text{-(BEDT-TTF)}_2\text{X}$ (where X denotes an anion).²¹¹

SmNiO_3 exhibits a colossal (8 orders in magnitude) resistance jump by hydrogenation. The SmNiO_3 channel with the solid state proton gate has demonstrated the electric base gated large ON/OFF switching.²⁰¹ The trigger for switching is based on the proton intercalation by electric field, and the DFT calculation explains the large gap opening by additional electron doping via protonation and is the origin for colossal resistance jump phenomena.²¹² These results indicate that the device using the metal – insulator (Mott) transition driven by the strong electron-electron correlation is powerful as well as appropriate for the switching devices.

Scalability has been demonstrated down $110 \times 110 \text{ nm}^2$ in Mott memristors consisting of NbO_2 that shows the temperature-driven Mott transition from a low-temperature insulator phase to a high-temperature metal phase. The switching speed, energies, and endurance of the NbO_2 -Mott memristors have been evaluated to be less than 2.3 ns, of the order of 100 fJ, and $>10^9$, respectively.^{205,206} The programing and read voltages reported so far are $<2 \text{ V}$ and $<0.2 \text{ V}$, respectively.²⁰³ The non-volatile resistive switching of AM_4X_8 single crystals was induced by the electric field of less than 10 kV/cm .^{207,208,209,210} This suggests that if the device consisting of a 10-nm-thick AM_4X_8 film is fabricated, the switching voltage will be less than 0.01 V.

Although non-volatile switching has been reported in the devices based on AM_4X_8 and $\theta\text{-(BEDT-TTF)}_2\text{X}$, their retention characteristics are not elucidated in detail.^{207,208,209,210,211} In addition, the NbO_2 -Mott memristors and VO_2 -based devices exhibit volatile switching.^{198,199,200,205,206} The retention is thus a major concern of Mott memory. In principle, the Mott transition can be driven even by a small amount of carrier doping to the integer-filling or half-filling valence states of the transition element.¹⁹³ However, because of disorders, defects, and spatial variation of chemical composition, a rather large amount of carriers of more than 10^{22} cm^{-3} are required to drive the Mott transition in actual correlated electron materials, resulting in a relatively large switching voltage required in the Mott memory. Therefore, one of the key challenges is the control of crystallinity and chemical-composition in the thin films of correlated electron materials, including the integration of the correlated electron materials onto Si platform. There are some theoretical mechanisms proposed for Mott memories such as the interfacial Mott transition¹⁹⁴ and the formation of conductive filament generated by local Mott transition.^{207,209,210} However, a thorough understanding of the mechanism has not been achieved yet. Therefore, the elucidation of detailed mechanism is also a major research challenge.

2.3. CANDIDATES DEVICES FOR ANALOG IN-MEMORY COMPUTING

Analog In-Memory Computing is an emerging research architecture that promises to enable unprecedented performance per watt capabilities for certain workloads, such as deep neural networks, by performing matrix operations directly on weights stored in memory. This architecture, which is discussed in detail in Section 4, creates different requirements for memory devices than when used for traditional information storage applications. In the following we briefly describe the memory device candidates and their key properties as they relate to analog in memory computing.

2.3.1. **ReRAM AND CBRAM**

ReRAM memory is very dense and can be integrated in the back end of the line (BEOL) since the process temperature of ReRAM is usually lower than 400°C, thus avoiding the use of transistor area for the memory. Several ReRAM crossbars have been demonstrated for inference.^{213,214,215} The conductance of CBRAM can be tuned as an effect analogous to the long-term potentiation of biological synapses, which enables analog memory and neural computing systems.²¹⁶ A key challenge is maintaining good analog properties and high resistance at the same time. Nanoscale oxide-based cross-bar memristors with analog properties at high resistance were demonstrated owing to a natural thermal-confinement-effect when reducing the cross-point area.^{217,218} ReRAM for training remains a challenge due to the non-ideal electrical characteristics of synaptic devices.

2.3.2. **PHASE CHANGE MEMORY**

Phase-change memory (PCM) offers a wide range of analog memory states due to the large contrast between the amorphous and crystalline phases.²¹⁹ For memory applications, PCM devices can be switched between a high-resistance RESET state, formed by melting and quenching an amorphous plug that blocks a narrow constriction within the device; and a low-resistance SET state created by a crystallizing voltage pulse, which frequently ramps down in amplitude over a long duration to produce an extremely low-resistance state.

In contrast, for VMM applications where “device history” is a desirable feature, PCM devices programmed into the RESET state can be slowly brought to a much lower-resistance SET state using many repeated partial-crystallization pulses.²²⁰ Careful choice of pulse condition can stretch this procedure out to many hundreds of pulses.²²¹ Some recent work has shown some evidence for gradual increases in resistance with multiple successive pulses,²²² although the operating regime must be carefully prepared, and the underlying mechanisms are not fully understood.

Many early VMM results using PCM focused on in-situ training.⁷³⁰ Challenges for training include the one-sided nature of PCM programming, the nonlinear evolution of conductance with partial crystallization pulses, and the stochastic nature of PCM programming.

More recent VMM results using PCM have turned to inference of previously trained weights. Key challenges for inference include accurate programming of synaptic state despite the inherent stochasticity observed in PCM device programming,^{223,224} reducing resistance drift due to long-term relaxation of the amorphous phase, and ensuring long retention at high operating temperatures. PCM unit-cell designs that sacrifice some of the inherently large resistance contrast in order to suppress resistance drift have been proposed and demonstrated, in both memory and VMM contexts.^{225,226,227} It turns out that intra-device (e.g., shot-to-shot) variability in the rate at which devices drift over time (the “drift coefficient”) is more problematic than the actual drift itself.²²⁸ This is because any highly-predictable signal loss can be compensated by signal amplification, at least until background noise becomes strongly amplified.

2.3.3. **ELECTROCHEMICAL MEMORY (ECRAM):**

Electrochemical random-access memory (ECRAM) has attracted much attention in recent years as a multi-level, non-volatile memory for its suitability for use in analog computing applications. ECRAM is a family of three-terminal devices composed of an electron-ion mixed conductor channel, an electrolyte layer, (possibly) an ion reservoir layer, and metal contacts. The resistance (or conductance) of the channel is modulated by redox reactions at the interface between the electrolyte and the channel when voltage is applied to a gate electrode relative to channel electrodes. As a result of the reversible insertion and extraction of active ions to a channel, ECRAM exhibits symmetric and linear potentiation and depression in channel conductance under applications of gate voltage pulses. This makes ECRAM attractive for employment as synaptic weights in the physical implementation of artificial neural networks. Current pulse operation or an access device is sometimes used to improve the retention characteristics of programmed states, in order to ensure a deterministic conductance change by injecting/extracting a constant amount of charges and prevent discharging through the electrolyte, respectively.²²⁹

ECRAM has been developed primarily using Li ions and protons as the active ion. Li ion-based ECRAMs have demonstrated repeatable and controlled switching by utilizing materials known from battery technology.²³⁰ However, the technical challenges still remain in the controlling of open circuit potential (OCP), which varies a range of several volts depending on the input voltage conditions, and in the semiconductor foundry compatibility associated with said battery materials. On the other hand, proton-based ECRAMs have proven to have a small driving force for inducing synaptic weight updates. Fast proton diffusion in liquid and/or organic electrolytes can be accompanied by lack of both retention and endurance characteristics, however, ultrafast, high energy efficient ECRAM, based on a glass electrolyte, has been shown to overcome such drawbacks.²³¹ Recently, another type of ECRAM was proposed, in which metal oxides are used for all layers, showing fab-friendly, negligible OCP, and sub-μs operations.^{232,233}

2.3.4. MAGNETIC NEURAL NETWORK DEVICES

Several types of magnetic circuits can be used to implement neural networks including spin-diffusion-based devices,²³⁴ charge-coupled spin logic (CSL),²³⁵ and domain wall logic (mLogic).²³⁶ The use of these different devices has been benchmarked.²³⁷

Magnet switching dynamics that follow the Landau-Lifshitz-Gilbert (LLG) equation with a spin-transfer-torque term are quite similar to the cell dynamics in a Cellular Neural Network (CeNN). CeNNs have been designed based on spin diffusion using all-spin logic (ASL) with PMA magnets as the basic building block. A CeNN cell can also be implemented by using MTJs as synapses and using spin Hall effect or domain wall propagation-based devices as the neuron. In all these cases, the read-out circuit consists of read and reference MTJs and an inverter that amplifies the voltage division between the two MTJs.

In contrast to Boolean circuits, spintronic devices are more attractive compared to charge-based devices. This is because a single magnet can mimic the functionality of a neuron, and these spintronic devices operate at a low supply voltage. The domain wall device provides the best performance, in terms of Energy-Delay Product (EDP), thanks to its low critical current requirement. The spin diffusion based CeNN with IMA magnets consumes more energy due to the large critical current required to switch the magnet.

For optimal circuit-level performance using spintronic devices, several properties are desired including: MTJs with a large TMR and a moderate resistance-area product, large spin injection coefficient β , large perpendicular anisotropy K_u for PMA magnets, large spin Hall angle θ for SHE materials, and small critical depinning current for domain wall magnets.

2.3.5. FLOATING GATE

Floating gate synapses (so-called “synaptic transistors”) were first developed in 1994.²³⁸ They are modified EEPROM devices which can be fabricated in a standard CMOS process and programmed to within 0.2% accuracy.²³⁹ A number of sophisticated systems have been developed based on the arrays of synaptic transistors.^{240,241,242} The main advantage of analog and mixed-signal VMMs based on floating gate memories are very high input and output impedances, which help reducing overhead of peripheral circuitry. The main drawback of synaptic transistor approach is the relatively large cell area, i.e., $> 1000F^2$, where F is the minimum feature size²⁴³, leading to higher interconnect capacitances and hence larger energy losses and time delays in analog computing circuits.

Recently, it was shown that much better area may be obtained re-designing, by simple re-wiring, the arrays of the ubiquitous NOR flash memories with their highly optimized cells.^{244,245} One representative example is Embedded SuperFlash (ESF) memory from Silicon Storage Technology, Inc.²⁴⁶ The areas of the modified arrays of the ESF1 and ESF3 NOR flash memories, with the latter technology scalable to $F = 28$ nm, are close to $120F^2$ and may be further reduced to $\sim 40F^2$. (Note that such areas are much smaller compared to the contemporary 1T1R ReRAM.) Modified 180 nm ESF1 technology was successfully utilized to demonstrate a medium-scale (28×28 -binary-input, 10-output, 2-layer, 101,780-synapse) network for pattern classification. The measured delay and energy dissipation compared very favorably with digital approaches, while the results for chip-to-chip statistics, long-term drift, and temperature sensitivity of the network were also very encouraging. Simulations have shown that similarly superior energy efficiency may also be reached in mixed-signal neuromorphic circuits based on industrial-grade SONOS memories.^{247,248}

Even higher density floating gate neuromorphic circuits can be achieved by utilizing NAND flash memories. 2D NAND memory devices designed for digital memory application are already capable of storing 4 bits (16 levels) in a single transistor of $100 \text{ nm} \times 100 \text{ nm}$ area in 32nm process.²⁴⁹ Commercial NAND manufacturers have shown devices at 15 nm²⁵⁰ and 19 nm.^{251,252} EEPROM devices are found at every CMOS IC node, including 7 nm and 11 nm nodes. At these nodes, we still expect very small capacitors to retain 100s of quantization levels (7-10 bits) limited by electron resolution. In practice, larger capacitors are used, resulting in sufficiently high potential resolution. One expects EEPROM linear scaling down to 10 nm process to result in a $30 \text{ nm} \times 30 \text{ nm}$ or smaller array pitch area. Perhaps, the most exciting opportunity is presented by the modern 3D NAND circuits. 3D NAND memories already feature 96 layers of floating gate cells.²⁵³ The number of layers is projected to further increase to 512 to enable 10 Tb/in^2 density,²⁵⁴ which will be essential for storing large-scale neuromorphic models. The very high density of 3D NAND circuits is achieved at the cost of certain restrictions at the circuit level, such as cells connected sequentially in strings and shared gate (word) voltages for the cells in the same level. The sequential structure of NAND flash memory can be efficiently exploited by using time multiplexed computations at the architecture level, in which one cell from a string being utilized at a particular time step in a distributed VMM circuit.²⁵⁵ Time-domain encoding of inputs was proposed to implement VMM circuits based on the existing 3D-NAND flash memory blocks with common word plane structure, not requiring any modification.²⁵⁶

2.3.6. CAPACITOR-ON-GATE

A recent proposal called for a small capacitor that can be programmed with standard CMOS devices to be tied to the gate of a read transistor.²⁵⁷ In contrast to DRAM, where the charge on the capacitor is transferred through a select transistor onto a bit-line

for readout, here the voltage on the capacitor modulates the conductance of a read transistor by direct attachment to its gate terminal. Although the charge leaks away with a time-constant of milliseconds, the training process can succeed if the time-per-example is at least $100,000\times$ smaller than the decay constant (e.g., 20 ms decay constant and 200 ns per training example).²⁵⁸ One method to obtain good update linearity, so that the amount of charge added and subtracted are balanced, is to use multiple large transistors to supply the current in each unit cell. Additional transistors are then added in order to ensure that, as per the weight-update algorithm, charge is added or subtracted only when both the upstream neuron (say, along the same row) and the downstream neuron (along the same column) agree that weight update should occur in a particular synapse shared by those two neurons. Recently, Ambrogio et al. introduced a combined PCM+capacitor-on-gate unit-cell, in which the PCM provided the non-volatile storage in a “higher significance” conductance, and the capacitor-on-gate devices provided high update linearity in a “lower significance” conductance, with periodic weight transfer from the lower to higher significance devices. The number of transistors associated with the capacitor could be reduced to three: The read transistor, an NFET for charge subtraction and a PFET for charge addition. This was made possible by giving the downstream neuron control over the NFET and PFET gates, and having the upstream neuron control the source contacts of these same two charge addition/subtraction transistors. Furthermore, variability between charge-addition and charge-subtraction due to process nonuniformity was compensated upon weight transfer from capacitor-on-gate cell to the PCM devices using a “polarity inversion” technique. Later, it was shown that this same technique could suppress fixed device variabilities in other kinds of lower-significance conductances, including PCM devices.²⁵⁹ This combined PCM+capacitor-on-gate unit-cell was shown to allow GPU-equivalent training accuracies, despite the known imperfections of PCM devices and typical fab-level CMOS variability in the capacitor-on-gate devices.

2.3.7. CHARGE-BASED ANALOG ARRAYS FOR VMM

Charge-based analog arrays are amenable to very low energy and high-density parallel implementations of vector-matrix multiplication (VMM). Efficient charge storage and weighting in array-based analog computing are achieved through the use of capacitive reactive elements or other charge-based linear weighting elements such as charge-coupled devices (CCDs) and charge-injection devices (CID). Their efficiency stems from inherent charge conservation throughout the computational cycle.

Charge-injection device (CID) arrays store each bit of the matrix element in a DRAM storage element. The charge for each bit in a weight is stored in one of two locations. If the input bit that is multiplying the weight bit is 1, the charge is non-destructively shifted between locations during readout causing a charge to be capacitively induced on the bit line. The charge induced by multiple weights can be summed and sensed allowing the entire matrix to be read out in a single operation. Multi-bit inputs are processed serially. Furthermore, charge is recycled during the computation, and so adiabatic techniques can be used to further lower the energy (at the cost of speed).

High-density mixed-signal adiabatic processors^{260,261,262} using CIDs have been developed using these principles. To optimize for resonant adiabatic energy recovery a stochastic encoding and decoding scheme can be used to ensure a constant capacitive load of the CID array. This has resulted in better than 1.1 TMACS/mW efficiency excluding on-chip digitization.

Alternatively, several approaches have combined a capacitive charge-based VMM with analog-to-digital conversion to maintain high overall system efficiencies. Many analog multiply-and-accumulate operations can be performed for each digitization. High precision implementations of capacitive charge based VMM have achieved low-pJ/MAC energy efficiencies,²⁶³ while low-precision versions have achieved efficiencies at the level of 100 fJ/MAC.²⁶⁴ Comparison of key metrics with the state-of-the-art in analog capacitive VMM ICs^{265,266,267,268} is given in Table BC4.2 above.

Table BC2.2 *Metrics for Analog Capacitive Vector-Matrix Multiply (VMM) ICs*

2.4. MEMORY SELECTOR DEVICE

The *capacity* (or *density*) is one of the most important parameters for memory systems. In a typical memory array, a memory cell can be viewed as being composed of two components: the ‘*storage node*’, which is usually characterized by an element with switchable states, and the ‘*selector*’, which allows the storage node to be selectively addressed for read and write. Both components impact scaling limits of memory. It should be noted that for several advanced concepts of resistance-based memories, the storage node could in principle be scaled down below 10 nm,²⁶⁹ and the memory density is often limited by the selector devices. Thus, the selector device represents a serious bottleneck for emerging memory scaling to 10 nm and beyond.

The most commonly used memory selector devices are transistors (e.g., FET or BJT), as in DRAM, FRAM, *etc.* Flash memory is an example of a storage node (floating gate) and a selector (transistor) combined in one device. Planar transistors typically have the footprint around $(6-8)F^2$. In order to reach the highest possible 2D memory density of $4F^2$, a vertical transistor selector may be used. However, transistors as selector devices are generally unsuitable for 3D memory architectures. Two-terminal memory selector devices are preferred for scalability and can be used in crossbar memory arrays to achieve $4F^2$ footprint.^{270,271}

The function of selector devices is essentially to minimize leakage through unselected paths (“sneak paths”). Two-terminal selector devices can achieve this through asymmetry (e.g., rectifying diodes) or nonlinearity (e.g., nonlinear devices).²⁷² Volatile switches with large on/off ratio can also be used as selector devices. Figure BC2.2 shows a taxonomy of memory selector devices.²⁷³ In addition to external selector devices, some storage elements may have inherent self-selecting properties (e.g., intrinsic nonlinearity or self-rectification), which may enable functional crossbar arrays without external selectors.

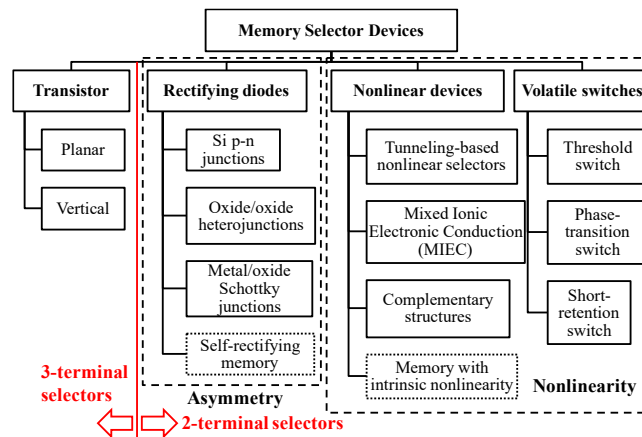


Figure BC2.2. Taxonomy of Memory Select Devices

Since the announcement of 3D cross-point (XP) memory by Intel and Micron in 2015, threshold switch selectors have become one of the most actively researched selector devices. With large on/off ratio and small turn-on voltage, threshold switch selectors can be effective in large crossbar arrays, especially for memory elements with small on/off ratio (e.g., STT-MRAM). As switching devices, threshold switch selectors also face challenges in cycling endurance. Another important challenge is the resistance and voltage balance between the memory element and the threshold switch selector, related to the voltage redistribution when the memory element and the selector, one or both, start switching. Historically, diode selectors were among the earliest two-terminal selectors in development; however, rectifying diodes are only suitable for unipolar memory elements. Nonlinear selectors usually rely on large resistance contrast in exponential transport characteristics to separate on and off states. They tend to have better endurance, but at the same time have non-negligible or even significant on-resistance that contribute to higher array operating voltage and power consumption.

Previous editions of the “Beyond-CMOS” chapter have described the selector devices in Figure BC2.2.²⁷⁴ No major update is included in the 2022 edition.

2.5. STORAGE CLASS MEMORY

Storage-class memory (SCM) describes a device category that combines the benefits of solid-state memory, such as high performance and robustness, with the archival capabilities and low cost of conventional hard-disk magnetic storage.^{275,276} Such a device requires a non-volatile memory (NVM) technology that could be manufactured at a very low cost per bit.

A number of suitable NVM candidate technologies have long received research attention, originally under the motivation of readying a “replacement” for NAND Flash, should that prove necessary. Yet the scaling roadmap for NAND Flash has progressed steadily so far, without needing any replacement by such technologies. So long as the established commodity continues to scale successfully, there would seem to be little need to gamble on implementing an unproven replacement technology instead.

However, while these NVM candidate technologies are still relatively unproven compared to Flash, there is a strong opportunity for one or more of them to find success in applications that do not involve simply “replacing” NAND Flash. Storage Class Memory can be thought of as the realization that many of these emerging alternative non-volatile memory technologies can potentially offer significantly *more* than Flash, in terms of higher endurance, significantly faster performance, and direct-byte access capabilities. In principle, SCM could engender two entirely new and distinct levels within the memory and storage hierarchy. These levels would be differentiated from each other by access time, with both levels located within more than two orders of magnitude between the latencies of off-chip DRAM (~80 ns) and NAND Flash (20 μs).

In July 2015, Intel and Micron jointly announced a new non-volatile memory technology, called “3D-Xpoint.” This technology is said to offer 1000x lower latency and 1000x higher endurance than NAND Flash, at a density that is 10x higher than DRAM.^{277,278} (Note that it is most likely that the latency referred to here is write latency rather than read latency, since NAND write latency is much slower than its read latency.) 3D-Xpoint technology, said to have been implemented at the 128Gbit chip

level, is based on a two-layer stacked crossbar array, with each intersection point containing a non-volatile memory device and a nonlinear access device.^{279,280} The projected array specifications and the target applications of 3D-Xpoint memory were, for all intents and purposes, indistinguishable from those described for SCM. Thus we can consider 3D-Xpoint as the first commercial implementation of the SCM concept.

Previous editions of the “Beyond-CMOS” chapter have detailed description of SCM types, specifications, device candidates, and architecture considerations. No major update is included in the 2022 edition.

3. EMERGING LOGIC AND ALTERNATIVE INFORMATION PROCESSING DEVICES

3.1. OVERVIEW

Recent developments in the global semiconductor industry documented in the More Moore chapter of this Roadmap and elsewhere point to near-future maturation of CMOS devices including lateral and vertical gate-all-around (GAA) transistors for extending Moore’s Law. Historically this chapter tracks the development of logic devices based on materials and mechanisms other than those engineered in conventional silicon transistors. The underlying assumption for this emphasis is the expectation that, as these many varieties of novel devices are subject to research and engineering efforts, some efforts will wane while others lead to technological maturation and transition to commercialization. Along such lines, this chapter categorizes emerging logic devices into four classes: (1) devices for CMOS extension, (2) charge-based devices beyond CMOS, (3) spin-based devices beyond CMOS, and (4) other beyond CMOS and alternative information processing devices.

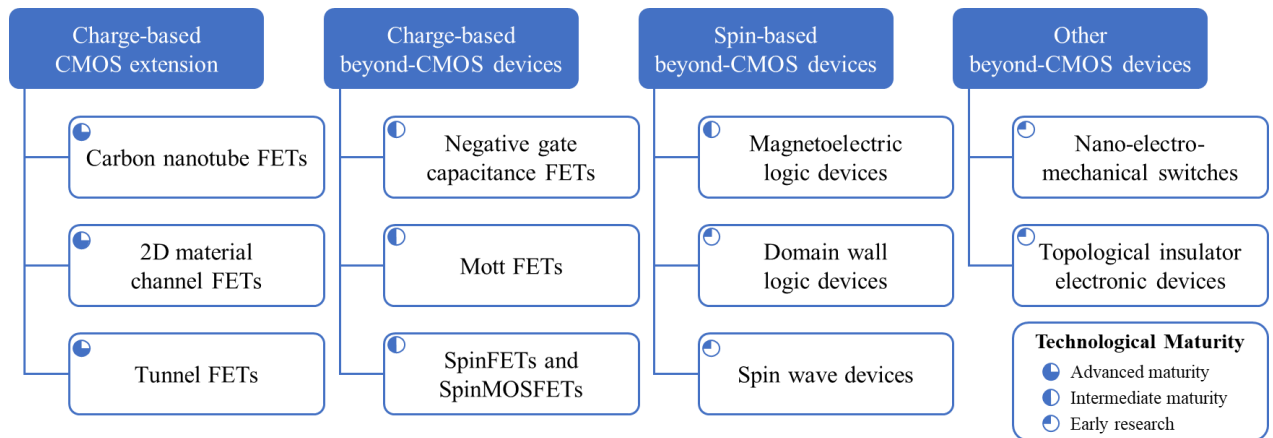


Figure BC3.1 Taxonomy of Beyond CMOS Logic Devices

As the end of the roadmap for conventional silicon-based CMOS comes into view, the focus of this section is refined to consider two technological maturity classes: (1) CMOS extension devices consisting of a small set of candidates that might be viable for industrial incorporation at the end of the CMOS roadmap; and (2) Beyond CMOS devices that are primarily research-grade in their maturity but depart in significant and novel ways from CMOS in their behavior or performance.

The CMOS extension candidates include carbon-nanotube devices, 2D material channel FETs, and tunneling transistors. The following section provides a detailed review of present status and prospects for each. The sections after provide short synopses of recent results for the wider variety of novel research candidates.

Table BC3.1 Beyond CMOS Devices for Logic and Computing

3.2. DEVICES FOR CMOS EXTENSION

3.2.1. CARBON NANOTUBE FETs

Semiconductor carbon nanotubes have a naturally ultra-thin body of around 1 nm and a bandgap large enough for room temperature operation of transistors, and both electrons and holes have extremely high mobility. Owing to these structural and electrical advantages, CNTs offer a solution when channel length is scaled down to the sub-10 nm dimensional scale. Furthermore,

they can be deposited using low-temperature processes, and it is also possible to integrate CNT integrated circuits three-dimensionally on top of Si VLSIs.

In the past several years, significant advances have been made in improving CNT thin film formation, in enhancing device performance in CNT FETs, and in integration. These include realizing (1) a purity of semiconductor CNTs of >99.9999%, (2) high-density aligned semiconductor CNT films of >100 CNTs/ μm on the entire surface of a 4-inch Si wafer, (3) precise control of positioning at a 10-nm pitch using DNA, (4) symmetrical characteristics in NMOS and PMOS via electrostatic doping with YO_x/AlN in semiconductor CNTs with a relatively large band gap, (5) a low-threshold voltage CMOS with a foundry-compatible manufacturing process, (6) single-CNT FETs of a 5-nm gate length with SS of 73 mV/dec. and estimated EDP of 6.95×10^{-31} Js, (7) CNT FETs with a high-density aligned CNT film with a current density of 2.24 A/mm, (8) 5-stage ring oscillator with an oscillation frequency of 5.5 GHz, (9) 50-nm CNT FET with $f_T = 540$ GHz and $f_{\text{max}} = 306$ GHz, (10) a 16-bit RISC-V microprocessors using CNT FETs, (11) three-dimensional integration of CNT ICs on Si logic ICs, and (12) manufacturing highly-uniform CNT FETs on a 200-mm wafer with commercial silicon manufacturing facilities.

There remains a need for further research toward improving other material- and device-level aspects, including methodology of purity evaluation for semiconductor CNTs, wafer-scale aligned CNT film with a controlled pitch, further reduction of contact resistance at small contact lengths, reduction in variability, control of gate dielectric interfaces and properties, and the experimental study of devices and circuits fabricated using the most scaled and relevant device structures and materials. In short, much work remains for CNT FETs, but they have some of the most substantial (and already demonstrated) potential in high-performance, low-voltage, sub-10 nm scaled transistor applications.

3.2.2. 2D MATERIAL CHANNEL FETs

Ultra-thin two-dimensional (2D) layered semiconductors, such as certain transition metal dichalcogenides (TMDs), are promising candidates as future channel materials in field-effect transistors (FETs) for very large-scale integration (VLSI)²⁸¹. TMDs, in general, are characterized by the chemical formula MX_2 , where M is a transition metal element and X is a chalcogen. They can be metallic, half-metallic, semiconducting, or superconducting depending on their compositions. MoS_2 , WS_2 , MoSe_2 , WSe_2 and MoTe_2 with thicknesses from 1 to 4 layers are the most popular 2D TMDs explored for FETs; thanks to their bandgaps ranging from 1.6 to 2 eV, these materials can (unlike graphene) achieve low currents in the off state²⁸². The synthesis of semiconducting graphene nanoribbons and carbon nanotubes has been also demonstrated, but their processing is much more complex, variability is much higher, and stability is significantly worse²⁸³. The ultra-thin body thicknesses (t_{body}) provided by the TMD channel (down to ~ 0.7 nm for monolayers)²⁸⁴ improves electrostatic control over the on/off switching process of the FET, leading to less vertical parasitic current towards the substrate, less horizontal phonon scattering, and reduced short channel effects²⁸⁵.

In atomically thin TMDs produced by mechanical exfoliation (which contain very few native defects), mobilities are consistently extracted in the range of 20 to 60 cm^2/Vs at room temperature from devices fabricated on traditional oxide substrates,^{286,287} while higher mobilities are predicted assuming limited scattering induced by phonon and defects.²⁸⁸ Note that some very high reported mobilities can be a result of using “incorrect” extraction methods as discussed in reference.²⁸⁹ Considering the ultra-thin t_{body} , these are outstanding transport properties that are unachievable in any three-dimensional crystal thinned to these dimensions. At the same time, for the same gate electrical field, the ultra-thin t_{body} allows for most aggressive channel length scaling because the entire volume of the semiconductor channel is more deeply influenced by the electrical field lines. A key parameter in this context is the so-called geometric screening length or electrostatic scale length²⁹⁰ (λ), which closely relates the electrostatic properties (permittivity and thickness) of the gate insulator and the semiconducting body. This parameter quantifies how the band bending inside the transistor channel along the source/drain direction can take over the control from the vertical direction (gate terminal). In other words, it is a measure of how short of a gate length (L_g) can be achieved without introducing short-channel effects, mostly dominated by drain-induced barrier lowering (DIBL) that degrades off-state leakage current and subthreshold slope (SS)²⁹¹. Overall, the smaller EOT (thinner oxide and/or higher dielectric permittivity) and t_{body} , thus the smaller λ and the more aggressively L_g can be scaled. However, quantitatively speaking, the final electrostatics are also largely dependent on device architecture, such as single or double gated devices and gated or doped source-drain contacts. Moreover, λ defines the band bending at the metal-to-TMD interface: at very small values of λ , highly intensified electric-fields take place at the TMD/metal junction of the source and drain contacts. Therefore, Schottky barriers in the source/drain contact region, contact geometry (namely, current crowding)²⁹², and metal/TMD interface quality²⁹³, define the contact resistance in device structures without a sophisticated doping profile. Last, the small λ allows for the introduction of novel device concepts such a 2D based tunneling field-effect transistors (TFETs), overcoming fundamental quantum limits as the 60 mV/dec SS), and also exploiting ballistic transport to obtain scattering-free, high effective carrier mobilities (at very short channel lengths below the electron mean free path), as discussed below. While much work has been done on multi-layer TMD transistors due to higher achievable mobilities, it is ultimately essential to focus on monolayer TMD FETs to benefit from the above discussed λ -scaling to the maximum extent. Therefore, in terms of device size, atomically thin dielectrics and semiconductor body are the ultimate geometric scaling limit, with multiple engineering tradeoffs around device architecture.

The first TMD FET was demonstrated in 2011^[294]; it used a monolayer MoS₂ channel produced by mechanical exfoliation, and it showed a large on/off current ratio ($\sim 10^8$), good SS (74 mV/decade) for a channel length of $L_g = 1.5 \mu\text{m}$ and a gate dielectric film thickness of 30 nm HfO₂. While studies based in mechanical exfoliation are useful to understand the fundamental limits of the material, this approach is not scalable to the wafer level. Scalable methods to produce 2D materials are molecular assembly, liquid phase exfoliation, and chemical vapor deposition (CVD) – and its derivatives, like metal-organic CVD (or MOCVD) and plasma enhanced CVD (or PECVD). For TMD growth, CVD and MOCVD are preferred because thickness can be controlled accurately, and the number of native defects is lower. In these approaches, precursor seeds are deposited on the sample, and at high temperatures they grow laterally until forming a continuous film. The CVD synthesis of MoS₂^{295,296}, MoSe₂²⁹⁷, WS₂²⁹⁸ and WSe₂²⁹⁹ has been achieved using different types of precursors (solid, liquid), temperatures (above 650 °C), synthesis times (usually above 30 minutes), and setups (tube furnace, shower head).

A lot of studies have focused on the fabrication of FETs using ultra-thin CVD/MOCVD-grown TMDs as channel. As these processes employ high temperatures of at least 650 °C, most studies grew the TMDs on empty SiO₂/Si substrates and built the devices on it. Unfortunately, most studies extracted conclusions from large devices with little statistical validity. Among proper studies, reference³⁰⁰ demonstrated monolayer back-gated CVD-grown MoS₂ FETs with mobilities of 30 to 45 cm²V⁻¹s⁻¹ – statistically validated in hundreds of devices over an area of 1 cm – but the channel length and width in this study were too large (above 4 and 11 μm , respectively), and the gate dielectric was 30 nm SiO₂. However, in long devices the effect of the source/drain contacts cannot be accurately extracted, while being ultimately limiting the device performance of ultra-scaled devices. Moreover, metal source/drain contacts that result in the formation of Schottky barriers at the metal/TMD interface do not give rise to a constant series resistance, since most back-gated device geometries impact the Schottky barriers thickness that exponentially controls the current injection when scanning the gate voltage, while modulating the channel simultaneously. This in turn can prevent the correct extraction of intrinsic properties such as the carrier concentration in the channel and its mobility unless scaling of devices is utilized to clearly distinguish between contact and channel effects. Most importantly, short-channel effects cannot be studied unless both channel lengths and dielectric film thicknesses are properly scaled. These are just a few reasons why there is the urgent need for more work on scaled devices, since making predictions based on devices with long channels and/or thick gate dielectrics is likely to be misleading. It is the full device and circuits from the same that needs to be studied, since TMD FETs are *not* behaving as conventional CMOS devices.

Reference³⁰¹ demonstrated hundreds of back-gated MoS₂ transistors (channel lengths ranging from 10 μm down to 29 nm, and channel widths down to 135 nm) with 4 nm HfO₂ gate dielectric on 2-inch wafers, and observed high on-state current densities up to 250 $\mu\text{A}/\mu\text{m}$ and excellent $\text{SS}_{\text{min}} = 80 \text{ mV/dec}$. The fact that the channel of the transistors was 3-4 layers thick indicates that there is room for improvement. State-of-the-art electrical performance in TMD FETs fabricated via CVD/MOCVD has been reported in reference³⁰², which measured hundreds of back-gate FETs with single-crystal monolayer MoS₂ as channel (length 20 nm, width 1 μm) and Sb source/drain contacts on HfO₂/highly doped Si wafers. The authors reported on-state currents up to 1.23 mA/ μm for $V_{\text{DS}} = 1 \text{ V}$, on/off ratios over 10^8 and an intrinsic delay of 74 femtoseconds, although the best SS reported was in average 180 mV/dec. These values are even higher than those achieved in equivalent (planar) silicon CMOS technologies and satisfy the 2028 roadmap target. The fact that these two studies³⁰¹⁻³⁰² (i) transferred the MoS₂ from an optimized c-plane sapphire template (for growth) using polymer scaffolds (which introduces impurities³⁰³) and (ii) used non-optimized gate dielectrics (HfO₂), indicates that there is significant room for improvements.

The state-of-the-art electronic properties of TMD FETs, as well as their yield, variability, and reliability, may be further enhanced if progress is achieved in mainly five directions: channel fabrication, doping, source/drain contact resistance, gate dielectric, and variability.

Regarding channel fabrication, a common challenge is to avoid the formation of domains with different crystal orientations, as the boundaries produce uncontrollable scattering that reduces mobility and increases variability. Multiple studies have claimed the synthesis of single-crystal monolayer TMDs, but most of them apply to crystals with lengths of few micrometers, while wafer level single-crystal quality is rarely obtained³⁰⁴. However, these studies constructed the devices using the transfer process, which is not only slow, but it also introduces impurities that increase variability. Significant effort needs to be placed on improving wafer scale transfer processes. Best demonstrations are based on using tapes with adhesive forces controlled by ultraviolet light³⁰⁵ and wafer de-bonding techniques³⁰⁶. Another option could be the growth of TMDs directly on CMOS processed silicon wafers using a CVD/MOCVD system with two temperature zones (one above 650 °C for the chemical reactions, and another one at 250°C for deposition on the CMOS wafer)³⁰⁷; but so far, the maximum on-state current densities reported using this method are $\sim 3 \mu\text{A}/\mu\text{m}^{-1}$ (1000 times lower than state-of-the-art MoS₂ FETs in reference³⁰²).

Regarding doping, this is a necessary process to fabricate CMOS circuits. As early as 2011, experimental circuit demonstrations³⁰⁸ accompanied the work on improving individual device characteristics. Interestingly, due to the relative ease of fabricating MoS₂ transistors at a sufficiently large scale, most circuits utilize exclusively n-type MoS₂ FETs^{309,310} and only very limited work has truly focused on CMOS integration based on n-type MoS₂ and p-type black phosphorus (BP)³¹¹ or n-type MoS₂ FETs and p-type

WSe₂ FETs.³¹² Using the same 2D material to create both, n-type and p-type devices and assemble the same into a circuit has only been accomplished so far using WSe₂³¹³ and BP.^{314,315}

MoS₂ and WS₂ exhibit n-type characteristics under typical gate field conditions due to a much preferred line-up of the source/drain metal Fermi levels with the conduction band edge, while WSe₂ allows for hole injection at sufficiently negative gate voltages due to a near mid-gap line-up of the source/drain metal Fermi levels.³¹⁶ This effect is not related to doping, but to the specifics of electron and/or hole injection from the contacts. The actual doping level determines the threshold voltage of the devices, as in any conventional transistor, but the absence of the so-called hole-branch in MoS₂ and WS₂ Schottky barrier FETs is a result of the suppressed hole injection from the source/drain contacts. This emphasizes the need to understand and control source/drain contacts to the TMD channel, which is key to lowering contact resistances in TMD FETs to unravel their intrinsic performance specs. Due to limited access to reliable doping approaches for TMDs, particular attention has been paid to engineering metal source/drain contacts to improve the current injection. Bismuth has recently been reported as enabling ~mA/□m on-currents at drain voltages of around 1V.³¹⁷ More work is needed to understand how to unlock the intrinsic channel properties, employing improved contact engineering schemes.

This leads to another important topic, namely the need for both, n-type and p-type transistors of similar performance from TMDs—or ideally one type of TMD. Since substitutional doping through growth processes to achieve degenerate doping is not yet available, alternative approaches such as “charge transfer doping” have been employed to achieve e.g., n-doping in MoS₂ using benzyl viologen molecules³¹⁸ and p-doping in WSe₂ using MoO₃,³¹⁹ WO_x,³²⁰ NO₂,³²¹ or NO.³²² While some of the above examples demonstrated a significantly enhanced on-current level due to improved contact resistances and shifted threshold voltages, the degenerate doping levels achieved resulted in the absence of a significant gate voltage response of the channel, i.e., the devices could not be turned off. It should be noted that unipolar transistor operation could be enabled if doping is only used in the contact regions rather than the entire TMD channel^{318,321}. Last, it should be noted that it is very desirable to extend the scaled device approach, as described in the last paragraph, towards doped scaled TMD FETs with a particular focus on threshold voltage control and additional attention on dopants’ air stability³²².

Regarding source/drain contact resistance, important progress has been achieved. State-of-the-art reports used semi-metals like Bi and Sb as contacts for N-type TMD FETs. In reference³¹⁷, the use of Bi lead to low contact resistances (R_C) down to 123 $\Omega \mu\text{m}^{-1}$ and on-state current densities of 1.35 mA/ μm ; similar values ($R_C=146 \Omega \mu\text{m}^{-1}$) have been achieved using Sb contacts³²³, with the advantage of a higher melting point that can withstand back-end-of-line (BEOL) processing temperatures. Values of R_C for P-type TMD FETs are significantly higher, with best numbers ($R_C=2.7 \text{ K}\Omega \mu\text{m}^{-1}$) reported for WSe₂ FETs with Ru contacts³²³. Note that in ultra-scaled nodes the contact length (L_C , defined as the overlap between the source/drain electrodes and the channel) ends up dominating R_C (and the total resistance in the on-state, R_{TOT}) in the TMD FET—in other words, it is pointless to fabricate a TMD FET with $L_g=5 \text{ nm}$ if later one needs a much higher L_C to reduce R_C . In this direction, non-conventional contact strategies such as the employment of carbon nanotubes (CNT) have recently shown promising results ($R_C \approx 500 \Omega \mu\text{m}^{-1}$, lacking statistical validation) at the scaling limit of the contact length (1–2 nm for 1D contacts)³²⁴—what would outperform standard metal approaches with Bi—although this has been only demonstrated in mechanically exfoliated TMD nanoflakes.

Regarding the gate dielectric this has not only implications in terms of device performance, but also on reliability. Early reports on back-gated TMD FETs (with exposed TMD channel) showed that scattering of charge carriers in the TMD channel with remote impurities via Coulomb interaction can result in a dependent mobility^{287,325}. In TMD FETs with a more realistic structure (channel fully wrapped by insulating materials) improved screening is expected to suppress impurity scattering in gate stacks that include high-k dielectrics,³²⁶ but at the same time high-k insulators have a detrimental impact on the TMD mobility through the interaction between soft phonon modes of the dielectric and charge carriers in the TMD channel²⁸⁸. The natural solution to this problem is to use a layered insulator that forms a van der Waals interface with the TMD channel, being hexagonal boron nitride (hBN) one of the few choices as wide bandgap 2D material (5.9eV)³²⁷. However, this would require either an additional painful transfer step or the optimization of CVD/MOCVD processes for direct h-BN growth on TMDs. Note that the permittivity of h-BN is between 2 and 4 (similar to SiO₂, 3.9)³²⁸, and many scientists have claimed that this results in high leakage current in applications requiring very low EOT. However, comparing EOTs in devices with same gate dielectric material but different semiconductor channels could be misleading due to the different physical mechanisms involved in the formation of the channel: in Si the channel is formed by activating minority charge carriers (inversion), while in monolayer TMDs it is related to the activation of majority carriers (accumulation), which requires different gate electrical field and could enable switching the TMD FETs on at different EOTs. Another problem of h-BN is that stacks of at least 3–4 layers are required (monolayer would lead to prohibitive gate leakage current), and CVD/MOCVD synthesis of multilayer h-BN with low amounts of native defects is much more challenging. Multiple alternatives to h-BN have been proposed, including the use of a monolayer molecular crystal (e.g., 10-perylene-tetracarboxylic dianhydride) as a seeding layer³²⁹, the use of 2D semiconductors that can be oxidized to form a high-k dielectric (e.g., Bi₂O₂Se $\rightarrow$ Bi₂SeO₅)³³⁰, and the use of ionic crystals like CaF₂ as the gate dielectric³³¹. However, all those methods face integration challenges and so far the industry has not adopted any of them. As learnt from the introduction of HK

dielectrics into Si-based technologies, in the short-to-mid terms a compromise could be targeted through bi-layered dielectric stacks. State-of-the-art demonstration³²³ of n-type and p-type channel TMD MOS capacitors employed a bi-layered stack of $\text{AlO}_x/\text{HfO}_2$, but the demonstrated transistors still feature a single HfO_2 dielectric. Meanwhile, the use of a van der Waals material as interface layer such as single crystal h-BN (reducing direct scattering) with a high-k dielectric in the stack³³² raises as an intrinsic material solution for 2D TMD devices. Statistical validation of the benefits of such an approach in deeply scaled devices is still lacking but worthy of further research. Naturally, remote phonon scattering remains a challenge, alongside trapping phenomena and stability issues related to the long-term reliability of the devices (just as those that are dealt with in modern silicon-based technologies).

Regarding variability, experimental studies have exhibited significant (usually prohibitive) time-zero and time-dependent variability of key device figures-of-merit like carriers' mobility, V_{TH} , SS, on-state maximum current ($I_{\text{ON,MAX}}$), off-state minimum current ($I_{\text{OFF,MIN}}$), $I_{\text{ON,MAX}}/I_{\text{ON,MIN}}$, indicating the presence of a high number of impurities at the interfaces between the TMD channel and the adjacent dielectrics, which can be charged when the device is polarized^{7,333}. These impurities are generated during different steps of the fabrication process, such as: (i) formation of native defects in the TMD during CVD/MOCVD growth (missing atoms, impurities like oxygen, strained bonds, wrinkles, multilayer nano-islands), (ii) introduction of impurities and pinholes during TMD transfer (mainly rests of polymer and atoms from the growth substrate), (iii) nano-perforation of the TMD during metal evaporation, and (iv) formation of dangling bonds, thickness inhomogeneities and even nano-islands during dielectric deposition. In ultra-scaled TMD FETs the variations of these figures-of-merit appear in a more scattered/stepped fashion, due to the larger influence of a single defect³³⁴. State-of-the-art TMD FETs still exhibit transfer characteristics with hysteresis at least 50 times higher than Si FETs with comparable size and gate dielectric. Future studies should include exhaustive electrical characterization and modelling to understand these phenomena, although real progress can only be achieved by improving atmosphere conditions, channel encapsulation and cleaning steps during TMD FET fabrication process.

All these values have been reported in planar transistors, and additional demonstrations in stacked nanosheet/nanoribbon transistors with gate-all-around structure are necessary. This is not only important from the perspective of device integration but also due to the significant impact that this architecture can have on critical device parameters (mainly, source/drain contact resistance) when compared to typical, bottom-gated, evaporated-electrode TMD FETs.

Beyond the work that focuses on TMD-based FETs, 2D materials have also been employed to demonstrate the feasibility of novel device concepts, in particular focusing on steep slope devices. Using MoS_2 as the active channel material, germanium as the source electrode and a solid polymer electrolyte as the gate, tunneling field-effect transistor (TFET) operation with a subthreshold swing as small as $\sim 31 \text{ mV/dec}$ over four decades of drain current at room temperature was demonstrated.³³⁵ In another approach, the so-called Dirac source (DS) FET demonstrated a subthreshold swing of $\sim 40 \text{ mV/dec}$ employing a 2D graphene injector and a carbon nanotube.³³⁶ The operation principle is similar to a TFET but uses graphene without a bandgap to create a low-pass filter by making use of graphene's low density of states (DOS) at the Dirac point. Another interesting device concept employing 2D materials with multiple gates demonstrated the opportunity to reconfigure the same device into an n-MOSFET, p-MOSFET, n-TFET and p-TFET,^{337,338} highlighting the unique device level opportunities that 2D materials offer in comparison to conventional, silicon-based CMOS. In all these cases, the reports are restricted to proof-of-concept demonstrations and more statistical data is necessary to evaluate potential impact in the semiconductors industry.

3.2.3. TUNNEL FETS

Tunneling Field Effect Transistors (TFETs) have attracted significant interest thanks to their ability to achieve low operating voltages by overcoming the thermally limited subthreshold swing voltage of 60 mV/decade by using tunneling as a switching mechanism.^{339,340,341} However, over a decade of TFET research has shown that, in spite of their steep sub-threshold slope, TFETs are often significantly limited in their utility by low on-state current. Accordingly, demonstrated devices have thus far been able to switch at speeds that are equivalent to CMOS only in the very best cases, and their on-off ratios expose TFET-based logic to more error than their CMOS-based counterparts. Consequently, while these limitations may be surmountable, TFET research interest has recently diminished, limiting TFET device development progress since the last IRDS Beyond CMOS update.

In its simplest form, a TFET is a gated, reverse-biased p-i-n diode with a gate controlled intrinsic channel. There are two mechanisms that can be used to achieve a low voltage turn on. The gate voltage can be used to modulate the thickness of the tunneling barrier at the source channel junction and thus modulate the tunneling probability.^{342,343,344,345} The thickness of the tunneling barrier is controlled by changing the electric field in the tunneling junction. Alternatively, it is possible to use energy filtering or density of states switching: if the conduction and valence band do not overlap at the tunneling junction, no current can flow; once they do overlap, current can flow. Simulations have predicted arbitrarily steep subthreshold swings when relying on density of states switching as the current is abruptly cutoff when the conduction band and valence band no longer overlap.³⁴¹ If phonons or short channel lengths are accounted for, simulated subthreshold swings on the order of $20\text{--}30 \text{ mV/decade}$ are typical.³⁴⁶ It is possible to use the tunneling switching mechanisms in series with the standard MOSFET thermal switching

mechanism to get an overall subthreshold swing that is steeper than 60 mV/decade when no individual mechanism is steeper than 60 mV/decade.³⁴⁷ The best experimental results to date have relied on a combination of thermal switching and density of states switching.³⁴⁸ So far, the experimental results are far worse than simulated device characteristics. The review by Lu and Seabaugh³³⁹ shows a comprehensive benchmarking of published experimental results prior to 2014. The benchmarking shows two problems with TFETs as a MOSFET replacement: 1) Devices are unable to achieve SS <60 mV/decade over a large range or at useful current levels and 2) The on-state current is too low for reasonable performance.

The review shows 14 reports of subthreshold swings below 60 mV/decade, and a few additional results have been published since. Most of the results are for group-IV materials such as Si,^{344,349,350,351,352,353} strained SiGe,³⁵⁴ Si/Ge,³⁵⁵ and strained Ge.³⁵⁶ Nanowire III-V TFETs have shown even steeper swings. An InP/GaAs heterojunction³⁵⁷ has shown 30 mV/decade at 1 pA/μm. The steepest result ever reported is in a Si/InAs heterojunction³⁵⁸ of approximately 20 mV/decade at 0.1 pA/μm. However, there are only a few data points defining this result. Even for low power applications, at least 1–10 μA/μm is needed.³⁵⁹ Recently, a promising InAs/GaAsSb/GaSb nanowire heterostructure TFET with a 48 mV/decade SS at 67 nA/μm and an I_{60} (current at 60 mV/decade) of 0.31 μA/μm was demonstrated.³⁴⁸

Researchers attempting to achieve higher on-current TFETs have traditionally relied on reducing the effective mass by using III-V's and reducing effective bandgap by using a heterostructure. While this has increased the current, the subthreshold swings and off currents have gotten worse. The increase in off-state current and subthreshold swing needs to be decoupled from the increase in on-state current. Unfortunately, this is a fundamental tradeoff when modulating the thickness of the tunneling barrier:³⁴⁰ barrier thickness modulation only gives a step subthreshold swing at low current densities.

An ideal density of states (DOS) switch would switch abruptly from zero-conductance to the desired on-conductance, thus displaying zero subthreshold swing voltage.³⁶⁰ Practically, the band edges are not perfectly sharp, so there is a finite density of states extending into the band gap. Optical measurements of intrinsic GaAs imply a band edge steepness of 17 meV/decade.³⁶¹ However, the electrically measured joint DOS in diodes has generally indicated a steepness >90 mV/decade.³⁶² This broadening is likely due to the spatial inhomogeneity and on heavy doping that appears in real devices. Effectively, there are many distinct channel thresholds in a macroscopic device, leading to threshold broadening. Fortunately, it has been experimentally demonstrated that a band edge worse than 60 mV/decade can be combined with a thermal switching mechanism to give an overall subthreshold swing better than 60 mV/decade.³⁴⁷

TFETs are reverse-biased diodes and, hence, are subject to generation in the depletion region. These generation events include but are not limited to bulk and interface trap-assisted Shockley-Read-Hall (SRH)^{363,364,365} and spontaneous and Auger generation.³⁶⁶ Calculations based upon these mechanisms show that these significantly degrade the subthreshold swing and increase the leakage currents but do not prevent TFETs from achieving sub-60 mV/decade subthreshold swing. Material defects and gate interface traps make these effects worse and result in worse band edges.

To overcome these challenges, better material perfection than ever before is needed. Every defect or dangling bond can create a trap that ruins the band edge or creates a parallel conduction path. The defects due to doping can be eliminated by electrostatically inducing carriers. Proof of concept devices can be made by making the device a few nanometers large so that there is a low probability of having a trap within the device. 2D transition metal dichalcogenides (TMD) heterostructures potentially have better electrostatic control and lower defects as there are ideally no dangling bonds at the semiconductor oxide interface.

3.3. CHARGE-BASED BEYOND-CMOS DEVICES

3.3.1. NEGATIVE GATE CAPACITANCE FETs

The Mott field-effect transistor (Mott FET) utilizes a phase change in a correlated electron system induced by a gate as the fundamental switching paradigm.^{367,368} Correlated electron materials can undergo Mott insulator-to-metal phase transitions under an applied electric field due to electrostatically doped carriers.^{369,370} Thus, Mott FETs can be engineered to have a similar structure to conventional three-terminal semiconductor FETs, with the semiconductor channel materials being replaced by correlated electron materials. Besides electric field excitation, the Mott phase transition can also be triggered by photo- and thermal-excitations for optical and thermal switches. Defects created by environmental exposure to chemicals or electrochemical reactions can also induce Mott transition via carrier doping. The critical threshold for inducing phase change can be tuned via stress.

Mott FET structures have been explored with different oxide channel materials.³⁶⁸ Among several correlated materials that could be explored as channel materials for Mott FET, vanadium dioxide (VO₂) has attracted much attention due the sharp metal-insulator transition near room temperature (nearly five orders in single crystals).³⁷¹ The phase transition time constant in VO₂ materials is in sub-picosecond range determined by optical pump-probe methods.³⁷² Device modeling indicates that the VO₂-channel-based Mott FET lower bound switching time is of the order of 0.5 ps at a power dissipation of 0.1 μW.³⁷³ The possibility of electrochemical reactions must also be carefully examined in these proof-of-principle devices due to the instability of the liquid-oxide interfaces and the ease of cations in such complex oxides to change valence state.^{374,375} On the other hand, unlike

traditional CMOS that is volatile and digital, electrochemically gated transistors exhibit non-volatile and analog behaviors, which can be utilized to demonstrate synaptic transistors³⁷⁶ and circuits³⁷⁷ that mimic neural activities in the animal brains. Voltage induced phase transitions in two-terminal Mott switches have also been implemented to realize neuron-like devices³⁷⁸ and steep-slope transistors.³⁷⁹ As a Mott device with purely electrostatic modulation in the solid-state base,³⁸⁰ the VO₂-FET with a high-k oxide/organic hybrid dielectric gate has been proposed.^{381,382} Their reversible as well as quick resistance switching upon an application of gate bias and the maximum resistance modulation at the Metal-Insulator transition temperature indicate the possibility of purely electrostatic field-induced metal-insulator (Mott) transition. The gate-tunable abrupt switching device based on a VO₂ microwire integrated monolithically with a two-dimensional tungsten diselenide semiconductor by van der Waals stacking has been reported.³⁸³ Nanofabrication engineering has been demonstrated to enhance the performance of Mott FET³⁸⁴. The 3-dimensional Fe₃O₄ nanowires on the length scale of 10 nm exhibited the remarkable Verwey transition at about 112 K, which was found to be 6 times larger than that for the thin-film configuration.³⁸⁵

Experimental challenges with correlated electron oxide Mott FETs include fundamental understanding of gate oxide-functional oxide interfaces and local band structure changes in the presence of electric fields. Methods to extract quantitative properties (such as defect density) of the interfaces are an important topic that have not been explored much to date. The relatively large intrinsic carrier density in many of the Mott insulators requires the growth of ultra-thin channel materials and smooth gate oxide-functional oxide interfaces for optimized device performance. It is also important to understand the origin of low room-temperature carrier mobility in these materials.³⁶⁹ Theoretical studies on the channel/dielectric interfacial electronic band structure are needed for the modeling of subthreshold behaviors of Mott FETs. Metal-insulator transition induced by resonant tunneling in double quantum well structures has been successfully demonstrated in strongly correlated oxides.³⁸⁶ Understanding the electronic transition mechanisms while de-coupling from structural Peierls (lattice) distortions is also of interest and important in the context of energy dissipation for switching.

While the electric field-induced transitions are typically explored with Mott FET, nanoscale thermal switches with Mott materials could also be of substantial interest. Recent simulation studies of “ON and “OFF” times for nanoscale two-terminal VO₂ switches indicate the possibility of sub-ns switching speeds in ultra-thin device elements in the vicinity of room temperature.^{387,388} Such correlated electron systems can be understood as ‘threshold materials’ where the conducting state can be rapidly switched by a slight external perturbation, lending themselves to applications in electronic devices.^{389, 390} Electronically driven transitions in perovskite-structured oxides such as rare-earth nickelates³⁹¹ with minimal lattice distortions would also be relevant in this regard. Three-terminal devices are being investigated using these materials and will likely be an area of growth.^{392,393, 394,395} SmNiO₃, with its metal-insulator transition temperature near 130°C and nearly hysteresis-free transition, is particularly interesting due to the possibility of direct integration onto CMOS platforms. Floating gate transistors have recently been demonstrated on silicon.³⁹⁶ It has been found that non-thermal electron doping in SmNiO₃ can lead to a colossal increase in its resistivity, which has been utilized to demonstrate a solid-state proton-gated transistor with large ON/OFF ratio.³⁹⁷

Early results of Mott device research offer a role for correlated oxide semiconductors in logic devices. While the doping process indicates slower dynamics than is possible with purely electrostatic carrier density modulation, the non-volatile nature of the Mott transition in 3-terminal devices suggests that combining logic and memory operations into a single device could provide computing enhancements and warrants further research. Architectural innovations that can create new computing modalities with slower switches but at lower power consumption can benefit in the near term with results to date while in the longer term transistor gate stacks need to be studied further for these classes of emerging semiconductors.

3.3.2. MOTT FETs

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3.3.3. SPINFETs AND SPINMOSFETs

Spin-transistors are classified as “non-conventional charge-based extended CMOS devices,”⁴²⁹ and can be further divided into two categories: the spin-FETs proposed by Datta and Das⁴³⁰ and spin-MOSFETs proposed by Sugahara and Tanaka.⁴³¹ The structures of both types of spin transistors consist of a ferromagnetic source and a ferromagnetic drain which act as a spin injector and a detector, respectively. Although the devices have similar structures, they have quite different operating principles.^{429,432} In spin-MOSFETs, the gate has the same current switching function as in ordinary transistors, whereas in the spin-FETs, the gate acts to control the spin direction via the Rashba spin-orbit interaction. Both types of devices behave as a transistor and function as a magnetoresistive device. The important features of spin transistors are that they allow a variable current to be controlled by the magnetization configuration of the ferromagnetic electrodes (spin-MOSFETs) or the spin direction of the carriers (spin-FETs), and they offer the capability for non-volatile information storage using the magnetization configurations. These features are very useful for energy-efficient, low-power circuit architectures that cannot be achieved by ordinary CMOS circuits. Non-volatile logic and reconfigurable logic circuits have been proposed using the spin-MOSFET and the pseudo-spin-MOSFETs, which are suitable for power-gating systems with low static energy.^{432,433,434,435,436,437,438}

The full operation suggested for spin-FETs^{432,439} and spin-MOSFETs^{432,440,441,442,443} have not yet been experimentally verified. For realizing fully functional spin transistors, some important progress in the underlying technologies such as electrical spin injection, spin detection, and spin manipulation in semiconductors (SCs) should be required.^{444,445,446} Lots of theories^{447,448,449,450} have predicted that the insertion of a tunnel barrier between the ferromagnet (FM) and SC is a promising method for highly

efficient electrical spin injection and detection. In particular, large spin signals induced by the efficient spin injection and detection were observed in Si-based lateral spin-valve devices with FM/MgO tunnel-barrier contacts even at room temperature.^{451,452,453,454} Also, by using a back-gated device structure with FM/MgO tunnel-barrier contacts,^{441,442,443} a basic read operation of Si-spin-MOSFETs was demonstrated at room temperature. These are important developments for Si-based spin-MOSFETs. However, if an insulator tunnel barrier such as MgO was utilized, the large parasitic resistance can cause the obstacle for the development of source and drain structures in the spin transistors. Another key development for highly efficient spin injection/detection in SCs is half-metallic FM contacts. Thus far, electrical spin injection, transport, and detection in SCs without using insulator tunnel barriers have been demonstrated in lateral spin-valve devices with Co₂-Heusler alloy/SC Schottky-tunnel-barrier contacts.^{455,456,457,458,459} To reduce the value of RA (interface resistance area product) at the source and drain structures in spin-MOSFETs, the delta-doping of dopant impurities near the Co₂-Heusler alloy/Ge Schottky-tunnel-barrier⁴⁶⁰ has been demonstrated, leading to the room-temperature spin transport⁴⁶¹ including local magnetoresistance effect in Ge-based lateral devices with RA values of less than 0.2 kΩμm².⁴⁶² Recently, the quality of the Co₂-Heusler alloy/Ge heterointerface was significantly improved by inserting Fe atomic layers between Co₂-Heusler alloy and Ge.^{463,464} This leads to the two-terminal magnetoresistance (MR) ratios of more than 0.1 % at room temperature even in the Ge-based lateral spin-valve devices.⁴⁶⁴ For enhancing the MR ratio in Ge-based spin-MOSFET structures, it is essential to further optimize the formation of L₂₁-ordered Co₂-Heusler alloy/Ge Schottky-tunnel contacts and to reduce the spin relaxation in the Ge channel.⁴⁶⁵ Recently, a low-temperature gate-stack for the Ge-based spin-MOSFET has been developed,⁴⁶⁶ leading to the expectation for the operation of a top-gate-type spin-MOSFET.

Alternative approaches for realizing spin-MOSFETs have also been proposed.^{432,437,440,467,468} Pseudo-spin-MOSFETs are circuits that reproduce the functions of spin-MOSFETs using an ordinary MOSFET and a magnetic tunnel junction (*MTJ*) which is connected to the MOSFET in a negative feedback configuration. Although pseudo-spin-MOSFETs offer the same functionality as spin transistors, such as the ability to drive variable current, pseudo-spin-MOSFETs have larger resistance than spin-FETs or spin-MOSFETs.

For spin manipulation in SCs, channel materials with a strong spin-orbit interaction, such as InGaAs, InAs and InSb, are required⁴³⁰ in order to sufficiently induce the Rashba spin-orbit interaction by an electric gate voltage. Using InAs⁴⁶⁹ and InGaAs⁴⁷⁰ 2DEG heterostructures with and without FM spin injector and detector, respectively, the spin manipulation was demonstrated by the electric field, meaning the operation of a spin-FET.

However, even in 2024, the operation temperature is limited to the low temperature less than 40 K.⁴⁷¹ The experimental proof of electrical spin injection, detection and manipulation in SCs with the strong spin-orbit interaction above room temperature is needed to create spin-FETs. Recently, it was found that the Rashba spin-orbit interaction is induced by strong electric fields applied to the Si-MOS interfaces at room temperature.⁴⁷² For realizing spin-FETs using Si, the changes in MR ratios or magneto-current ratios in a Si-MOS structure should be explored at room temperature.

3.4. SPIN-BASED BEYOND CMOS DEVICES

3.4.1. MAGNETOELECTRIC LOGIC DEVICES

Among the now demonstrated magneto-electric devices is the magneto-electric FET (ME-FET) utilizing magneto-electric antiferromagnetic chromia as the gate dielectric and graphene as the channel material.⁴⁷³ Improved performance, over this recent device demonstration, can be expected with increases in the spin-orbit coupling in the 2D channel material.⁴⁷⁴ The essential operating principle of the ME-FET is to utilize boundary spin polarization of the magnetoelectric surface to induce, via proximity coupling, a significant spin polarization in a 2D channel that is heterogeneously integrated.^{474,475} If a gate potential is applied to reverse (via ME coupling) the direction of the boundary magnetism at this heterointerface, the spin populations in the channel can be substantially modulated, generating a large and non-volatile conductance change.^{474,476,477} These include the realization of robust non-volatile states, that can be switched quickly (<20 ps/full adder).^{476,478,479,480} At the same time, the strongly dielectric nature of chromia should allow for write operations at low energy cost (<20 aJ/full adder), with negligible leakage current.^{476,478,479,480}

The ME-FET schemes could serve as "beyond CMOS" 'plug-in' replacement logic.^{476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488} There are also some benchmarking efforts,^{476,478,479,480} where there has been an effort to compare the most competitive magneto-electric devices with CMOS. The result is that it is now increasingly clear that magneto-electric field effect transistors are more likely to be competitive or surpass CMOS.^{476,480} Indeed, because of the nonvolatile nature of the magneto-electric transistor, a full adder may require only 8 device elements instead of the typical 28 needed for CMOS.⁴⁷⁶ Similarly, the silicon CMOS majority gate requires 13 components. The magneto-electric FET majority gate requires, including clocking, 6 components. This represents an area improvement of over 50%, assuming similar size transistors.⁴⁷⁶ This is equivalent to greater than one process node. If we split the magnetoelectric side of the gate so that the channel can independently be spin polarized up or down, this results in a component reduction from the previous best for the MEFET of six, down to four components, a further 50% reduction over the

standard ME-FET circuit, and a reduction to less than 30% in area compared to CMOS.⁴⁷⁶ Recent simulations benchmarking the performance of cache-memory architectures, based upon a CMOS + X memory-cell design that integrates the ME-FET with a pair of MOSFETs, have also established several important advantages over CMOS.^{478,479} These attributes also make the ME-FET superior to other spintronic devices concepts,^{476,478,479,480,489,490} by avoiding the costly switching energetics, and slow domain evolution, associated with ferromagnets.

The magneto-electric chromia (Cr_2O_3) has been identified as being of particular interest for such applications, being a room-temperature antiferromagnet that exhibits significant spin polarization – or **boundary magnetism** – that is roughness insensitive^{491,492,493,494} thus easy to manufacture into devices. Recently, it has been demonstrated that doping of chromia with boron (B-doped chromia) will increase the Néel temperature of chromia to above 400 K^{495,496,497} and also activates a hitherto unexplored switching mechanism which eliminates the need for an applied H field.^{496,497} This discovery makes the magnetoelectric-FET a much more realistic competitor to CMOS that is both nonvolatile, and which may even surpass CMOS^{476,478,479,480,489,490} significantly in terms of energy efficiency and functionality.

The challenges in pushing forward these technologies extends not only to the fabrication and characterization of a new generation of nonvolatile magneto-electric devices, but also to ascertaining the optimal implementation of CMOS plug in replacement circuits. Questions that need to be resolved include demonstration that the magneto-electric devices can be scaled to less than 10 nm, and this includes finding a suitable 2D channel material that can be polarized by exchange coupling with the boundary polarization of the magneto-electric and yet does not suffer from large scale edge scattering. Experimental demonstration of limits to the switching speeds of any antiferromagnetic magneto-electric still remain absent but scaling studies⁴⁹⁸ provide some promise the switching speed could be very fast (≤ 10 ps).

The demonstration of a magneto-electric transistor⁴⁷³ and magneto-electric devices with boron doped chromia^{496,497} has been complemented by the demonstration multiferroic transistors made from molecular heterolayers^{499,500,501} which reliably to 350 K⁵⁰⁰ and may be scalable to the 3.6 nm node. With the multiferroic molecular heterolayer transistors, the questions are the device impedance and endurance, which have not been fully explored.⁵⁰¹

3.4.2. DOMAIN WALL LOGIC DEVICES

The domain wall-magnetic tunnel junction (DW-MTJ) or three-terminal magnetic tunnel junction (3T-MTJ) operates as an in-memory computing nonvolatile logic device through current or voltage driven manipulation of a single domain wall in a magnetic patterned wire,^{502,503,504} with readout performed using a magnetic tunnel junction. It can be considered as an extension of racetrack memory to a single domain wall racetrack for compute-in-memory, and it also has applications for computing in high radiation environments such as space since nanomagnets are naturally rad-hard^{505,506}. There has also been extensive development on using DW-MTJ devices for neuromorphic computing and brain-inspired computing.

The domain wall track is composed of heavy metal/magnet/oxide thin films patterned into a wire shape. Traditionally the magnetic layer is a ferromagnet, but antiferromagnets and ferrimagnets can also be used similar to racetrack memory⁵⁰⁷. Standard materials examples are Ta/CoFeB/MgO. On top of the track is a patterned MTJ hard reference layer with additional related thin films to promote high switching field of the hard layer compared to the magnetic track.

The ends of the ferromagnetic track can be exchange-biased in opposing directions using antiferromagnets, and/or an additional electrode (called an Oersted field line) can be used to ensure a single domain wall is electrically created in the track, and/or an additional magnetic tunnel junction can be placed to nucleate the domain wall electrically, and/or lithographically-defined pinning notches can be fabricated to keep the domain wall in the track.

The simplest form of the device has three terminals: input (IN) and clock (CLK) contacting the ferromagnetic track, and output (OUT) contacting the top of the MTJ. During the write operation, a voltage applied between IN and CLK drives a current and moves the domain wall using either spin transfer torque (STT) and/or spin orbit torque (SOT), and the DW motion can be assisted by voltage-controlled magnetic anisotropy (VCMA)⁵¹³. During the read operation, a voltage applied between CLK and OUT measures the resistance state of the MTJ relative to the domain wall, which will determine the amount of current to drive subsequent devices. The device can act as an analog universal NAND gate, in addition to other basic logic gates: if IN is connected to the OUT of two previous devices, and only if both are in a low resistance state (logic output 1) will there be sufficient current to depin and move the domain wall to the other side of the MTJ, changing the output of the device from a low resistance state (logic output 1) to high resistance state (logic output 0). The DW-MTJ can also exhibit fanout: a fanout of 2 has been partially demonstrated experimentally⁵⁰⁴, and theoretically higher fanout is possible.

A related device is the mLogic four-terminal version, which has an additional non-magnetic, non-conducting spacer on top of the ferromagnetic track that couples the current-manipulated domain wall to a domain wall or nanomagnet in an electrically-separated layer, which then alters the output MTJ resistance.^{508,509,510,511,512} The additional terminal comes from two terminals connected to the output MTJ. The four-terminal version provides complete input/output isolation.

The device operation has been modeled in micromagnetics including NAND, shift register⁵⁰³, 32-bit full adder⁵¹⁹, 4-bit and 8-bit multiply-accumulate (MAC), and a systolic array⁵¹³. Benchmarking of the STT and SOT-driven device type was performed for a 32-bit full adder simulation^{514,515,516}, followed by benchmarking against state-of-the-art CMOS neural network accelerators for VCMA-assisted DW-MTJs⁵¹³. A SPICE model has been developed to enable larger scale circuit simulations.⁵¹⁷ Single device operation as well as two-device and three-device circuit operation has been shown in experimental prototypes, including inverter, buffer, and concatenation operation^{505,504}. The prototypes showed in experiment one of the first examples of concatenable magnetic logic devices for building larger circuits. The four-terminal mLogic spacer coupling has also been demonstrated^{508,509,510,511}.

Larger scale system-level simulations have been performed to determine the benefits and drawbacks of DW-MTJ logic. One circuit-level energy-performance analysis showed that while very low voltage can be used to operate the essentially all-metallic devices, it comes with increased extrinsic domain wall pinning effects and thermal noise.⁵¹⁸ They predicted the energy reduction from increasing the tunnel magnetoresistance (TMR) will saturate when $TMR > 100\%$, but further increasing TMR can mitigate predicted thermal noise limits. Another work simulated a 32-bit adder communicating with registers with all DW-MTJ devices and shows SOT switching can make the technology competitive with a comparable CMOS sub-processor component.⁵¹⁹ A follow-up work introduced VCMA contacts to create an energy well for the DW to settle into on each side of the device⁵¹³. The simulations show 100% reliability is possible for all combinations of DW-MTJ logic states; interestingly, there is a range of TMR with 100% correctness, and too high TMR creates errors due to too much current between devices in the circuit for some combinations. The device design was also modified to have constant footprint with varying fanout. Large scale simulations were performed, up to a systolic array processing of matrix vector multiplication (MVM) using 4-bit MAC units. The simulations show the dense pipelining of DW-MTJs makes the MAC operation on par with CMOS delay times, overcoming the slower magnetization switching. The energy per MAC operation was estimated to be on par with state-of-the-art CMOS accelerators. The systolic array was found to be a good use case for DW-MTJs since it already has high activity factor.

Seminal works made advances for experimental implementation of DW-MTJ computing. Prototypes were shown that benefit from perpendicular magnetic anisotropy (PMA) and SOT domain wall switching, therefore lowering the switching current density compared to previously demonstrated prototypes by over $10\times$ ⁵²⁰, down to $1 - 5 \times 10^{11} \text{ A/m}^2$. One challenge for DW-MTJs has been the precise etching needed to define both the MTJ and the domain wall track, without damaging the delicate thin film layers, which has previously lowered the device TMR after patterning to between 10-40%. Fabrication advancements in this work show that the TMR and resistance-area product of the unpatterned film stack can be maintained after patterning, and showed a TMR of 170%, around the highest that can be obtained to-date for PMA MTJs. The work also showed all-electrical operation of the DW-MTJ, including domain wall nucleation, except for an external DC bias field to overcome coupling of the domain wall track layer to the pinned layers. With these modernized prototypes that combine lower switching current density and high TMR, cycle-to-cycle studies of the domain wall operation were able to be performed, showing cycling variation of 7-10% and that the domain wall can be pushed back and forth in the device for continuous operation without the domain wall exiting the track. A two-device circuit was demonstrated showing an inverter operation. A 3-weight and 5-weight artificial synapse⁵²¹ and a stochastically-driven artificial neuron⁵²² were demonstrated in prototypes.

A series of works also were released that apply to both DW-MTJs and domain wall-based majority logic^{523,524}. These works also showed full integration of PMA, SOT, and MTJs into the device. Current through an MTJ is used to electrically initialize the domain wall. A hybrid free layer is designed and implemented to separate the delicate domain wall-containing track from the etching ions and therefore preserve the domain wall movement properties. Domain wall velocity is measured and show to be as high as 42 m/s.

These prototype results are a critical advancement since they provide heretofore unknown metrics for DW-MTJs, such as switching current density, switching speed, TMR, resistance-area product, cycle-to-cycle variation, device-to-device variation, scaling behavior, etc. These metrics can now be put into models to predict circuit operation and guide future development: for example, the 10% cycling variation with $TMR = 165\%$, when put into a model of a full adder, predicts 90% accuracy, which could be increased to near-100% accuracy if the variation could be decreased to under 5%.

Here we will explain the numbers used in the comparison table for the DW logic device. The cell size of a DW-MTJ based NAND has been calculated to be $18F^2$ where F is the feature size⁵⁰³, which is a $20\times$ improvement compared to a CMOS-based NAND cell size of $360F^2$. This is because a single DW-MTJ can perform NAND. To accommodate the contacts, MTJ, room for the domain wall to switch, and spacing between devices, the minimum needed device pitch is $6F$, which is 90 nm for $F = 15 \text{ nm}$, and the minimum device length is $4.5F$, which is $\sim 70 \text{ nm}$ for $F = 15 \text{ nm}$. A feature size down to $F = 50 \text{ nm}$ has been demonstrated in a fully functional device⁵²⁴. Using a NAND feature size of $18F^2$, for $F = 15 \text{ nm}$ this is a projected density of $2.5 \times 10^{10} \text{ devices/cm}^2$.

Recent results in new materials have shown experimentally that domain walls can obtain speeds up to 250 m/s in Pt-Bi/Co-Gd⁵²⁵ and up to 5700 m/s in ferrimagnetic CoGd alloys⁵²⁶. Assuming the above scaled feature sizes, this would correspond to predicted switching speeds of 3.5-10 GHz, but these materials have not yet been integrated into a full device. The 42 m/s measured in a full device prototype would correspond to 0.6 GHz in a scaled device.

Circuit speed has been shown in simulation to be 33 MHz (30 ns) for a 1-bit full adder and 2 MHz (495 ns) for a 32-bit full adder; circuit energy has been shown in simulation to be 0.5 fJ for a 1-bit full adder and 31 fJ for a 32-bit full adder⁵¹⁹. These numbers are considering a conventional CoFeB ferromagnetic track, and much higher speeds could be obtained using the new materials described above. Circuit speed and energy have not yet been demonstrated experimentally.

Scaled device switching energy is predicted to be approximately 5×10^{-18} Joules for $F = 15$ nm⁵⁰³. In experimental prototypes, the switching current density was measured $\sim 1 \times 10^{11}$ A/m² for $F = 250$ -450 nm⁵²⁰. Assuming the switching current density stays size-independent as feature size scales down, and conservatively assuming a scaled $F = 15$ nm device would use switching voltage $V = 100$ mV and switching time $t = 5$ ns, this would correspond to approximately $E = IVt = 7.5 \times 10^{-15}$ Joules. Thus the number cited as demonstrated is based off of the measured switching current density in prototypes, but translated to a scaled energy with the described assumptions.

Subthreshold slope is not a good metric for DW logic, since thermionic emission is not the main source of switching error. For high enough switching current above the domain wall depinning energy, the switch can be considered as a collective switch of the magnetic object as it jumps out of its potential well and moves with momentum across the track. When the switching current density is close to or below the depinning energy, stochastic depinning will dominate: this stochasticity has been simulated to be useful for deep neural network applications⁵⁴¹.

The DW-MTJ is being studied by many in the community for non-Boolean applications in analog and neuromorphic circuits,^{527,528,529,530,531} including synapses^{532,533,521} and neurons^{534,535,522}. Since the DW-MTJ can act as both an artificial synapse and an artificial neuron, it provides a monolithic platform for neuromorphic computing. By exploiting the dynamical behavior of the domain wall, or an analogous device with skyrmion position being read by a tunnel junction, it can be used to mimic bio-inspired dynamical effects that are essential for learning and operation in the human brain^{536,537}. Some example device-inherent behaviors have been shown to have system-level benefits, such as using magnetic stray field interactions between the DW-MTJs to implement lateral inhibition and winner take all⁵³⁸, manipulating the resetting of the domain wall to mimic the leaking and edginess properties of neurons^{539,540}, and engineering the DW-MTJ shape to have necessary dynamics for training of deep neural networks⁵⁴¹ or for online learning⁵³⁷.

While the cycle-to-cycle variation challenges are starting to be measured and addressed, Major challenges still exist in understanding and improving the device-to-device variation of the devices, which arises from variation in the domain wall location and pinning landscape. A discussion of influence of device variability on circuit performance is presented in Xiao et al⁵¹⁹. Better understanding of experimental viability of the technology is needed, given TMR variability and constraints and scaling-induced variability and errors.⁵⁴² There is some initial work addressing variability through using VCMA⁵¹³. Thin film stack growth and engineering needs to be done to adapt the traditional MRAM-like stack for DW-MTJs, including minimizing stray magnetic fields from the pinning layers to remove the need for a DC bias field when the device is operated. Example ideal applications of the device technology are still needed, with some potential areas being radiation-hard environments, lower latency hardware accelerators, and low-area, low-energy needs of edge-computing internet of things devices. Experimental demonstrations of larger operating circuits and systems is still lacking. The experimental evidence of applications to bio-inspired neuromorphic computing is nascent with a few prototypes demonstrated, but many of the above predicted effects are only shown in simulation.

3.4.3. SPIN WAVE DEVICES

Spin Wave Device (SWD) is a type of magnetic logic devices exploiting collective spin oscillation (spin waves) for information transmission and processing.^{543,544,545,546,547,548} The basic elements of the SWD include: (i) magneto-electric cells (e.g., multiferroic elements) aimed to convert voltage pulses into the spin waves and vice versa;^{549,550} (ii) magnetic waveguides – spin wave buses for spin wave signal propagation between the magneto-electric cells,⁵⁵¹ (iii) magnetic junctions to couple two or several waveguides,⁵⁵² (iv) spin wave amplifiers,^{553,554} (v) phase shifters to control the phase of the propagating spin waves,⁵⁵⁵ (vi) spin wave phase error correctors,⁵⁵⁶ and (vi) non-volatile memory elements controlled by spin waves⁵⁵⁷. SWD converts input voltage signals into the spin waves, computes with spin waves, and converts the output spin waves into the voltage signals. Computing with spin waves usually utilizes spin wave interference and control of magnetic ground states by spin waves, which enables functional nanometer scale logic devices. Since the first proposal on spin wave logic,⁵⁴³ SWD concept has evolved in different ways encompassing volatile⁵⁵⁸ and non-volatile,⁵⁵⁹ Boolean^{559,560} and non-Boolean,⁵⁶¹ single-frequency and multi-frequency circuits.⁵⁶² The primary expected advantages of SWD over Si CMOS are the following: (i) the ability to utilize phase in addition to amplitude for building logic devices with a fewer number of elements than required for transistor-based approach;

(ii) power consumption minimization by exploiting the intrinsically low energy of spin waves in ferromagnets (~ 10 μeV) and antiferromagnets (~ 1 meV) as well as built-in non-volatile magnetic memory⁵⁵⁷ and magnetic reconfigurability⁵⁶³, and (iii) parallel data processing on multiple frequencies in a single core structure by exploiting each frequency as a distinct information channel.

Micrometer-scale SWD MAJ gate has been experimentally demonstrated.⁵⁶⁴ It is based on ferromagnetic $\text{Ni}_{81}\text{Fe}_{19}$ structure, operates within 1-3 GHz frequency range, and exhibits signal-to-noise ratio of approximately 10 at room temperature.⁵⁶⁴ The internal delay of SWD is defined by the spin wave group velocity (e.g., 3×10^6 cm/s in $\text{Ni}_{81}\text{Fe}_{19}$ waveguides). Power dissipation in SWD is mainly defined by the efficiency of the spin wave excitation mechanism. Recent experiments with synthetic multiferroics comprising piezoelectric (lead magnesium niobate-lead titanate PMN-PT) and magnetostrictive (Ni) materials have demonstrated spin wave generation by relatively low electric field (e.g., 0.6 MV/m for PMN-PT/Ni).⁵⁶⁵ The later translates in ultra-low power consumption (e.g., 1 aJ per multiferroic switching). Recent experimental demonstration of parametric spin wave excitation and amplification⁵⁵⁰ by voltage-controlled magnetic anisotropy (VCMA) is another promising route towards energy-efficient generation and control of spin waves.

A SWD-based magnonic holographic memory (MHM) has been proposed.⁵⁶¹ The principle of operation of MHM is similar to optical holographic memory, while spin waves are utilized instead of light waves. The first 2-bit MHM prototype based on yttrium iron garnet structure has been demonstrated.⁵⁶⁶ MHM also possesses unique capabilities for pattern recognition by exploiting correlation between the phases of the input waves and the output interference pattern. Pattern recognition using MHM has been recently demonstrated.⁵⁶⁷ Magnonic holograms have potential for high information density (about 1 Tb/cm²) due to the nanometer scale wavelength of spin waves. According to estimates, the functional throughput of magnonic holographic devices may exceed 10^{18} bits/s/cm.^{2,561} Another interesting recent direction of research is development of energy-efficient compact spin-wave Ising machines for solving NP-hard problems.⁵⁶⁸

There are two important milestones crucial for further SWD development: (i) nanomagnet switching by spin waves^{557,569} and (ii) integration of several magneto-electric cells on a single spin wave bus. In order to have an advantage over CMOS in functional throughput, the operational wavelength of SWDs should be scaled down below 100 nm.^{559,570,571} The success of the SWD will also depend on the ability to restore/amplify spin waves (e.g., by multiferroic elements⁵⁷², by magneto-electric pumping⁵⁷³ or by anti-damping spin-transfer and spin-orbit torques^{574,575}).

A very important recent direction of research is antiferromagnetic (AFM) SWD⁵⁷⁶ that offers a thousand-time increase in operation speed due to the THz-scale frequencies of the AFM spin waves. The frequency-momentum dispersion of AFM spin waves is linear, which results in high group velocity of AFM spin waves and faster information transfer within SWD-based circuits. In addition, AFM spin waves usually exhibit degeneracy due to the presence of multiple (typically two) sublattices. This degeneracy provides an additional degree of freedom for encoding information carried by AFM spin waves. For example, AFM spin wave polarization can be used to encode information⁵⁷⁷. Alternatively, information can be encoded in the phase shift between the degenerate spin wave modes, which can be used in topologically protected spin wave gates⁵⁷⁸. Challenges in AFM SWD development include: (i) efficient interconversion between spin wave signals and electric signals^{579,580} and (ii) efficient methods to control the AFM ground state, including the AFM domain control. Development of reliable AFM-based magnetic tunnel junctions with high tunnel magneto-resistance⁵⁸¹ is needed to address the former challenge. Magneto-electric control of AFM domains is a promising approach to the latter challenge. More work on characterization of the speed and bandwidth of spin-wave based devices⁵⁸² is needed.

An intriguing recent direction of research is SWDs based on two-dimensional van der Waals (2D vdW) materials⁵⁸³. Spin waves in both ferromagnetic^{583,584} and antiferromagnetic⁵⁸⁵ 2D vdW materials have been detected. The potential advantages of the 2D vdW materials are: (i) the ease of spin wave control by out-of-plane electric fields⁵⁸⁶ and (ii) the ability to easily assemble designer magnetic heterostructures. Major challenge in this field is identifying materials with low magnetic damping for long-range spin wave propagation.

3.5. OTHER BEYOND CMOS DEVICES

3.5.1. NANO-ELECTRO-MECHANICAL SWITCHES

Nano-Electro-Mechanical (NEM) switches use electrostatic force to mechanically actuate a movable structure to make or break physical contact between current-conducting *source* and *drain* electrodes. When the electrodes are separated physically by an air gap, no current flows across the gap, resulting in **zero OFF-state current**. The NEM switch undergoes an abrupt change in current conduction ability between non-contacting and contacting states, with nearly **zero subthreshold swing**.⁵⁸⁷ While zero OFF-state current provides for zero standby power dissipation, zero subthreshold swing enables very low operating voltages for low dynamic power dissipation as well. Moreover, a NEM switch can be operated with either positive or negative voltage polarity due to the ambipolar nature of the electrostatic force, so that an electrostatically actuated NEM relay can be configured to turn on either with increasing control (*gate*) voltage or with decreasing *gate* voltage and can serve as either a pull-down device (passing

a low voltage – ground – from the *source* to the *drain*, i.e., discharging the voltage at the *drain*) or a pull-up device (passing a high voltage – V_{DD} – from the *source* to the *drain*, i.e., charging the voltage at the *drain*).⁵⁸⁸ Additional advantages of mechanical devices include robust operation across a wide temperature range, down to cryogenic temperatures,⁵⁸⁹ and immunity to ionizing radiation⁵⁹⁰. NEM switches can be monolithically integrated with CMOS circuitry by a modular post-CMOS fabrication process with relatively low thermal budget.⁵⁹¹ Since state-of-the-art CMOS technology incorporates air-gapped interconnects,⁵⁹² NEM switches can be implemented using multiple interconnect layers formed in the back-end-of-line (BEOL) process.⁵⁹³ Potential applications for hybrid NEMS-CMOS technology include CMOS power gating,⁵⁹⁴ configuration of FPGAs,⁵⁹⁵ non-volatile back-up storage of information in SRAM and CAM⁵⁹⁶ cells, and energy-efficient, fast and reconfigurable look-up tables.⁵⁹⁷

Conventional planar processing techniques (i.e., thin-film deposition, lithography and etch processes) can be employed to fabricate the conducting electrodes of a mechanical switch. The air-gaps between electrodes are formed with a final “release” etch step in which a sacrificial material such as silicon dioxide, photoresist, polyimide or silicon is selectively removed. The switching delay and operating voltage of a NEM switch can be reduced by scaling down the size of these air-gaps. The smallest air-gap demonstrated to date for a functional NEM structure fabricated using a top-down approach is 4 nm, resulting in an actuation voltage of approximately 0.4 V.⁵⁹⁸ As expected, it exhibited very low OFF-state current and sub-threshold swing; however, it is only a 2-terminal device, not suitable for logic switch application. SiC nanowires have been used as the movable structure for NEM switches that are suitable for high-temperature operation. Functioning 2-terminal SiC switches with air-gaps as small as 10 nm and switching voltage as low as 1V have been demonstrated.⁵⁹⁹ 3-terminal devices and corresponding logic gates also were demonstrated.⁶⁰⁰ Piezoelectric materials have been incorporated in NEM devices to enable sub-1V switching⁶⁰¹.

The operating speed of a NEM switch is much slower than that of a transistor because it is dominated by mechanical delay related to the physical motion of the movable structure rather than the electrical (charging/discharging) delay; therefore an optimized relay-based IC design should arrange for all mechanical movements to happen simultaneously, so that the delay per operation is essentially one mechanical delay.⁷ For a pass-gate circuit topology, multiple switches are connected together in series to drive the output signal. This means that the *source* voltage can vary between the reference voltage (ground) and the supply voltage (V_{DD}). For proper pass-gate circuit operation, the state of the switch cannot be dependent on the *source* voltage; therefore a fourth electrode is necessary to provide a constant reference voltage, such that the voltage applied between the control (*gate*) electrode and the reference (*body*) electrode determines the state of the switch, i.e., whether a current path is established between the *source* and *drain* electrodes.⁶⁰² The contact and actuation air-gaps can be reduced by biasing the *body* electrode to reduce the magnitude of the *gate* voltage required to switch the device ON/OFF. Switches with multiple gate terminals have been implemented, forming single-device logic gates for higher functional density.^{603,604} A constant-field scaling methodology can be applied to miniaturize NEM switches for reduced footprint, switching delay and switching energy.⁶⁰⁵ With the aid of body biasing, the minimum switching energy for a nanoscale relay is anticipated to be on the order of 10 aJ, which compares well against the switching energy for an ultimately scaled MOSFET.^{606,607} Piezoelectric NEM devices have demonstrated 10 mV switching operation using body bias, providing for very low energy dissipation per switching cycle (23 aJ), and an extremely small subthreshold slope (0.013 mV/decade).¹⁵ A variety of relay-based computational and memory building blocks have been experimentally demonstrated to date.^{608,609}

The prospective system-level benefits of mechanical logic have been analyzed by performing simulation-based assessments of VLSI circuit blocks implemented with NEM switches. These indicate that relays can provide for more than 10x reduction in energy per operation as compared with MOSFETs and can reach clock speeds in the GHz regime.⁶¹⁰ Thus, a major potential advantage of NEM switch technology is **improved energy efficiency**. Moreover, the contact adhesive force and structural stiffness can be engineered to achieve bi-stable operation, which makes mechanical switches attractive for embedded non-volatile memory applications.⁶¹¹ Recently hybrid CMOS-NEMS circuits have been demonstrated, with non-volatile NEM switches operating at the same V_{DD} as for the CMOS circuitry.⁶¹² The BEOL metallic layers used to form interconnects in a conventional CMOS process can be leveraged to implement compact NV-NEM switches for dynamically reconfigurable circuit functionality⁶¹³. Monolithically integrated hybrid CMOS-NEM circuitry has been fabricated and circuit functionalities have been demonstrated utilizing BEOL NEM switches. Arrays of non-volatile NEM switches have been leveraged to demonstrate look-up table functionality⁶¹⁴ and data search operation.⁶¹⁵ These hybrid circuits consequentially have **improved area efficiency** due to the vertical integration of BEOL NEM switches on top of FEOL transistors. Recently, a CMOS+NEM CAM array fabricated using a 65-nm CMOS manufacturing process demonstrated 3X improvement in power consumption and 9X improvement in area relative to a traditional CAM cell.⁶¹⁶ Moreover, hybrid CMOS+NEM circuits with an all-CMOS signal path were demonstrated, that eliminate issues arising from the NEM switch contact resistance and still provide for improved power and area efficiency as compared with traditional CMOS implementations.⁶¹⁷ Reprogrammable NEM switches, with air gaps as small as 32 nm, have been fabricated using a standard 16-nm CMOS manufacturing process.¹⁶

For all of the aforementioned applications, the NEM switches need to operate reliably and consistently for at least 10^9 cycles. Due to the extremely small mass of the movable electrode (less than 1 ng), neither gravitational acceleration nor mechanical

vibration substantially affects their operation. Structural fatigue or creep can be easily avoided by designing the movable structure/anchor such that the maximum induced strain is well below the yield strength. However, permanent stiction can be a mode of device failure: for soft electrode materials such as gold and platinum, Joule heating at the contacting asperities can lead to atomic diffusion (welding). This issue can be mitigated by using a refractory electrode material such as tungsten to minimize contact wear and by reducing the peak voltage difference between the *source* and *drain* electrodes (V_{DD}) when they are in contact. NEM switches with tungsten electrodes have been demonstrated to have endurance up to 1 billion ON/OFF cycles at 2.5 Volts for a relatively large load capacitance of 300 pF (i.e., exaggerated electrical delay), and endurance exceeding 10^{16} ON/OFF cycles is projected for voltages below 1 Volt and load capacitance below 1 pF.⁶¹⁸ Novel one- and two-dimensional materials have recently been explored to improve cycling endurance as well. A self-assembled carbon nanotube array was implemented as a NEM contact for high temperature applications, enabling over one million switching cycles at 550 °C⁶¹⁹. The remaining reliability challenge for NEM logic switches is degradation (dramatic increase) of on-state resistance due to electrode surface oxidation or the formation of friction polymers over the lifetime of device operation. Alternative contacting electrode materials such as Ru/RuO₂ and/or hermetic packaging are potential solutions to this issue.

Adhesive force between the contacting electrodes dictates the minimum required stiffness of the movable structure, which in turn determines the minimum *gate-to-body* voltage required to switch the NEM relay. The adhesion energy is determined by metallic bonding (at the contacting asperities) and by van der Waals force (in the non-contacting regions of the electrodes). Self-assembled molecular (SAM) coating of hydrophobic materials has been demonstrated to reduce the adhesive force and thereby enable switching operation at lower *gate* voltage.⁶²⁰ Sub-50 mV operation of relay integrated circuits demonstrating OR, AND, and XOR gate functionality have been demonstrated with body-biased SAM-coated NEM switches.⁶²¹ Operation of NEM switches and integrated circuits at temperatures down to 4K were demonstrated recently for the first time.⁶²² Due to reduced adhesion energy and elimination of contact oxidation, relays can be operated reliably with voltages as low as 25 mV for more than 10^8 cycles at cryogenic temperatures, showing promise for monolithic integration of multiplexing control circuitry with qubits. Most recently, coupled MEM relays have been proposed to implement an Ising machine utilizing the bias-dependent oscillatory behavior of the relays.⁶²³ In conclusion, the negligible OFF-state current and ultra-low-voltage operation capability of NEM switches make them a compelling option for ultra-low-power digital computing applications such as the Internet of Things, particularly where resilience to extreme temperatures and/or immunity to radiation are required. Furthermore, hybrid CMOS-NEM technology shows promise for enhanced chip functionality, e.g., with dynamically reconfigurable interconnects. Remaining challenges to realizing the promise of mechanical computing include materials and process optimization to achieve stably low contact resistance with minimal contact adhesive force. Further work is also needed to integrate NEM switches on the most advanced CMOS platforms and improve their reliability for future IoT device applications of hybrid CMOS-NEM circuits.

3.5.2. TOPOLOGICAL INSULATOR ELECTRONIC DEVICES

Topological insulators are recently discovered materials that possess a bandgap in their interior, however the topology of their electronic states necessitates that the existence of gapless, conducting modes on their boundaries – one dimensional (1D) edges in the case of two-dimensional (2D) topological insulators, and 2D surfaces in the case of three-dimensional (3D) topological insulators^{624,625,626,627}. These conducting modes are protected from backscattering by symmetry, and in the case of 1D edge modes of 2D topological insulators, are expected to be ballistic conductors with conductance $e^2/h = 38.7 \mu\text{S}$ per edge, which has been confirmed experimentally in several materials^{628,629,630}.

Systematic searches of materials databases have found that topological materials are commonplace, representing a significant fraction of all known materials^{631,632,633}. Two-dimensional topological insulators have been realized with very large bandgaps of 360 meV (Na₃Bi⁶³⁴) and 800 meV (bismuthene⁶³⁵), significantly exceeding the thermal energy at room temperature (25 meV), which indicates that topological phenomena may be robust at room temperature in suitable materials.

Numerous proposals have been put forward to exploit topological materials in transistor design^{636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648}. One topological transistor design envisions switching a 2D material from topological insulator (“on” with ballistic edge channels) to a conventional insulator (“off”). This switching may be accomplished via electric field through several mechanisms such as Rashba effect^{637,647}, staggered sublattice potential^{636,637,640,641}, inversion symmetry breaking⁶³⁸, or Stark effect^{634, 639, 642, 643, 646}.

Electric field effect switching has been proposed in a number of materials including graphene⁶³⁷, two-dimensional Xenes (e.g., germanene, stanene, etc.)^{636,641,647}, monolayer transition metal dichalcogenides in 1T' phase⁶⁴³, SnTe and Pb_{1-x}Sn_xSe(Te)⁶³⁸, topological semimetals such as Cd₂As₃ and Na₃Bi^{634,642}, and phosphorene⁶³⁹. Electric-field switching of the topological state has been demonstrated in at least two materials, Na₃Bi⁶³⁴ and germanene⁶⁴⁹. Other transistor proposals have focused on electric-field switching of tunnelling between topological edges⁶⁵⁰, or using strong disorder to produce an off state in a bulk conducting topological insulator⁶⁴⁴.

Electric field effect switching in strongly spin-orbit coupled topological materials has been shown to lead to low-voltage switching in a topological transistor⁶⁴⁷. For staggered honeycomb lattice topological insulators in the Xene family, electric field acts in two ways; inducing a staggered sublattice potential and creating a Rashba spin-orbit coupling. Both effects act to open a bandgap in the topological insulator. A device termed a topological quantum field-effect transistor (TQFET) exploits this combined electric field effect on the bandgap, which is used as the barrier to electron flow in the “off” state of the transistor. Due to the combined effects of sublattice potential and Rashba spin-orbit coupling on the bandgap, the TQFET can overcome Boltzmann’s tyranny with a subthreshold swing lower than $kT\ln 10/e \approx 60$ mV/decade at room temperature^{498,651}.

Further reductions in switching voltage can be achieved by combining the TQFET concept with a negative capacitor to further reduce the subthreshold swing. Negative capacitance (realized, for example, in a ferroelectric layer) in series with a positive capacitance has previously been proposed to reduce the subthreshold swing in conventional MOSFETs⁶⁵². However, in a well-designed MOSFET in the subthreshold region (in which the gate capacitance dominates over stray capacitances to the channel), the gain due to negative capacitance is negligible⁶⁵³. This is because the negative capacitor in series with a positive capacitor acts to amplify the electric field in the positive capacitor at a given voltage. However, in an ideal conventional MOSFET in the subthreshold regime, the gate electric field is negligible as the channel electric potential follows that of the gate, hence the device cannot benefit from amplification of the electric field in the channel.

A TQFET switches by a qualitatively different mechanism, applying an electric field across the insulating channel, rather than changing the electrostatic potential of the channel as in a MOSFET. Thus, a TQFET can benefit maximally from electric field amplification in a negative capacitance structure. A negative capacitance topological quantum field-effect transistor (NC-TQFET) incorporates a stack consisting of gate electrode, ferroelectric, topological channel, ferroelectric, ground plane; with the ferroelectric layers having an effective negative capacitance⁶⁴⁸. The entire stack is designed to have a net positive capacitance to ensure absence of hysteresis and amplification of the electric field in the topological channel layer.

Benchmarking of a hypothetical NC-TQFET using a bismuthene topological channel and La-doped HfO₂ ferroelectric layers demonstrated a 10X reduction in switching voltage and 8X reduction in switching energy compared to a GAAFET CMOS LV⁶⁵⁴, with high on-current⁶⁴⁸. Further gains appear possible by using ferroelectrics with higher remnant polarization. While the NC-TQFET design is promising for low-voltage devices, many challenges remain to realize such a device. An appropriate topological channel material in the Xene family must be demonstrated experimentally and integrated with an appropriate ferroelectric layer of precisely controlled thickness.

In addition to switching via electric field, topology may also be controlled by magnetic fields^{655,656}, strain⁶⁵⁷, temperature⁶⁵⁸, electric polarization⁶⁵⁹, or time-dependent electric fields (such as light)^{660,661}, hence topological transistors have the prospect of adding additional new functionality for “More than Moore” devices.

The deep connection between topological insulators and spin-orbit coupling also suggests further synergies between topological electronics and spin electronics. Indeed, 2D topological insulators exhibit a quantized spin Hall effect and can be used to produce completely polarized spin current⁶³⁷, of potential use in spintronics devices. Magnetic 2D topological insulators (quantum anomalous Hall effect) may produce perfectly spin polarized current with spin direction determined by magnetization direction^{627,662}, opening new possibilities for spin transistors and non-volatile random-access memory⁶⁶³. Near-perfect ballistic conduction in the quantum anomalous Hall (QAH) regime with current-induced magnetization switching at very low currents (1 nA) was recently demonstrated in graphene/boron nitride heterostructures⁶⁶⁴ albeit at cryogenic temperatures. Recent theoretical work has shown that electric fields may be used to accomplish a topological transition from QAH to insulator in a MnBi₂Se₄/Bi₂Se₃ heterostructure⁶⁶⁵, while other work has shown that electric field could be used to control the magnetic order in similar MnBi₂Te₄ monolayers⁶⁶⁶, suggesting that electric-field controlled magnetic logic and memory devices could be feasible in an appropriate magnetic topological insulator platform. Analogous topological effects may be realized for other degrees of freedom such as the valley degree of freedom in materials with multiple conduction valleys, with analogous quantum valley Hall effects switchable by electric field^{667,668,669}.

Many challenges remain to realizing low-voltage topological transistors. While numerous topological materials have been identified^{631, 632, 633}, searches have not focused on key materials parameters necessary for efficient topological transistor operation, such as a large accessible modulation of bandgap (particularly in the non-topological “off” state which will determine the subthreshold characteristics). While some 2D topological insulator materials have been grown on large-area substrates, device-level demonstrations over large areas have not been demonstrated yet, nor integration with conventional silicon substrates and processes. Integration of topological materials with other functional materials such as ferroelectrics or ferromagnets also remains a challenge.

4. EMERGING DEVICE-ARCHITECTURE INTERACTION

4.1. INTRODUCTION

As the pursuit of Moore's Law scaling faces increasing physical and economic challenges, the trajectory of computing beyond traditional CMOS paradigms demands a fundamental rethinking of how devices and architectures interact. Future advancements will not stem solely from isolated improvements in either domain, but rather from their synergistic co-design and co-optimization.

Many new emerging Beyond-CMOS devices will require co-design between devices and higher levels of computer design (e.g., circuit, architecture and application). These emerging devices are not intended simply as “drop-in” replacements for standard CMOS devices, but will require new types of circuit designs, new functional module architectures, and even new software to best utilize the new devices' capabilities.

Novel design issues spanning the device and architecture levels especially need to be considered when adopting new low-level computing paradigms. In such situations, devices may be organized in radically new ways to carry out computation in a very different manner from what we may consider the most “conventional” computing paradigm, which has relied on von Neumann architectures using standard combinational and sequential irreversible Boolean logic. Examples of unconventional or alternative computing architectures and systems that comprise new computing paradigms are listed in Figure BC4.1 and include neuro-inspired systems, dynamical systems, probabilistic systems, reversible systems and digital in-memory systems. These systems are composed of new computational functions and enabling devices as shown in Figure BC4.2.

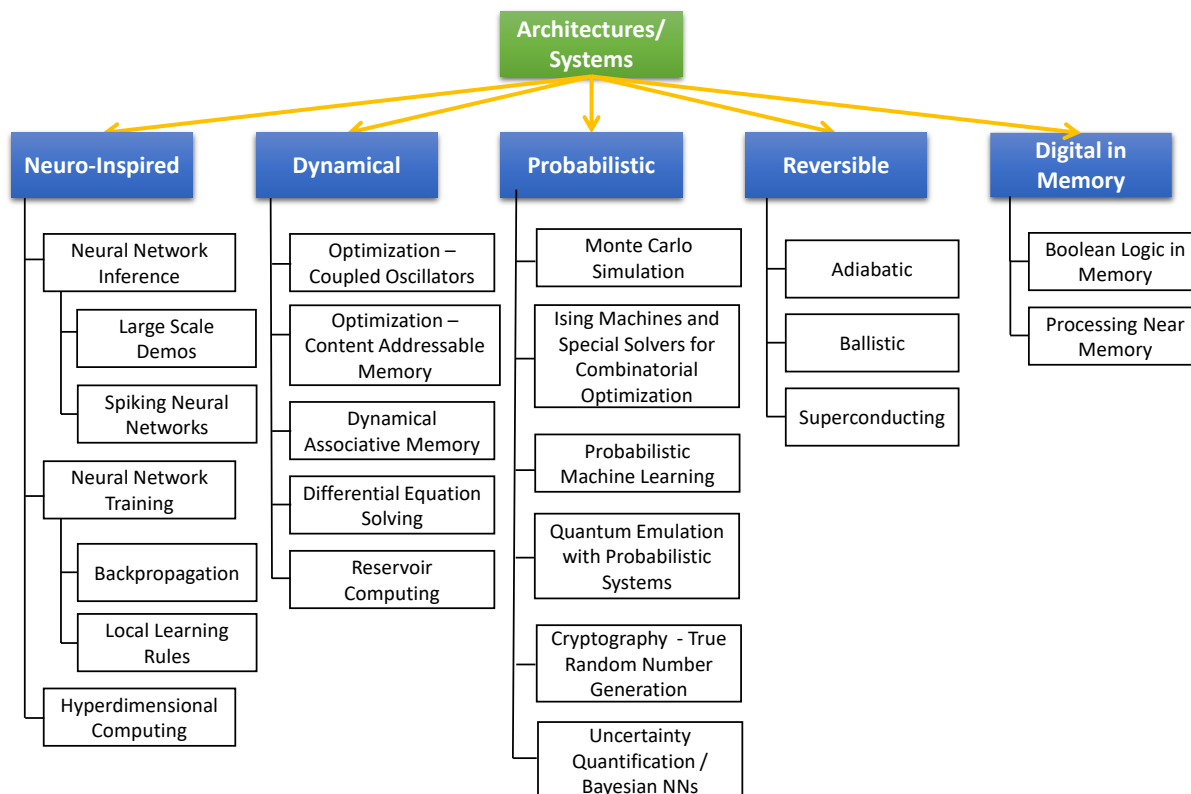


Figure BC4.1. Architectures/Systems for Novel Computing Paradigms Discussed in This Chapter Requiring Codesign with Emerging Devices

Neuro-inspired, dynamical, and stochastic systems all typically leverage analog computing. Analog computing attempts to “let physics do the computation” by using physical processes directly (as opposed to, by going through the traditional digital abstraction barrier) to compute complex functions. Historically, this required inefficient analog circuitry for all elements, and expensive analog to digital conversion, resulting in limited applications, specifically those requiring analog signal processing. Recently, new analog devices have enabled a new generation of efficient analog architectures. This is especially true for hybrid

analog and digital systems where efficient designs may exploit analog preprocessing and computation prior to digitization. Analog preprocessing can reduce the required A/D precision and therefore reduce the system energy consumption by orders of magnitude. Additionally, new architectures can be used for ultra-low-power co-processors for conventional CMOS designs. A key challenge is that analog signals are typically low precision, with energy and latencies increasing exponentially with higher bit precision. Fortunately, many machine learning and other applications are being developed that can tolerate such lower precision computation.

All these architectures except for reversible computing are special purpose accelerators designed to accelerate a specific function. Reversible computing has the unique property that it can do general digital computing at lower energy than is possible with conventional methods.

The reader should note that the material in this section is not intended to comprise an exhaustive list of *all* possible new computing paradigms, new devices, new circuits, or new architectures. It is only intended to serve as a representative sample of several new general computing paradigms and specific technology concepts.

In this section we first describe enabling computational functions (§4.2), then systems and architectures exhibiting novel computing paradigms (§4.3), and finally we discuss some of the enabling devices (§4.4).

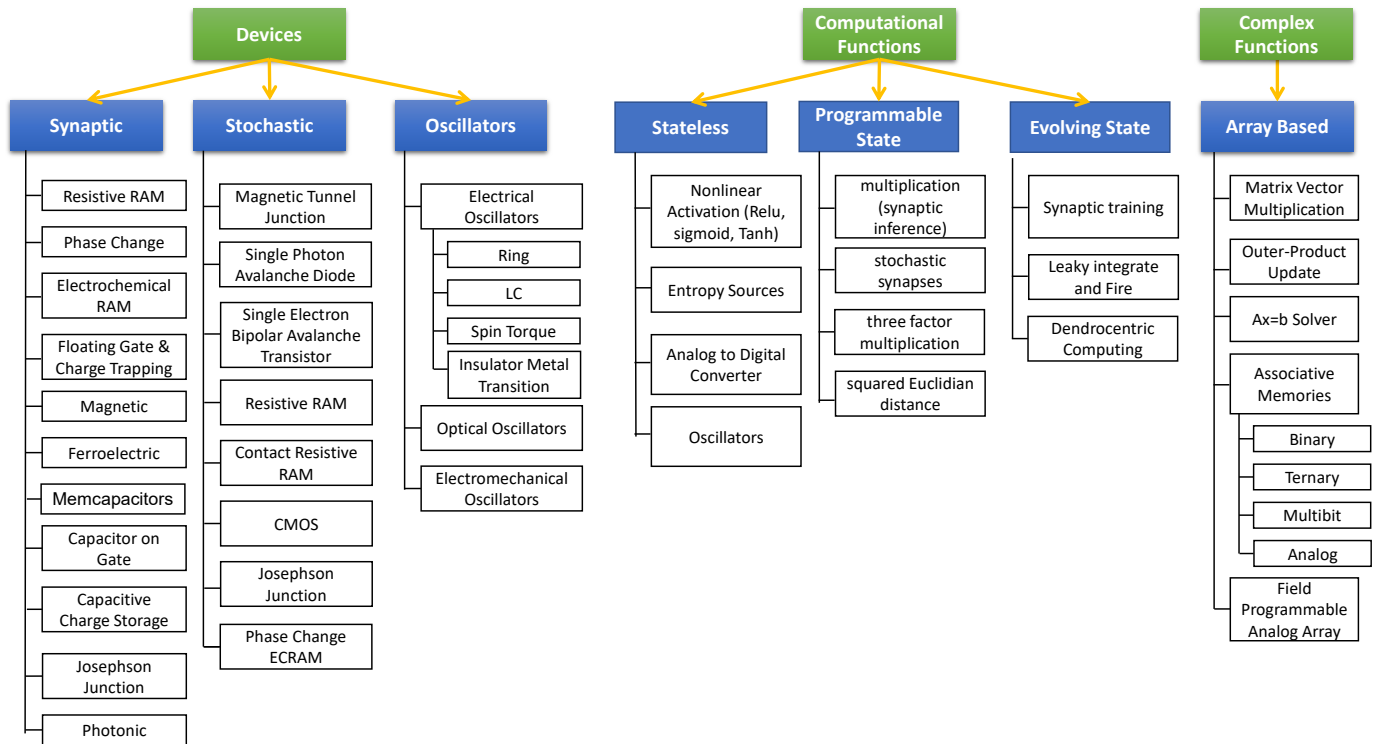


Figure BC4.2. *Emerging Devices and Computational Kernels Requiring Codesign between the Device Layer and Higher Layers of the Technology Stack*

4.2. COMPUTATIONAL FUNCTIONS

Emerging computing systems are composed of enabling functions that perform some novel computational function more efficiently than a corresponding digital implementation. At the device level these functions can be divided into three categories: stateless functions, non-volatile state functions, and evolving state functions. Stateless functions take an input and transform it without using any internal state. Programmable state functions have an internal programmable state that is used to modulate function that is computed with the input. Often programming this state is expensive and done infrequently. Evolving state functions have a dynamic internal state that changes with every input.

These functions can then be composed into larger array-based kernels that perform larger scale computations (§4.2.4).

4.2.1. STATELESS FUNCTIONS

Stateless functions take an input and transform it without using any internal state.

In simpler neural architectures, these often function as neurons. This allows for a complex expensive function to be computed using a single or few devices in an energy efficient manner, obviating the need for time- and energy-consuming ADC/DAC in the circuits. Neuron kernels are also often active devices that restore a signal and inject energy into a system. If a passive device is used in large systems, it is often paired with an active driver circuit to power downstream synaptic kernels. Just as in digital logic, energy periodically needs to be injected into a system or the signal will decay. In many applications neurons provide a natural place to do so.

The benchmarks used to evaluate electronic neurons generally measure the energy per operation, the fabrication cost, or the chip area of the integrated functional block, and the fidelity to the desired neuron function (e.g., integrate and fire). High reliability and low variation of devices are two key factors for the viability of a neuron technology. Device failure will require a lot more circuitry for error detection and correction.⁶⁷⁰ Large variation increases the difficulty for designing peripheral circuits and degrades the adaptability of the block.

4.2.1.1. NONLINEAR ACTIVATION (THRESHOLD, RELU, SIGMOID, TANH)

One of the most basic and common neuron functionalities is to provide a nonlinear activation function. In many neural networks, after a linear multiply accumulate is performed on a set of inputs, or in a hidden layer, a non-linear activation or transfer function is applied. The presence of this function prevents the network from mathematically collapsing into a single linear equation, which helps improve the computational capabilities as the number of layers increases.

This can be as simple as a binary threshold that outputs a 1 if the input crosses a predefined threshold or a 0 otherwise. A binary threshold can be implemented both with an analog comparator or a new device. In more conventional neural networks, functions like rectified linear units, sigmoid functions and Tanh functions are common. There are many device candidates that can approximate these functions. If a device transfer function does not exactly model the intended numerical function, the rest of the system can often be adapted to compensate (i.e., a neural network can be trained to use the transfer function available)

4.2.1.2. ENTROPY SOURCES

Probabilistic systems can have randomness be injected in many ways. If the randomness is intrinsic or only tuned by an input such as in a p-bit, it is a stateless entropy source. Possible stochastic devices are surveyed in §4.4.3.

4.2.1.3. ANALOG-TO-DIGITAL CONVERTER

While conventional analog-to-digital converters are often used by many architectures. They allow for flexibility for many different subsequent computations, but are costly. Nevertheless, for many systems, the bulk of the computation is accelerated by the array-based kernels. Consequently, it is often possible to average out the cost of an analog-to-digital converter and still provide system level advantages. Using digital signals between analog array kernels also allows for the use of flexible digital routing to enable more flexible computing accelerators.

4.2.1.4. OSCILLATORS

Many dynamical systems (§4.3.2) are based on coupled oscillators. Possible device implementations are surveyed in §4.4.2.

4.2.2. PROGRAMMABLE STATE FUNCTIONS

Programmable state functions typically represent the memory in a novel system and must have a programmable internal state and use that state to modulate an input, most commonly by multiplying the internal state with the input. By merging memory with a simple computation, synaptic devices allow emerging systems to overcome the von-Neumann bottleneck by merging processing and memory. Programmable state functions are also typically passive devices that modulate an incoming signal. In neural architectures these would represent synapses. As the state is infrequently updated, it's acceptable for devices to have moderate write endurance and for the write process to be more costly than the read process.

4.2.2.1. MULTIPLICATION (SYNAPTIC INFERENCE)

The most common kernel is synaptic inference where a stored state is used to modulate an input. The simplest form of this is to use an analog memory device as a programmable resistor and to do multiplication using Ohm's law $V = I \times R$, or equivalently $I = V \times G$. Many variations on this are possible. Nonlinear devices can be used to implement non-linear functions and multiple devices can be combined to provide pattern matching or associative memory functionality (§0). Enabling devices are surveyed in §4.4.1.

4.2.2.2. STOCHASTIC SYNAPSES

For probabilistic systems (§4.3.3), having a good source of randomness is critical. Often a stochastic synapse will have a programmable mean state that varies stochastically around that mean state. For many applications it is also critical to be able to tune the probability distribution or standard deviation of the variation around the mean. This functionality can be implemented in a single device or in a combination of devices (one to tune the mean state and one to tune the standard deviation). Possible stochastic devices are surveyed in §4.4.3.

4.2.2.3. THREE FACTOR MULTIPLICATION

A critical challenge in polynomial based algorithms is multiplying three quantities together. An internal state needs to be multiplied by two operands. This can be done by encoding inputs using different state variables such as time and voltage. An example of this is in 671.

4.2.2.4. SQUARED EUCLIDEAN DISTANCE

The squared Euclidean distance is the most common distance measure used to determine the distance between 2 points. Circuits have been recently proposed to compute this in memory where the output current is proportional to the square difference between an input voltage and a programmed transistor threshold voltage ⁶⁷²

4.2.3. EVOLVING STATE FUNCTIONS

The last type of function is an evolving state function where the stored state is updated often, typically at every computation. Devices that implement these functions need to have extremely high write endurance as the state will be constantly updated. The write energy also needs to be low, on the same order as the read energy. This often limits the type of devices that can be used.

4.2.3.1. SYNAPTIC TRAINING

When training a neural network, the weights need to be constantly and efficiently updated. Synaptic devices in a training accelerator need to both use a stored value to modulate an input and the stored value needs to be programmable. If the values can be programmed in parallel, the devices can also implement training functionality like outer product updates (§4.2.4.2), three factor local learning rules (§4.3.1.2.2) and spike timing dependent plasticity.

4.2.3.2. (LEAKY) INTEGRATE AND FIRE

For systems that operate using temporal dynamics (leaky) integrate and fire neurons are common. An input signal is integrated until a threshold is reached and then a binary spike is created. Mott memristors,⁶⁷³ phase change based memristive switches,⁶⁷⁴ and chalcogenide threshold switches⁶⁷⁵ have all been reported to be capable of performing temporal voltage signal integration in which the effects of non-simultaneous unitary post-synaptic potentials add in time. Leaky integrate and fire functionality can also be implemented using standard analog circuitry.

4.2.3.3. DENDROCENTRIC COMPUTING

In neural systems dendrites often perform several different complex computations integrating multiple sources of information. One approach called dendrocentric learning works by ordering inputs along a short stretch of dendrite⁶⁷⁶. This means that the hardware implementing the dendrite needs to be able to store the order in which inputs arrived. One physical realization of this utilizes the ferroelectric layer in a segmented multi-gate FeFET design to detect a consecutive sequence of input pulses⁶⁷⁷.

4.2.4. ARRAY-BASED KERNELS

Analog crossbars or memory arrays can perform low-precision matrix operations in parallel, by processing analog data directly at each memory element. Because a digital memory must individually access each memory cell and move the data to a separate computation unit, digital systems consume more energy and incur longer latencies. Computing on larger crossbars/matrices allows for any analog overhead to be averaged out over many matrix elements. Any two- or three-terminal device that features a modifiable internal physical state variable (which might be, for example, a variable resistance, a variable capacitance, a stored charge, or a stored magnetic field) that modifies the device's behavior can be used as a building block for analog operations. Several different types of array architectures are summarized in the following sections.

4.2.4.1. MATRIX VECTOR MULTIPLICATION (MVM) AND VECTOR MATRIX MULTIPLICATION (VMM)

MVM and VMM are key computational kernels underlying many different algorithms. Although these terms are sometimes used interchangeably, we refer to MVM as the operation Ax , where A is a weight matrix and x is the input vector. VMM uses the transpose of the same matrix: $A^T x$. In an analog accelerator, MVM is used during the forward propagation step (inference) while VMM is needed during the backpropagation steps (training).

There are several approaches to accelerating these kernels. Any programmable resistor such as a two-terminal resistive memory or a three-terminal floating gate cell can be used.⁶⁷⁸ Alternatively, a capacitive MVM can be designed by adding charge from capacitive memory elements.⁶⁷⁹

For many algorithms such as neural network inference (of an already-trained network), accelerating MVM accelerates the bulk of the computation,⁶⁸⁰ allowing for large system level efficiency gains. An $N \times N$ crossbar accelerates $O(N^2)$ operations, leaving only $O(N)$ inputs and outputs that need to be processed and communicated. This allows each unit of communication and processing cost (such as analog-to-digital conversion) to be amortized over N memory elements. This changes the tradeoff for some neural-network algorithms as larger crossbars are more efficient than smaller ones, provided that the algorithm can make use of them. There are ultimately several factors that limit how large a crossbar can be made, including: (1) accumulation of device errors on a column^{695,681}, (2) accumulation of voltage drops induced by array parasitic resistance,⁶⁹¹ and (3) ability to isolate individual devices during programming.⁶⁹²

Analog MVMs have been used for experimental demonstration of threshold logic,⁶⁸² compressed sensing initial filtering,⁶⁸³ robotic navigation and control,⁶⁸⁴ adaptive filtering,⁶⁸⁵ Fourier transforms⁶⁸⁶ and more. Additionally, Analog MVM techniques have been used for ultra-low power classification and neural networks.⁶⁸⁷

4.2.4.1.1. RESISTIVE MVM AND VMM

With a few exceptions, resistive MVMs are based on using Ohm's law, $V = I \times R$, to perform multiplication, and Kirchhoff's current law to perform addition by summing currents. Programmable resistors are used to program the weights, but the resistive element does not need to be a two-terminal resistor (like ReRAM or PCM); devices with three or more terminals (such as flash memory, FeFET, or ECRAM) can be used as programmable resistors as long as the conductance between two of the terminals can be modulated. Arranging the memory elements in an array allows for the entire operation to be performed in a single parallel step, giving a fundamental $O(N)$ energy and latency advantage over a standard digital memory that, at best, must access a memory array one row at a time.⁶⁸⁸ An MVM and the transpose VMM can be performed on the same memory array, by changing whether the rows of an array are driven and the columns are read, or vice versa.^{689,690}

The key metrics for a resistive MVM are 1) the energy per multiply-and-accumulate (MAC) operation, 2) latency per MVM, 3) crossbar and supporting circuitry area per matrix element, 4) crossbar dimensions, 5) input/output bit precision for digitally driven MVMs, and 6) the standard deviation of the noise or error per conductance when programmed as a percentage of the absolute conductance range. Relating to #6, the absolute error is more important than the error relative to a given conductance level. This is because in an MVM, it is the absolute conductance errors that are summed by Kirchhoff's law along a column of the array, not the relative errors. For example, a 10% error in a 1 μS device contributes less error to the MVM than a 1% error in a 100 μS device.

If high-resistance memory elements ($R_{\text{on}} = 100 \text{ M}\Omega$) with good analog properties are developed, one ReRAM based crossbar design projects that each multiply and accumulate operation will require 12 fJ when using 8-bit A/Ds and only 0.4 fJ when using 2-bit A/Ds.⁶⁹⁰ The latency for a 1024×1024 MVM will only be 384 ns or 11 ns for 8-bit or 2-bit A/Ds, respectively. This is over 100 $\times$ better than an optimized SRAM-based accelerator, which would require 2,700 fJ and 4,000 ns for 8 bits. The area per weight for the 8-bit A/D ReRAM accelerator is 0.05 μm^2 , 16 $\times$ better than the 0.8 μm^2 needed for an SRAM accelerator. The energy and latency are dominated by the A/D circuitry and not by the crossbar itself, with the A/D converters and digital circuitry occupying 10 $\times$ the area of the ReRAM array itself.

To allow for large arrays and minimize parasitic resistance drops along the metal interconnects, high resistance ($\sim 100 \text{ M}\Omega$) memory elements are needed. The higher the resistance, the larger the array possible, and the more any A/D costs and system level communication costs are amortized out.⁶⁹¹ However, such high resistances would prolong and potentially complicate the process of programming each conductance value accurately to encode already-trained neural network weights or matrix element values.

A key design choice is the bit precision of the inputs and outputs to the crossbar. The fewer bits are needed by an algorithm, the more efficient the crossbar is. If analog or binary inputs/outputs can be used, the A/D costs can also be avoided. The inputs to the crossbar can be encoded in voltage, time, or digitally. Voltage encoding applies different voltages to represent different analog input values. This requires circuitry to create different input voltages, and it requires that the memory elements have a linear I-V relationship, greatly complicating the use of nonlinear access devices.⁶⁹² Encoding inputs in variable length pulses requires longer reads and an integrator to sum the resulting current. Digital encoding applies each bit of the input sequentially and then combines the result digitally.⁶⁷⁹ For digital encoding, the usefulness of the lower-order bits in the input is limited by the noise/errors on the highest-order bit. To save on ADC costs, each bit position can also be combined in analog using successive integration and rescaling⁶⁹³.

The precision with which each resistor needs to be programmed depends on the application. Neural networks trained for simple image classification tasks such as MNIST digit recognition can often tolerate as much as a 15% error or noise on their conductance values before losing accuracy. The requirement is much more stringent for the ImageNet dataset, which is much more complex (100× larger images, 100× more classes than MNIST) and is considered more representative of a real-world application. For example, the ResNet50 neural network, commonly used for benchmarking digital accelerators⁶⁹⁴, tolerates only about 1% weight error and requires 8-bit activations and ADCs⁶⁹⁵.

For many applications, it is useful not to treat the required resistor programming precision as a single number, but to instead consider the desired precision as a function of the programmed resistance (or conductance). This is because the actual weight values programmed into the resistive array may not be uniformly distributed, and thus certain conductance ranges may be much more heavily utilized than others. The precision of the more heavily used weight values generally have a larger effect on the end-to-end algorithm accuracy. A good example is neural network inference. It is well known that for many neural network models, the weights have a skewed distribution where values near zero are the most abundant.⁶⁹⁶ Therefore, for this application, the resistances which encode low weight values should be made as precise as possible, while precision for the high weight values is less important because these tend to be outliers.⁶⁹⁵ One class of devices that has high precision at low conductance is flash transistors operated in the subthreshold regime.⁶⁹⁷

To extend the precision of computation beyond the limits of reliable programming, a technique called bit slicing can be used.⁶⁹⁸ With bit slicing, a matrix of wide operands is striped across multiple crossbars, enabling the crossbars to collectively perform computation on arbitrarily wide operands at the cost of additional digital circuitry to reduce partial results from multiple bit slices. When bit slicing is combined with digital input encoding, each bit of the input must be applied to each bit slice, analogous to multibit scalar multiplication. Leveraging bit slicing, accelerators for a wide range of applications have been proposed, including combinatorial optimization,⁶⁹⁸ neural network inference,^{699,700} graph analytics,⁷⁰¹ and scientific computing.⁷⁰²

The resistive MVM scheme described above is sometimes called the current-mode MVM, because the dot products are accumulated as currents. A notable alternative is the voltage-mode MVM, which also uses programmable resistors but has both voltage inputs and voltage outputs.⁷⁰³ In this scheme, the columns are not held at a virtual ground, and instead are left floating. The output voltage on the columns settles to a value that is defined by the voltage divider network that is formed between the various input voltages and programmed resistance values. This can be shown to *nearly* implement an MVM operation, but with a column-dependent normalization factor: $V_{out,j} \sim \sum_i G_{i,j} V_i / \sum_i G_{i,j}$. To obtain the desired dot product, the output can be de-normalized by the sum of conductances on the column, which is typically done in the digital domain. The main advantage of this approach is that it removes the need for a virtual ground, which needs to be supplied by an amplifier circuit. Also, since V_{out} is converted to digital only after the circuit has settled, the effect of transients can be reduced compared to a current integration approach. The voltage-mode MVM approach requires similar requirements on the memory device to achieve high precision, but additionally requires that the memory devices have linear I-V characteristics. Because the column voltage is floating, the voltage across each resistor can change continuously, even if the input voltages are binary, as when digital encoding is used.

4.2.4.1.2. CAPACITIVE MVM

Some analog accelerators propose crossbar-based parallel MVMs using volatile capacitive memories, where a weight value (binary or multi-bit) is stored as charge on a capacitor. Analog multiplication of the capacitive weight and a binary voltage input can be carried out using a multiplying digital-to-analog converter (MDAC) circuit; the resulting currents can be summed on a wire as in the resistive MVM case.⁷⁰⁴ Another implementation uses a binary input to control the redistribution of charge between two capacitors. This charge movement induces a voltage change on a metal lines, and the voltage changes induced by all the capacitive cells on a column are summed to form a dot product.⁶⁷⁹

Some other techniques combine capacitive and resistive MVM. For example, an analog weight can be stored as charge on a capacitor, which is also connected to the gate of a transistor. The amount of charge then controls the source-drain resistance of the transistor.^{705,706} One challenge of these schemes is the fact that capacitor charge is not truly non-volatile and can decay within milliseconds to seconds.

4.2.4.1.3. SRAM-BASED COMPUTE-IN-MEMORY MVM

SRAM-based Compute-in-Memory (CIM)⁷⁰⁷ designs have gained traction as they address the limitations of conventional architectures by allowing in-memory computations. Among these computations, Matrix-Vector Multiplication (MVM) is particularly critical in AI/ML workloads, as it forms the backbone of numerous workloads, including deep neural networks (DNNs). The ability to perform MVM directly within the memory array accelerates processing and significantly reduces power consumption, making SRAM-CIM⁷⁰⁸ an attractive option for edge computing applications.

SRAM technology is a promising choice for CIM implementations due to its robustness, reliable integration with CMOS technology, high endurance, and suitability for small to medium-capacity systems. Despite not offering the same area density as

embedded non-volatile memory (eNVM) based CIM solutions⁷⁰⁹, SRAM's superior attributes make it an ideal candidate for CIM macros. eNVM-based CIMs lag behind SRAM-CIM and digital accelerators in energy efficiency and computational density,⁷¹⁰ although ROM-based CIMs have shown promising results. Several works have attempted to benchmark the performance of SRAM-CIMs and compare them with other CIM macros.⁷¹¹

SRAM CIM macros can be implemented using analog or digital techniques, and several critical factors, including the type of SRAM cells used, influence their design. Integral to the architecture, these cells vary in their transistor count, with standard configurations including 6T⁷¹², 8T⁷¹³, and 10T⁷¹⁴ cells. 8T-cells are favored for their minimal read-disturb issues and high noise margin, enabling more parallel MAC operations and better energy efficiency. Although 6T-cells are more compact, they require additional circuit techniques to avoid read-disturb issues during CIM computations. Various circuit techniques have been introduced to address the read-disturb issue in 6T-based CIM macros. For instance, a local computing block⁷¹⁵ is added to a group of 6T cells sharing the duplicate BL. Multiple groups exist within a column, with each group's cells not sharing the duplicate BL. To prevent read disturbance, only one 6T cell per group is activated during computation.

Another critical aspect of CIM macros is how input data is managed and utilized for computation. The data can be fed via read-word lines (RWL)⁷¹⁶ as an analog voltage value, or the inputs can be applied as pulse-width modulated (PWM)⁷¹⁷ values on RWLs. The inputs can also be applied digitally to RWL⁷¹⁸. Another way is to provide input activations to the bitlines (BLs) in the SRAM-CIM macros either directly as pseudo-digital reference voltages⁷¹⁹ or using digital-to-analog converters (DACs) to convert the digital inputs to analog voltages, which are then applied to the BLs⁷²⁰.

In digital SRAM-CIM architectures, multiply and accumulate operations are executed using bit-serial logic gates in the digital domain. These operations are typically managed by logic circuits located in the peripheral circuitry of the CIM macro. The extent of computational parallelism in these digital CIM systems is primarily determined by the amount of logic resources integrated into the macro. However, increasing the number of logic circuits can lead to substantial area overhead, creating a trade-off between performance, energy efficiency, and area utilization. Analog CIMs, however, activate all rows simultaneously, with each SRAM cell multiplying a binary weight by an input along the wordline. Analog CIMs offer higher computational density and energy efficiency, but only when the costs of analog-to-digital and digital-to-analog conversions are spread across a large array and the array is highly utilized.

Recent benchmarking results have brought to light specific critical SRAM-CIM trends and insights⁷²¹. Contrary to expectations, SRAM-CIMs benefit from technology scaling, with improved compute density and energy efficiency observed in advanced nodes. Designs utilizing 6+ transistors in SRAM bitcells (e.g., 8T, 10T) enhance energy efficiency but often at the expense of compute density. This is primarily because 8T/10T bitcells can execute pseudo-digital (full-swing)⁷²² and have robust signal-to-noise ratio (SNR) computations with high signal margins after charge redistribution on BLs. This also reduces the need for very high-precision ADCs, reducing the area footprint of ADCs and increasing overall throughput via efficient parallelism. The compute SNR is a crucial but often neglected metric in CIM designs. It plays a significant role in determining energy efficiency and accuracy, particularly in analog-based computations. Optimizing SNR at the ADC column level can help ensure high network-level accuracy. 6T bitcells with embedded digital logic (6T+D)⁷²³ achieve high throughput by performing summing in the digital domain, avoiding the voltage limitations of analog summing. These hybrid designs could become strong contenders in the CIM space if digital computation's energy and area costs are minimized.

Benchmarking studies on CIM accelerators highlight that SRAM-CIMs, particularly those using bank-level design methods, excel in energy efficiency and compute density, outperforming digital accelerators at the core level by 13x and 6.6x, respectively. SRAM-CIMs are one of the best macros in terms of energy efficiency, and they also outperform eNVM-based CIMs⁷²⁴ by 4.8x and eDRAM-based CIMs⁷²⁵ by 2.44x on average. For eNVM-based CIMs, high-row parallelism is difficult to realize because it limits the achievable compute SNR.

A criticism of the SRAM-CIMs can be seen from the fact that even though at the macro level, they outperform digital accelerators in energy efficiency, at the bank level, this gap has been reduced, especially in the advanced process technology nodes. Hence, it is imperative to develop efficient scale-up methods that maintain bank-level efficiency while ensuring network and application-level accuracy for various AI workloads and leveraging circuits, microarchitecture, and statistical analysis. Further, to capitalize on the strengths of analog SRAM-CIMs and a certain degree of noise tolerance inherent to the machine learning applications, hybrid analog-digital architectures should be created to achieve high SNR and energy efficiency targets. As an aside, novel training algorithms that incorporate the devices' non-linearities can be incorporated into the training regime of these AI workloads.

Lastly, the performance of CIM primitives also depends on the types of workloads⁷²⁶. As the input shape varies with changes in matrix size, energy efficiency in SRAM-CIM architectures increases until the weight matrix size reaches the on-chip memory capacity, after which it plateaus due to the increased access energy costs. Beyond this, further increases in weight size lead to re-writing weights from the main memory, limiting reuse opportunities and maintaining constant energy efficiency. Throughput continues to rise with weight matrix size until full utilization is reached, with maximum performance achieved when all CIM

units operate in parallel. Keeping these results in mind, it would be imperative to design SRAM-CIM macros that are agnostic to the type of workload and can work at near-peak energy efficiency and throughputs for a wide variety of AI workloads. Future research will likely focus on improving the scalability of both analog and digital SRAM-CIM designs while addressing the inherent trade-offs between energy efficiency, computational density, and accuracy. Optimizing or eliminating peripheral circuits, such as DACs and ADCs, and exploring hybrid analog-digital approaches may offer pathways to overcoming current limitations.

4.2.4.2. OUTER PRODUCT UPDATE (OPU)

Analog resistive memory crossbars can also perform a parallel write or an outer product, rank 1 update where all the weights are incremented by the outer product of a vector applied to the rows and a different vector applied to the columns. This is a key kernel for many learning algorithms such as backpropagation⁷²⁷ and sparse coding.^{688,728} Row inputs are encoded in time and column inputs are encoded in either time^{728,729,728} or voltage.⁶⁹⁰

When an MVM, VMM and OPU are combined on the same crossbar, extremely efficient learning accelerators can be designed,^{690,730,731} with the potential to be 100–1000× more energy efficient and faster than an optimized digital ReRAM or SRAM based accelerator.⁶⁹⁰

The same figures of merit and design considerations for a VMM apply to the OPU. Additionally, the 1) write noise and 2) asymmetric write nonlinearity are important for determining how well a learning algorithm will perform. The 3) ability to withstand failures, and 4) endurance are also important for training systems. To have an efficient learning accelerator, parallel blind updates are needed where weights are updated without knowing the previous value and without verifying that the correct value is written. To obtain ideal accuracies, a low write noise is needed, less than 0.4% of the weight range. Even more important is having low asymmetries in the write process. The change in conductance for a positive pulse should be the same as that for a negative pulse for all starting states.⁷³² Often the conductance will saturate near a maximum where a positive pulse will not change the conductance, while a single negative pulse will cause a large decrease in conductance. This significantly lowers accuracy as it only takes a single negative write pulse to cancel many positive pulses.

One requirement of the parallel OPU is that the update has to be rank 1. This is satisfied by stochastic gradient descent updates but not batch or minibatch gradient descent which are also commonly used. One way to make a batch update computable with the parallel OPU is to approximate it as a series of M rank-1 updates where M is much smaller than the batch size. These approximations can be built using streaming principal component analysis, though there is a tradeoff between efficiency and the approximation quality, which determines the final accuracy.⁷³³

4.2.4.3. RESISTIVE MEMORY CROSSBAR SOLVER

Resistive memory crossbars can be used to solve matrix problems, such as the linear system of equations $Ax = b$, where x is the unknown vector, represented by output voltage, and b is the known vector represented by input current.^{734,735} Each resistive memory element is a programmable resistor that represents an element in the coefficient matrix A . (Alternatively, any programmable analog element can also be used.) The equation $Ax = b$ can be mapped to Ohm's law, $\sum G_{ij}(V_j - V_i) = I_i$. Op-amps are used to set the V_i to zero by using the Op-amp virtual ground. Currents, I_i , are applied to the crossbar and the resulting voltages, V_j , are measured. The op-amps provide feedback allowing the V_j to be determined. In effect, the op-amp feedback causes the nodal voltages of the crossbars to represent A^{-1} and the applied vector is multiplied by this A^{-1} representation performing a matrix solve.

Similarly, eigenvectors of a matrix A can be calculated. To solve problems with positive/negative coefficients in A , two crossbars can be used. In all cases, crossbar solvers yield their solution in one computational step, without any digital iteration, and the solution is generally obtained in less than 1 μ s, depending on the poles of the analogue feedback circuit.⁷³⁶ The same scheme can be extended to one-shot learning by linear/logistic regression.⁷³⁷

The biggest challenge in taking advantage of analog solvers for HPC is that analog operations only offer low precision, ~8 bits fixed point, while HPC applications often demand 32 or more bits of floating-point precision. This can be potentially overcome by hybrid analog/digital systems where the computationally intense parts of a calculation can be done in analog, while the required precision can be achieved by refining the solution in digital using a method with lower computational complexity.⁷³⁸ This allows for some digital computation, while still getting a reduction in the overall computational complexity. The precision can potentially be improved by using iterative refinement or by using the crossbar to initialize a digital solver. The analog solution can also be used as a preconditioner within a Krylov method like CG or GMRES⁷³⁹. Large matrices can be broken down into smaller blocks compatible with the accelerator and scaling can be used to compensate for finite on/off ranges. Work is still needed to show how noisy crossbar solvers can be used with ill-conditioned matrices. In general, iterative refinement will only converge if the noise is less than $1/\kappa$ where κ is the condition number of a matrix.

4.2.4.4. ASSOCIATIVE MEMORIES

Associative memories (AMs), which efficiently “associate” input queries with appropriate data words/locations in the memory, are powerful in-memory-computing cores. AMs search for data words/locations in the memory in a highly parallel fashion according to (i) an input query and (ii) the desired association (matching) function.⁷⁴⁰ AMs are widely used for network routers⁷⁴¹, and recently have been exploited to accelerate machine learning models such as memory augmented neural networks (MANNs)⁷⁴², clustering, and bioinformatics tasks.

CMOS AM or CAM designs suffer from several fundamental challenges:

- (i) The density of CMOS AMs, especially ternary CAM (TCAMs)—where “don’t care” states as well as “1s” and “0s” need to be stored/searched—is about half of the SRAM density.
- (ii) The standby power of CMOS AMs is high.
- (iii) CMOS AMs are typically limited to implementing Hamming (HM) distance-based matching.

Recent research on AMs based on beyond-CMOS devices suggests new directions for addressing the above challenges. For example, a ferroelectric FET (FeFET) based multi-bit CAM (MCAM)⁷⁴³ implements nearest-neighbor (NN) search according to a sigmoid-like (SG) distance function and offers significant performance/energy advantages versus competing designs for memory augmented neural networks (MANNs). This, as well as other recent work suggest that harnessing the benefits of AM for data intensive applications requires cross-layer efforts spanning from devices and circuits to architectures and applications.

Different non-volatile memory (NVM) technologies (such as RRAM, FeFET memory and Flash) can be exploited to implement various AMs (e.g., exact and best match searches, multi-level and analog data representations, and different distance functions). Like with the much-studied resistive crossbar array architecture, the large performance benefit for using CAMs for in-memory computation comes from the vast reduction of data movement for target applications with large numbers of compare or lookup operations. We classify AMs according to (i) the data representation (binary, ternary, multi-bit and analog) in a cell, and (ii) the matching type (Table BC4.1). Representative AM designs based on different technologies are discussed below.

FeFET AM cell designs—were originally proposed to accomplish exact match (EX), best match (BE), and threshold match (TH) TCAM based on HM distance. Different match types employ different sensing circuits⁷⁴². The same cell design can also function as a EX/BE/TH-MCAM when FeFETs are programmed to store multiple bits⁷⁴³, which improves storage density. The FeFET BE/TH-MCAM can implement SG and squared Euclidean (SE) distance functions⁷⁴⁴, which is useful for machine learning applications. This design can also function as an EX-ACAM where the threshold voltage (V_{th}) values in FeFETs define either upper or lower bounds, and an analog input matches stored cell data if it is within the bounds defined by the FeFETs⁷⁴⁵. ACAMs can encode more information per cell than MCAMs but may suffer more from noise and variation effects. Other FeFET AM cell designs have also been considered for reducing search latency, lowering search energy, simplifying peripherals, and/or improving scalability^{746,747}.

Table BC4.1 Classification of associative memory based on the representation and matching function. Match-line (ML_i) functions are defined based on input query (q_j) and memory content (C_{ij})

	Binary CAM (BCAM)	Ternary CAM (TCAM)	Multi-bit CAM (MCAM)	Analog CAM (ACAM)
Data type of q_j and C_{ij}	0 1	0 1 X	S0 S1 S2 S3	
Exact match (EX-)	$ML_i = 1$ if $C_{ij} == q_j$ for all j / 0 otherwise			$ML_i = 1$ if $q_j \cup C_{ij}$ for all j / 0 otherwise
Best match (BE-)	$ML_i = 1$ if $\sum_j d(C_{ij}, q_j)$ is the smallest among all ML_i s, where $d(C_{ij}, q_j)$ is the distance between C_{ij} and q_j			
Threshold match (TH-)	$ML_i = 1$ if $\sum_j d(C_{ij}, q_j) \leq Thr$, where $d(C_{ij}, q_j)$ is the distance between C_{ij} and q_j and Thr is the threshold			

RRAM and PCM AM Designs—Resistive memory (RRAM), has been widely used for crossbar type IMC core designs. RRAM has also been used to realize TCAMs, MCAMs, and ACAMs. There are a variety of RRAM TCAM designs (2T2R, 5T2R, 3T1R, and 2.5T1R)⁷⁴⁶ that support EX match and BE/TH match based on the HM distance. A 6T2R EX-ACAM⁷⁴⁸ was proposed and can directly process analog inputs. This design has high static power consumption and does not support BE/TH search. Phase-change memory (PCM) has similar characteristics to RRAM. A PCM EX-TCAM design was proposed and evaluated in Ref⁷⁴⁹. This design can perform BE/TH searches based on HM distance with appropriate sensing circuits.

Flash AM Designs—Floating-gate MOSFET (flash) is a mature NVM technology that is suitable for AM design.⁷⁵⁰ and⁷⁵¹ have proposed EX-TCAM/MCAM designs based on 3D NAND Flash. The same 2-transistor design the FeFET approach was implemented in Ref.⁷⁴⁴ with flash technology where it can perform EX match and BE/TH match based on SG and SE distance functions. The main drawbacks of flash technology for AM design are high write voltages, low endurance, and scalability.

MTJ AM Designs—Magnetic tunnel junction (MTJ) memory (MRAM) offers fast write operations and high endurance. EX-TCAMs based on voltage-dividers (9T-2MTJ) and latches (15T-4MTJ) were proposed in Ref⁷⁵² and Ref⁷⁵³, respectively. MTJs cannot offer high R_{OFF}/R_{ON} ratios and usually require several transistors to cope with this issue. Hence it may not be a good candidate for MCAM or ACAM.

While new technologies may enable more efficient CAMs, and enable new/existing distance functions to be computed directly within the memory, said solutions may also *fundamentally change algorithms* developed and deployed by the ML community. Data precision might change due to the multi-bit capacity of NVMs, different distance metrics might be used for an original ML algorithm (e.g., cosine instead of HM distance), etc. To ensure technology-enabled CAM solutions have value, evaluations must extend to the application level (e.g., to ensure iso-accuracy for ML applications). Recent work has proposed using associative memories as in-memory computation blocks for applications such as associative processing,^{754,755,756} approximate computing,⁷⁵⁷ spiking NNs,⁷⁵⁸ string matching,⁷⁵⁹ and regular expression matching finite state machines.⁷⁶⁰ Given the broad range of application spaces – from machine learning models like MANNs which reduce training costs, to deep random forests (i.e., interpretable AI)⁷⁶¹, to bioinformatics⁷⁴¹, to security⁷⁶², etc., further study is warranted.

4.2.4.5. LARGE-SCALE FIELD PROGRAMMABLE ANALOG ARRAYS (FPAAs)

A field programmable analog array has configurable analog components, digital components, configurable interconnects between those components and off-chip communications.^{763,764,765} FPAAs allow users to build analog applications without having expertise in IC design. FPAA I/O lines can transmit or receive analog signals, digital signals and create direct connection lines typical of analog circuits. The routing between analog and digital blocks can occur between the blocks of devices, with converters between these blocks, or more finely connected heterogeneous analog and digital component populations. The components are often organized into regions called computational analog blocks (CABs). CAB components vary considerably between implementations but often include nFET and pFET transistors, transconductance amplifiers [TA or operational transconductance amplifier (OTA)], other amplifiers, passives (e.g., capacitors), as well as more complicated elements (e.g., multipliers). The most advanced FPAAs to date utilize Floating-Gate (FG) devices, dramatically improving the analog parameter storage and therefore the resulting computational capability.⁷⁶⁶ FPAAs include aspects of digital computation, such as FPGA blocks or shift registers or microprocessors, to complete the full end-to-end configurable system.⁷⁶⁶

The fundamental breakthrough was recognizing that a switch matrix of single floating gate elements could be used for analog computation. The routing crossbar networks were, in fact, crossbar networks that could support VMM and other computations.⁷⁶⁷ Routing was no longer dead weight, as perceived for FPGA architectures. The floating gate cells could also allow for mismatch calibration at the mismatch source.^{768,769} The density for VMM in FPAA architectures is nearly the level of custom IC design. These analog computations can be made robust to temperature fluctuations.⁷⁷⁰ These techniques have been utilized by a number of students in university courses.^{771,772} FPAA based VMMs can be scaled to small geometry (e.g., 40 nm and smaller) and operated at RF frequencies.^{773,774} FPAAs have been used for command-word recognition in less than 23 μ W with standard digital interfaces.⁷⁶⁵ The full classification results in less than 1 μ J per classification (or inference), which has 1000 $\times$ improvement over similar digital neuromorphic solutions requiring roughly 1 mJ or higher for just an inference.⁷⁷⁵

4.3. EMERGING ARCHITECTURES AND SYSTEMS

4.3.1. NEURAL-INSPIRED COMPUTING

Neural inspired systems include both systems that accelerate conventional machine learning and systems that are more biologically inspired. There are several characteristics of how the brain computes that have been proposed for efficient computing technologies. Specifically, inspiration is often taken from three different features of the brain: 1) processing in memory is analogous to the analog computation that occurs at synapses within the brain 2) spiking communication is analogous to neuronal

spiking and is leveraged to minimize communication costs and 3) specialized computational devices are analogous to the diversity of neurons and mechanisms in neural systems that perform specific computations.

As discussed in §4.2.4, analog crossbars can be used as building blocks for conventional neural networks to give significant improvements in energy, latency and area over digital accelerators. For accelerators specialized to a particular algorithm, various analog neurons can be used to process the crossbar outputs and avoid the need for and high cost of analog-to-digital conversion.

There is also a lot of work on more biologically inspired neural hardware. For biologically inspired neurons, connections are often designed to be persistent on short timescales, but (depending on the model) may exhibit mutability/plasticity in their strength and/or topology on longer timescales to facilitate, for example, adaptive in-situ learning. Connections between neurons can be modeled as discrete events or spikes (often implemented using an address event representation) or continuous-valued analog signals such as voltage or current. A key challenge for more biologically inspired architectures is the need for algorithmic co-design. For many neuro-inspired computational models, further research is needed to demonstrate state of the art machine-learning performance.

4.3.1.1. NEURAL NETWORK INFERENCE

In-memory analog computing has been extensively explored as energy-efficient implementations of deep learning algorithms, or deep neural networks (DNNs). To be useful for a particular task, a DNN must first be trained so that the values of its internal parameters or weights are optimized to yield high accuracy. After training, the DNN can be deployed for the intended application, where it may encounter unseen data that deviate significantly from the training data. During this phase, called inference, the weights are left fixed. Emerging accelerators for neural network inference are a more near-term goal compared to training accelerators. During inference, there is no need to apply frequent updates to the memory device conductances, or to propagate information backward through the network, both of which greatly simplify the architecture. Since MVMs are the dominant computational kernel used during inference,⁶⁸⁰ large system-level energy savings are possible by efficiently computing the MVMs in the analog domain, as described in Section 4.2.4.1. Furthermore, executing MVMs inside memory arrays largely eliminates the very significant energy cost in conventional digital systems of moving weight matrix data between memory and the processor.^{776,1091}

The basic architecture of an analog in-memory inference accelerator consists of resistive arrays that accelerate MVMs, combined with digital elements to perform a small amount of computation and handle the data routing. Since weight values are directly encoded in physical device resistance values, the computation is inherently weight stationary: weight values stay inside their MVM arrays, but the digital activation values are moved between processing elements as the computation proceeds from layer to layer.⁷⁷⁷ The MVM array receives digital inputs at a specified precision (e.g., 8 bits) and the analog MVM result is digitized by an ADC. These outputs are then further processed digitally (e.g., activation function, pooling) and digitally routed to the MVM array of the next neural network layer. For all but the simplest multi-layer perception (MLP) networks, digital memory arrays are needed to buffer these intermediate values between layers.

Inference accelerators have been demonstrated fully in hardware using flash memory⁷⁷⁸ and ReRAM.⁷⁷⁹ So far, these demonstrations have shown high accuracy on simple benchmark tasks such as MNIST. There are many more proposed inference accelerator architectures, whose performance has been evaluated in simulation on larger-scale machine learning problems such as ImageNet.^{699,698,697,681,780,700,781,782,783} Physically demonstrating these larger-scale systems is challenging, due to the very large number of resistive devices needed, and the sensitivity of these more difficult tasks to analog errors. Nonetheless, these systems can be realistically modeled by combining measured device- or array-level properties with system-level simulation.⁷⁸² A common architectural choice in these proposed large-scale architectures is to group a set of analog and digital circuit elements into a modular unit called a *tile*.^{698,699,780,697} Many tiles are connected together, e.g., using a concentrated mesh interconnection topology, to implement the many layers of a deep neural network. Since the tiles have the same structure, this type of system can scale to larger networks by adding more tiles.

The main challenge with analog in-memory inference accelerators is to ensure high accuracy along with the high energy efficiency. Analog computing differs fundamentally from digital in that (non-catastrophic) errors at the device and circuit level can directly affect the inference prediction. In a DNN, these errors can propagate and grow from one layer to the next. At the same time, nonlinear operations such as activation functions, ADC quantization, and the argmax operation (at the network output) can help suppress the propagation of these errors. Given these complex mechanics, it is difficult to know how much precision is needed at the device level, or even the MVM level. Therefore, analog systems should be designed with the *end-to-end* neural network accuracy as the figure of merit, rather than simply the MVM precision. A common, alternative design choice is the full-precision guarantee: the system is parameterized (via the array size, ADC resolution, number of bit slices, etc.) such that each analog MVM result, after being digitized, can match the precision of an N -bit digital processor.^{699,698,681} This choice often leads to an MVM-level precision that is far more than sufficient for high end-to-end inference accuracy, and thus can cause energy and area inefficiency.⁶⁹⁵

The sources of error that affect inference accuracy in an analog system are the same ones described in Section 4.2.4.1 that degrade the accuracy of an MVM. These include: (1) errors in device resistance programming due to process variations or device/circuit noise in the write process, (2) noise in the device resistance during an MVM, (3) resolution and noise in the ADC, (4) parasitic voltage drops due to array interconnect resistance, (5) drift over time in the programmed device resistances, and (6) noise, process variation, and transients in the peripheral analog circuits, and (7) temperature dependence.

For neural network inference specifically, the effect of these errors (especially #1-4 above) can depend very strongly on the distribution of the data values in the neural network. A near-universal property of neural networks is the fact that the distribution of weight values is not uniform, but rather heavily concentrated near zero. This property is partly why pruning (eliminating low-valued weights) has been so successful in compressing neural networks, making them easier to process by digital inference accelerators.⁶⁹⁶ For analog inference accelerators, one implication is that the low-valued weights need to be more precise than the much less common high-valued weights. As a result, the method that is used to map weight values to the device resistances can have a very large effect on the accuracy. If weights with low absolute value correspond to low conductance, higher accuracy to be achieved by devices that have *low error at low conductance*.⁶⁹⁵ This also leads to *high On/Off ratio* as a desirable property for neural network inference. Since low-valued weights are abundant, an improvement in the On/Off ratio can greatly reduce the array currents, reduce the parasitic voltage drops,⁶⁹¹ and enable larger arrays with higher energy efficiency. This is an important example of how the device and system architecture (e.g., mapping method) are designed with the algorithm in mind to ensure robustness to errors, as well as high efficiency.

Table BC4.2 Representative list of large-scale analog MVM based AI inference accelerators.

Paper	(Fick et al., 2022) ⁷⁸⁴	(Agrawal et al., 2022) ⁷⁹⁹	(Khwa et al., 2022) ⁷⁸⁵	(Ambrogio et al., 2023) ⁷⁸⁶	(Le Gallo et al., 2023) ⁷⁹⁸	(Wan et al., 2022) ⁷⁰³	(W. Song et al., 2024) ⁸⁰⁰
Memory Type	Flash	Flash	PCM	PCM	PCM	ReRAM	ReRAM
# Cores Array Size (In x Out)	76 Cores 1024×256 ^v	1024×1024	8 Cores 256×1024	34 Cores 512 x 2048	64 Cores 256×256	48 Cores 256×256	10 Cores 256×256
TOPS	16.6 – 59.3		0.475		16.1	0.3 – 2.1 (estimated from plot)	
TOPS/W	5.2–18.5 (Array) 3.3–10.9 (System)	55 (Modeled)	20.5	2.77 – 12.4	2.48	7 – 42 (estimated from plot)	160 (Array only, calculated)
Area Metric (die size or compute density)	16.5 x 11.5 mm (die size)		18 mm ²		0.4 TOPS/mm ²	10.8 x 14.7 mm	
Benchmark	Pose Detection (COCO Dataset)	CIFAR-100 w/ RESNET-50 ^{vi}	1) CIFAR-10 2) CIFAR-100	1) Keyword Spotting 2) Recurrent Neural Network Transducer	1) CIFAR-10 2) LSTM 3) Captioning	1) CIFAR-10 w/ RESNET-20 2) MNIST 3) LSTM	Poisson Solver
Performance Metric	29.3% mAP ⁵⁰⁻⁹⁵	73.01% (vs 73.89% digital)	1) 91.89% 2) 67.53%	1) 86.14% vs 86.75% digital 2) 9.475% word-error rate vs 7.452% digital	1) 92.81% vs 93.67% digital 2) 1.433 vs 1.336 digital bits-per-char 3) 0.544 vs 0.534 digital (BLEU-1 score)	1) 85.66% vs 87.81% digital 2) 99% vs 99.2% digital 3) 84.7% vs 90.7% digital	~520 iterations vs 400 digital ^{vii}

There are two other prominently used methods to reach high accuracy even with very limited precision in the memory device technology. One of these is bit slicing,⁶⁹⁹ described in Section 4.2.4.1.1, where the N bits of weight precision are “sliced” across

^v Each array consists of four 1024x256 banks (to approximate a 1024x1024 array); however, the banks are operated sequentially.

^{vi} The last layer of the RESNET-50 model is programmed into the silicon-oxide-nitride-oxide-silicon (SONOS) array; the proceeding layers are performed in digital.

^{vii} Both the analog and digital implementation converged to the same relative residual error; however, the number of iterations was larger in the analog case.

multiple devices, and the MVM results from these slices are aggregated digitally. Bit slicing can be helpful, or even necessary, when device precision is limited. However, there are two important caveats: (1) representing weight values with high fidelity does not imply MVMs are computed with high fidelity, because errors accumulate across many devices within a slice, and (2) bit slicing incurs significant energy and area overheads, due to requiring a separate MVM for each slice.⁶⁹⁵

Neural network retraining is also commonly used to compensate for device errors. The most common style of compensation is to inject noise during the forward and/or backward propagation phase of training, by an amount that emulates the amount of error or noise in the device conductance.^{782,787,788,789,790} The drawbacks of these methods are that: (1) the retrained network that compensates for device errors may not have the same accuracy as the original network and (2) the device compensation method may be difficult to integrate with state-of-the-art training workflows. In a similar vein, methods have been developed to optimize the mapping of weights to conductances to mitigate the effects of parasitic voltage drops.^{781,788,791,792}

Conductance drift is a concern for inference accelerators and arises from physical changes in the memory devices that occur over time. This causes a drift in the weight values; if this drift is predictable, it can be compensated.⁷⁹³ However, the physical processes underlying drift are often thermally driven, and result not only in weight drift but also an increase in the weight variance over time.⁶⁹⁷ Eventually, the accuracy may be degraded to the point where the device resistances must be refreshed from a digital reference: the exact refresh frequency depends on the device and the application.

4.3.1.1.1. LARGE SCALE DEMONSTRATIONS

DNNs have shown immense promise in many application areas; however, due to the need for billions of multiply-accumulate (MAC) operations, they have often been limited to cloud computing⁷⁹⁴. As the number of operations required for these DNN inferences continues to grow, the need to manage power budgets becomes critical. Additionally, the increasing popularity of edge computing requires careful consideration for power^{794,795}. Recently, accelerators based on analog MVM operations in resistive crossbars have shown immense promise for tackling DNN problems at lower power than digital operations^{678,795,796}. Large-scale MVMs are the driving force behind the analog in-memory computing based formulations for DNNs. The most large-scale demonstrations have so far used three primary hardware technologies: flash memory, ReRAM, and PCM. A table summarizing the recent large-scale AIMC demonstrations is included below. Additionally, M.-K. Song et al.⁷⁹⁷ provides an excellent review of recent materials, devices, and techniques for DNNs and other neuromorphic computing applications. Additionally, while not a review, Le Gallo et al.⁷⁹⁸ compares their work to a number of other AIMC devices, some of which are included below. While not reported in the respective papers, the chips described below are at different levels of readiness for use as an analog accelerator. For example, neither Agrawal et al.⁷⁹⁹ nor W. Song et al.⁸⁰⁰ report measured TOPS/W values, suggesting those devices do not yet have optimized supporting circuitry.

For a more extensive list of analog in-memory computing hardware demonstrations, we refer the reader to a recent review.⁸⁰¹

4.3.1.1.2. SPIKING-BASED NEURAL NETWORKS (SNNs)

Inspired by the brain which couples both processing-in-memory and spike-based communication for maximal space and energy efficiency, a good spiking neural network needs to couple both. Taking inspiration from biological brains, SNN designs have components that are named to match similar biological functions. At a minimum, this includes a synapse, which applies a weight and optional delay to incoming spike events, and a neuron, which accumulates incoming charge from the connected synapses to produce outgoing spike events. At an architecture level, these spikes are then routed between these components, allowing the evaluation of a Spiking Neural Network. These Spiking Neural Networks can be represented as graphs with the nodes of the graphs representing neurons and the edges representing Synapses. More advanced and biologically realistic designs include additional components inspired by biology, including dendritic trees and axons. The exact functionality and effect of these more detailed components are still a source of active research.

Table BC4.3 Neural Networks: Architectural-level Metrics of Design

Metric	Target Values	Notes & references
Network Time Resolution	1 kHz is biologically realistic	This refers to the network clock or real-time resolution of events. Network time can be fixed or varying compared to real time.
Spiking Rate of Neurons	1 kHz is biologically realistic	The maximum rate at which neurons can fire. Also known as the neuron's refractory period.
Connectivity Constraints	Any-to-any (with biological realistic connectivity)	Biologically realistic is any-to-any connections possible with dense local connections and sparser global connections.
Number of Neurons	~100 Billion	Total number of neurons in a biologically realistic network.
Synapses per Neuron (Fan-in/Fan-out)	~1,000–10,000	Biologically realistic neurons have around 1,000 to 10,000 connections per neuron.
Synaptic Latency (min)	1	Minimum time required to route a spike between neurons via the synapse.
Synaptic/Axonal Delay (min-max)	1-?	Delay range supported for synaptic (edge specified) or axonal (node specified) delay.
Energy per spike	~pJ	Energy required for a neuron to fire.
Energy per packet	~pJ	Energy required for an AER packet to be routed.
Binary or weighted spikes	?	Neurons can optionally have a weight value when they fire. Traditionally SNNs have binary fire events.
Resolution/range for configurable parameters	Depends on parameters	Ranges and resolutions for parameters like synapse weight, neuron threshold, delay, and leak.
Leaky and type of leak	Exponential	Is leak implemented, and how is the leak function defined?
STDP	Yes	Is Spike-timing-dependent plasticity (STDP) implemented?
Online-Learning	?	Are additional online learning algorithms supported by the design?
Accumulation/activation function	Non-linear	What accumulation method and activation function is used by the neuron to accumulate charge and decide when to fire?
Other biologically relevant features?	?	Are other effects modeled, such as dendritic tree structures, dendritic computing, stochastic operation, etc.?

Ultimately, the spiking operation exhibited by neurons has a few features that must be captured by proposed devices and built into neurons at the circuit level: its non-linearity and in-memory processing. Core to its operation, it must accomplish the analog-to-spiking conversion, converting charge received from synapses into a membrane potential which will generate spikes when the charge grows large enough. The threshold at which the neuron activates and the accumulated charge must be stored locally in the neuron circuit, combining the storage with the spiking operation. When in its simplest form, a simplistic neuron is a compact 1-bit analog-to-digital converter, but ideally, it would also enable holding some additional state or history from the analog inputs as membrane potential or neuron charge. There have been many proposed devices for building spiking neuron circuits including neuristors,⁸⁰² spin-torque-based devices,⁸⁰³ stochastic phase-change neurons,⁸⁰⁴ superconducting neurons,⁸⁰⁵ and others.

Other crucial SNN features must be captured at the architecture level, which builds up a complete design from the neuron and synaptic circuits constructed using novel devices. These features include local and global spiking communication with the ability to scale to biologically realistic sizes. A survey of routing spike events both locally and globally can be found in *A Review of Spiking Neuromorphic Hardware Communication Systems*.⁸⁰⁶ One target for a biologically realistic scale is to target billions of neurons with each neuron having thousands of synapses for a total of trillions of synapses. The scalability of this spike routing is enabled by the majority of the connections being local, with relatively fewer long-distance connections. Currently, CMOS systems rely on event-driven communication of Address-Event Representation (AER) packets that contain some source or destination address relevant to routing the spike to appropriate destinations. Thus, CMOS systems do benefit from the relative rarity of the communication (only transmit when an event occurs), and are already achieving considerable savings from that, but do not benefit significantly from the theoretical 1-bit precision of the spike as a multi-bit address is needed. In this respect, more efficient mechanisms for direct point-to-point communication, such as superconducting systems, 3D nanowires, or perhaps optical interconnects are needed. The challenge in these systems is how to achieve the necessary level of fan-in/fan-out (i.e., number of synapses per neuron) between non-local regions and how to deliver that information in a suitable form for processing in the analog memory circuits used as synapses. Metrics used to describe spiking systems are summarized in Table BC4.3.

4.3.1.2. NEURAL NETWORK TRAINING

In-memory accelerators for neural network training is inherently more difficult than inference accelerators, due to the need to support frequent updates to the conductances of the memory devices. These conductance updates must not only be frequent, but also must be both blind and sufficiently accurate. Having to verify the conductance after each programming step is costly and would eliminate the energy benefits of an in-memory training accelerator.

For deep neural networks, the most ubiquitously used training algorithm is gradient-descent-based learning with backpropagation. In backpropagation, a loss function is computed at the output of the network and the weights in each layer are tuned to optimize the value of the global loss function.

Alternatively, local learning rules are also being that only require local information to train weights. Keeping all computation is more energy efficient, but algorithm research is still on-going to try to match the accuracies provided by back-propagation based training.

4.3.1.2.1. BACKPROPAGATION

In backpropagation, training data is first fed in the forward direction through a neural network and a loss function is computed at the output. The gradient of the loss function with respect to each layer (the layer-wise error) is then computed for each layer. These errors are computed from the last layer backward to the first, using successive VMMs between the layer-wise error and the weight matrix. Within a layer, the optimal weight update is an outer product between two vectors: the layer-wise error and the layer's activations. Compared to an analog inference accelerator, supporting the backpropagation of errors within a training accelerator requires additional digital logic and memory, since activations computed during the forward pass must be saved to compute the correct weight update.⁶⁷⁸ For high efficiency, the same array used for MVM should be used to compute the VMM (transpose operation) by allowing the inputs to drive either the rows or the columns of the array.⁶⁹⁰

The in-memory OPU computational primitive is very useful for accelerating backpropagation, since the update needed for each weight in the matrix can be computed in analog by the array itself: see Section 4.2.4.2. This reduces the digital processing overhead and is faster and more efficient than programming row by row.⁶⁸⁸ However, one limitation of the method is that OPU updates assume a batch size of 1 during training. Larger batch sizes are often desired, which can reduce wear on the devices, enable greater pipelining, and improve training convergence.⁶⁷⁸ Thus, some works have proposed computing the weight updates within a batch in digital hardware, then transferring these updates to the memory array at the end of a batch. This mixed-signal technique can also be used to account for the low bit precision of a device update, by digitally storing the residual of the update and accumulating over this residual in the next batch.⁸⁰⁷ Another method is to construct an optimal rank-1 approximation of a

batch update that allows the efficiency of the in-memory OPU to be leveraged while also retaining the convergence and endurance benefits of batched training.⁷³³

The greatest challenge for analog in-memory training accelerators is ensuring accurate, blind conductance updates. This accuracy is affected by: (1) cycle-to-cycle write noise or limited programming resolution, (2) linearity in the update response, e.g., the conductance change in response to a pulse should be the same regardless of the starting conductance, (3) symmetry in the update response, e.g., the conductance change in response to a pulse should be the same regardless of the update polarity, and (4) device-to-device variations in the update response.⁸⁰⁸ In general, it has been found that an asymmetric nonlinearity is much more harmful than a symmetric nonlinearity, because it tends to cause the weight to settle toward zero after a sequence of updates.^{732,809} In addition to meeting these requirements, the resistive device should further have high endurance, low write latency, and low write energy. Low write currents are also desired to reduce parasitic voltage drops during write, which can further distort the desired weight update.⁶⁷⁸

Several devices have been examined for neural network training, including phase change memory,⁷³⁰ resistive memory^{690,810} and novel Electro-Chemical RAM devices⁸¹¹. Currently no devices meet all the ideal requirements for training (high resistance $>10\text{ M}\Omega$, low write noise $<0.4\%$, low write asymmetry⁷³²). Nevertheless, algorithmic approaches such as periodic carry,⁸¹² Local Gains,⁸¹³ Tiki-Taka,⁸¹⁴ or the inclusion of semi-volatile capacitor-on-gate devices⁸¹⁵ can be used to help compensate and achieve ideal accuracies. Several co-design tools have been developed to model the impact of device level properties on algorithmic performance,^{730,816,817} which have allowed for the algorithmic development needed.

4.3.1.2.2. LOCAL LEARNING RULES

Spiking neural networks can be viewed as a type of recurrent neural network, where activities are binary and recurrence is both due to explicit connections and internal dynamics (referred to as implicit connections)^{818,819}. This analogy enables the transfer of learning algorithms based on gradient descent to local synaptic plasticity. The synaptic plasticity dynamics that result from these derivations are “three-factor rules.” Three factor rules are popular among computational neuroscientists for reward-based learning⁸²⁰. Because gradient-based rules vastly outperform other traditional learning rules borrowed from neuroscience on industry relevant benchmarks, our focus here is on gradient-based learning.

As in deep learning, estimating the gradients with respect to parameters embedded in deeper layers requires solving a spatial and temporal credit assignment problem. Through approximations that rely on ignoring gradients caused by explicit connections, it is possible to solve the temporal credit assignment problem. The accuracy cost of this approximation compared to the exact gradients depends on the used neuron and synapse type, which can be negligible to catastrophic.⁸²¹ Even in the ideal case, accuracy is generally not better than non-local learning. The improvements from local learning instead lie in the energy and latency cost per update, compared to updating the parameters by means of an external computer (e.g., few-shot continual learning tasks⁸²²).

Solving the spatial credit assignment problem in a local fashion remain a challenge, however. This is because all interlayer connections are explicit and ignoring explicit connections is equivalent to ignoring the effect of deeper layers. Solving the spatial credit assignment problem is under heavy investigation, ranging from approximating gradient back-propagation⁸²³, predictive coding^{824,825}, self-supervised contrastive learning^{826,827} and prospective coding⁸²⁸. Understanding how such approximations in the context of novel in-memory architectures is key to local learning using both conventional and spiking neural networks.

The machine learning community has been dominated so far by GPUs high bandwidth access to global memory, making gradient backpropagation like rules possible. However, the demands on scale and power are increasingly pushing to distribute computations across multiple nodes and processors with more limited access to global memory⁸²⁹. As a result, one can expect an increase in output on the topics of local learning from the machine learning community. Furthermore, local learn rules that keep all data local naturally map to in-memory computing technologies with primarily local interconnects⁸³⁰.

Local learning often implies learning online. Online learning is subject to constraints that do not verify the assumptions of deep learning and statistical learning theory: data is potentially independent, and identically distributed (i.i.d.), causing catastrophic forgetting; data consists of a unique stream (“batch size is one”), leading to long training times and the requirement of impractically small learning rates; and large amounts of data are required. These problems are compounded by the realities of limited precision and hardware non-idealities, especially in the case of emerging nanotechnologies. A result of these challenges is that hybrid learning schemes that combine offline pre-training and online learning need to be explored. Recent work leveraging meta-learning or bi-level optimization are showing promise to overcome certain non-idealities of conductance updates and accelerate online learning in spiking neural networks^{831,832}.

Another approach to local learning rules is the forward-forward algorithm by Geoffrey Hinton⁸³³. The Forward-Forward algorithm replaces the forward and backward passes of backpropagation by two forward passes, one with positive (i.e., real) data and the

other with negative data which could be generated by the network itself. This algorithm has been implemented with hardware in Ref. 834.

4.3.1.3. HYPERDIMENSIONAL COMPUTING

Hyperdimensional (HD) computing is a cognitive computing model based on the high-dimensional properties of neural circuits in the brain.⁸³⁵ Information is encoded into high-dimensional distributed representations in the form of high-dimensional vectors or *hypervectors*.^{835,836,837} A hypervector distributes information uniformly across all of its dimensions, resulting in a distributed or *holographic* representation.^{838,839} This contrasts with conventional *positional representations* in which different digits or bits can convey vastly different amounts of information depending on their position. Consequently, incurring bit errors in particular positions can result in *catastrophic failure*. In contrast, the distributive nature of hypervectors combined with high dimensionality provides a robustness to bit errors: errors that occur in any dimension result in the same information loss, and this information loss is small due to the large number of dimensions. Thus, HD computers exhibit *graceful degradation* as hardware components fail, analogous to in the brain when neurons die^{839,840}.

HD computing relies on the algebraic properties of their key operations to incorporate the advantages of structured symbolic and vector distributed representations. It incorporates learning capability along with typical memory functions of storing/loading information.⁸³⁵ It mimics important functionalities of the human memory model with vector operations, which are computationally tractable and mathematically rigorous in describing human cognition.⁸⁴¹ HD computing operates over a well-defined and hardware-friendly set of mathematics: *Binding* is well suited for associating two hypervectors and used for variable-value association. *Bundling* is a memorization function that keeps the information of input data into a bundled vector. *Permutation* is an operation to represent sequences by creating a near-orthogonal but reversible hypervector. *Reasoning* is done by measuring the similarity of hypervectors.

HD computing has shown several advantages as the next generation of cognitive machines. First, its training capability in one or few shots, where partial data are learned from one or few examples as opposed to many iterations.^{840,842,843} Second, HD operations are highly parallel and do not require frequent weight updates, thus empowering online learning on *processing in-memory*.^{842,844} Third, HD computing has natural robustness to noise and bit errors provided by the high-dimensional distributed representations. This enables greater tolerance for device variability and unreliable emerging technologies, such as non-volatile memory^{845,846,847} or nanoscale devices with low signal-to-noise ratios.^{844,848,849} In fact, HD computing enables further size and energy scaling by abandoning the deterministic requirement set by the variability and reliability of the composing devices. Finally, HD computing has been shown great potential to enable lightweight privacy and security.^{850,851,852,853}

4.3.2. COMPUTING WITH DYNAMICAL SYSTEMS

In computing with dynamical systems, the built-in dynamical behavior of a physical system exhibiting continuous degrees of freedom is used to compute. The entire computational process can be analog, with only the results being digitized. The following subsections give a few examples of different types of dynamical systems-based approaches.

Although some of the below methods target NP-hard problems and provide approximate solutions, it's important to note that, to date, no general physical computing method has yet been clearly demonstrated to be capable of exactly solving NP-hard problems without requiring exponential physical resources (energy and/or time) to be invested in the physical process performing the computation.⁸⁵⁴

4.3.2.1. COUPLED-OSCILLATOR BASED OPTIMIZATION

Coupled-oscillator machines are a class of analog accelerators for combinatorial optimization that are composed of a network of decentralized nonlinear oscillators, and the programmable strength of the coupling between them encodes the specific problem to be solved. These networks have been proposed and demonstrated with electrical,^{855,856,857,858,859} optical,^{860,861,862} and electromechanical⁸⁶³ oscillators. These systems are often called “Ising machines”, because they map readily to the Ising graph optimization problem, with each oscillator representing one bistable spin. Any combinatorial optimization problem can be converted into an equivalent Ising problem that is programmed onto and solved by the machine. The fixed points of the equations of motion of such networks correspond to the solutions of an NP-hard combinatorial optimization problem and can model other NP-hard problems as well.

Networks of coupled oscillators have been shown to embed the energy (cost function) of the Ising problem in their physical dynamics and relax to configurations that minimize this energy. Networks of coupled parametric or second-harmonic injection locked nonlinear oscillators can be designed such that the network physical dynamics implement the method of Lagrange multipliers on the Ising problem.⁸⁶⁴ They can thus rapidly sample the local minima of a problem, and can potentially also be used to arrive at solutions close to the global minimum. The sampling speed is determined fundamentally by the oscillator frequency and the quality factor of the oscillator network. The accuracy of the different architectures has been benchmarked using large

instances of the Ising or Max-Cut problems generated by the operations research community. Solutions have been found that match those obtained by state-of-the-art digital algorithms. Energy and speed (time-to-solution) benchmarking is in its early stages, with current hardware prototypes not yet competitive with conventional digital-electronic solutions.

Of the proposed coupled oscillator optimization schemes, the systems that use electrical (*LC*) or electronic (ring oscillator) oscillators are the most compatible with CMOS technology. The Ising problem is specified by a connectivity matrix, which can have mostly non-zero entries, i.e., represent dense connectivity between spins; the oscillators in an Ising machine can be densely connected using a crossbar.⁸⁶⁵ In these networks, each oscillator takes on a discrete binary nature by synchronizing to its associated second-harmonic pump oscillator through a circuit nonlinearity. In an N -spin Ising problem, the bistable voltage signals of the N fundamental oscillators play the role of the N Ising spins while the N second-harmonic pump voltage signals play the role of the corresponding Lagrange multipliers that enforce bistable constraints on the spins.⁸⁶⁶ Novel nonlinear dynamical system approaches that go beyond Lagrange multiplier optimization and exceed state-of-the-art performance on certain benchmark problems have been proposed⁸⁶⁷ and their physical implementation is an area of ongoing research. The nonlinearities in all these approaches can be easily implemented by semiconductor components such as diodes, MOS capacitors, or amplifiers. In fact, phase-based Boolean computing using such nonlinear oscillator circuits was first proposed decades ago.^{868,869,870,871}

The physical limits of coupled oscillator systems are governed by the achievable level of weight precision and circuit delay. The resistive connections must be linear, programmable to high precision, and have minimal drift. The precision and retention of the resistive connections limit the accuracy to which a problem can be programmed onto the hardware, and the precision requirements for an adequate representation increase with problem size. For very large problems, device or process limitations will impose an upper bound on the quality of the solution; the target error bound depends on the application, but it must be superior to bounds that are guaranteed by digital approximation algorithms. Since a coupled oscillator network relies on synchronization between the oscillators, signal delays can also impact performance, especially in high-frequency circuits needed for rapid optimization. To this end, architectures have been proposed that separate the oscillators in time rather than space using optics, but this comes at the loss of continuous-time communication among the oscillators, leading to slower convergence.

Related constraint satisfaction systems have been built.⁸⁷² An interesting approach based on memory co-processors was introduced as Memcomputing.⁸⁷³ Useful insights can also be obtained by looking into dynamical systems like iterated maps,⁸⁷⁴ and 0-1 continuous reformulations of discrete optimization problems.⁸⁷⁵

Ultimately the goal of optimization accelerators, including ones based on coupled oscillators, is to be able to perform optimization on problem instances of practical interest faster or more energy-efficiently (or both) than conventional solutions. Most likely it will make sense for accelerators to be built that are at least somewhat tailored to a certain class of problem rather than trying to make an accelerator that is best for all possible optimization problems. Depending on what problems are being addressed, different choices can be made for a variety of machine specifications to aid in optimizing the accelerator's speed and energy efficiency, including:

- Total number of optimization variables.
- Variable connectivity (all-to-all; hierarchical; if only sparse, how reconfigurable is it).
- Variable types (binary – for Ising machines; integer; continuous), and how many of each type, or how reconfigurable the hardware is to assign different variables different types.
- Precision of optimization variable states.
- Problem “order” supported (2-body - for Ising machines; 3-body – for directly supporting 3SAT problems; 4-body; ...).
- Accuracy of programming problem coefficients (e.g., in the case of Ising problems: how accurately you can program the h_i and J_{ij} coefficients) Time to load new problem instances and time to read out solutions.

4.3.2.2. CONTENT ADDRESSABLE MEMORIES FOR OPTIMIZATION

Associative Memories (AM), and their implementations in Content Addressable Memories (CAM), perform a parallel data search operation within a stored memory array. As noted in Section 0, the AMs perform a parallel search operation for a given input query and a given matching function on the memory array. There can be significant flexibility in the matching function, including equality tests, inequality relations, logical relations, etc. With traditional binary CAMs, the matching function needed to involve binary values, but the extension to analog CAMs allows for matching functions on non-binary values as well. Such flexibility opens up the use of CAMs for solving challenging optimization problems, where a generic objective function $E_{Obj}(\)$ is to be minimized or maximized. Such problems are at the core of many computationally-intensive tasks such as cryptography⁸⁷⁶, software, and hardware verification⁸⁷⁷. A challenging example is Boolean Satisfiability (SAT), which is presented in conjunctive normal form (CNF),

$$CNF = \bigvee_i c_i = \bigvee_i \bigwedge_{j \in c_i} x_j$$

with AND operations performed between clauses c_i and an OR operation between each literal x_j inside the clause. Each clause can have up to k literals, with k specifying the order of the problem, such as $k = 3$ for 3-SAT. A SAT solver can aim to demonstrate that a variable assignment exists (solves all clauses), or seek to maximize the number of satisfied clauses. At the same time, different algorithms can be used such as an exact solver based on backtracking, which assigns one variable at a time until a clause is violated leading to reverting the previous operation(s)⁸⁷⁸, or Stochastic Local Search solvers which assigns all the variables at once and compute gradients to select which input variables to change⁸⁷⁹. Nevertheless, most SAT solvers require computing at each iteration which clauses are violated and which are satisfied, in order to propose a new candidate solution or to stop. Computing the index of violated and satisfied clauses in parallel is challenging and leads to significant hardware overhead. These operations can be performed in CAMs, thanks to the flexible matching function described above, which can be easily compatible with the CNF form. Proposals have been developed for the massively parallel lookup of violated/satisfied clauses in Ternary CAMs and analog CAMs^{880,881,882}. Here, given an input vector to search, each output returns a match if the input is equal to the stored word. A wildcard (or don't care) is used for matching both 1s and 0s either in stored words or inputs. By mapping each (inverted) clause to a TCAM word, using the wildcard (or don't care) to map the variables that do not belong to a clause, a match would represent the violation of the clause. This parallel search operation leads to significant improvements compared to conventional digital hardware and other approaches for optimization solvers using emerging memory such as Ising Machines⁸⁸³ and Hopfield Neural Networks⁸⁸⁴.

Further extending the capabilities of CAMs and Associative Memories, the use of emerging memory technologies (FeFET, RRAM, PCM, MTJ) offer great potential (see section 0). While most requirements such as density and power consumption are relevant for using CAMs in optimization applications, modern SAT solvers⁸⁸⁵ can require new clauses to be learned iteratively, thus frequent memory updates are needed during problem solving. As such, this application adds new challenges in programming speed, reliability, and endurance for CAMs based on emerging memories. However, compared to neural network training where the weights are multiple bits and all are frequently updated, for optimization the stored values can often be binary, with only a small fraction of the clauses updated at a time (e.g., when a clause with a single satisfied literal is found).

4.3.2.3. DYNAMICAL ASSOCIATIVE MEMORIES

Hopfield networks are attractor networks proposed for associative memories⁸⁸⁶ where the fixed points (or stable states) of the system correspond to memories, and the dynamics of the network is such that the system settles to the fixed point, which is closest to the initial state the system starts from. These networks can be implemented with coupled oscillators.^{887,888,889}

The associative memory application is widely used in the tasks of voice and image recognition, which can be performed in a cellular neural network architecture.^{890,891} In this work, five decimal digits, '1'–'5', are associated with the other five digits, '6'–'0'. Hebbian learning is used for storing patterns.⁸⁹² Patterns with noisy input pixels can still be recalled. The delay per cellular neural network operation is dependent on the input pattern, input noise, and thermal noise.

A key challenge for oscillator-based associative memories is storage capacity. A fully connected net with N units can only store around $0.15N$ memories while requiring N^2 weights, resulting in a poor memory density.⁸⁹³

4.3.2.4. DIFFERENTIAL EQUATION MODELING

The circuit dynamics of CMOS systems can serve as computational primitives for solving differential equations. The premise of this style of computing is to map continuous-valued and continuous-time solutions to ordinary differential equations onto the currents and voltages of analog electronic circuits, the dynamics of which are also described by ODEs. This basic idea of analog computing for solving ODEs originated in mid-20th Century analog computers and was extended to combine analog and digital representations to offer a wider tradeoff of dynamic range and precision.⁸⁹⁴ The idea of analog computing has since been revisited in CMOS; the implementations in modern device technologies demonstrate reconfigurability, scalability, and energy efficiency.^{895,896,897,898}

The fundamental operation of solving ODEs is integration, which in analog computers is carried out via capacitors that integrate current over time. To form arbitrarily complex systems of ODEs, the analog computer can include constant coefficient multipliers, variable-variable multipliers, and summers to construct polynomial functions, and non-polynomial functions can also be made via hybrid techniques that involve lookup tables.^{899, 900} An important extension to ODEs in the form of stochastic differential equations can be tackled by introducing controllable noise sources.⁹⁰¹

Using analog systems to solve ODEs confers several benefits. 1) Deeply embedded cyber-physical systems can take advantage of analog computation to eliminate the need to digitize analog sensory inputs and actuator outputs. 2) Analog integration avoids some difficulties in algorithmic numerical integration on digital architectures such as round-off error due to digital representations and truncation error due to numerical approximations.⁹⁰² 3) With appropriate design of the analog circuits, analog computing can handle numerical difficulties due to stiffness in the ODEs being solved.

Potential downsides include: 1) Difficulty in realizing more advanced ODE solving algorithms such as high-order quadrature. 2) Difficulty in offering both high dynamic range and high precision due to physical costs in area and power consumption. 3) Difficulty in implementing in device technologies that are not optimized for mixed-signal design.

Solving partial differential equations (PDEs) is even more important to modern scientific computation than solving ODEs. In PDEs, problem variables span space in addition to time. This class of problems introduces a wide range of options in terms of whether continuous space, time, and variable values are discretized. A guiding principle for research in this area is to survey the taxonomy of PDE types and demonstrate analog computer implementations of key benchmark solution methods and problems, such as the finite-difference time-domain approach to Maxwell's equations.^{903,904} Given the sheer diversity in PDE problem types and numerical methods supported by digital high-performance computers, a fruitful approach has been to retain the space and time discretization steps in numerical methods for PDEs, and only map the core ODE solving kernels to analog co-processors.

An even broader use case for analog computing is to use the circuit dynamics to solve algebraic and satisfiability problems entirely outside of differential equations.⁹⁰⁵ The key idea here is to view the differential equations as a continuous interpolation of problem variables as they get updated over the course of discrete algorithm steps. For example, steepest gradient descent algorithms for linear optimization can be rephrased as ODEs such that analog computers can tackle linear algebra.^{906,907} Likewise, Newton methods can be rephrased such that analog computers can provide initial seeds for solving nonlinear systems of equations.⁹⁰⁸

4.3.2.5. **HARDWARE-BASED RESERVOIR COMPUTING**

Reservoir computing (RC) remains an appealing machine learning paradigm for processing time-dependent signals due to its high computational power combined with a relatively simple learning protocol⁹⁰⁹. In software-based RC, the reservoir typically consists of a recurrent neural network with randomly weighted connections. Recently, there have been suggestions to replace the traditional reservoir with an ad-hoc nonlinear embedding of the input signals, forming the basis of what is now known as next-generation RC⁹¹⁰. Such next-generation reservoir computing implementations typically require small amounts of data for training, but ongoing research continues to test whether they can fully replace classical approaches.

Hardware-based RC implementations have emerged as competitive alternatives for information processing, offering superior performance in power consumption, latency, and computing speed with minimal optimization⁹¹¹. Advances in this area include the development of quantum RC, which uses quantum mechanical properties to enhance computational capabilities^{912,913}. Additionally, researchers are exploring the extension of RC to multi-layer or deep RC implementations, which involve stacking multiple layers of reservoirs to process information in a hierarchical manner. This approach has the potential to increase the complexity and power of the computations, enabling RC to tackle more complex tasks^{914,915}.

Various technological platforms are being explored to optimize RC for energy efficiency and processing rates. These platforms include Photonic Systems^{916,917}, which utilize light to process information, and can offer high-speed data processing with low latencies, Electronic Systems^{918,919}, which benefit from compatibility to mature fabrication technologies, Spintronic Systems^{920,921}, which exploit the spin of electrons in materials to provide non-volatile storage and potentially faster processing speeds, Mechanical Systems⁹²², which use mechanical vibrations or movements to perform computations and can be beneficial in specific contexts where electronic systems might fail.

RC shares many similarities with dynamical systems computing and has inherited some perspectives from analog computing⁹²³. Recent advancements suggest a promising future for RC in applications such as real-time signal processing, adaptive control systems, edge computing, and beyond. Continued innovation is essential to enhance the robustness, scalability, and integration of hardware-based RC with existing technologies.

4.3.3. **PROBABILISTIC SYSTEMS**

Traditionally, conventional computational processes are designed to be deterministic, with computational results determined by the machine's initial state and inputs. Nevertheless, computations that are *intentionally* designed to behave randomly or stochastically, even at the level of individual bit-operations, are of interest and can have many useful applications in *naturally* probabilistic tasks. Applications of this approach include powerful statistical sampling algorithms used in Monte Carlo simulation (Markov Chain Monte Carlo, direct sampling, simulated annealing, parallel Tempering, etc.), machine learning and artificial intelligence algorithms (Bayesian networks, Boltzmann machines, deep Boltzmann machines), randomized algorithms^{924,925} as studied in computational complexity theory, and cryptographically secure random number generation for generating secure private keys. Noise in biological neurons is beneficial for information processing in nonlinear systems and is essential for computation and learning in cortical microcircuits.^{926,927,928,929}

Obtaining randomness in traditional CMOS is difficult and typically relies on costly pseudo-random number generators. This requires a large circuit block and significant computational effort to obtain high quality random numbers. Despite this computational difficulty, industry and academia have implemented purely digital CMOS-based probabilistic systems due to the

importance of randomized and probabilistic algorithms. These digital systems are readily available and easy to scale to an extent though the upper limits imposed still restrict very large-scale experiments.

Several new devices have been proposed to obtain true randomness as discussed in §4.4.3. These allow for a random bit to be generated with a single device, often leveraging the intrinsic noise out of natural phenomena. Chaotic devices can be used to turn poor quality randomness into high quality random numbers⁹³⁰.

One example of probabilistic computation is achieved by p-bits. The main function of a p-bit is to provide tunable randomness of a digitized voltage at its output terminal controlled by an analog input terminal⁹³¹. The main advantage of a device-based p-bit comes from its compact, low-power implementation of a complex functionality (tunable randomness) compared to digital implementations. In the context of machine learning, this functionality is an approximate hardware representation of a binary stochastic neuron, allowing a natural mapping of powerful algorithms developed for such stochastic neural networks. Some metrics for probabilistic Computing with p-bits is shown in Table BC4.4.

Table BC4.4 Metrics for Probabilistic Computing with p-bits

Metric	Target Values	Notes & references
Energy per random bit	MTJ-based: ~ 2 fJ all-digital: ~ 300 fJ	Energy consumption of the hardware required to generate a single random bit [i]
Time per random bit	~ 1 ns	Time needed for probabilistic hardware to generate a single random bit
Probabilistic bits/samples per nanosecond	MTJ-based: ~ 1e6 [i] all-digital: ~ 8-150 [ii]	System-level metric taking into account the total number of random bit generators along with the time per random bit (see Synapse time requirement metric) [ii]-[v]
Graph connectivity	32-bit Factorizer and 3SAT (Density ~ 0.2-1.5%) [ii] problem-dependent	Details regarding whether the graphs used are sparse or all-to-all along with mention of graph density and number of “colors” used [ii].
Synapse time requirement	$\tau_{\text{syn}} < \tau_{\text{neuron}}$	Need to have $\tau_{\text{syn}} < \tau_{\text{neuron}}$ for correct functionality of dynamical equations [vi]
Synapse Energy	Per random bit: ~ 10 μ W Total: ~ 10 W	Energy consumed by the synaptic component corresponding to a single random bit [iii]
Samples to Solution	XORSAT: 10^{-8} to 10^{-4} s [vii] (problem-dependent)	This problem-dependent metric can also be represented as Time to Solution (TTS) and is plotted against the problem size N (say for an N-city TSP or an N-bit factorization problem) [vii]

4.3.3.1. MONTE CARLO SIMULATION

Probabilistic circuits have also been discussed in the context of traditional Monte Carlo simulation⁹³². A broad application of Monte Carlo simulation is that of integration. At high dimensions, Monte Carlo integration is known to scale better than deterministic numerical integration techniques⁹³³ and probabilistic circuits accelerating Monte Carlo simulations can potentially result in energy and area savings with massively parallel probabilistic bits in hardware.

4.3.3.2. ISING MACHINES AND SPECIAL SOLVERS FOR COMBINATORIAL OPTIMIZATION

Many of the optimization problems that are found in modern operations research—such as routing, scheduling, and other types of resource allocation—are intractably hard. Consider this example: finding an optimal route among three cities can be done using the digits on two hands, but with 12 45-cities, we are left with more than 40 billion routes to choose from. As the size of the problem grows, the resources needed to solve the problem increases exponentially. Finding even approximate solutions to large combinatorial optimization problems are prohibitively resource-intensive with the best supercomputers we have. Exact solutions to these problems are known in computational complexity theory to be NP-hard (non-deterministic polynomial time hard), meaning that it is too hard for any computer, analog or digital, to solve exactly, in general, and specifically at large scale. The most common way to address these large-scale problems with the classical computers is using the variants of Monte Carlo methods. Yet, with the stagnation of the Moore’s law, alternative realizations of unconventional computing methodologies sit on the center of attention, with a possibility of replacing with ‘natural solvers’ for such optimization problems.

These special purpose computing systems, sometimes, called Ising Machines map the underlying mathematics of generalized Ising Models in their computing architecture to identify the lowest energy configurations of the Ising model. This model is particularly valuable in computational science because it can represent any NP-class problem with only polynomial overhead. In real-world scenarios of these combinatorial optimization problems are highly relevant to a broad range of applications, including planning, manufacturing, financial management, computer vision, artificial intelligence, drug design, and various other challenges⁹³⁴. Solving hard combinatorial optimization problems with powerful hardware and systems such as simulated bifurcation⁹³⁵, memcomputing⁹³⁶, oscillatory networks^{937 938}, coherent Ising machines⁹³⁹ has received significant attention. Many special CMOS solvers from companies (Hitachi⁹⁴⁰, Fujitsu⁹⁴¹, Toshiba⁹³⁵, and others) and academic laboratories^{942,943,944} to solve hard optimization problems have been designed and implemented to solve select optimization problems. A crucial point to emphasize is what is meant by solving the Ising model. There are many ways to compute approximate solutions, meaning finding a good solution but not necessarily the best one. In recognition of this, there have been many approaches that exploit the inherent computational ability and parallelism in physical processes to solve these hard optimization problems. Such generalized Ising Models have been shown to map to many combinatorial optimization problems⁹⁴⁵ and performances have been compared to D-Wave, despite the technical difference between D-Wave's simulated quantum annealing machines.

Ising models can employ probabilistic activation units to benefit noise perturbations in the energy landscape to discover alternative paths to ground state preventing local-minima-convergences. In a series of recent papers, Camsari, Datta and collaborators proposed a type of probabilistic computing model introducing the concept of p -bits and p -circuits.^{946,947} The authors explored how p -bits can be compactly realized by leveraging existing magnetoresistive RAM (MRAM) technology⁹⁴⁸ and showed different applications of p -circuits including image recognition (inference),⁹⁴⁹ combinatorial optimization^{950,951}, enhanced type of Boolean logic that allows invertible operation^{952,953}, and different algorithms such as quantum annealing, quantum monte carlo, adaptive parallel tempering, higher order interacting units (see Shuvro et. al.⁹³¹ and references therein). The generic p -circuit consists of autonomously operating p -bits without any digital clocking circuitry, leading to a massively parallel architecture whose performance increases with the number of p -bits in the system.⁹⁵⁴

Another example that utilize energy minimization using a deterministic decision mechanism is Hopfield network (§4.3.2.2).⁹⁵⁵ A Hopfield network is a popular neural network with its output being calculated via a simple decision system (e.g., thresholding of input), which is then weighted and fed back to its input. The feedback weights define an energy landscape based on values emerging from the output. If a Hopfield network is initialized to a particular value, the network will follow a trajectory that will take it to a minimum, or local minimum, of the energy landscape.

As an example, consider using a Hopfield network to solve the Traveling Salesman Problem. The Traveling Salesman Problem is to find the best route for a salesman that needs to visit a series of cities, each pair separated by some distance. The salesman seeks the shortest route that visits each city exactly once and then returns to the starting city. In an analog system the weights are set to encode the intercity distances. The system drives the outputs to a starting point for the salesman's route. The system will settle into a candidate salesman's route in an amount of time equal to a few time constants of the feedback loop.

The method described above may find the ideal solution, or just a better but suboptimal solution. To improve the odds of finding the best solution, the circuit can include either a true random noise generator or a chaotic pseudo-random noise generator. Under control of an external digital computer, the Hopfield network is driven to a random starting point and released many times in a cycle. The randomness causes many of the starting points to be different, making it more likely that the system will find the global minimum. The digital computer collects all the results, checking each to see which is best.

Such a system leverages multiple new circuit blocks including both a crossbar to encode the n intercity distances (that can be built using memristors), an analog neuron to run the Hopfield network, and either chaos or a noise generator to get randomness. As the Traveling Salesman problem is NP-hard, no solution method can solve it exactly at scale. Nevertheless, the analog solution seems to be at least comparable in efficiency with some software algorithms. For instance, a memristor based Hopfield network has been built.⁹⁵⁶

Another important application for combinatorial optimization, specifically graph coloring, was described⁹⁵⁷ and developed theoretically with support from experimental demonstrations using relaxation oscillators based on phase-change IMT materials. An architecture based on non-repeating phase relations⁹⁵⁸ between fabricated CMOS oscillators tries to emulate stochastic local search (SLS) for constraint satisfaction problems.

4.3.3.3. PROBABILISTIC MACHINE LEARNING

Ising Machines have been implemented in various platforms and show great promise for solving combinatorial optimization problems. Despite the diversity in hardware from photonic⁹⁵⁹, electronic oscillators⁹⁶⁰, analog⁹⁶¹ and digital CMOS^{962,950} the application domain of Ising Machines has so far been limited to combinatorial optimization problems. Recent advancements have

extended the potential of Ising Machines far beyond their traditional role in combinatorial optimization. Researchers have started to explore how to combine Ising Machines with probabilistic machine learning, opening a new paradigm in this AI-driven.

One remarkable innovation is the nanoscale implementation of a Boltzmann machine using individual atoms on a surface⁹⁶³. This atomic-scale device has been proven to perform machine learning tasks and self-adaptation to optimize its performance. More recently, the "probabilistic bits" (p-bits) based Ising Machines have been introduced to train deep generative artificial intelligence (AI) models on probabilistic hardware, indicating a significant advancement in the field of probabilistic machine learning⁹⁶⁴. The p-bit community has extended the potential application of p-circuits to Bayesian networks by developing autonomous hardware architectures using embedded magnetoresistive RAM (MRAM) technology⁹⁶⁵. Furthermore, in situ learning using probabilistic circuits with stochastic magnetic tunnel junctions has been explored to enhance the performance and efficiency of machine learning applications⁹⁶⁶.

Stochastic computing techniques have also been demonstrated to implement binarized neural networks in hardware for machine learning tasks⁹⁶⁷. Moreover, the natural variability of memristors has been exploited for in situ learning through Markov chain Monte Carlo sampling, optimizing machine learning models directly on hardware⁹⁶⁸. In addition, a method for optimizing Ising machines using equilibrium propagation has been presented, which improves their ability to solve machine learning problems with improved performance⁹⁶⁹.

Recent breakthroughs in the modern MRAM industry have led to production-ready integrated chips with up to 1 Gb cell densities, thus leveraging this technology could lead to application-specific probabilistic coprocessors with broad applications for the active fields of Quantum Computing and Machine Learning⁹⁷⁰.

4.3.3.4. QUANTUM EMULATION WITH PROBABILISTIC SYSTEMS

Quantum Monte Carlo (QMC) techniques are essential for understanding quantum many-body systems and exploring various quantum phenomena in the absence of general-purpose quantum computers. Recent advancements in hardware technologies have enabled significant improvements in classical computing performance, especially using probabilistic bits (p-bits). A mapping to emulate stoquastic quantum Hamiltonians with p-bits was proposed by Camsari et. al.⁹⁷¹ which makes the use of discrete-time (DT) Path Integral Monte Carlo (PIMC) approach based on the Suzuki-Trotter transformation. This was later extended to emulate non-stoquastic Hamiltonians⁹⁷² and even to emulate gate-based quantum computers⁹⁷³ by incorporating sign correction. FPGA based Ising computers were developed by Okuyama et. al.⁹⁷⁴ and Waidyasooriya et. al.⁹⁷⁵ to perform simulated quantum annealing (SQA). Grimaldi et. al.⁹⁷⁶, provided a performance evaluation of SQA with MTJ-based p-bits augmented with an FPGA based p-computer. In a recent study by Chowdhury et. al.⁹⁷⁷, an FPGA-based processor based on p-bits is employed to accelerate QMC simulations of a 2D square-octagonal quantum lattice. The p-bit-based system leverages massive parallelism and efficient synapses to maximize computational sweeps per clock cycle, achieving a three-order-of-magnitude improvement in convergence time on a programmable gate array (FPGA) compared to traditional CPUs. Further speedups are anticipated through SPICE simulations, suggesting potential acceleration of up to five to six orders of magnitude for simulating transverse field Ising models (TFIM). This hardware-accelerated approach offers a viable classical counterpart to quantum annealers, significantly advancing the simulation capabilities for quantum systems.

The integration of machine learning (ML) techniques with quantum many-body systems holds great promise, particularly when leveraging probabilistic systems. By training stochastic neural networks, such as Boltzmann machines (BMs), to approximate solutions to quantum many-body problems, significant advancements can be made in simulating these complex systems. Scaled p-computers, which utilize millions of magnetic p-bits, enable massively parallel and energy-efficient hardware implementations of deep BMs, facilitating the simulation of practical quantum systems. Chowdhury et. al., proposed that using p-bits in sparse, hardware-aware graphs can, in principle, allow for the efficient learning of the ground state of quantum Hamiltonians^{978,931}. This approach combines probabilistic sampling with iterative weight updates, enhancing the quality of the variational guesses for the quantum system's wave function. As a result, the use of p-computers in machine learning applications paves the way for scalable and efficient simulations of quantum many-body systems, addressing the limitations of classical ML techniques and overcoming the practical challenges of quantum computing.

4.3.3.5. CRYPTOGRAPHY

Digital computers often rely on pseudo-random number generators that produce random numbers deterministically given an input seed. For cryptographic applications, certifiable true random number generation (TRNGs) is an important requirement and many stochastic noise sources can be used to generate TRNGs. Among others, optoelectronic (single photon avalanche diodes⁹⁷⁹), magnetic (magnetic tunnel junctions⁹⁸⁰) and metal-insulator phase transition⁹⁸¹ (stochasticity in VO₂ transitions) based true random number generators have been proposed and tested against statistical suites to test the quality of randomness.

4.3.3.6. UNCERTAINTY QUANTIFICATION WITH BAYESIAN NEURAL NETWORKS

Bayesian neural network (BNN) inference is extremely computationally intensive as each weight is sampled from a probability distribution and the network must be sampled multiple times to provide an uncertainty estimate. To efficiently perform this computation, analog matrix vector multiplications with weights that have tunable noise characteristics are needed. The critical device property needed for BNNs is the ability to vary the standard deviation of the noise (σ) over a large range. This is measured by the ratio of the maximum and minimum σ . It's been shown that only the programmable standard deviation of the noise is critical and not the shape of the noise distribution⁹⁸². Several device approaches for developing a programmable noise distribution have been explored including programmable Vanadium Oxide⁹⁸³ and magnetic tunnel junctions⁹⁸⁴.

4.3.4. REVERSIBLE COMPUTING

In reversible computing, the aim is for physical processes to be both logically reversible (no loss of known information) and thermodynamically reversible (minimal increase in physical entropy per operation, ideally approaching zero).

Rolf Landauer's 1961 work⁹⁸⁵ established a fundamental limit on the energy efficiency of conventional, irreversible digital operations. This Landauer's limit states that for every bit of correlated information lost from a computational state, at least $kT \ln 2$ of energy (where k is Boltzmann's constant and T is the heat bath temperature) must be dissipated as heat. This limit applies only to correlated information; uncorrelated random bits can be transferred without thermodynamic cost.

An important caveat to be aware of is that Landauer's limit only applies to computational information that is *correlated* with other available information, as opposed to independent random information⁹⁸⁶. A computational bit that bears *no* correlations with other available bits is, in effect, *already* entropy, and thus it can be transferred back and forth between a stable, digital form in a computer and a rapidly-fluctuating physical form in a thermal environment with asymptotically zero net increase in total entropy.

Since most bits in a computer are correlated, avoiding Landauer's limit in deterministic computation requires avoiding many-to-one transformations such as destructively overwriting or erasing a bit. As shown by Bennett⁹⁸⁷, any irreversible computation can be embedded into an equivalent reversible one. While this may incur algorithmic overheads in time or space^{988,989}, the potential energy savings can ultimately outweigh these costs. Therefore, reversible computing is considered the only path for general digital computation to continue increasing efficiency indefinitely, without fundamental limits.

In current adiabatic implementations, there's typically a linear tradeoff between energy dissipation (D) and delay (T) for a given operation, meaning their product, the dissipation-delay product ($D \times T$), is constant ($D \times T = \text{constant}$). This constant is also called the energy coefficient. While some analyses suggest this linear tradeoff might be fundamental⁹⁹⁰, others propose that exponential energy reduction with delay could be possible in certain quantum regimes^{991,992}. However, even among cases where the linear relation still applies, there are no known *technology-independent* lower bounds on the *value* of the dissipation-delay constant.

Even with thermal limitations, reversible computing can boost the overall performance of parallel processors. For instance, in 3D stacked chips where cooling is a bottleneck, adiabatic technologies enhance per-area performance proportionally to the square root of the processor's thickness.⁹⁹³ Similarly, for highly parallel applications with fixed power, the performance gains from adiabatic computing can scale significantly with improved energy efficiency.

Realizing the energy efficiency promise of reversible computing at a system level remains a significant engineering hurdle. Despite various proposed adiabatic and ballistic designs, achieving substantial gains in practice is challenging due to design complexity and parasitic energy losses (See §4.3.4.3 for further discussion.). No single "magic bullet" solution exists to address all energy loss mechanisms. While logical reversibility is necessary for physical reversibility in deterministic digital computations, it's not sufficient.

Despite these difficulties, physically reversible computing is the only path for general digital computing to advance indefinitely, unlike conventional methods limited by Landauer's principle. For example, adiabatic CMOS circuits could offer energy efficiency benefits for power-constrained applications like spacecraft if high-Q resonators are developed. Existing reversible superconducting logic (e.g., RQFP, nSQUID) already shows potential to dissipate less energy than the Landauer limit. For cryogenic systems, moving power supply dissipation to a higher-temperature environment can significantly reduce internal power consumption. Superconducting technologies' extremely low signal energies also ease AC supply design for high-performance computing. See §4.3.4.1 below for further discussion of superconducting reversible computing technologies.

Reversible computing can also integrate with probabilistic computing (§4.3.3). By absorbing environmental entropy for random bits, randomized reversible computations can return these bits without net thermodynamic cost, relaxing reversibility requirements for non-deterministic operations.

Beyond digital, the principles of reversible computing extend to analog computing. Adiabatic techniques have demonstrated record energy efficiencies in analog vector-matrix multiplication. The microscopic reversibility of physics suggests that carefully

engineered analog computers could achieve macroscopic reversibility and unlimited energy efficiency improvements. This would require isolating analog computational degrees of freedom from thermal ones and managing chaotic behavior, potentially by harnessing it for computational purposes, as seen in chaotic logic or reservoir computing.

REVERSIBLE ADIABATIC CMOS

As mentioned above, currently the most well-developed implementation technologies for reversible computing are those that utilize classical quasi-adiabatic transformations to carry out digital state transitions. Among such technologies, the most well-developed class of them at present has been referred to variously as *adiabatic CMOS*, *adiabatic transistor circuits*, or just *adiabatic circuits*. In traditional (irreversible, non-adiabatic) CMOS circuits, the full digital circuit-node signal energy of $\frac{1}{2}CV^2$ is dissipated to heat on every digital switching event. In contrast, the use of classical quasi-adiabatic transitions for switching reduces the associated local dissipation by a factor of $\sim t/2RC$, where t is the transition time for a linear voltage ramp and R is the resistance of the charging path. This reduction yields a linear tradeoff between speed and energy dissipation per operation over a certain range of frequencies, where the *dissipation-delay product* or *energy coefficient* scales as $c_E \propto C^2V^2R$.

The earliest complete circuit families for sequential, pipelined reversible computing with adiabatic CMOS were developed in the 1990s in Tom Knight's group at MIT.^{994,995,996,997,998} These methods did not gain widespread traction at the time, perhaps in part because, to save energy, adiabatic CMOS must operate relatively slowly compared to the inherent RC propagation delay of the gates. However, in the period since the end of Dennard scaling in ~ 2005 , multi-core processor performance has become increasingly limited by power dissipation rather than by raw gate delays (witness the increasing amounts of “dark silicon” in modern processor designs), so, revisiting the adiabatic energy-delay tradeoff appears timely at present. Adiabatic switching holds promise as a design technique for the future, since it offers a means by which the dissipation-delay pareto frontier of any given CMOS technology might be expanded beyond the limits of what can be achieved using more conventional low-power design techniques, such as subthreshold operation.

Subsequent to the original MIT adiabatic logic families cited above, a number of other adiabatic logic design styles were also explored, such as *two-level adiabatic logic* (2LAL),⁹⁹⁹ *positive feedback logic* (PFAL),^{1000,1001} and *efficient charge recovery logic* (ECRL).^{1002,1003} In addition, applications to the design of secure circuits have also been explored, since the unique electrical behavior of adiabatic CMOS circuits can help to prevent non-invasive side channel attacks.^{1004,1005} And, general-purpose computing has been pursued in the design of adiabatic microprocessors.^{1006,1007} Simulation results¹⁰⁰⁷ indicate that, when operated adiabatically, advanced CMOS nodes can dissipate 1-2 orders of magnitude less energy per cycle than irreversible operation at relatively high frequencies (1-5 GHz).

Another very recent development is the introduction of fully adiabatic CMOS logic families that are also *fully static*,^{1008,1009} meaning that all nodes are at all times connected to a supply reference, which eliminates voltage-level drift from leakage and capacitively-induced voltage sag, which could otherwise occur on floating nodes and contribute to non-adiabatic losses. In principle, as leakage is reduced, these “perfectly adiabatic” logic styles are predicted to be capable of exceeding the energy efficiency of *any other* semiconductor-based form of digital logic.

One preliminary study carried out in 2022–23 analyzed the improvements in aggregate package-level performance that could theoretically be attained in future CMOS technology generations using adiabatic switching, and concluded that raw switching throughput density could be improved by up to $\sim 400\times$ using these techniques, given the process technology specifications outlined in the More Moore chapter of the 2022 roadmap.¹⁰¹⁰ However, that analysis did not yet include the design complexity overheads from reversible logic, and more detailed study is needed.

4.3.4.1. REVERSIBLE ADIABATIC SUPERCONDUCTING LOGIC

After adiabatic CMOS, currently the second most well-developed type of hardware technology for reversible computing based on classical adiabatic transformations is the class of adiabatic superconducting logic families. A significant motivation for the consideration of superconducting circuits for energy-efficient computation is the lossless nature of charge transport in Josephson junctions and superconducting wires, which act as switching elements and interconnects, respectively. Also, the naturally discrete phenomenon of *flux quantization* facilitates the restoration and stabilization of digital signals.

Two basic families of superconducting reversible digital elements, the parametric quantron (PQ) and quantum flux parametron (QFP)¹⁰¹¹ were proposed in early studies. Both approaches were based conceptually on the abstract physical model of adiabatic digital operations introduced by Landauer,⁹⁸⁵ in which the potential energy function of the digital element is transformed adiabatically between single-well and double-well configurations in the course of an operation. As with adiabatic CMOS, superconducting reversible logic gates utilizing this approach require AC driving waveforms, in this case to provide a time-dependent flux bias to each gate. More recently, a DC-powered superconducting reversible logic gate based on a negative-inductance SQUID (nSQUID) was proposed,¹⁰¹² and its energy dissipation was estimated to be a few kT per reversible bit-operation.

Recently, a further reduction of the energy dissipation of the QFP approach was achieved by appropriately optimizing the circuit parameters for the adiabatic mode of operation and eliminating the junction shunt resistance.¹⁰¹³ The energy dissipation of this improved adiabatic QFP (AQFP) was investigated numerically, taking thermal noise into account,¹⁰¹⁴ and found to be well below kT with a low error rate. The energy dissipation of a single AQFP gate was estimated to be 10 zJ per gate at 5 GHz by measuring the scattering parameters of a superconducting resonator coupled to an AQFP gate.¹⁰¹⁵ The energy dissipation per operation of an (irreversible) AQFP 8-b carry-lookahead adder was experimentally evaluated to be 24 kT per Josephson junction.¹⁰¹⁶

The first demonstration of logically and physically reversible operation of superconducting logic was performed using a newer reversible QFP (RQFP) design style. The basic RQFP element is a logic gate having three binary inputs x_0, x_1, x_2 and three outputs y_0, y_1, y_2 that are related by

$$(y_0, y_1, y_2) = (\text{MAJ}(\bar{x}_0, x_1, x_2), \text{MAJ}(x_0, \bar{x}_1, x_2), \text{MAJ}(x_0, x_1, \bar{x}_2))$$

where $\text{MAJ}(i, j, k) = (i \wedge j) \vee (j \wedge k) \vee (k \wedge i)$. This logically reversible element is composed of three AQFP splitter gates and three AQFP majority gates. The bidirectionality and time reversal symmetry of the RQFP gate were investigated, revealing the cause of the energy dissipation in logically irreversible AQFP logic. Using RQFP gates, the functionality of 1-bit and 8-bit¹⁰¹⁷ reversible full adders and an 8-word by 1-bit RQFP register file¹⁰¹⁸ were demonstrated, and their energy dissipation was numerically calculated, while accounting for thermal noise. It was found that the energy dissipation of the reversible full adder is much lower than that of the irreversible full adder; it becomes lower than the kT thermal energy at 4.2 K at frequencies below 20 MHz. The numerical simulation of a reversible flip-flop comprising RQFP demonstrates that both combinational and sequential logic operations can be executed without energy dissipation in the quasi-static limit, and that the minimum energy dissipation is determined by the amount of data erased.¹⁰¹⁹

4.3.4.2. OTHER REVERSIBLE TECHNOLOGY CONCEPTS

Beyond the adiabatic semiconductor/superconductor technologies discussed in §§0–4.3.4.1 above, over the years, a rather wide variety of disparate aspirational concepts for physical implementation technologies for adiabatic reversible computing have been described in the literature, although typically without accompanying physical demonstrations as of yet.

In the 1980s, the pioneering nanotechnology visionary K. Eric Drexler at MIT outlined a variety of concepts for nanoscale computing technologies, including adiabatic reversible versions, that were based on nanoscale *mechanical*, rather than electrical, interactions.^{1020, 1021, 1022} More recently, a group led by Ralph Merkle at the Institute for Molecular Manufacturing (IMM) has been developing an even more advanced nanomechanical reversible logic concept based on (single-atomic-bond) rotary bearings, which were analyzed to dissipate as little as $\sim 4 \times 10^{-26}$ J per operation at room temperature at speeds of 100 MHz. This is roughly $74,000\times$ lower energy than the Landauer limit at 300 K, and roughly $10^6\times$ smaller dissipation-delay product (DdP) than even *end-of-roadmap* CMOS. Although we do not yet have a nanomanufacturing technology that is capable of fabricating the atomically precise nanostructures envisioned in the IMM designs, this analysis nevertheless suggests how much farther the dissipation-delay frontier might someday be extended, beyond what is possible in today's semiconductor- and superconductor-based technologies.

Meanwhile, back in the electronic domain, since the 1990s, a number of adiabatic reversible computing concepts based on *single-electron devices* have been proposed.^{1023, 991, 1024} Notable among these is the *quantum-dot cellular automata* (QCA or QDCA) technology concept pioneered at Notre Dame, which has been taken up to the level of complete simulated processor designs.^{1025, 1026} However, there is not, as of yet, a viable manufacturing process for fabricating scalable QCA-based processors.

To conclude our review of the adiabatic approaches, we mention in passing an interesting concept for *adiabatic capacitive logic*,^{1027, 1028, 1029} which avoids the limitations on the efficiency of adiabatic CMOS due to leakage.

In addition to the various adiabatic approaches to reversible computing, there are also a number of *ballistic* reversible computing concepts. These are based on a rather different picture of the basic physical mechanism of reversible computing than the adiabatic approach suggested by Landauer.⁹⁸⁵ In the adiabatic approach, some *external* system (i.e., separate from the logic circuits) drives the adiabatic transformations of the computing system that carry out transitions between digital states. Whereas, in the *ballistic* picture, first conceived by Ed Fredkin, the physically reversible dynamics of the system is instead *self-contained*; in other words, individual entities (such as particles or pulses) carrying information-bearing degrees of freedom evolve forwards reversibly under their own (generalized) inertia, as it were, with no direct external influence.

We should note that the distinction between the two classes of approaches is not a perfectly crisp one, since, even in the adiabatic approach, the driving system (such as a resonant oscillator) can be viewed as evolving ballistically, and even in the ballistic approach, the interactions (e.g., elastic collisions) between individual ballistically-propagating information-bearing entities can be analyzed, on a sufficiently fine timescale, as adiabatic processes. So, to some extent, the distinction between the approaches is primarily just one of perspective and emphasis. However, generally speaking, the *adiabatic* approaches are characterized by a large-scale *separation* of the ballistic driving systems from the adiabatic logic transitions being driven, whereas in the *ballistic*

approaches, the ballistic properties are *distributed* throughout the system, and are associated to the lowest-level information-bearing entities themselves. We can also imagine that other, future approaches could interpolate between these two extremes. *E.g.*, one could imagine systems comprising large numbers of small ballistic oscillators, each driving just a small region of local adiabatic logic, with the various subsystems communicating timing information and data to each other via elastic interactions transmitted via (short- or long-range) couplings between individual oscillators.

In terms of practical realizations of a (fully-distributed) ballistic approach to reversible computing, the approaches to this that have been developed most intensively to date are based on superconducting electronics. This is a particularly convenient technology for ballistic computing, because, unlike in semiconductors, superconductors exhibit the phenomenon of naturally-discrete *single flux quanta* (SFQ), which can propagate near-ballistically along interconnects consisting of *passive transmission lines* (PTLs)¹⁰³⁰ or *long Josephson junctions* (LJJs).¹⁰³¹ Currently active efforts to develop reversible computing technologies focused on SFQ-based approaches include the *synchronous ballistic* approach, which has been explored since around 2010 at U. Maryland, and the *asynchronous ballistic* approach, which has been in development since 2016 at Sandia National Laboratories.¹⁰³²

4.3.4.3. CHALLENGES FOR REVERSIBLE COMPUTING

Despite the great long-term promise of reversible computing, many fundamental engineering challenges associated with the development of a practical reversible computing technology remain to be solved at this time. These include the following:

- Even at the level of very basic physics, a more complete understanding is needed of the fundamental (technology-independent) physical limitations of important cost metrics for reversible computing, such as the dissipation-delay product (DdP), or, more generally, energy dissipation as a function of delay, $E_{\text{diss}}(t_{\text{del}})$. An important question is: Are there universal lower bounds on this quantity that we can derive based on parameters such as temperature, or the length scale of devices, or perhaps based on some kind of generalized viscosity characteristics, or on other fundamental physical or materials-dependent parameters?^{1033,1034}
- New, more complete abstract (but still realistic) physical models of reversible computing should be crafted to illustrate how we might more closely saturate the above fundamental limits in real artifacts, pointing the way to new device and circuit concepts for reversible computing. Are there quantum-mechanical approaches or phenomena that could be usefully harnessed, such as shortcuts to adiabaticity (STA),¹⁰³⁵ topological invariants, dynamical variations of the quantum Zeno effect (QZE)^{1036,1037,990} or others, to help reversible computing technologies to further suppress the rate of entropy increase while still operating as quickly as possible?
- Facilitated by fundamental advances such as the above, new device and circuit concepts for reversible computing need to be developed that significantly reduce $E_{\text{diss}}(t_{\text{del}})$ at useful operating speeds while still being inexpensively manufacturable. Novel physical mechanisms for computing need to be developed *with reversible operation in mind from the start*.
- Meanwhile, to advance the achievable energy efficiency of adiabatic CMOS for cryogenic applications, novel FET device structures that are optimized to minimize leakage at particular cryogenic temperatures of interest with minimal impact on device performance (expressed in terms of, say, DdP) need to be developed.
- For adiabatic reversible computing technologies operating at room temperature, the logic signal energy (*e.g.*, $\frac{1}{2}CV^2$ in CMOS) remains a concern, since it still exists even in adiabatic circuits, and is merely transferred dynamically to the power-clock generator system, rather than being dissipated locally within the logic. Thus, to achieve significant overall energy savings at the system level, compared to the corresponding irreversible technology, this generator must be designed to efficiently recover a large fraction of this signal energy, *e.g.*, by comprising a resonant oscillator with a high *quality factor* (Q). Designing extremely high- Q resonators and clock distribution networks already demands advanced, high-precision engineering. Further, as RF designers know, achieving high Q implies narrow bandwidth. This in turn implies that the returned clock waveform must be extremely pristine—*e.g.*, any data-dependent back-action from the logic must be avoided. Thus, we must maintain a careful load balancing discipline, *e.g.*, via complementary signaling. And if bulk semiconductors are used, this adds another level of challenges relating to time-varying loads during transitions, since device capacitances are more voltage-dependent when depletion regions are not structurally constrained. Thus, fully depleted SOI, thin-film, or gate-all-around (GAA) nanosheet/nanowire FET geometries may be preferred.
- At higher levels, many advances in areas such as reversible architectures, EDA tool enhancements to support reversible design styles, reversible algorithms and so forth still need to be developed. As useful reversible computing hardware technologies emerge and develop, systems engineering practice will also need to evolve to best leverage the opportunities and tradeoffs offered by reversible design. However, all of these R&D areas remain in their infancy at this time.

4.3.5. DIGITAL IN-MEMORY SYSTEMS

4.3.5.1. BOOLEAN LOGIC OPERATIONS USING MEMORY

Boolean operations are another class of operations that can be performed *in situ* using memory cells. Memories that can perform Boolean operations *in situ* typically share two key properties: (1) they can enable multiple cells and/or multiple rows/columns to be active simultaneously, (2) the analog interaction of the stored values in two or more of the memory's cells produces an output that is a Boolean function of the cells. While it is not a strict requirement, Boolean-capable memories typically arrange their memory cells into two-dimensional arrays, which enables them to perform *in situ* Boolean operations on input memory cells that sit in the same row and/or column of the array as each other, and to store the result in an output memory cell that also sits in the same row and/or column as the input memory cells. The specific family of Boolean operations that can be performed using memory are highly dependent on the underlying memory technology and the architecture, as discussed below.

Boolean operations can be performed using conventional bitline-based volatile memories (e.g., DRAM^{1038,1039,1040,1041}, SRAM^{1042,1043,1044,1045,1046}) by taking advantage of, and making minor modifications to, the access circuitry. As a motivating example, Boolean AND and OR operations can be performed between the DRAM cells and the sense amplifier that share a bitline in a conventional 1T1C DRAM¹⁰⁴⁷. By simultaneously activating three cells on the same bitline, *charge sharing* shifts the bitline's voltage slightly towards the *average* of the voltages stored in the three cells, causing the sense amplifier to drive a voltage that represents the *majority* (MAJ) function of the three cells. This 3-input MAJ function can be converted into 2-input AND or OR by presetting one of the three cells to 0 or 1, respectively. Additional support is required in the array to provide Boolean completeness, such as the inclusion of dedicated rows of DRAM cells with special inverted bitlines to enable NOT^{1039,1048}. An alternative approach uses 3T1C DRAM cells to perform in-DRAM NOR operations¹⁰⁴⁰. Similar architectures can be used to perform Boolean operations using SRAM^{1049,1045,1046}, by taking advantage of the dual bitlines (one for the true value and one for the inverted value) connected to each SRAM cell and making minor modifications to the sensing circuitry to support multi-row reading.

Recent works propose to perform Boolean operations using crossbar-based non-volatile memories (e.g., ReRAM^{1050,1051,1052}, PCM¹⁰⁵³, STT-MRAM^{1054,1055}, SOT-MRAM¹⁰⁵⁶). While the precise implementations differ based on the underlying memory technology, we discuss one such example, MAGIC¹⁰⁵², which can perform NOR operations in a ReRAM crossbar. For two input cells and one output cell connected to the same row line in a crossbar, MAGIC performs 2-input NOR by applying a fixed voltage by applying a column selection voltage of V_{nor} to the column lines connected to the two input cells, while keeping the row selection voltage of the row line with the three cells floating. (Note that we can also perform NOR on two input cells connected to the same column, but the voltages can differ.) By floating the row selection voltage, the current on the row line settles to a value that is equivalent to the NOR function of the two cells. Setting a column selection voltage of GND to the column line connected to the output cell stores the NOR value in the output cell.

One downside of performing Boolean logic operations in memory is the long latency of performing common operations such as addition and multiplication, which often need to be performed in a bit-serial manner (or require additional arithmetic units in memory). To amortize these latencies, Boolean-capable memories typically perform bit-parallel operations on an entire row or column of data at a time^{1042,1048,1057}. Recent works provide various architectural abstractions and/or frameworks that expose bit-parallel operations as vector operations^{1042,1039,1057} or as single-instruction multiple-thread (SIMT; e.g., GPU) instructions¹⁰⁴³. A key limiting factor of bit-parallel Boolean operations is the current carrying capacity of the crossbar wires¹⁰⁵⁷. For example, in ReRAM, this limits practically achievable column-wide NOR to arrays where the column length is less than approximately 200 cells, requiring other techniques (e.g., pipelining¹⁰⁵⁷) to fully amortize the overheads of bit serialization.

The Boolean-capable operations described above have been proposed for single-level cell memories (i.e., memories that store only one bit of data per cell). While this limitation reduces the information density compared to other in-memory functions that can operate on multi-level cells, it eases the ability to fabricate logic-capable memories, by reducing the need to be sensitive to non-linear cell functions that can be difficult to control with peripheral circuitry.

4.3.5.2. PROCESSING-NEAR-MEMORY CIRCUITS

An alternative to *in situ* processing using memory cells is *processing-near-memory* (PNM) or *near-memory processing*, where dedicated logic elements for compute are integrated near (but not inside) a memory array¹⁰⁵⁸. Logic elements for processing-near-memory are made up of digital transistor-based circuits. While any arbitrary CMOS-like logic can be implemented, PNM logic elements tend to be area-constrained and/or power-constrained, and thus tend to consist of simpler logic than modern conventional CPUs. There are three main approaches to integrating PNM circuits with memory arrays.

The first approach, *near-bank acceleration*^{1059,1060,1061,1062,1063} integrates logic elements close to each memory bank, with the logic sitting adjacent to the peripheral circuitry of a bank within the same die as the memory array. Near-bank acceleration avoids the need for data to go through (semi-)global I/O drivers during PNM compute and allows the number of logic elements to scale

linearly with the number of memory banks. However, the complexity of logic that can be implemented near the bank is significantly limited, in order to avoid significant disruption of the peripheral circuit design.

The second approach, *logic layer acceleration*^{1064,1065,1066}, takes advantage of the design of modern 3D-stacked memory chips. The High-Bandwidth Memory (HBM) specification¹⁰⁶⁷ includes an optional logic layer known as a base logic die, which is integrated as the lowest die in the vertical 3D stack. (The Hybrid Memory Cube (HMC)¹⁰⁶⁸, which is not currently in active production, also includes a logic layer.) The logic layer can be used to implement global I/O circuits and logic elements, which have access to the through-silicon vias (TSVs) that span the chip. Logic elements implemented in the logic layer can (but do not need to) be distributed across the chip-level memory partitions (e.g., pseudo channels, vaults) that exist in HBM and HMC chips. One challenge of logic layer acceleration is that the logic layer may be fabricated using a DRAM-optimized CMOS manufacturing process technology and may be suboptimal for implementing logic transistors.

The third approach, *discrete die acceleration*, is similar to logic layer acceleration, but allows for entirely separate dies/chips containing the logic elements to be tightly integrated with memory dies/chips. One method for discrete die acceleration makes use of 2.5D/3D integration to connect discrete CMOS logic dies to the internal pins/TSVs of a memory^{1069,1070}. Silicon interposers can facilitate 2.5D integration and can potentially allow for relatively large logic dies that make use of a logic-optimized CMOS manufacturing process technology. However, logic dies attached with silicon interposers must be physically placed side-by-side with the memory, which can create power delivery and/or thermal dissipation challenges for large dies. A second method for discrete die acceleration uses buffer chips that are mounted on certain memory modules to house the logic elements¹⁰⁷¹. While implementing PNM circuits in buffer chips allows for easy and cost-effective integration with commodity memory chips, communication between the buffer chip and the memory chips can often still traverse pin-limited buses that consume significant energy and have tight bandwidth constraints.

To date, there are two commercial PNM-based products that are available, both of which employ near-bank acceleration. UPMEM produces PIM-enabled DIMMs (dual inline memory modules), which include PIM chips that contain a DDR4 DRAM array integrated with general-purpose in-order CPU cores^{1072,1073}. Samsung produces HBM-PIM, where application-specific PNM logic is integrated with HBM arrays^{1060,1061, 1074}. Currently available HBM-PIM products support deep neural network (DNN) computation, but products with logic elements for other applications are expected to be released in the future.

4.4. ENABLING DEVICES

4.4.1. SYNAPTIC DEVICES

This section lists some specific device technologies that are useful in analog computing.

4.4.1.1. MEMORY BASED DEVICES

In general, synaptic devices also need to store a synaptic state and are therefore often based on a memory element. Consequently, all but the single flux quantum and photonic based systems are described in the memory section, §2. The memory-based synaptic devices surveyed include:

- ReRAM
- Phase Change Memory
- Ion-Insertion Redox Transistor:
- Floating Gate
- Capacitor-on-Gate
- Charge-Based Analog Arrays for VMM
- Single Flux Quantum Neural Network Inference
- Magnetic Neural Network Devices

4.4.1.2. SINGLE FLUX QUANTUM NEURAL NETWORK INFERENCE

Josephson junctions assembled into single flux quantum (SFQ) circuits form a natural neuromorphic system.¹⁰⁷⁵ In this technology, SFQ pulses act as the action potential for a spiking neuron, and superconducting transmission lines act as axons. The synaptic or weighting function can be implemented with a flux-biased Josephson junction, quantum phase-slip junctions, or magnetic nano-clustered junctions.^{1076, 1077,1078, 1079,1080} These elements represent the basic functionality of an artificial spiking neural network (SNN).

Superconducting and magnetic device technologies are relatively mature among emerging devices. On the magnetic device side, large non-volatile memories based on magnetic tunnel junctions, using spin-polarized currents to write, have been commercialized.^{1081,1082} On the superconducting device side, high speed microprocessors and communication systems have been developed, based mostly on superconducting tunnel junctions.^{1083,1084}

When implementing the synaptic function with a magnetic nanocluster Josephson junction, the superconducting order parameter is modulated by the magnetic state of the nanoclusters in the barrier. Josephson junction base neural networks have potential operating frequencies above 100 GHz, while simultaneously have the potential for spiking and training energies of roughly 10^{-20} J and 10^{-18} J, respectively. Further energy savings can potentially be found with the related adiabatic quantum flux parametron implementation of neuromorphic circuits.¹⁰⁸⁵ High speed and low-power operation are promising for direct hardware implementation of neural networks that could perform inference at higher speeds and potentially lower power than alternatives even when including the cooling overhead to operate at 4 K.

Some of the key metrics for Josephson junction based computing are summarized in Table BC4.5

Table BC4.5 Metrics for Josephson junction-based inference

Metric	Target Values	Notes & references
Spiking energy value	$\sim 10^{-19}$ J	The lower the critical current value the lower the spiking energy. Values below 25 μ A ($\sim 0.5 \times 10^{-19}$ J) will be more susceptible to thermal noise at 4 K.
Analog synaptic range	$\sim 10 / 2$	A factor of 10 change in the current amplitude of an SFQ spike being passed into a soma circuit is the goal and should be achievable with magnetic nano-clustered junctions. Biased SQUID synapses may be limited to changes of about a factor of 2, similar to biological synaptic range.
Circuit size	$\sim 10^6$	The largest single chip Josephson junction based circuits are on the order of 1 million junctions. While inter connecting multiple chips is one way to continue scaling, junction count is a limiting factor in the size of problems available to this technology in its current form.
Spiking speed	~ 2 ps	Superconductor insulator superconductor type junctions can regularly achieve 2 ps pulse widths.
Timing jitter limitations	~ 10 ps	While this value depends significantly on the architecture, clock rates approaching 100 GHz is a target goal. Target rates of 10 GHz is more representative of current inference networks.
Analog fan-in	~ 100	In many superconducting neuromorphic architectures, there is an analog fan-in component. These are typically limited to 100 to 1 or less.

4.4.1.3. PHOTONIC VECTOR-MATRIX MULTIPLY (VMM)

There are currently two major bottlenecks in the energy efficiency of artificial intelligence accelerators: data movement, and the performance of multiply-accumulate (MAC) operations, or matrix multiplications. Light is an established communication medium and has traditionally been used to address data movement on a larger scale. As photonic links are scaled smaller and some of their practical problems addressed, photonic devices have the potential to address both of these bottlenecks on-chip simultaneously. Such photonic systems have been proposed in various configurations to accelerate neural network operations (see ^{1086,1087,1088}). However, their main advantage comes from addressing MAC operations directly. Here, we will look at the advantages of a simple matrix vector multiplication (MVM) unit made of integrated photonic components, in which inputs and outputs are encoded as light signals, and analog matrix multiplications are performed using a passive optical array.

In one possible instantiation of a photonic MVMs, power or phase can be used to encode information, while wavelength or phase selectivity can be used to program the network into a desired configuration. Wavelength division multiplexing (WDM) can further increasing the compute density of the approach. Classic examples include arrays of resonator weight banks^{1087,1089,1090} or Mach Zehnder interferometers¹⁰⁸⁸. The most important metrics are *energy efficiency (energy/MAC)*, *throughput per unit area (MACs/s/mm²)*, *speed (MVM/s)*, and *latency (s)*, where both speed and latency are measured across an entire matrix-vector

(MVM) operation. In CMOS, MVM operations are typically instantiated using systolic arrays¹⁰⁹¹ or SIMD units,¹⁰⁹² although there are some other architectures that use aspects of both.¹⁰⁹³ Digital systems are limited by the use of many transistors to represent simple operations and require machinery to coordinate the data movement involved in both weights and activations. The state-of-the-art values typically hover around 0.5–1 pJ/MAC, 0.5–1 TMACs/mm², 0.5–1 GMVM/s, and 1–2 μ s, respectively. In contrast, photonics MVM units could perform in range 2–10 fJ/MAC, 50 TMACs/mm², and $\sim$ 3 ps (1 clock cycle) per MVM operation. This performance depends on solving a number of practical problems which are possible to address in the short term. These are discussed below.

The largest bottleneck in efficient photonic MVM operations is the use of heaters for coarse tuning. Typically, the thermo-optic coefficient (dn/dT) is the strongest effect in most materials of interest (i.e., silicon), leading to heavy use of heaters in almost any tunable passive photonic system. There are several ways these can be eradicated, via the use of post-fabrication trimming^{1094,1095} or devices with an enhanced electro-optic coefficient (dn/dT , $d\alpha/dT$) such that heaters are not as necessary.^{1096,1097} The second largest problem is fabrication variation, which can result in parameter drifts for devices in an array. Resonators, for example, are highly sensitive to such variation, particularly across a wafer. This can also be remedied by enhancing the electro-optic coefficient of devices and some other tricks (see ^{1098,1099} for resonators). Third, the signal-to-noise ratio of the output must be optimized by reducing the intrinsic loss of photonic components together with the noise on the receiver. There are a variety of technologies that can address this—for example, lasers can be coupled on-chip with < 1 dB of loss,¹¹⁰⁰ photonic devices in state-of-the-art silicon foundries can be designed with low scattering,¹¹⁰¹ while detectors such as avalanche photodiodes,¹¹⁰² can reduce the relative contribution of thermal noise to the signal at the receiver.

Photonic arrays ultimately have very similar limits to analog electronic crossbar arrays, as analyzed in Ref. ¹¹⁰³: single-digit aJ/MAC efficiencies, and 100s of PMACs/s/mm² compute densities. However, photonic MVMs garner an advantage for larger MVM units, both in the size of the matrix and in the physical footprint of the core. Generally speaking, optimized photonic systems tend to perform worse than their electrical counterparts for smaller arrays (distances approximately < 100 μ m), but perform better for larger arrays (distances approximately > 100 μ m)¹¹⁰³. In that sense, photonic MVM arrays have a similar profile to photonic communication channels, with better performance over larger distances. However, photonic systems tend to have worse signal-to-noise ratios, as a result of several factors: (1) photonic channels are ultimately shot noise limited, which is more than an order of magnitude greater than the thermal noise limits on resistors,¹¹⁰³ and (2) to achieve similar compute densities to electronics, photonic MVMs must run faster to compensate for their larger device sizes, and noise is speed dependent. That being said, there are some architectural tricks to reduce this issue—for example, optical unitary operations¹⁰⁸⁸ can conserve the variation of the input and output signals, in contrast to other approaches such as resistive crossbar arrays,⁶⁸⁸ which by default, see a $\sqrt{N}$ decrease in effective signal variation from input to output for an $N \times N$ matrix operation.

Although photonic arrays exhibit some fundamental advantages over analog electronics (particularly for large matrix sizes or large physical sizes), a more important question is whether photonics arrays are practical. Thankfully, the transceiver industry has created a silicon photonic ecosystem fully compatible with high volume manufacturing (HVM). Compared to CMOS chips, photonics has costlier packaging, largely because light generation cannot be done easily in silicon—in fact, the cost of a production photonic chip is dominated by packaging. In addition, the tools required for the design and testing of large-scale photonic systems ($> 10k$ components) are still early in early development—analogue photonic systems must grapple with the challenge of addressing yield, variability, precision, and tunability. Nonetheless, the total cost to produce a photonic chip package at high volume is dipping below one hundred dollars, and it is expected that the trend will continue¹¹⁰⁴. The orders of magnitude advantages offered by photonics, and its potential for HVM scalability, makes it a viable inroad for the breakneck performance and innovation required by artificial intelligence algorithms in the years to come.

4.4.2. OSCILLATOR DEVICES

The principles that underlie the mechanism of combinatorial optimization in coupled oscillator systems can be considered universal, present in many different oscillator technologies. These general physical principles include parametric oscillation, injection locking, and sub-harmonic injection locking. While the same properties have been exhibited in a number of highly differentiated technologies, these solutions can differ considerably in their scalability, coupling scheme, energy efficiency, susceptibility to process variations or parasitics, and foundry compatibility.

4.4.2.1. ELECTRICAL OSCILLATORS:

Electrical oscillators have the advantage that they can be coupled electrically, which is relatively straightforward to do in a CMOS process. For ring oscillators, LC oscillators, and other electrical oscillators, positive and negative coupling coefficients can be implemented with resistors.¹¹⁰⁵ Arbitrary continuous-valued coupling coefficients can be realized using precisely programmed resistances. This allows an array of synaptic memory devices (such as ReRAM, flash, PCM) to be leveraged to connect together the oscillators, so as long as they are compatible with the oscillators' operating voltage and frequency. Such an array can provide

all-to-all connectivity among a large number of oscillators. High programming precision in these resistors enables scaling to larger problem sizes while retaining high solution quality.

4.4.2.1.1. RING OSCILLATORS:

Digital ring oscillators, based on CMOS inverter feedback loops, are perhaps the most readily implemented coupled oscillator system in today's foundry process; a large network of ring oscillators was recently demonstrated that solves combinatorial optimization problems.¹¹⁰⁵

4.4.2.1.2. LC OSCILLATORS:

Electrical oscillators can also be built from transistor-driven LC oscillator circuits, with CMOS cross-coupled pairs supplying the gain needed to sustain the oscillations. Gain can also be supplied using the negative differential resistance of a tunnel diode.¹¹⁰⁶ The oscillator amplitudes remain fixed during optimization but their phases are induced into bistability (separated by 180°) through sub-harmonic injection locking. Alternatively, phase bistability can be induced in an LC oscillator by modulating the capacitance with a pump at twice the resonance frequency.¹¹⁰⁷ This can be done using the nonlinear capacitance-voltage characteristics of a semiconductor varactor. Pumping the varactor introduces phase-sensitive gain, and a phase-bistable parametric oscillation grows from noise. In each case above, which of the two bistable phases is chosen is random for an isolated oscillator, depending on thermal noise and phase jitter. Coupling to other oscillators changes the probability distribution, but the process remains stochastic. This allows the system to find a different local minimum of the optimization problem when re-initialized. For scalability of on-chip solutions, LC oscillator systems can leverage the fact that at high frequencies (>10 GHz), integrated inductors have about the same area as the capacitor, so the oscillator can remain compact.^{1108,1109}

4.4.2.1.3. SPIN TORQUE OSCILLATORS:

Among electrical oscillators, one prominent effort is the use of spin torque oscillators (STOs) for providing a computational platform for machine learning, spiking neural networks, optimization, and other applications.^{1110,1111,1112,1113,1114,1115} STOs are based on the phenomenon that when a spin-polarized current passes through a ferromagnetic material, precession of the film's magnetization can be induced. This magnetic precession (which has ~1-10 GHz resonance) can be converted to an oscillating electrical signal by incorporating the ferromagnetic inside a magnetic tunnel junction, which would have an oscillating tunnel resistance when driven by a DC current. STOs can be coupled together resistively and sub-harmonic injection locking can be used to solve discrete optimization problems.¹¹¹⁶ However, the high current densities of STOs and the limited range of spin diffusion currents continue to pose serious challenges in creating coupled networks of such oscillators.

4.4.2.1.4. INSULATOR-METAL TRANSITION OSCILLATORS:

Another promising non-silicon technology for very compact oscillators is the IMT (insulator-metal transition) material-based oscillator technology.^{1117,1118,1119,1120,1121,1122} IMT devices, using materials such as VO₂, have a resistance that can change abruptly with the applied electric field (voltage). Connecting the device to a load circuit can induce self-sustained oscillations between the two resistance states, and sub-harmonic injection locking can be used to induce phase-bistable oscillations.^{1123,1124,1125,1126} IMT oscillators can operate at room temperature, and the IMT transition can occur on nanosecond time scales. As the oscillation mechanism is completely electrical, the coupling of oscillators can be done easily using electrical components. There have been other implementation efforts for electrical oscillators^{1127,1128,1129,1130,1131} but the focus has been to build high frequency and low power individual oscillators, as opposed to the demonstration of coupled systems of oscillators, or the generation of interesting dynamics for computing.

4.4.2.2. OPTICAL OSCILLATORS:

Optical oscillators have been studied^{1132,1133} and used for computing, but challenges include bulky components, difficult interfacing between the electrical and optical domains, and lack of programmability to enable an optical computing apparatus. Optical parametric oscillators are particularly useful for discrete optimization applications. Like their electrical counterpart, rely on a second-harmonic pump to induce a phase-bistable coherent oscillation from noise. This has been shown to solve the Ising problem with success, but coupling the oscillators together is more practically difficult in the optical domain. The Coherent Ising Machine implemented these oscillators as pulses traveling along a long optical fiber, but the coupling had to be accomplished using electronic circuits that read out optical signals, digitally applied coupling weights, and re-injected pulses into the fiber at precise times to achieve coupling.

4.4.2.3. ELECTROMECHANICAL OSCILLATORS:

Parametric oscillation has also been demonstrated in electromechanical systems.¹¹³⁴ For binary optimization, two spins can be encoded in the symmetric and anti-symmetric vibrational modes of a single mechanical resonator, which are independent. Modulating the spring constant of the resonator material at the appropriate frequency can implement specific coupling schemes between two spins. The modulation can be achieved using a voltage signal by exploiting the piezoelectric effect.

4.4.3. STOCHASTIC DEVICE TECHNOLOGIES FOR RANDOM BIT GENERATION

New devices based on memristors, avalanche breakdown, and magnetic tunnel junctions and other technologies have been proposed for generating random bits. A key enabling functionality for some architectures like probabilistic (p)-logic is the ability to tunably control the probability of a zero or one based on an input current or voltage. Several proposed devices are listed below.

4.4.3.1. MAGNETIC TUNNEL JUNCTIONS (MTJ)

Existing Embedded MRAM technology can be used to create a tunable random bit, provided that the Magnetic Tunnel Junctions are engineered to be thermally unstable. Such thermally unstable magnets have been experimentally observed. As MTJ dimensions are scaled, keeping them thermally stable becomes a hard challenge for memories, therefore destabilizing them in a controllable manner should be feasible in current technology.

Low-barrier MTJs can convert ambient thermal noise on nanomagnets into a fluctuating resistance, which is then used to build a device with tunable randomness when integrated with minimal CMOS periphery. The fluctuating resistance change due to thermal magnetic noise in MTJs can be measured by Tunneling Magneto resistance (TMR). State-of-the-art TMR values range from upwards of 100% to 600% demonstrated by the Tohoku Group,¹¹³⁵ and commercial STT-MRAM devices exhibit >100% TMR. A large TMR would enable a robust functional unit for controllable randomness. The theoretical limit for TMR in MgO-based MTJs has been reported¹¹³⁶ to be 1,000% and can presumably be larger. There is currently intense research activity in half-metallic ferromagnets to increase TMR.

Several key experimental breakthroughs on MTJs have been demonstrated showing nanosecond (GHz) fluctuations of low-barrier magnets, effectively proposing a true-random-number-generation in hardware at standard ambient temperature and pressure (SATP). These breakthroughs are shown with utilizing in-plane (IMA) magnet tunnel junctions by researchers from IBM¹¹³⁷, Tohoku University^{1138,1139}, Gutenberg University¹¹⁴⁰ and utilizing perpendicular magnetized tunnel junctions (PMA) from CEA-SPINTEC¹¹⁴¹. Another experimental breakthrough on chip integration for stochastic MTJs from Purdue University¹¹⁴² which could find use in probabilistic circuits integrated with transistor technology considering process compatibility.

Another focus point on addressing the existing issues of MTJs on the way of scalability and compatibility in read-write type of architectures. Typically, magnetic tunnel junctions are composed of a free layer with a fixed (reference) layer that is forming the effective resistance of the junction. Yet, these particular junctions show a significant voltage bias dependency which is a bottleneck in the scaling up due to the possible read disturbance in such MRAM type architectures and the necessity of fine tuning. A theoretical work has shown to design MTJs with no fixed layers¹¹⁴³, where symmetric free layers receive spin-currents with opposing signs, leading to nearly uniform fluctuations and almost no voltage-bias dependence. Recent experiments employing SAF-based MTJs from IBM¹¹⁴⁴ and double-free-layer MTJs from Tohoku University demonstrated the key features of bias-independence and nanosecond fluctuations. Further improving the double-free-layer MTJ designs by mitigating the unfavorable magnetostatic coupling addressed by recent theoretical work from UCSB with double-SAF-free MTJs proposed to be magnetic field and voltage bias inert devices for new generation of scalable building blocks¹¹⁴⁵.

4.4.3.2. SINGLE PHOTON AVALANCHE DIODES (SPAD)

The single photon avalanche diode (SPAD) is a reverse-biased diode operating beyond its breakdown voltage, that enables converting single photon absorption into amplified detectable pulse signal. When the source of uncorrelated photons are continues, SPAD provide output pulses of a Poisson process, making these devices eligible to be used as true-random number generators. These devices are controlled with incident illumination and reverse bias to achieve tunability with various architectures shown to be CMOS compatible. In a recent work, SPAD circuits¹¹⁴⁶ designed in a way to implement architectures for Ising and Potts models¹¹⁴⁷.

4.4.3.3. SINGLE-ELECTRON BIPOLAR AVALANCHE TRANSISTOR (SEBAT)

The single-electron bipolar avalanche transistor (SEBAT) is a novel Geiger-mode avalanche bipolar transistor structure.^{1148,1149} The device generates Poisson-distributed digital output pulses at rates between 1kHz and 20MHz. The pulse rate is linearly proportional to the emitted current. A MOS transistor is also formed within the base region of the device, allowing for voltage control of the pulse rate. The device is fully compatible with low-voltage CMOS circuits and standard digital process steps.

4.4.3.4. MEMRISTORS/RESISTIVE RAM

The intrinsic variability of memristive switching, particularly the switching delay time of memristors, can be a good source of stochasticity.^{1150,1151,1152} Such stochasticity originates from the ionic dynamics within the memristors.¹¹⁵³

4.4.3.5. CONTACT-RESISTIVE RANDOM ACCESS MEMORY (CRRAM)

CRRAM can be used for random number generation.¹¹⁵⁴ A CRRAM device may be based on a layer of silicon dioxide that is sandwiched between two electrodes; the bottom electrode could simply be the drain of a CMOS transistor.¹¹⁵⁵ During operation, the current flowing in a filament channel will be (randomly) impacted by any electrons trapped in the insulating layer. If a high voltage is applied to a device, the current in the filament channel will be large and not impacted by trapped electrons. However, with the application of a lower voltage, the width of a filament will shrink, and the trapped electrons will (randomly) influence output current.

4.4.3.6. CMOS

There are different ways to obtain random number generators (RNG) in CMOS using different physical noise sources, one being “jitter” in ring oscillators.¹¹⁵⁶ These TRNGs can be tuned into tunable random number generators as required but require significant amounts of area when compared to single device alternatives.

4.4.3.7. STOCHASTIC JOSEPHSON JUNCTION

Single flux quantum (SFQ) logic relies on voltage pulses generated by 2π phase slips of the superconducting order parameter across a Josephson junction. These voltage pulses have a time-integrated amplitude given by the flux quantum $\Phi_0 = 2 \times 10^{-15}$ Wb. A standard circuit model of a Josephson junction is the parallel connection of a supercurrent up to I_c the critical current, a normal state resistor, a capacitor, and a channel for the thermal noise term. The resulting dynamics are the same as a forced damped pendulum. The energy barrier is given by $(I_c \Phi_0)/(2\pi)$.

For stochastic operation one can operate with junctions that meet the condition $\delta = (2\pi k_B T)/(I_c \Phi_0) \sim 10$, where δ is the stochasticity, k_B is the Boltzmann constant and T is the temperature in kelvin. The exact value of the stochasticity is an important circuit parameter as it determines the frequency of spiking events. In this regime, the energy in a single flux quantum spike is sub-attojoule while the frequency can be greater than 100 GHz.¹¹⁵⁷

For values of the stochasticity less than 10, the dynamics are basically deterministic, whereas when the stochasticity is larger there is a significant stochastic component. The value of the stochasticity can be effectively tuned between the deterministic state and the stochastic state by changing the temperature of the circuit by a few degrees.¹¹⁵⁸ For example, stochastic Josephson junctions have been shown to make effective pseudo-sigmoid generators,¹¹⁵⁹ and stochastic Josephson junction spiking can perform the neural accumulate operation at speeds up to 70 GHz.¹¹⁶⁰

In addition to fast stochastic operation, the dynamics of Josephson junctions can simulation dynamic neuron behavior. Coupled Josephson junction neurons have been shown to exhibit a variety of dynamic neuron behaviors.¹¹⁶¹ Including phase-flip bifurcation and chaotic dynamics.^{1162,1163}

4.4.3.8. PHASE CHANGE ECRAM

Realizing tunability in most physical TRNGs is challenging because the conditions under which stochastic switching occurs are rather narrow. Electrochemical random access memory (ECRAM) could be a particularly attractive solution to realize device where noise could be dynamically programmed in a non-volatile manner, thus potentially enabling the merging of learning and stochastic neural networks. To this end, Oh et al. recently used oxygen vacancy tuning in a vanadium oxide Mott insulator to program electronic noise originating from stochastic switching during an IMT, and to show that this approach was effective for BNN implementation.¹¹⁶⁴ The basic mechanism involves modulation of the IMT temperature, and thus the way the lattice thermal fluctuations are amplified by the IMT. It is possible that a similar approach could be extended to other oxides such as NbO_2 and LaCoO_3 ,¹¹⁶⁵ that exhibit IMT to enable TRNGs that could operate over a wide temperature range¹¹⁶⁶.

5. BEYOND CMOS DEVICES FOR MORE-THAN-MOORE APPLICATIONS

5.1. EMERGING DEVICES FOR SECURITY APPLICATIONS

5.1.1. INTRODUCTION

Like performance, power, and reliability, hardware security is becoming a critical design consideration. Hardware security threats in the IC supply chain, include 1) counterfeiting of semiconductor components, 2) side-channel attacks, 3) invasive/semi-invasive reverse engineering, and 4) IP piracy. A rapid growth in the “Internet of Things” (IoT) only exacerbates problems. While hardware security enhancements and circuit protection methods can mitigate security threats in protected components, they often incur a high cost with respect to performance, power and/or cost.

Advances in emerging, post-CMOS technologies may provide hardware security researchers with new opportunities to change the passive role that CMOS technology currently plays in security applications. While many emerging technologies aim to sustain Moore's Law-based performance scaling and/or to improve energy efficiency,^{1167,1168,1169} emerging technologies also demonstrate unique features that could drastically simplify circuit structures for protection against hardware security threats. Security applications could not only benefit from the non-traditional I-V characteristics of some emerging devices, but also help shape research at the device level by raising security measures to the level of other design metrics.

At present, many emerging technologies being studied in the context of hardware security applications are related to designing physically unclonable functions (PUFs). Many post-CMOS devices^{1170,1171,1172} have been suggested as a pathway to a PUF design. (More detailed reviews are also available.¹¹⁷³) With a PUF, challenge/response pairs are mapped (typically in a trusted environment). Responses are derived from natural/random variations and disorders in an integrated circuit that cannot be copied (or cloned) by an adversary. PUFs have been employed for tasks such as device authentication,¹¹⁷³ to securely extract software,¹¹⁷⁴ in trusted Field Programmable Gate Arrays (FPGAs),¹¹⁷⁵ and for encrypted storage.¹¹⁷⁴ Post-CMOS devices also find utility as random number generators (RNGs) that may be employed for secure communication channels (e.g., to generate session keys¹¹⁷³). That said, while intriguing, PUFs and RNGs may only cover a small part of the hardware security landscape. (Furthermore, one must be careful that PUF designs based on emerging technologies do not depend on device characteristics that a designer would like to eliminate when considering utility for logic or memory.)

Given the many emerging devices being studied¹¹⁶⁷ and that few if any devices were proposed with hardware security as a “killer application,” this document also reports initial efforts as to how the unique I-V characteristics of emerging transistors that are not found in traditional MOSFETs could benefit hardware security applications.

Below, we review the efforts described above, beginning with efforts to design PUFs and RNGs with emerging technologies. How device characteristics can enable novel circuits to achieve hardware security-centric ends such as IP protection, logic locking, and the prevention of side channel attacks are also discussed.

5.1.2. PHYSICALLY UNCLONABLE FUNCTIONS (PUFs) AND EMERGING TECHNOLOGIES

A variety of different emerging logic and memory technologies have been considered in the context of PUFs. As has been reviewed,¹¹⁷³ variations in the required write time in spin torque transfer random access memory (STT-RAM) was proposed to create a domain wall memory PUF.¹¹⁷⁰ Other structures based on magnetic tunnel junctions have also been proposed.^{1176,1177} Variations in write times have also been exploited to produce unique responses in phase change memory (PCM) arrays.¹¹⁷⁸ The variability of ReRAM presents a natural opportunity for PUF implementation, and array demonstration has been reported.^{1179, 1180} At the array-level, variations in diode resistivity have also been used to derive challenge/response pairs from crossbar structures.¹¹⁸¹ PUFs based on graphene¹¹⁸² and carbon nanotubes have also been proposed/considered.¹¹⁸³

As a more representative case study, prior work¹¹⁷³ considers an array structure based on process variation in memristors^{1184,1185} to create a PUF structure (referred to as *NanoPUF*¹¹⁷³). NanoPUF is based on 1) a crossbar with memristors. 2) A challenge is applied to the memristor array by using a row decoder to apply a voltage amplitude (V_{dd}) to a given row that can vary in duration; a column decoder connects a given column to a resistance R_{load} . All other rows and columns remain floating. 3) A response circuit (to collect outputs to different challenges) would consist of R_{load} and a current comparator that compares I_{out} from a given column to a reference current I_{ref} . A logic 1 might be recorded if $I_{out} > I_{ref}$, while a logic 0 might be recorded if $I_{out} < I_{ref}$. With respect to PUF functionality, when a write pulse is applied, natural process variations will cause some memristors to turn on (leading to a logic 1), and others to remain off (leading to a logic 0). While the time of the right pulse serves as one variable,¹¹⁸⁴ the pulse's duration and amplitude may also be varied.¹¹⁸⁵

5.1.3. RANDOM NUMBER GENERATORS (RNGs) AND EMERGING TECHNOLOGIES

The inherent randomness in emerging devices can also be used to generate random numbers.¹¹⁷³ As a representative case study, prior work¹¹⁷³ explores an approach based on contact-resistive random access memory (CRRAM).¹¹⁸⁶ (Note that a CRRAM device may be based on a layer of silicon dioxide that is sandwiched between two electrodes; the bottom electrode could simply be the drain of a CMOS transistor—which in turn suggests that RNGs based on emerging technologies can be CMOS compatible.¹¹⁸⁷)

During operation, the current flowing in a filament channel will be (randomly) impacted by any electrons trapped in the insulating layer. If a high voltage is applied to a device, the current in the filament channel will be large and not impacted by trapped electrons. However, with the application of a lower voltage, the width of a filament will shrink, and the trapped electrons *will* (randomly) influence output current.¹¹⁸⁷ Indeed, RNGs based on emerging devices¹¹⁸⁷ can successfully pass randomness tests such as those provided by the National Institute of Standards and Technology (NIST).

As random number are derived from current passing through filaments, memristors, PCM, and RRAM devices can also be leveraged to build similar RNGs.¹¹⁷³

5.1.4. OTHER HARDWARE SECURITY PRIMITIVES BASED ON EMERGING TECHNOLOGIES

Below, other security-centric primitives (non-PUFs and non-RNGs) based on emerging technologies are also discussed. How new devices might be employed for IP protection and to prevent side channel attacks are considered. In each section, device characteristics of interest are discussed first. Subsequent discussions then consider how device characteristics can be employed to achieve a security centric end.

5.1.4.1. EMERGING TECHNOLOGIES FOR IP PROTECTION

Tunable Polarity—In many nanoscale FETs (45 nm and below), the superposition of n-type and p-type carriers is observable under normal bias conditions. The ambipolarity phenomenon exists in various materials such as silicon,¹¹⁸⁸ carbon nanotubes¹¹⁸⁹ and graphene.¹¹⁹⁰ By controlling ambipolarity, device polarity can be adjusted/tuned post-deployment. Transistors with a configurable polarity—e.g., carbon nanotubes,¹¹⁹¹ graphene,¹¹⁹² silicon nanowires (SiNWs),¹¹⁹³ and transition metal dichalcogenides (TMDs)¹¹⁹⁴—have already been experimentally demonstrated.

As more detailed examples, SiNW FETs have an ultra-thin body structure and lightly-doped channel, which provides the ability to change the carrier type in the channel by means of a gate. FET operation is enabled by the regulation of Schottky barriers at the source/drain junctions. The control gate (CG) acts conventionally by turning the device on and off via a gate voltage. The polarity gate (PG) acts on the side regions of the device, in proximity to the source/drain (S/D) Schottky junctions, switching the device polarity dynamically between n- and p-type. The input and output voltage levels are compatible, enabling directly cascable logic gates.¹¹⁹⁵

Ambipolarity is an inherent property of TFETs due to the use of different doping types for drain and source if an n/i/p doping profile is employed.¹¹⁹⁶ By properly biasing the n-doped and p-doped regions as well as the gate, a TFET can function either as an n- or p-type device, and no polarity gate is needed. As the magnitude of ambipolar current can be tuned (i.e., reduced) via doping or by increasing the drain extension length,¹¹⁹⁶ one can envision fabricating devices that could be better suited for logic as well as security-related applications. Given that the screening length in TMD devices scales with their body thickness, one can achieve substantial tunneling currents.

Polymorphic logic gates—The ability to dynamically change the polarity of a transistor opens the door to define the functionality of a layout or a netlist post fabrication. Though one may use field programmable gate arrays (FPGAs) to achieve the same goal, FPGAs cannot compete with ASICs in terms of performance and power, and an FPGA's reliance on configuration bits being stored in memory introduces another vulnerability. Security primitives to be discussed can serve as building blocks for IP protection, IP piracy prevention, and to counter hardware Trojan attacks.

Polymorphic logic circuits provide an effective way for logic encryption such that attackers cannot easily identify circuit functionality even though the entire netlist/layout is available. However, polymorphic logic gates have never been widely used in CMOS circuits mainly due to the difficulties in designing such circuits using CMOS technology.

SiNW FET based polymorphic gates to prevent IP piracy have been introduced.^{1197,1198} If the control gate (CG) of a SiNW FET is connected to a normal input while the polarity gate (PG) is treated as the polymorphic control input, we can easily change the circuit functionality through different configurations on the polymorphic control inputs without a performance penalty. For example, a SiNW FET-based NAND gate can be converted to a NOR gate, whereas a CMOS-based NAND cannot be converted to a fully functioning NOR by switching power and ground.

TFET-based polymorphic logic circuits have also recently been developed.¹¹⁹⁹ By properly biasing the gate, the n-doped region, and the p-doped region, a TFET device can function either as an n-type transistor or p-type transistor. If the n-doped region of the two parallel TFETs is connected to V_{DD} , and the p-doped region of the bottom TFET is connected to GND, the circuit behaves like a NAND gate. If the n-doped region of the two parallel TFETs is connected to GND and the p-doped region of the bottom TFET is connected to V_{dd} , the circuit behaves as a NOR gate. By using two MUXes (one at the top and the other at the bottom) to select between the two types of connections, the circuit then functions as a polymorphic gate where the control to the MUXes forms a 1-bit key.¹¹⁹⁹

One can readily design polymorphic functional modules using the low-cost polymorphic logic gates built from either SiNW FETs or TFETs that only perform a desired computation if properly configured. If some key components (e.g., the datapath) in an ASIC are designed in this manner, the chip is thus encrypted such that a key, i.e., the correct circuit configuration, is required to unlock the circuit functionality. Without the key, invalid users or attackers cannot use the circuit. Thus, IP cloning and IP piracy can be prevented with extremely low performance overhead. A 32-bit polymorphic adder using SiNW FETs has been designed and simulated. Two pairs of configuration bits (with up to 32-bits in length) are introduced and the adder can only perform addition functionality if the correct configuration bits are provided.

Camouflaging Layout—Split manufacturing and IC camouflaging are used to secure the CMOS fabrication process, albeit with high overhead and decreased circuit reliability. With CMOS camouflaging layouts, both power and area would increase significantly in order to achieve high levels of protection.¹²⁰⁰ A CMOS camouflaging layout that can function either as an XOR, NAND or NOR gate requires at least 12 transistors. Emerging technologies help reduce the area overhead. Recent work has demonstrated that only four SiNW FETs with tunable polarity are required to build a camouflaging layout that can perform NAND, NOR, XOR or XNOR functionality.^{1201,1202} Again, the SiNW FET based camouflaging layout has more functionality and requires less area than CMOS counterparts and could offer higher levels of protection to circuit designs.

Security Analysis—Logic obfuscation is subject to brute-force attacks. If there are N polymorphic gates incorporated in the design, it would take 2^N trials for an attacker to determine the exact functionality of the circuit. As the value of N increases, the probability of successfully mounting a brute-force attack becomes extremely low. In a preliminary implementation of 32-bit adder, the incorporated key size is 32 bit.¹¹⁹⁹ The probability that an attacker can retrieve the correct key becomes $1/2^{32}$ (2.33×10^{-10}). Obviously, polymorphic based logic obfuscation techniques are resistant to a conventional brute-force attack. With respect to camouflaging layouts, given that our proposed SiNW based camouflaging layout can perform four different functions, the probability that an attacker can retrieve the correct layout is 25%. Therefore, if N SiNW FET camouflaging layouts are incorporated in a design, the attacker has to compute up to 4^N times to resolve the correct layout design. Compared to polymorphic gate-based logic obfuscation, camouflaging layout embraces higher security level but with larger area overhead.

5.1.4.2. EMERGING TECHNOLOGIES TO PREVENT SIDE-CHANNEL ATTACKS

Many post-CMOS transistors aim to achieve steeper subthreshold swing, which in turn enables lower operating voltage and power. Many devices in this space also exhibit I-V characteristics that are not representative of a conventional MOSFET. An example of how to exploit said characteristics for designing hardware security primitives is discussed.

Steep slope transistors—TFETs have been exploited to design current mode logic (CML) style light-weight ciphers.^{1203,1204} The high energy carriers in TFETs can be filtered by the gate-voltage-controlled tunneling such that a sub-60 mV/decade subthreshold swing is achievable at room temperature.¹²⁰⁵ With improved steep slope and high on-current at a low supply voltage, TFETs could enable supply voltage scaling to address challenges such as undesirable leakage currents, threshold voltage reduction, etc. Different types of TFETs have been developed and fabricated.^{1205,1206}

Bell-Shaped I-Vs—Emerging transistor technologies may also exhibit bell-shaped I-V curves. Symmetric graphene FETs (SymFETs) and ThinTFETs are representatives of this group. In a SymFET, tunneling occurs between two, 2-D materials separated by a thin insulator. The I_{DS} - V_{GS} relationship exhibits a strong, negative differential resistance (NDR) region. The I-V characteristics of the device are “bell-shaped,” and the device can remain off even at higher values of V_{DS} . The magnitude of the current peak and the position of the peak are tunable via the top gate (V_{TG}) and back gate (V_{BG}) voltages of the device.¹¹⁹⁷ Such behavior has been observed experimentally.^{1207,1208} More specifically, V_{TG} and V_{BG} change the carrier type/density of the drain and source graphene layers by the electrostatic field, which can modulate I_{DS} . ITFETs or ThinTFETs may exhibit similar I-V characteristics.¹²⁰⁹

Preventing fault injection—Side-channel analysis, such as fault injection, power, and timing, allows attackers to learn about internal circuit signals without destroying the fabricated chips. Countermeasures have been proposed to balance the delay and power consumption when performing encryption/decryption at either the algorithm or circuit levels.¹²¹⁰ These methods often cause higher power consumption and longer computation time in order to balance the side-channel signals under different conditions. Thus, an important goal is to prevent fault injection and to counter side-channel analysis by introducing low-cost, on-

chip voltage/current monitors and protectors. Graphene SymFETs, which have a voltage-controlled unique peak current can be used to build low-cost, high-sensitivity circuit protectors through supply voltage monitoring.

Recent work has developed a SymFET-based power supply protector.^{1197,1198} With only two SymFETs, the power supply protector can easily monitor the supply voltage to ensure that the supply voltage to the circuit-under-protection is within a predefined range. In the event of a fault injection, the decreased supply voltage will power down the circuit rather than injecting a single-bit fault,¹¹⁹⁸ and can thus protect the circuit from fault injection attacks. If one uses V_{out} as the power supply to a circuit under protection (e.g., an adder), due to the bell-shaped I-V characteristic of the SymFET, an intentional lowering of V_{DD} cuts off the power supply. Thus, the sum and carry-out of the full adder output is '0', and no delay related faults are induced. A similar CMOS power supply protector would require op-amps for voltage comparison. As a result of the voltage/current monitors developed thus far, voltage/current-based fault injections can be largely prevented. By inserting the protectors in the critical components of a given circuit design, the power supply to these components can be monitored and protected.¹¹⁹⁹ (SymFET-based Boolean logic is also possible.¹²¹¹)

Preventing differential power analysis (DPA)—As an advanced side-channel attack scheme, DPA employs analysis of statistic power consumption measurements from a crypto system to obtain secret keys. Since the introduction of DPA¹²¹³, there has been many efforts to develop low-cost and efficient countermeasures. Countermeasures are generally classified into two categories: 1) algorithm-level solutions and 2) hardware-level solutions.

Algorithm-level solutions—aim to design cryptographic algorithms that can withstand a certain amount of information leakage,¹²¹² e.g., frequently changing the keys to prevent the attacker from collecting enough power traces¹²¹³ or using masking bits during the internal stages to limit information leakage.¹²¹⁴

A more practical circuit-level method for preventing DPA attack leverages a *sense amplifier-based logic* (SABL) or *current mode logic* (CML) for cryptographic algorithm implementations.¹²¹⁵ A CML gate includes a tail current source, a current steering core and a differential load. A CML gate will switch the constant current through the differential network of input transistors, utilizing the reduced voltage swing on the two load devices as the output. Although CML is not widely used in mainstream circuit design, its unique features, namely low latency and stable power consumption, can be leveraged to serve as a countermeasure against a DPA attack.

The strength of the CML-based approach is the constant power consumption of differential logic which can counter power-based attacks as operation power is independent of processed data. The drawback with these (mostly CMOS-based) logic designs, is their large area and power consumption when compared to static single ended logic. When considering hardware for the IoT (where the systems can be severely power constrained), system designers are presented with a dilemma in which they need to choose either high security or low power consumption. Emerging transistor technologies could help mitigate risks of DPA attacks while maintaining low power consumption.

Recent work has implemented a standard cell library of TFET CML gates and conducted a detailed study of their performance, power and area with respect to CMOS equivalents.¹²¹⁶ Standard cells were used to implement and evaluate TFET-based CML on a 32-bit KATAN cipher (a light-weight block cipher). All KATAN ciphers share the same key schedule with the key size of 80 bits as well as the 254-round iteration with the same non-linear function units.¹²¹⁷

The two CML implementations consume less gate equivalents and area compared to the two static counterparts given that the majority of KATAN32 is made up by the D flip flops. The area of TFET CML KATAN32 is $1.441 \mu\text{m}^2$, which is about 60% less than the Static TFET KATAN32. The power consumption of TFET CML ($9.76 \mu\text{W}$) is slightly lower than static CMOS ($9.96 \mu\text{W}$). It also outperforms CMOS CML.

Moreover, the correlation coefficient of a TFET static KATAN32 reaches its highest when the correct keys are applied. By comparison, the correlation coefficient of TFET CML KATAN32 is much more scattered, and all four hypothetical keys are equally distributed. Thus, the TFET CML KATAN32 implementation can successfully counteract CPA. Because the power consumption is mainly determined by AND/XOR logic gates of two nonlinear functions—and the effect of CPA is maximized—the correlation coefficients for KATAN32 are higher on average than other block ciphers, e.g., CPA on S-box¹²¹⁸.

6. EMERGING MATERIALS INTEGRATION

6.1. INTRODUCTION AND SCOPE

6.1.1. CURRENT STATE OF TECHNOLOGY

The semiconductor industry was historically driven by a strong correlation between technology scaling and performance of most integrated circuits. The PC market required more complex and faster microprocessors that largely drove the development and scaling of transistors and memory. These devices required new materials and processes such as strained silicon, high-k gate dielectrics and metal gate electrodes that are now widely, and will continue, to be used in IC manufacturing. In the past decade, a completely new ecosystem has emerged. New system integrators, from mobile to data centers to the Internet of Everything, have appeared with new and complex technology requirements. These system integrators will have impact that includes microprocessors, but extends towards new applications including medicine, energy and the environment.

6.1.2. DRIVERS AND TECHNOLOGY TARGETS

As transistors and memory begin to run out of horizontal space and ICs continue to be limited by power, device technologies will enter a phase characterized by vertical integration and performance specifications driven towards reduction of power. New transistor, memory, interconnect, lithography materials and processes will be required to enable this new More Moore scaling paradigm. As conventional information processing and storage technology reaches its ultimate limits, entirely new non-CMOS logic and memory devices and even new, non-Von Neumann circuit architectures are potential Beyond CMOS solutions. Such solutions ideally can be integrated onto the Si-based platform to take advantage of the established processing infrastructure, as well as being able to include Si devices such as memories onto the same chip. However, while these technologies will likely be integrated on a Si-based platform, the vast majority of these Beyond CMOS technologies are based on entirely new materials and physics. Finally, new system integrators require materials that enable potentially trans-disciplinary advances in monolithically integrated complex functionality, i.e., functional scaling. Significant challenges must be overcome for these emerging materials to provide viable solutions for future integrated circuit technologies. To deliver these capabilities, enhanced Metrology will be needed to accelerate material evaluation, improvement and capabilities. The ultimate goal is to provide timely guidance on emerging material and process performance, cost, reliability, and sustainability options that will drive breakthrough advances in future manufacturing technology.

6.1.3. SCOPE

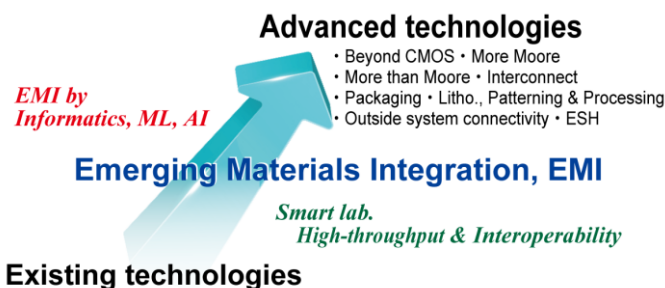


Figure EMII Emerging Material Integration Promotes the Advancement of Existing Technologies

The IRDS represents a strategic repositioning of the community's scope, needs, and set of emergent opportunities. In alignment with this new perspective, this edition of the emerging materials integration (EMI) sub-chapter represents a work in transition with a primary goal of aligning with the needs of related IRDS working groups. Much of the associated information in the detailed requirements and solutions tables comes from prior ERM chapters and input from current IRDS working groups, and will be updated in future editions. The chapter emphasizes strategic difficult challenges and/or enabling of novel, breakthrough and potentially disruptive opportunities for emerging material properties, synthetic methods, and metrology, organized in the following areas:

1. **Scaled technology materials needs for More Moore:** transistors, memory, interconnects, lithography, heterogeneous integration, assembly and packaging.
2. **Novel materials for Beyond CMOS:** emerging logic and information processing devices, emerging memory and storage devices, and novel computational paradigms and architectures.
3. **Potentially disruptive material opportunities for functional scaling and convergent applications:** Heterogeneous components, outside system connectivity, and high impact application areas such as energy, environment, agriculture, health, medical, etc.

For all areas, the advancement requires an intergration of emerging materials as illustrated in Figure EMI 1.

6.2. CHALLENGES

6.2.1. NEAR-TERM CHALLENGES

Table EMI1 *Near-term Difficult Challenges*

<i>Near-Term Challenges: 2022–2028</i>	<i>Description</i>
<i>Materials and processes that achieve performance and power scaling of lateral fin- and nanowire FETs (Si, SiGe, Ge, III-V).</i>	Integrated high k dielectrics with EOT <0.5nm and low leakage. Integrated contact structures that have ultralow contact resistivity. Achieving high hole mobility in III-V materials in FET structures. Achieving high electron mobility in Ge with low contact resistivity in FET structures. Processes for achieving low dislocations and anti-phase boundary generating interface between Ge/III-V channel materials and Si. Dopant placement and activation i.e., deterministic doping with desired number at precise location for V _{th} control and S/D formation in Si as well as alternate materials.
<i>Materials and processes that improve copper interconnect resistance and reliability</i>	Mitigate impact of size effects in interconnect structures. Patterning, cleaning, and filling at nano dimensions. Cu wiring barrier materials must prevent Cu diffusion into the adjacent dielectric but also must form a suitable, high quality interface with Cu to limit vacancy diffusion and achieve acceptable electromigration lifetimes. Reduction of the k value of inter-metal dielectrics.
<i>Materials and processes for continued scaling of DRAM/SRAM and embedded NVM</i>	Low temperature materials for high performance vertical transistor memory select structures. High-k, low leakage DRAM dielectrics. Processes for stacking of 3D flash.
<i>Materials and processes that improve AI and Quantum computing</i>	Integrate ultra-low power consumption AI devices for advanced edge computing. Development of process technology for introducing various new-principle computing materials, including liquids and biomaterials, into existing electronics. Elucidation of device physics at low temperatures to realize low-temperature electronics.
<i>Materials and processes that extend lithography to sub-10 nm dimensions with reproducible properties</i>	Novel resists to extend 193 nm lithography and support EUV lithography. Directed self-assembly (DSA) with materials such as block-copolymers to potentially extend lithography though pattern rectification and pattern density multiplication.
<i>Materials for heterogeneous integration of multi-chip, multi-function packages.</i>	Materials to modify polymer properties to enable increased product reliability. Novel electrical attaching materials to allow lower assembly temperatures and improved product reliability. Simultaneously achieve package polymer CTE, modulus, electrical and thermal properties, with moisture and ion diffusion barriers. Nanosolders compatible with <200C assembly, multiple reflows, high strength, and high electromigration resistance. Nanoinks that can be printed as die attach adhesives with required electrical, mechanical, thermal, and reliability properties.

6.2.2. LONG-TERM CHALLENGES

Table EMI2 *Long-term Difficult Challenges*

<i>Long-term Challenges: 2029–2037</i>	<i>Description</i>
<i>Materials and processes that achieve 3D monolithic and vertical integration of high mobility and steep subthreshold transistors</i>	Processes for sequential 3D vertical integration of transistors. Methods to lower the synthesis temperature of vertical semiconductor nanowires. Methods to dope and contact vertical semiconductor nanowire transistors. Lithography-free and low-temperature methods to achieve gate stack on vertical transistors.
<i>Materials and processes that replace copper interconnects with improved reliability and electromagnetic performance at the nanoscale</i>	Synthesis or assembly of CNTs in predefined locations and directions with controlled diameters, chirality and site-density. Carbon and collective excitations. Novel interlayer dielectrics: Metal Organic Framework (MOF) and Carbon Organic Framework (COF). Metals with less size effects such as silicides.
<i>Materials and processes for charge-based and non-charge-based beyond CMOS logic that replaces or extends CMOS</i>	Achieving a bandgap and full interfaces control in graphene in FET structures and alternative FETs (TFETs etc). Synthesis of CNTs with tight distribution of bandgap and mobility. Complex metal oxides with low defect density. High mobility transition metal dichalcogenides with low defect density and low resistance ohmic contacts. Spin materials: characterization of spin, magnetic and electrical properties and correlation to nanostructure. Topological materials: large bandgaps much greater than kT at room temperature, ability to modulate bandgap efficiently with electric field. BiSFET heterostructures: achieving exciton condensation at room temperature.
<i>Materials and processes for emerging memory and select devices to replace DRAM/NVM.</i>	Multiferroic with Curie temperature >400K and high remnant magnetization to >400K. Ferromagnetic semiconductor with Curie temperature >400K. Complex Oxides: Control of oxygen vacancy formation at metal interfaces and interactions of electrodes with oxygen and vacancies. Switching mechanism of atomic switch: Improvements in switching speed, cyclic

<i>Long-term Challenges: 2029–2037</i>	<i>Description</i>
	endurance, uniformity of the switching bias voltage and resistances both for the on-state and the off-state.
<i>Materials and processes that enable monolithically 3D integrated complex functionality including thermal and yield challenges</i>	Integration on CMOS Platforms. Integration with flexible electronics. Biocompatible functional materials. Leveraging convergent materials expertise in adjacent sectors, including More than Moore functionalities (photonics, optics/metamaterials, outside connectivity, energy transfer/storage, power circuits).
<i>Green and sustainable fabrication</i>	To realize an integration process that is environmentally friendly and economically viable. Utilizing AI, ML, and informatics to develop production processes that can introduce green new materials.

6.3. TECHNOLOGY REQUIREMENTS AND POTENTIAL SOLUTIONS

6.3.1. SUMMARY

The IRDS seeks a framework for managing the convergence of scaled information processing and storage, i.e., More Moore (MM) and Beyond CMOS (BC), with the next emerging era of monolithically integrated systems that achieve enhanced overall functional density. The trend towards the convergence of monolithically integrated functional diversification with miniaturization manifests as increasing complexity in the road-mapping process. The IRDS reflects this growing complexity, with an increasing number of projected roadmap parameters and requirements associated with new functionalities. While EMI continues to support the evolutionary, and semiconductor centric needs of the traditional semiconductor community, emerging architectures could benefit from new device functionality, which may require new materials and new physical mechanisms. New waves of emerging materials technologies may represent potentially disruptive opportunities.

Candidate EMI materials and processes exhibit unique and useful properties that may require atomic level structural, interface, defect and compositional control. In some cases, current synthetic or manufacturing technologies are not yet capable of producing such materials with the required level of control. The difficulties could be due to: 1) The inability of a research environment to produce materials with the required level of control that would express the desired properties; or 2) scaling up the synthetic and fabrication processes to satisfy commercial manufacturing requirements. In some cases, current materials growth processes effect unacceptable levels of defect formation, which drive the need for new and more robust fabrication methods. In other cases, synthetic methods exist for producing high quality materials, but these processes cannot be scaled to the higher growth rates, yields, or purity needed for insertion into viable commercial applications. While these materials may provide proof of concept and suggest a potential solution, new cost-effective fabrication technologies may be required to warrant a candidate material's insertion into high volume manufacturing. It will be an extremely important issue to be able to quickly verify the integration process suitability of emerging materials in large-scale semiconductor factories. For this reason, it is important to establish standardized indices for evaluating the characteristics of new materials and new devices. We should also consider using the international standards being conducted at TC113 (Technical committee No.113, Nanotechnology for electrotechnical products and systems) of the International Electrotechnical Commission (IEC).

6.3.2. SCALED TECHNOLOGY MATERIALS FOR MORE MOORE

As described in the More Moore chapter, after 2027 there is no headroom for 2D geometry scaling and 3D VLSI integration of circuits and systems using sequential/stacked integration approaches will likely begin. Whether one is considering 2D geometry scaling or 3D integration, there are numerous materials challenges to achieving increasing device density and integrated performance. The following outlines key materials challenges for transistor scaling and integration, lithography, interconnects, heterogenous integration, assembly and packaging, and outside system connectivity.

6.3.2.1. MATERIALS FOR TRANSISTOR SCALING AND INTEGRATION

Continued increases in transistor device density require a variety of new materials and processes including new channels (Ge, III-V), improved doping techniques, gate stacks and contacting structures. Table EMI3 provides a set of materials and processes priorities for transistor scaling and integration.

Table EMI3

Materials for Transistor Scaling and Integration

6.3.2.2. MATERIALS FOR LITHOGRAPHY AND PATTERNING

The future of scaled technologies depends upon emerging patterning materials (resist or self-assembled) to enable extensible lithographic capabilities. New resist materials must concurrently exhibit higher resolution, higher sensitivity, reduced line edge

roughness, and sufficient etch resistance to enable robust pattern transfer. 193nm and EUV extension materials are being developed, which can improve LWR, pattern shrink materials, and topcoats for EUV to ameliorate issues with out-of-band optical flare and outgassing. Evolutionary approaches for enhancing positive, negative, and chemically amplified families of resists will continue to be evaluated. Leading process approaches to pitch division include multiple patterning (MP) and spacer patterning (SP) as options for extending 193 nm immersion lithography. Alternate technologies are utilizing patterning materials to create guide patterns for directed self-assembly, which can include resists to form chemoepitaxy and graphoepitaxy guides, or directly patternable brushes and SAMs. Directed self-assembly (DSA) with block-copolymers or polymer pairs has made significant progress in characterizing sources of defect formation and in applications such as contact rectification, fin patterning, and pattern density multiplication. Table EMI4 provides a set of materials and processes priorities for lithography and patterning.

Table EMI4 Materials for Lithography and Patterning

6.3.2.3. INTERCONNECT MATERIALS

Key challenges for continued increased performance of future integrated circuit interconnects consist of maintaining reductions of RC time constants for delivery of signals and power with high reliability. For copper interconnects, the sidewall copper barrier thickness must continue to be reduced, which is a significant challenge. For post copper interconnect scaling, novel interconnects, such as carbon nanotubes, are being explored. Several elemental metals have been studied by simulation and experiments to identify potential candidates for post-Cu interconnect materials at highly scaled dimension.¹²¹⁹ The interconnect material space becomes significantly larger when alloys and compounds are considered, and machine learning assisted data analysis has been employed to filter this vast material space. Also, lower dielectric constant (κ for both intra and inter level dielectric are needed; however, each of these emerging families of materials must overcome significant challenges for them to warrant adoption. Airgap, another approach to reducing the effective κ , places additional requirements on barrier layers or novel interconnects. Table EMI5 provides a set of materials and processes priorities for interconnects.

Table EMI5 Interconnect Materials

6.3.2.4. HETEROGENEOUS INTEGRATION, ASSEMBLY AND PACKAGING MATERIALS

The EMI and Heterogeneous Integration teams are in the process of prioritizing key heterogeneous integration and assembly & packaging EMI challenges, which include:

- New engineered materials: substrate, mold, underfill, wafer bond alloys, solder alloys
- Conductors: Nanomaterials (CNT, graphene, NWs), metals (Cu, Al, W, Ag, etc.), composites
- Dielectrics: Oxides, polymers, porous materials, composites
- Semiconductors: Elemental (Si, Ge), Compounds (SiC, III-V, II-VI, tertiary), polymers
- Critical factors: Cost, CTE differential, thermal conductivity, fracture toughness, modulus, processing temperature, interfacial adhesion, operating temperature, and breakdown field strength

Table EMI6 provides a set of heterogeneous integration and assembly and packaging priorities for EMI.

Table EMI6 Heterogeneous Integration, Assembly and Packaging Materials

6.3.2.5. MATERIALS CHALLENGES FOR OUTSIDE SYSTEM CONNECTIVITY

Table EMI7 provides a set of top Outside System Connectivity material priorities for EMI.

Table EMI7 Emerging Research Materials Needs for Outside System Connectivity

6.3.3. EMERGING MATERIALS FOR MEMORY, BEYOND CMOS LOGIC AND COMPUTING

Beyond 2030, MOSFET scaling will likely become ineffective and/or very costly. As described in this chapter, completely new, non-CMOS types of memory, logic devices and maybe even new circuit architectures are potential solutions. Such solutions ideally can be integrated onto the Si-based platform to take advantage of the established processing infrastructure, as well as being able to include Si devices such as memories onto the same chip. The following outlines key materials challenges for emerging materials for memory, beyond CMOS logic and alternative information processing.

6.3.3.1. EMERGING MATERIALS FOR MEMORY

Emerging memory devices includes capacitive memories (Fe FET), and resistive memories including ferroelectric devices, resistance change devices, devices based on Mott transitions and novel magnetic memories. Another key requirement for memory technology is the development of corresponding select devices that access only the selected memory cell of interest without perturbing non-selected cells. Table EMI8 provides a set of materials and associated challenges for emerging memory materials, and Table EMI9 provides materials and associated challenges for memory select.

Table EMI8 Emerging Materials for Memory

Table EMI9 Emerging Materials for Memory Select

6.3.3.2. EMERGING MATERIALS FOR ADVANCED AND BEYOND-CMOS LOGIC DEVICES

There are generally two classes of devices/materials for advanced and beyond-CMOS logic devices. The first are those that do not involve spin or magnetism such as ferroelectric FETs, nanoelectromechanical (NEM) switches, topological FETs, and transistors based on collective electron phenomena such as Mott effect or exciton condensation (BiSFET). The second are those based on spin and magnetism that each uses a variety of materials. Negative-capacitance topological quantum FET was proposed to achieve extremely low switching voltages and energies.¹²²⁰ Table EMI10 contains materials and associated challenges for the non-spin devices. Table EMI11 maps various spin device concepts to associated material types and Table EMI12 describes the requirements of these materials.

Table EMI10 Emerging Materials for Advanced and Beyond-CMOS Logic Devices

Table EMI11 Spin Devices versus Materials

Table EMI12 Spin Material Requirements and Properties

6.3.3.3. EMERGING MATERIALS FOR NOVEL COMPUTING

Emerging materials spur developments of novel computing, such as neuromorphic computing, reinforcement learning, topological quantum computing, and reversible computing, and probabilistic computing. Since the performance of the computing is considered to be dependent on the intrinsic properties of the emerging material, the material research will further boost the performance, such as the energy efficiency^{1221,1222,1223,1224,1225,1226}. Two-dimensional materials are also expected to play an important role in memristors for neuromorphic computing.^{1227, 1228}

6.3.3.4. NANOPARTICLE-BASED COMPUTING

The implementation of complex computation in particle systems, especially in nanoparticles, remains challenging. However, nanoparticle-based computing has a wide range of potential applications that benefit from algorithmically controlling their unique and potentially useful intrinsic properties (photonic, plasmonic, catalytic, photothermal, optoelectronic, electrical, magnetic, material properties, etc.) without human intervention^{1229, 1230,1231,1232}.

6.3.4. METROLOGY NEEDS AND CHALLENGES FOR EMERGING RESEARCH MATERIALS

Metrology is needed to characterize composition, properties, and understand structure of emerging research materials, at nanometer dimensions and below. The difficult EMI metrology challenges would be those associated with the introduction of directed self-assembly (DSA), such as evaluating critical material properties, size and location of features, registration, and defects. Also needed are non-destructive methods for characterizing embedded materials and interfaces defects in nano-scale devices,¹²³³ as well as platforms that enable simultaneous measurement of complex nanoscopic properties. Table EMI13 summarizes the current set of continuing and prioritized metrology related EMI challenges and needs.

Table EMI13 Metrology Needs and Challenges for Emerging Research Materials

6.3.5. NEEDS AND CHALLENGES OF EMERGING RESEARCH MATERIALS FOR ESHS

Research and development of Beyond CMOS devices will be stout under limitation of Green Materials. Therefore, the close liaison with Environment, Safety, Health and Sustainability (ESHS) will continue to be an important issue.

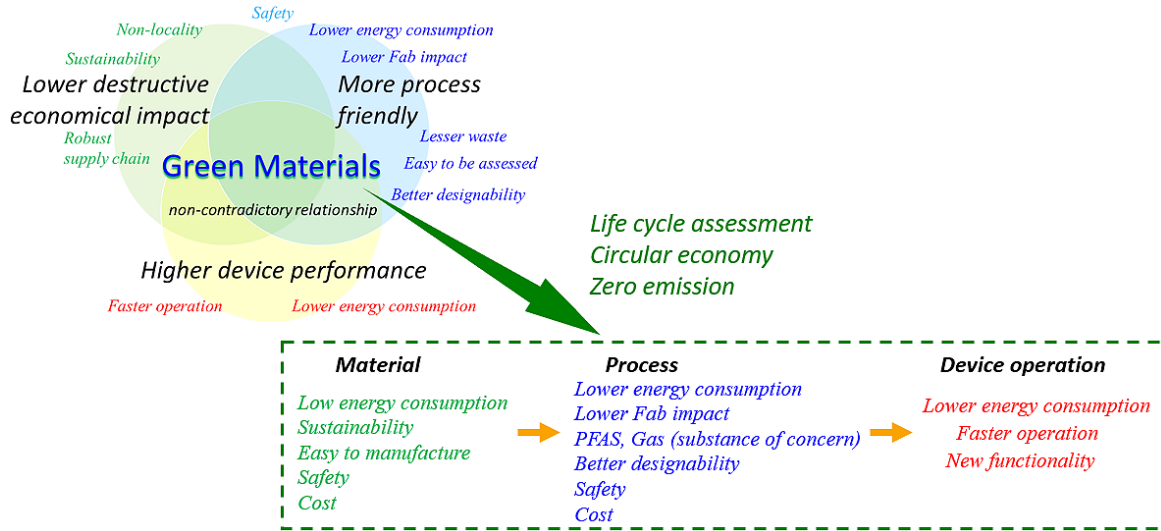


Figure EMI2 Liaison concept of EMI and ESHS

6.4. EMERGING/DISRUPTIVE CONCEPTS AND TECHNOLOGIES

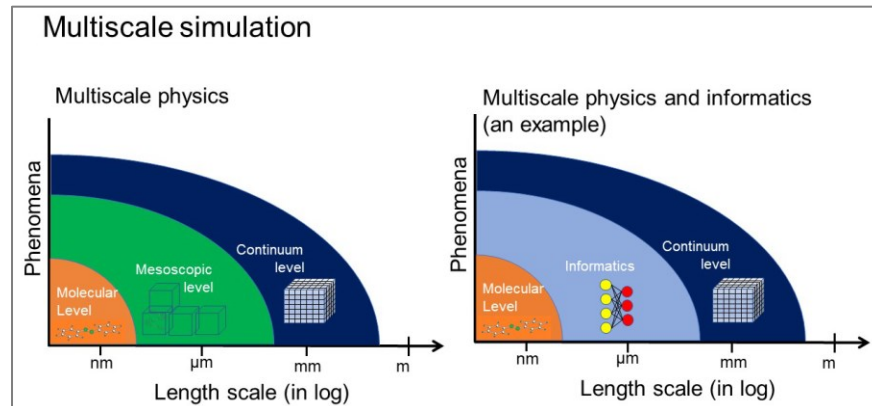
As mentioned at the beginning of the chapter, new system integrators, from mobile to data centers to the Internet of Everything, have appeared with new and complex technology requirements. These application domains require a highly interdisciplinary set of expertise, e.g., electrical and mechanical engineering; as well as materials, biological, medical, energy, aerospace, transportation, communication, and sustainability sciences. The trend towards the convergence of monolithically integrated functional diversification with miniaturization manifests as increasing complexity in the road-mapping process. Collaborative transdisciplinary research is needed to identify materials and processes that catalyze breakthrough and convergent advances in these technologies. Initiatives that leverage the expertise of colleagues in adjacent spaces who know the local environment, e.g., biology, energy, etc., will help to drive novel approaches and more optimal materials, process, manufacturing, and performance solutions to emerging IoT challenges than can be achieved by semiconductor centric approaches. As examples, (6.4.1) transient concept and (6.4.2) development aided by machine learning (Table EMI14) are shown. Table EMI15 identifies several emerging application opportunities that will drive and enhance future EMI working group activities.

6.4.1. TRANSIENT ELECTRONICS

Transient electronics is an emerging field that requires materials, devices, and systems to be capable of disappearing with minimal or non-traceable remains in a controllable period of time. The spontaneous and transient function appears after the stable operation. This emerging electronics with the disintegrating capability will bring about intelligent applications in various scenes, such as bioelectronics, environmentally friendly electronics^{1234,1235,1236,1237}. The conductance change with the controllable decay characteristics was demonstrated in molecular gap atomic switches^{1238,1239}.

6.4.2. MODELING AND SIMULATION FOR EMERGING MATERIALS DEVELOPMENTS

Simulation for designing material structures matching to targeted performance needs hierarchical understanding of material from nanometer scale to centimeter scale. First-principles approaches are developed to provide theoretical insight into emerging materials and in silico exploration of as-yet-unreported materials.¹²⁴⁰ Oppositely, despite its importance for the future design of device structures, the development of simulation for emerging devices is just beginning.¹²⁴¹ This requires multiscale understanding in space and time for phenomena in materials. For instance, the individual atom-atom interaction derived from chemical bonding given by details of electronic structures can be scaled up to mechanical response of many atoms against macroscopic load, which gives understanding of elastic behavior of crystals¹²⁴². Meanwhile, a performance of fuel cell can be derived from macro-model originated from the rate equation of chemical reactions of individual molecules¹²⁴³. Multiscale view of carrier transport in organic devices were treated by considering molecular orbital levels to mesoscopic scale of electron-hopping^{1244,1245,1246}. Beside the direct approach on multiscale physics, aid of machine learning can bridge the difference scales phenomena in composite material¹²⁴⁷. These approaches will accelerate the simulation, while proper modeling with understanding of natural physics in difference scales of time and space are still necessary. Finite element analysis (FEA) models will have to be further developed so that they can adequately represent 2D and other nanomaterials.

Table EMII4 *Modeling and Simulation*Figure EMII3 *An Example of the Role of Machine Learning in the Multiscale Simulation*

Note: The left shows conventional multiscale physics where physics to explain mesoscopic level phenomena is required to link molecular and continuum level phenomena. The right shows a potential role of machine learning that provides output of continuum level phenomena from inputs of molecular level phenomena.

Table EMII5 *Summary of Potentially Disruptive Emerging Research Materials Application Opportunities*

6.4.3. CVD/ALD PROCESS, PRECURSORS, AND AREA SELECTIVE DEPOSITION

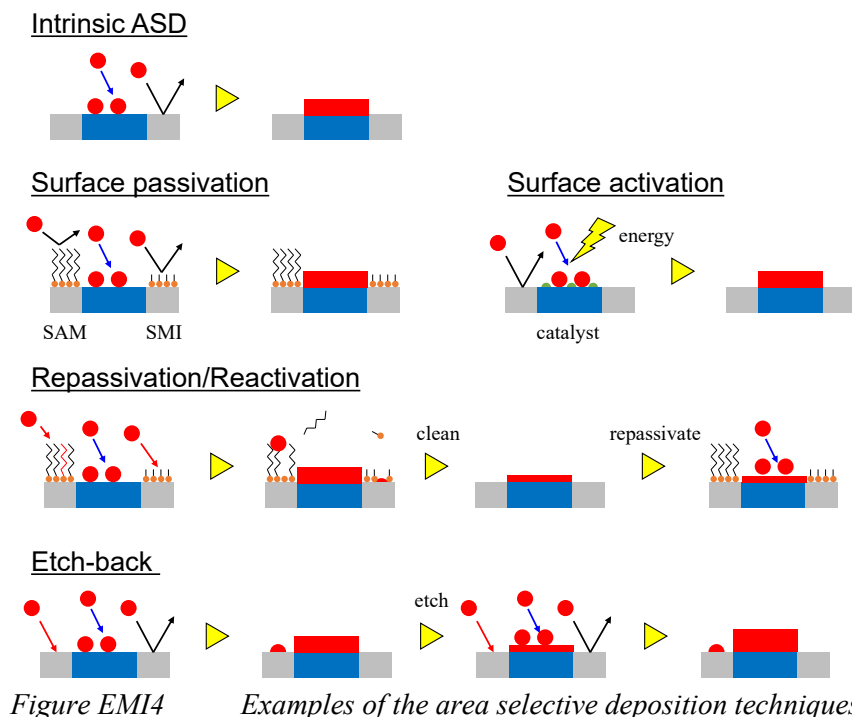
Recently, as the miniaturization of advanced electronic devices requires the formation of ultra-thin films in a limited space, Physical Vapor Deposition (PVD) processes have been partially replaced by Chemical Vapor Deposition (CVD) or Atomic Layer Deposition (ALD) processes. In CVD processes, volatile precursor compounds are vaporized and continuously drawn into the deposition chamber to form a thin film by chemical reactions in the gas phase or on a heated substrate. The ALD process is similar to the CVD process, but it enables the decomposition of self-limitingly adsorbed precursor compound on the substrate surface by separating the gas-phase precursor with inert gas purge, allowing more precise control of the thickness of the deposited film at the atomic level.

In the CVD/ALD deposition process, the various physical and chemical properties of the precursor compounds play an important role. A basic requirement for these precursors is that they must be sufficiently volatile to be drawn into the deposition chamber as a gas phase. They must also be thermally stable to avoid unwanted decomposition during the volatilization, delivery, and adsorption processes on the substrate surface. Evaluation of volatility and thermal stability by thermal analysis has been studied.¹²⁴⁸ On the other hand, precursor compounds must be reasonably chemically active and react appropriately with substrate surfaces and reaction gases; the design of precursor compounds is challenging considering the various requirements to control chemical reactions during the CVD and ALD processes. Some of the existing precursor compounds are no longer able to meet the requirements of state-of-the-art device fabrication processes, and the introduction of new compounds is being investigated to solve this problem.¹²⁴⁹

In recent years, Area Selective Deposition (ASD) has been intensively studied as an efficient thin film formation process.¹²⁵⁰ In ASD, the process is tailored to form a film in the desired area of the surface while avoiding deposition in adjacent areas. ASD utilizes the chemical properties of the reactants and the deposition source to control pattern growth on the surface, rather than using physical lithographic mask. As the miniaturization scale of semiconductors approaches the size of molecules, ASD solutions are becoming increasingly important, especially at lower deposition temperatures.

While the intrinsic selective adsorption nature of the precursor compounds is one of the most important parameters in the ASD process, the current ASD process is actually a combination of several techniques (Figure EMII4). Self-Assembled Monolayers (SAM) of organic molecules can be formed on the substrate surface to passivate the adsorption of precursor compounds on the substrate surface. In the nanoscale structure of devices, organic molecules larger than 1 nm can cause undesirable effect on the adjacent structures, Small Molecule Inhibitors (SMI) are being developed for the deposition on the limited surface. ASD can also be achieved by activating the substrate with local energy delivery or by introducing catalysts. Since the effects of these passivation

and activation techniques decay gradually during the deposition process, appropriate repassivation and reactivation are usually required. If small amounts of film are formed in undesirable locations, they can be removed by the etch-back process. In any combination of techniques, the area selective deposition property of the precursor compounds will play an important role in the success of the deposition process. ASD processes are classified into several groups according to the combination of substrates (metals/ dielectrics/ semiconductors) to activate and passivate the deposition: dielectric on dielectric (DoD),¹²⁵¹ metal on metal (MoM),¹²⁵² metal on dielectric (MoD),¹²⁵³ dielectric on semiconductor (DoS), etc. The ASD process will help reduce takt time and waste through efficient formation of, for example, contacts, hard masks, and via-fills.



6.5. CONCLUSIONS AND RECOMMENDATIONS

The IRDS represents a strategic repositioning of the community's scope, needs, and set of emergent opportunities. In alignment with this new perspective, this edition of the EMI chapter represents a work in transition that has aligned difficult challenges with the needs of related IRDS working groups. Much of the information in the detailed tables comes from prior ERM chapters and current IRDS working groups. Future editions of EMI will provide additional detailed descriptions and continue to adapt its scope to engage with a new set of EMIs, many of which will be identified by the IRDS working groups.

7. ASSESSMENT

7.1. INTRODUCTION

It is important to evaluate beyond-CMOS devices considered in this chapter against current CMOS technologies. Two methods of assessments have been reported: a "survey-based assessment" conducted previously by the ITRS ERD working group and a "quantitative device benchmarking" conducted by the Nanoelectronics Research Initiative (NRI).

Up to the 2013 ERD Chapter, a survey-based critical review was conducted based on eight criteria to compare emerging devices against CMOS. Spider chart has been used to visualize the perceived potential of these technology entries. However, the limited number of survey results sometimes raises questions of the accuracy of this survey. The most recent "survey-based assessment" was conducted in the 2014 ERD Emerging Memory and Logic Device Assessment Workshops (Albuquerque, NM). The survey collects voting on emerging technologies evaluated in the workshops in the categories of the "most promising" and the "most need of resources" to assess the potential of these technology entries perceived by ERD experts. A summary of previous survey-based assessments is included in section 7.3 "Archive".

In the "NRI benchmarking", each emerging device is evaluated by its operation in conventional Boolean Logic circuits, e.g., a unity gain inverter, a 2-input NAND gate, and a 32-bit shift register. Metrics evaluated include speed, areal footprint, power dissipation, etc. Each parameter is compared with the performance projected for high-performance and low-power 5 nm CMOS.

Most of the proposed beyond-CMOS devices are very different from their CMOS counterparts, and often pass computational state variables (or tokens) other than charge. Alternative state variables include collective or single spins, excitons, plasmons, photons, magnetic domains, polarization, etc. With the multiplicity of programs characterizing the physics of proposed new structures, it is necessary to find ways to benchmark the technologies effectively. This requires a combination of existing metrics used for CMOS and new metrics which take into consideration the idiosyncrasies of the new device behavior. Even more challenging is to extend this process to consider new circuits and architectures beyond the Boolean logic used by CMOS today, which may enable these devices to complete transactions more effectively.

7.1.1. ARCHITECTURAL REQUIREMENTS FOR A COMPETITIVE LOGIC DEVICE

The circuit designers and architects depend on the logic switch to exhibit specific desired characteristics in order to successfully realize a wide range of applications. These characteristics,¹²⁵⁴ which have since been supplemented in the literature, include:

- Inversion and flexibility (can form an infinite number of logic functions)
- Isolation (output does not affect input)
- Logic gain (output may drive more than one following gate and provides a high I_{on}/I_{off} ratio)
- Logical completeness (the device is capable of realizing any arbitrary logic function)
- Self-restoring / stable (signal quality restored in each gate)
- Low cost manufacturability (acceptable process tolerance)
- Reliability (aging, wear-out, radiation immunity)
- Performance (transaction throughput improvement)
- Span of control (measures number of devices that may be reached within a characteristic delay of the switch¹²⁵⁵)

Devices with intrinsic properties supporting the above features will be adopted more readily by the industry. Moreover, devices which enable architectures that address emerging concerns such as computational efficiency, complexity management, self-organized reliability and serviceability, and intrinsic cyber-security¹²⁵⁶ are particularly valuable.

7.2. NRI BEYOND-CMOS BENCHMARKING

The Nanoelectronics Research Initiative (<https://www.src.org/program/nri/>) has quantitatively evaluated quite a few beyond-CMOS technologies.^{1257, 1258, 1259} Several NRI devices have been described in detail in the Logic and Emerging Information Processing Device Section. While beyond-CMOS device benchmarking is still very much a work in progress – and no concrete decisions have been made on which devices should be chosen or eliminated as candidates to significantly extend or augment CMOS – this section summarizes some of the data and insights gained from these studies. Further benchmarks may alter some of the conclusions here and the outlook on some of these devices, but the overall message on the challenges of beyond-CMOS devices remains valid.

7.2.1. QUANTITATIVE RESULTS

NRI benchmarking analyzes the potential of major emerging switches using a variety of information tokens and communication transport mechanisms. Specifically, the projected effectiveness of these devices used in a number of logic gate configurations was evaluated and normalized to CMOS at the 5nm generation (projection). The initial work has focused on “standard” Boolean logic architecture, since the CMOS equivalent is readily available for comparison. It should be noted that the majority of devices are evaluated via simulations since many of them have not yet been built, so it should be considered only a “snapshot in time” of the potential of any given device. Data on all of them are still evolving.

At a high level, the data from these studies corroborates qualitative insights from earlier works, suggesting that many new logic switch structures may have some advantages over CMOS in terms of power or energy, but they are also inferior to CMOS in delay. This is perhaps not surprising; the primary goal for nanoelectronics and NRI is to find a lower power device¹²⁶⁰ since power density is a primary concern for future CMOS scaling. The power-speed tradeoffs commonly observed in CMOS also extend to beyond-CMOS devices. It is important to understand the impact of transport delay for the different information tokens these devices employ. Communication with many non-charge tokens can be significantly slower than moving charge, although this may be balanced in some cases with lower energy for transport. The combination of the new balance between switch speed, switch area, and interconnect speed can lead to advantages in the span of control for a given technology. For some of the technologies (e.g., nanomagnetic logic), there is no strong distinction between the switch and the interconnect, indicating the need for novel architecture to exploit unique attributes of these technologies.

A simplified 32-bit arithmetic logic unit (ALU) was built from these devices to evaluate their performance, as shown in Figure BC7.1(a)¹²⁶¹. While tunneling devices show limited advantages over CMOS in terms of energy-delay product, most beyond-

CMOS devices are inferior to CMOS in energy and/or delay. For example, spintronic devices tend to be slower than CMOS and also show no energy advantage.

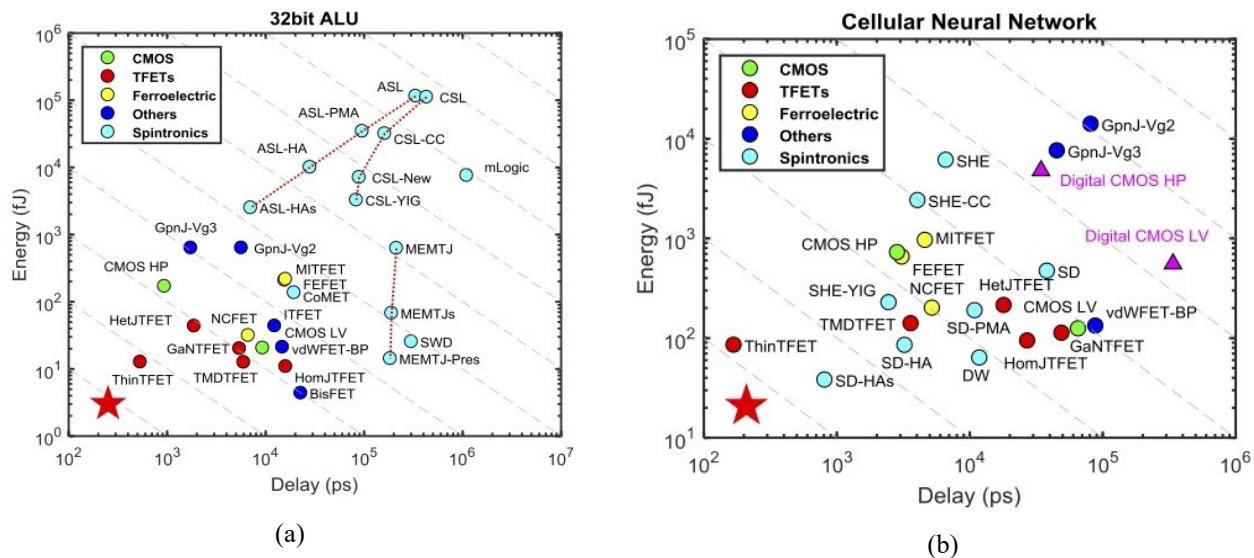


Figure BC7.1 (a) Energy versus Delay of a 32-bit ALU for a Variety of Charge- and Spin-based Devices;
(b) Energy versus Delay per Memory Association Operation Using Cellular Neural Network (CNN) for a Variety of Charge- and Spin-based Devices¹²⁶¹

At the architecture level, the ability to speculate on how these devices will perform is still in its infancy. While the ultimate goal is to compare at a very high level—e.g., how many MIPS can be produced for 100 mW on 1 mm²?—the current work must extrapolate from only very primitive gate structures. One initial attempt to start this process has been to look at the relative “logical effort”¹²⁶² for these technologies, a figure of merit that ties fundamental technology to a resulting logic transaction. Several of the devices appear to offer advantage over CMOS in logical effort, particular for more complex functions, which increases the urgency of doing more joint device-architecture co-design for these emerging technologies.

The direction of device-architecture co-optimization has driven NRI benchmarking to explore non-Boolean applications of beyond-CMOS devices. Cellular neural network (CNN) has been utilized as a benchmarking model that has been implemented with various novel devices.¹²⁶¹ The energy and delay of CNN based on beyond-CMOS devices are compared with CMOS-based CNN in Figure BC7.1(b). Tunneling devices have significant performance improvement because of their steep subthreshold slopes and large driving current at ultra-low supply voltage. Interestingly, spintronic devices are much closer to the preferred corner in CNN implementation in comparison with 32-bit ALU. This is because some characteristics of spintronic devices (e.g., spin diffusion, domain wall motion) may mimic the functionality of a neuron (e.g., integration) more naturally in a single device.

7.2.2. OBSERVATIONS

A number of common themes have emerged from these benchmarking studies and in the observations made during recent studies of beyond-CMOS devices.

1. The low voltage energy-delay tradeoff conundrum will continue to be a challenge for all devices. Getting to low voltage must remain a priority for achieving low power, but new approaches to achieve sufficient throughput with “slow” devices must be developed.
2. Most of the architectures that have been considered to date in the context of new devices utilize binary logic to implement von Neumann computing structures. In this area, CMOS implementations are difficult to supplant because they are very competitive across the spectrum of energy, delay and area – not surprising since these architectures have evolved over several decades to exploit the properties of CMOS most effectively. Novel electron-based devices—which can include devices that take advantage of collective and non-equilibrium effects—appear to be the best candidates as a drop-in replacement for CMOS for binary logic applications.
3. As the behavior of other emerging research devices becomes better understood, work on novel architectures that leverage these features will be increasingly important. A device that may not be competitive at doing a simple NAND function may have advantages in doing a complex adder or multiplier instead. Understanding the right building blocks for each

device to maximize throughput of the system will be critical. This may be best accomplished by thinking about the high-level metric a system or core is designed to achieve (e.g., computation, pattern recognition, FFT, etc.) and finding the best match between the device and circuit for maximizing this metric.

4. Patterning, precision layer deposition, material purity, dopant placement, and alignment precision critical to CMOS will continue to be important in the realization of architectures using these new switches.
5. Assessment of novel architectures using new switches must also include the transport mechanism for the information tokens. Fundamental relationships connecting information generation with information communication spatially and temporally will dictate CMOS' successor.

Based on the current data and observations, it is clear that CMOS will remain the primary basis for IC chips for the coming years. While it is unlikely that any of the current emerging devices could entirely replace CMOS, several do seem to offer advantages, such as ultra-low power or nonvolatility, which could be utilized to augment CMOS or to enable better performance in specific application spaces. One potential area for entry is special-purpose accelerators that could off-load specific computations from the primary general-purpose processor and provide overall improvement in system performance. If scaling slows in delivering the historically expected performance improvements in future generations, heterogeneous multi-core chips may be a more attractive option. These would include specific, custom-designed cores dedicated to accelerate high-value functions, such as accelerators already widely used today in CMOS (e.g., encryption/decryption, compression/decompression, floating point units, digital signal processors, etc.), as well as potentially new, higher-level functions (e.g., voice recognition). While integrating dissimilar technologies and materials is a big challenge, advances in packaging and 3D integration may make this more feasible over time.

An accelerator using a non-CMOS technology would likely need to offer an order of magnitude performance improvement relative to its CMOS implementation to be considered worthwhile. That is a high bar, but there may be instances where the unique characteristics of emerging devices, combined with a complementary architecture, could be advantageous in implementing a particular function. At the same time, the changing landscape of electronics (moving from uniform, general-purpose computing devices to a spectrum of devices with varying purposes, performance, and power constraints) and the changing landscape of workloads and processing needs (e.g., big data, unstructured information, real-time computing, 3D rich graphics) are increasing the need for new computing solutions. Therefore, future beyond-CMOS research should focus on specific emerging functions and device-architecture co-optimization to achieve solutions that can break through the current power/performance limits.

7.3. ARCHIVE

The survey-based emerging device assessment has not been continued after the 2015 International Technology Roadmap for Semiconductors (ITRS) Emerging Research Devices (ERD) chapter. Previous survey-based assessments are summarized here for references.

7.3.1. EMERGING DEVICE ASSESSMENT IN 2014 ERD WORKSHOPS

In August 2014, ERD organized an “Emerging Memory Device Assessment Workshop” and an “Emerging Logic Device Assessment Workshop”, where nine memory devices and fourteen logic devices were evaluated. A survey was conducted in the workshops for the experts to vote on the “most promising” devices and devices “needing more resources”. Figure BC7.2 shows the relative number of votes received by emerging devices in these two categories, ranked from high to low in the “most promising” category (red color bars).

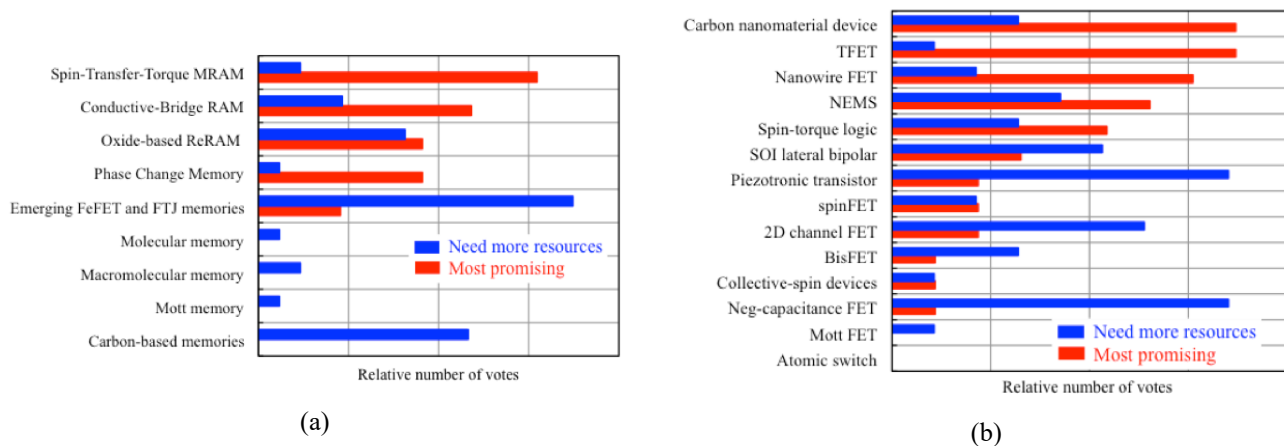


Figure BC7.2 (a) Survey of Emerging Memory Devices and (b) Survey of Emerging Logic Devices in 2014 ERD Emerging Logic Workshop (Albuquerque, NM)

In the “most promising” memory device category, the vote clearly accumulated to a few well-known memory devices: STTRAM, ReRAM (including CBRAM and oxide-based ReRAM), and PCM, ranked from high to low. Some memory devices received few vote, due to lack of progress. Results in this category reflect consensus among experts based on R&D status of these devices. The “need more resources” category reflects perceived value of these devices and experts’ consideration of R&D resource allocation for these devices. For example, with heavy R&D investment on STTRAM that is considered most promising, it is not surprising that it ranks low in the need of resources. The strong interest in emerging FeFET memory is closely linked to the discovery of ferroelectricity in doped HfO_x . Among emerging logic devices, “carbon nanomaterial device” (mainly carbon nanotube FET), tunnel FET, and nanowire FET were ranked as one of the most promising emerging logic devices. Notice that they are all charge-based devices, but involve novel materials, structures, and mechanisms. “Piezotronic transistors”, “negative-capacitance FET”, and “2D channel FET” were considered top choices for enhanced research investment.

7.3.2. 2013 ERD SURVEY CRITERIA, METHODOLOGY, AND RESULTS

In the traditional survey-based assessment conducted by ERD, a set of relevance or evaluation criteria are used to parameterize the extent to which “CMOS Extension” and “Beyond CMOS” technologies are applicable to memory or information processing applications. These criteria are: 1) Scalability, 2) Speed, 3) Energy Efficiency, 4) Gain (Logic) or ON/OFF Ratio (Memory), 5) Operational Reliability, 6) Operational Temperature, 7) CMOS Technological Compatibility, and 8) CMOS Architectural Compatibility. Description of each criterion can be found in 2013 ERD chapter.

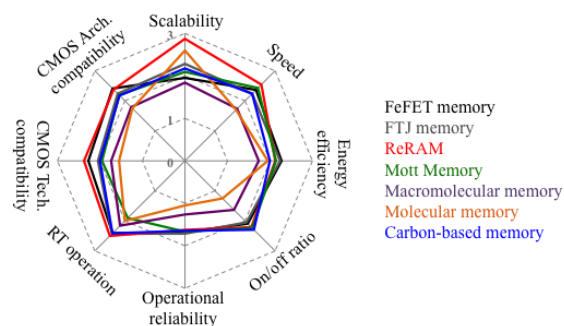


Figure BC7.3 Comparison of Emerging Memory Devices Based on 2013 Critical Review

Each CMOS extension and beyond-CMOS emerging memory and logic device technology is evaluated against these criteria according to a single factor. For logic, this factor relates to the *projected potential performance* of a nanoscale device technology, assuming its successful development to maturity, *compared to that for silicon CMOS scaled to the end of the Roadmap*. For memory, this factor relates the *projected potential performance* of each nanoscale memory device technology, assuming its successful development to maturity, *compared to that for ultimately scaled silicon memory technology which the new memory would displace*. Performance potential for each criterion is assigned a value from 1–3, with “3” substantially exceeding ultimately-

scaled CMOS, and “1” substantially inferior to CMOS or a comparable existing memory technology. This evaluation is determined by a survey of the ERD Working Group members composed of individuals representing a broad range of technical backgrounds and expertise. Details of the assessment results can be found in the 2013 ERD chapter.

Although this survey-based critical review has been conducted in ERD for several versions and has been widely cited in literatures, the decreasing number of votes of some less popular devices has raised concerns about the accuracy of some of the results. Figures BC7.3 and BC7.4 summarize the last critical review conducted in 2013 for emerging memory devices and emerging logic devices, respectively. Notice that the technology entries in these figures are based on the 2013 ERD chapter, while some of them have been removed in this chapter (e.g., molecular memory, atomic switch, etc.).

Since “3” represents the best result and “1” the worst in the spider chart, devices with larger circle area represent more promising devices. In Figure BC7.4 for emerging logic devices, the perceived potential of “beyond-CMOS devices” is generally poorer than “CMOS-extension devices”. Within “beyond-CMOS devices”, “non-charge-based devices” area perceived slightly less promising than “charge-based devices”. The general trend is consistent with the quantitative NRI assessment in section 7.2. Multiple factors contribute to this result, including the strength of CMOS as a platform technology, the challenges of beyond-CMOS devices in materials and fabrication, the lack of memory and interconnect solutions for beyond-CMOS devices, *etc.*

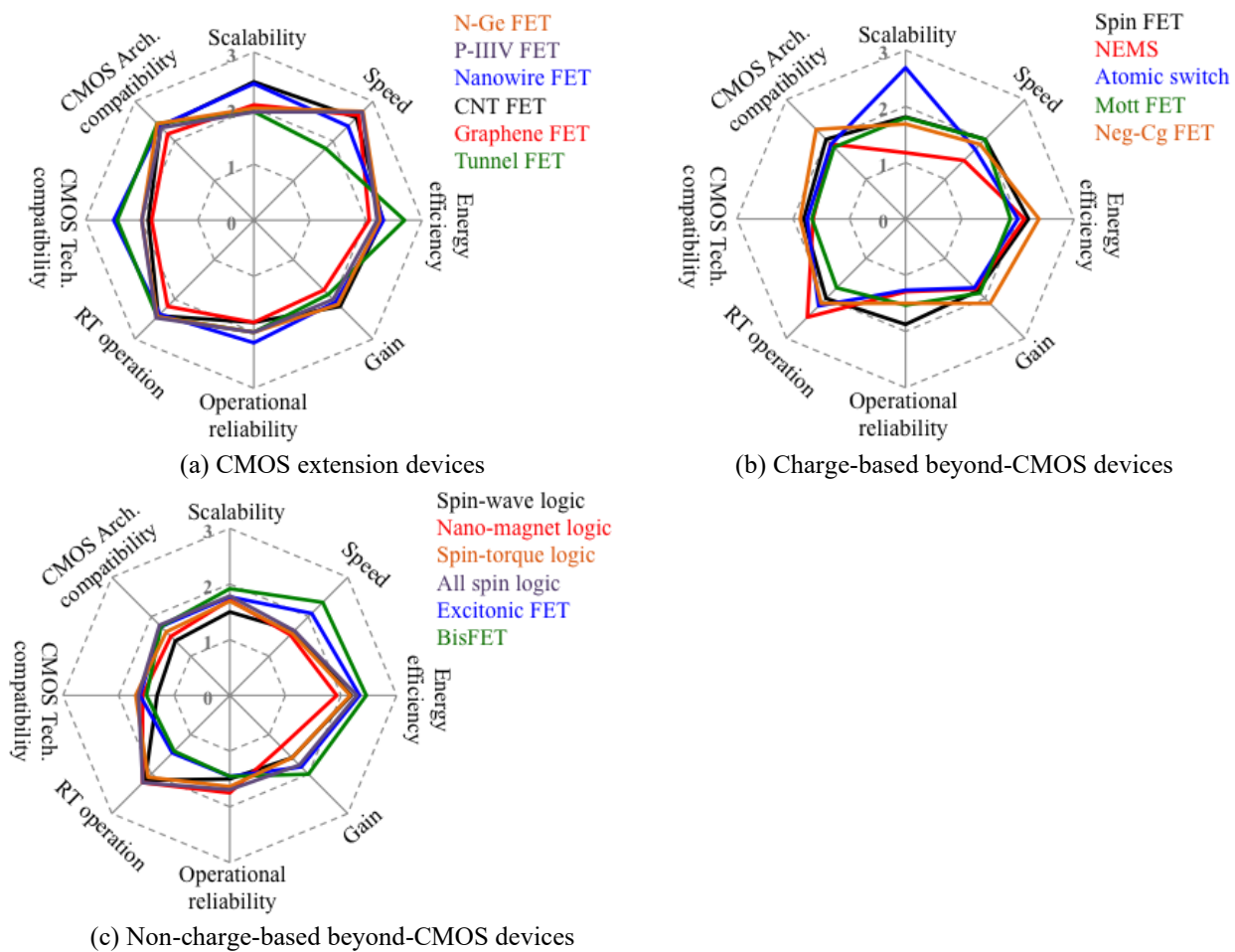


Figure BC7.4 Comparison of Emerging Logic Devices Based on 2013 ITRS ERD Critical Review: (a) CMOS Extension Devices; (b) Charge-based Beyond-CMOS Devices; (c) Non-charge-based Beyond-CMOS Devices

8. SUMMARY

The “Beyond CMOS” chapter systematically surveys emerging memory and logic devices (sections 2 and 3), novel technologies (section 4), and alternative architectures and computing paradigms (section 5), to explore potential solutions beyond the conventional scaling of CMOS technologies. Although high performance at low power consumption has been a primary objective of beyond-CMOS devices, novel functionalities and applications have become increasingly important. The recent emergence of energy-efficient data-intensive cognitive applications is also shifting the emphasis from high-precision computing solutions to novel computing paradigms with massive parallelism and bio-inspired mechanisms. Research opportunities exist in the co-optimization of beyond-CMOS devices and architectures to explore unique device characteristics and architectural designs.

Although a beyond-CMOS device competitive against CMOS FET has not been identified, beyond-CMOS devices with dramatically enhanced scalability and performance while simultaneously reducing the energy dissipation per functional operation would still be fundamentally important and a worthwhile research objective. In considering the many disparate new approaches proposed to provide order of magnitude scaling of information processing beyond that attainable with ultimately scaled CMOS, the following set of guiding principles are proposed to provide a useful structure for directing research on “Beyond CMOS” information processing technology.

- **Computational State Variable(s) other than Solely Electron Charge**

These include spin, phase, multipole orientation, mechanical position, polarity, orbital symmetry, magnetic flux quanta, molecular configuration, and other quantum states. The estimated performance comparison of alternative state variable devices to ultimately scaled CMOS should be made as early in a program as possible to down-select and identify key trade-offs.

- **Non-thermal Equilibrium Systems**

These are systems that are out of equilibrium with the ambient thermal environment for some period of their operation, thereby reducing the perturbations of stored information energy in the system caused by thermal interactions with the environment. The purpose is to allow lower energy computational processing while maintaining information integrity.

- **Novel Energy Transfer Interactions**

These interactions would provide the interconnect function between communicating information processing elements. Energy transfer mechanisms for device interconnection could be based on short range interactions, including, for example, quantum exchange and double exchange interactions, electron hopping, Förster coupling (dipole–dipole coupling), tunneling and coherent phonons.

- **Nanoscale Thermal Management**

This could be accomplished by manipulating lattice phonons for constructive energy transport and heat removal.

- **Sub-lithographic Manufacturing Process**

One example of this principle is directed self-assembly of complex structures composed of nanoscale building blocks. These self-assembly approaches should address non-regular, hierarchically organized structures, be tied to specific device ideas, and be consistent with high volume manufacturing processes.

- **Alternative Architectures**

In this case, architecture is the functional arrangement on a single chip of interconnected devices that includes embedded computational components. These architectures could utilize, for special purposes, novel devices other than CMOS to perform unique functions.

9. ENDNOTES/REFERENCES

- ¹V. V. Zhirmov, R. K. Cavin, S. Menzel, E. Linn, S. Schmelzer, D. Bräuhäus, C. Schindler and R. Waser, "Memory Devices: Energy-Space-Time Trade-offs," *Proc. IEEE*, vol. 98, pp. 2185-2200, Dec. 2010.
- ²K. Lee and S. H. Kang, "Development of Embedded STT-MRAM for Mobile System-on-Chips," *Ieee Transactions on Magnetics*, vol. 47, pp. 131-136, Jan 2011.
- ³S. Mangin, D. Ravelosona, J. A. Katine, M. J. Carey, B. D. Terris, and E. E. Fullerton, "Current-induced magnetization reversal in nanopillars with perpendicular anisotropy," *Nature Materials*, vol. 5, pp. 210-215, 2006.
- ⁴D. Saida, S. Kashiwada, M. Yakabe, T. Daibou, M. Fukumoto, S. Miwa, Y. Suzuki, K. Abe, H. Noguchi, J. Ito, and S. Fujita, "1x-to 2x-nm perpendicular MTJ Switching at Sub-3-ns Pulses Below 100 μ A for High-Performance Embedded STT-MRAM for Sub-20-nm CMOS," *Ieee Transactions on Electron Devices*, vol. 64, pp. 427-431, Feb 2017.
- ⁵G. Jan, L. Thomas, S. Le, Y. Lee, H. Liu, J. Zhu, J. Iwata-Harms, S. Patel, R. Tong, V. Sundar, S. Serrano-Guisan, D. Shen, R. He, J. Haq, Z. J. Teng, V. Lam, Y. Yang, Y. Wang, T. Zhong, H. Fukuzawa, and P. Wang, "Demonstration of Ultra-Low Voltage and Ultra Low Power STT-MRAM designed for compatibility with 0x node embedded LLC applications," in *2018 IEEE Symposium on VLSI Technology*, 2018, pp. 65-66.
- ⁶S. Sakhare, M. Perumkunnil, T. H. Bao, S. Rao, W. Kim, D. Crotti, F. Yasin, S. Couet, J. Swerts, S. Kundu, D. Yakimets, R. Baert, H. Oh, A. Spessot, A. Mocuta, G. S. Kar, and A. Furnemont, "Enablement of STT-MRAM as last level cache for the high performance computing domain at the 5nm node," in *2018 IEEE International Electron Devices Meeting (IEDM)*, 2018, pp. 18.3.1-18.3.4.
- ⁷L. Thomas, G. Jan, S. Serrano-Guisan, H. Liu, J. Zhu, Y. Lee, S. Le, J. Iwata-Harms, R. Tong, S. Patel, V. Sundar, D. Shen, Y. Yang, R. He, J. Haq, Z. Teng, V. Lam, P. Liu, Y. Wang, T. Zhong, H. Fukuzawa, and P. Wang, "STT-MRAM devices with low damping and moment optimized for LLC applications at 0x nodes," in *2018 IEEE International Electron Devices Meeting (IEDM)*, 2018, pp. 27.3.1-27.3.4.
- ⁸G. Hu, J. H. Lee, J. J. Nowak, J. Z. Sun, J. Harms, A. Annunziata, S. Brown, W. Chen, Y. H. Kim, G. Lauer, L. Liu, N. Marchack, S. Murthy, E. J. O. Sullivan, J. H. Park, M. Reuter, R. P. Robertazzi, P. L. Trouilloud, Y. Zhu, and D. C. Worledge, "STT-MRAM with double magnetic tunnel junctions," in *2015 IEEE International Electron Devices Meeting (IEDM)*, 2015, pp. 26.3.1-26.3.4.
- ⁹H. Sato, M. Yamanouchi, S. Ikeda, S. Fukami, F. Matsukura, and H. Ohno, "Perpendicular-anisotropy CoFeB-MgO magnetic tunnel junctions with a MgO/CoFeB/Ta/CoFeB/MgO recording structure," *Applied Physics Letters*, vol. 101, p. 022414, 2012.
- ¹⁰K. Watanabe, B. Jinnai, S. Fukami, H. Sato, and H. Ohno, "Shape anisotropy revisited in single-digit nanometer magnetic tunnel junctions," *Nature Communications*, vol. 9, p. 663, 2018/02/14 2018.
- ¹¹M. M. S. Aly, M. Gao, G. Hills, C. S. Lee, G. Pitner, M. M. Shulaker, T. F. Wu, M. Asheghi, J. Bokor, F. Franchetti, K. E. Goodson, C. Kozyrakakis, I. Markov, K. Olukotun, L. Pileggi, E. Pop, J. Rabaey, R. C. H. S. P. Wong, and S. Mitra, "Energy-Efficient Abundant-Data Computing: The N3XT 1,000x," *Computer*, vol. 48, pp. 24-33, 2015.
- ¹²S. Fujita, H. Noguchi, K. Ikegami, S. Takeda, K. Nomura, and K. Abe, "Novel memory hierarchy with e-STT-MRAM for near-future applications," in *2017 International Symposium on VLSI Design, Automation and Test (VLSI-DAT)*, 2017, pp. 1-2.
- ¹³H. Yang, X. Hao, Z. Wang, R. Malmhall, H. Gan, K. Satoh, J. Zhang, D. H. Jung, X. Wang, Y. Zhou, B. K. Yen, and Y. Huai, "Threshold switching selector and 1S1R integration development for 3D cross-point STT-MRAM," in *2017 IEEE International Electron Devices Meeting (IEDM)*, 2017, pp. 38.1.1-38.1.4.
- ¹⁴S. Fukami, T. Anekawa, A. Ohkawara, Z. Chaoliang, and H. Ohno, "A sub-ns three-terminal spin-orbit torque induced switching device," in *2016 IEEE Symposium on VLSI Technology*, 2016, pp. 1-2.
- ¹⁵S. Shi, Y. Ou, S. V. Aradhya, D. C. Ralph, and R. A. Buhrman, "Fast Low-Current Spin-Orbit-Torque Switching of Magnetic Tunnel Junctions through Atomic Modifications of the Free-Layer Interfaces," *Physical Review Applied*, vol. 9, p. 011002, 01/30/ 2018.
- ¹⁶E. Kitagawa, S. Fujita, K. Nomura, H. Noguchi, K. Abe, K. Ikegami, T. Daibou, Y. Kato, C. Kamata, S. Kashiwada, N. Shimomura, J. Ito, and H. Yoda, "Impact of ultra low power and fast write operation of advanced perpendicular MTJ on power reduction for high-performance mobile CPU," *2012 Ieee International Electron Devices Meeting (Iedm)*, 2012.
- ¹⁷L. Liu, C. F. Pai, Y. Li, H. W. Tseng, D. C. Ralph, and R. A. Buhrman, "Spin-torque switching with the giant spin Hall effect of tantalum," *Science*, vol. 336, pp. 555-8, May 4 2012.
- ¹⁸I. M. Miron, K. Garello, G. Gaudin, P. J. Zermatten, M. V. Costache, S. Auffret, S. Bandiera, B. Rodmacq, A. Schuhl, and P. Gambardella, "Perpendicular switching of a single ferromagnetic layer induced by in-plane current injection," *Nature*, vol. 476, pp. 189-93, Aug 11 2011.
- ¹⁹A. R. Mellnik, J. S. Lee, A. Richardella, J. L. Grab, P. J. Mintun, M. H. Fischer, A. Vaezi, A. Manchon, E. A. Kim, N. Samarth, and D. C. Ralph, "Spin-transfer torque generated by a topological insulator," *Nature*, vol. 511, pp. 449-51, Jul 24 2014.
- ²⁰K. Garello, F. Yasin, H. Hody, S. Couet, L. Souriau, S. H. Sharifi, J. Swerts, R. Carpenter, S. Rao, W. Kim, J. Wu, K. K. V. Sethu, M. Pak, N. Jossart, D. Crotti, A. Furnemont, and G. S. Kar, "Manufacturable 300mm platform solution for Field-Free Switching SOT-MRAM," in *2019 Symposium on VLSI Circuits*, 2019, pp. T194-T195.
- ²¹L. J. Zhu, D. C. Ralph, and R. A. Buhrman, "Highly Efficient Spin-Current Generation by the Spin Hall Effect in Au1-xPt_x," *Physical Review Applied*, vol. 10, Sep 6 2018.
- ²²Y. Fan, P. Upadhyaya, X. Kou, M. Lang, S. Takei, Z. Wang, J. Tang, L. He, L. T. Chang, M. Montazeri, G. Yu, W. Jiang, T. Nie, R. N. Schwartz, Y. Tserkovnyak, and K. L. Wang, "Magnetization switching through giant spin-orbit torque in a magnetically doped topological insulator heterostructure," *Nat Mater*, vol. 13, pp. 699-704, Jul 2014.

- ²³ S. Shi, S. Liang, Z. Zhu, K. Cai, S. D. Pollard, Y. Wang, J. Wang, Q. Wang, P. He, J. Yu, G. Eda, G. Liang, and H. Yang, "All-electric magnetization switching and Dzyaloshinskii-Moriya interaction in WTe₂/ferromagnet heterostructures," *Nat Nanotechnol*, vol. 14, pp. 945-949, Oct 2019.
- ²⁴ S. Fukami, C. Zhang, S. DuttaGupta, A. Kurenkov, and H. Ohno, "Magnetization switching by spin-orbit torque in an antiferromagnet-ferromagnet bilayer system," *Nat Mater*, vol. 15, pp. 535-41, May 2016.
- ²⁵ P. Wadley, B. Howells, J. Železný, C. Andrews, V. Hills, R. P. Campion, V. Novák, K. Olejník, F. Maccherozzi, S. S. Dhesi, S. Y. Martin, T. Wagner, J. Wunderlich, F. Freimuth, Y. Mokrousov, J. Kuneš, J. S. Chauhan, M. J. Grzybowski, A. W. Rushforth, K. W. Edmonds, B. L. Gallagher, and T. Jungwirth, "Electrical switching of an antiferromagnet," *Science*, vol. 351, pp. 587-590, 2016.
- ²⁶ Y. Wang, R. Ramaswamy, M. Motapothula, K. Narayanapillai, D. Zhu, J. Yu, T. Venkatesan, and H. Yang, "Room-Temperature Giant Charge-to-Spin Conversion at the SrTiO₃-LaAlO₃ Oxide Interface," *Nano Letters*, vol. 17, pp. 7659-7664, 2017/12/13 2017.
- ²⁷ L. Liu, Q. Qin, W. Lin, C. Li, Q. Xie, S. He, X. Shu, C. Zhou, Z. Lim, J. Yu, W. Lu, M. Li, X. Yan, S. J. Pennycook, and J. Chen, "Current-induced magnetization switching in all-oxide heterostructures," *Nature Nanotechnology*, 2019/09/09 2019.
- ²⁸ S. Fukami, T. Anekawa, C. Zhang, and H. Ohno, "A spin-orbit torque switching scheme with collinear magnetic easy axis and current configuration," *Nature Nanotechnology*, vol. 11, p. 621, 03/21/online 2016.
- ²⁹ F. Oboril, R. Bishnoi, M. Ebrahimi, and M. B. Tahoori, "Evaluation of Hybrid Memory Technologies Using SOT-MRAM for On-Chip Cache Hierarchy," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 34, pp. 367-380, 2015.
- ³⁰ N. Sato, F. Xue, R. M. White, C. Bi, and S. X. Wang, "Two-terminal spin-orbit torque magnetoresistive random access memory," *Nature Electronics*, vol. 1, pp. 508-511, 2018/09/01 2018.
- ³¹ H. Yoda, N. Shimomura, Y. Ohsawa, S. Shirotori, Y. Kato, T. Inokuchi, Y. Kamiguchi, B. Altansargai, Y. Saito, K. Koi, H. Sugiyama, S. Oikawa, M. Shimizu, M. Ishikawa, K. Ikegami, and A. Kurobe, "Voltage-control spintronics memory (VoCSM) having potentials of ultra-low energy-consumption and high-density," in *2016 IEEE International Electron Devices Meeting (IEDM)*, 2016, pp. 27.6.1-27.6.4.
- ³² S. V. Aradhya, G. E. Rowlands, J. Oh, D. C. Ralph, and R. A. Buhrman, "Nanosecond-Timescale Low Energy Switching of In-Plane Magnetic Tunnel Junctions through Dynamic Oersted-Field-Assisted Spin Hall Effect," *Nano Letters*, vol. 16, pp. 5987-5992, Oct 2016.
- ³³ G. Yu, P. Upadhyaya, Y. Fan, J. G. Alzate, W. Jiang, K. L. Wong, S. Takei, S. A. Bender, L. T. Chang, Y. Jiang, M. Lang, J. Tang, Y. Wang, Y. Tserkovnyak, P. K. Amiri, and K. L. Wang, "Switching of perpendicular magnetization by spin-orbit torques in the absence of external magnetic fields," *Nat Nanotechnol*, vol. 9, pp. 548-54, Jul 2014.
- ³⁴ L. You, O. Lee, D. Bhowmik, D. Labanowski, J. Hong, J. Bokor, and S. Salahuddin, "Switching of perpendicularly polarized nanomagnets with spin orbit torque without an external magnetic field by engineering a tilted anisotropy," *Proceedings of the National Academy of Sciences*, vol. 112, pp. 10310-10315, 2015.
- ³⁵ Y.-C. Lau, D. Betto, K. Rode, J. M. D. Coey, and P. Stamenov, "Spin-orbit torque switching without an external field using interlayer exchange coupling," *Nature Nanotechnology*, vol. 11, p. 758, 05/30/online 2016.
- ³⁶ Z. Zhao, A. K. Smith, M. Jamali, and J.-p. Wang, "External-field-free spin Hall switching of perpendicular magnetic nanopillar with a dipole-coupled composite structure," *arXiv preprint arXiv:1603.09624*, 2016.
- ³⁷ D. MacNeill, G. M. Stiehl, M. H. D. Guimaraes, R. A. Buhrman, J. Park, and D. C. Ralph, "Control of spin-orbit torques through crystal symmetry in WTe₂/ferromagnet bilayers," *Nature Physics*, vol. 13, p. 300, 11/07/online 2016.
- ³⁸ S. H. C. Baek, V. P. Amin, Y. W. Oh, G. Go, S. J. Lee, G. H. Lee, K. J. Kim, M. D. Stiles, B. G. Park, and K. J. Lee, "Spin currents and spin-orbit torques in ferromagnetic trilayers," *Nature Materials*, vol. 17, pp. 509-+, Jun 2018.
- ³⁹ A. M. Humphries, T. Wang, E. R. J. Edwards, S. R. Allen, J. M. Shaw, H. T. Nembach, J. Q. Xiao, T. J. Silva, and X. Fan, "Observation of spin-orbit effects with spin rotation symmetry," *Nature Communications*, vol. 8, p. 911, 2017/10/13 2017.
- ⁴⁰ T. Maruyama, Y. Shiota, T. Nozaki, K. Ohta, N. Toda, M. Mizuguchi, A. A. Tulapurkar, T. Shinjo, M. Shiraishi, S. Mizukami, Y. Ando, and Y. Suzuki, "Large voltage-induced magnetic anisotropy change in a few atomic layers of iron," *Nat Nanotechnol*, vol. 4, pp. 158-61, Mar 2009.
- ⁴¹ W. G. Wang, M. Li, S. Hageman, and C. L. Chien, "Electric-field-assisted switching in magnetic tunnel junctions," *Nat Mater*, vol. 11, pp. 64-8, Jan 2012.
- ⁴² C.-G. Duan, J. Velez, R. Sabirianov, Z. Zhu, J. Chu, S. Jaswal, and E. Tsymlal, "Surface Magnetoelectric Effect in Ferromagnetic Metal Films," *Physical Review Letters*, vol. 101, p. 137201, Sep 26 2008.
- ⁴³ K. Nakamura, R. Shimabukuro, Y. Fujiwara, T. Akiyama, T. Ito, and A. J. Freeman, "Giant Modification of the Magnetocrystalline Anisotropy in Transition-Metal Monolayers by an External Electric Field," *Physical Review Letters*, vol. 102, 2009.
- ⁴⁴ Y. Shiota, T. Nozaki, F. Bonell, S. Murakami, T. Shinjo, and Y. Suzuki, "Induction of coherent magnetization switching in a few atomic layers of FeCo using voltage pulses," *Nat Mater*, vol. 11, pp. 39-43, Jan 2012.
- ⁴⁵ C. Grezes, F. Ebrahimi, J. G. Alzate, X. Cai, J. A. Katine, J. Langer, B. Ocker, P. Khalili Amiri, and K. L. Wang, "Ultra-low switching energy and scaling in electric-field-controlled nanoscale magnetic tunnel junctions with high resistance-area product," *Applied Physics Letters*, vol. 108, p. 012403, 2016.
- ⁴⁶ T. Yamamoto, T. Nozaki, H. Imamura, Y. Shiota, S. Tamaru, K. Yakushiji, H. Kubota, A. Fukushima, Y. Suzuki, and S. Yuasa, "Improvement of write error rate in voltage-driven magnetization switching," *Journal of Physics D-Applied Physics*, vol. 52, Apr 17 2019.
- ⁴⁷ L. Liu, C.-F. Pai, D. Ralph, and R. Buhrman, "Gate voltage modulation of spin-Hall-torque-driven magnetic switching," *arXiv preprint arXiv:1209.0962*, 2012.

- ⁴⁸ T. Inokuchi, H. Yoda, K. Koi, N. Shimomura, Y. Ohsawa, Y. Kato, S. Shirotori, M. Shimizu, H. Sugiyama, S. Oikawa, B. Altansargai, and A. Kurobe, "Real-time observation of fast and highly reliable magnetization switching in Voltage-Control Spintronics Memory (VoCSM)," *Applied Physics Letters*, vol. 114, p. 192404, 2019.
- ⁴⁹ Y. Kato, H. Yoda, T. Inokuchi, M. Shimizu, Y. Ohsawa, K. Fujii, M. Yoshiki, S. Oikawa, S. Shirotori, K. Koi, N. Shimomura, B. Altansargai, H. Sugiyama, and A. Kurobe, "Low write current and strong durability in high-speed spintronics memory (spin-Hall MRAM and VoCSM) through development of a shunt-free design process and W spin-Hall electrode," *Journal of Magnetism and Magnetic Materials*, vol. 491, p. 165536, 2019/12/01/ 2019.
- ⁵⁰ X. Li, A. Lee, S. A. Razavi, H. Wu, and K. L. Wang, "Voltage-controlled magnetoelectric memory and logic devices," *MRS Bulletin*, vol. 43, pp. 970-977, 2018.
- ⁵¹ C. Grezes, X. Li, K. Wong, F. Ebrahimi, P. Khalili Amiri, and K. L. Wang, "Voltage-controlled magnetic tunnel junctions with synthetic ferromagnet free layer sandwiched by asymmetric double MgO barriers," *Journal of Physics D: Applied Physics*, 2019/09/26 2019.
- ⁵² T. Nozaki, A. Koziol-Rachwał, W. Skowroński, V. Zayets, Y. Shiota, S. Tamaru, H. Kubota, A. Fukushima, S. Yuasa, and Y. Suzuki, "Large Voltage-Induced Changes in the Perpendicular Magnetic Anisotropy of an MgO-Based Tunnel Junction with an Ultrathin Fe Layer," *Physical Review Applied*, vol. 5, p. 044006, 04/15/ 2016.
- ⁵³ Y. Kato, H. Yoda, Y. Saito, S. Oikawa, K. Fujii, M. Yoshiki, K. Koi, H. Sugiyama, M. Ishikawa, T. Inokuchi, N. Shimomura, M. Shimizu, S. Shirotori, B. Altansargai, Y. Ohsawa, K. Ikegami, A. Tiwari, and A. Kurobe, "Giant voltage-controlled magnetic anisotropy effect in a crystallographically strained CoFe system," *Applied Physics Express*, vol. 11, p. 053007, 2018.
- ⁵⁴ X. Li, T. Sasaki, C. Grezes, D. Wu, K. L. Wong, C. Bi, P.-V. Ong, F. Ebrahimi, G. Yu, N. Kiousis, W. Wang, T. Ohkubo, P. Khalili Amiri, and K. L. Wang, "Predictive Materials Design of Magnetic Random-Access Memory Based on Nanoscale Atomic Structure and Element Distribution," *Nano Letters*, vol. 19, pp. 8621-8629, 2019/11/07 2019.
- ⁵⁵ C. Grezes, H. Lee, A. Lee, S. Wang, F. Ebrahimi, X. Li, K. Wong, J. A. Katine, B. Ocker, J. Langer, P. Gupta, P. Khalili Amiri, and K. L. Wang, "Write Error Rate and Read Disturbance in Electric-Field-Controlled Magnetic Random-Access Memory," *IEEE Magnetics Letters*, vol. 8, pp. 1-5, 2017.
- ⁵⁶ H. Lee, A. Lee, S. D. Wang, F. Ebrahimi, P. Gupta, P. K. Amiri, and K. L. Wang, "A Word Line Pulse Circuit Technique for Reliable Magnetoelectric Random Access Memory," *Ieee Transactions on Very Large Scale Integration (Vlsi) Systems*, vol. 25, pp. 2027-2034, Jul 2017.
- ⁵⁷ T. Ikeura, T. Nozaki, Y. Shiota, T. Yamamoto, H. Imamura, H. Kubota, A. Fukushima, Y. Suzuki, and S. Yuasa, "Reduction in the write error rate of voltage-induced dynamic magnetization switching using the reverse bias method," *Japanese Journal of Applied Physics*, vol. 57, p. 040311, 2018.
- ⁵⁸ S. Kanai, Y. Nakatani, M. Yamanouchi, S. Ikeda, H. Sato, F. Matsukura, and H. Ohno, "Magnetization switching in a CoFeB/MgO magnetic tunnel junction by combining spin-transfer torque and electric field-effect," *Applied Physics Letters*, vol. 104, p. 212406, 2014.
- ⁵⁹ R. Waser, R. Dittman, G. Staikov, and K. Szot, "Redox-based resistive switching memories – nanoionic mechanisms, prospects, and challenges," *Adv. Mat.* 21 (2009) 2632-2663
- ⁶⁰ H. Akinaga and H. Shima, "Resistive random access memory (ReRAM) based on metal oxides," *Proc. IEEE* vol. 98, pp. 2237–2251, 2010.
- ⁶¹ R. Waser, R. Bruchhaus, and S. Menzel, "Redox-based resistive switching memories," *Nanoelectronics and Information Technology*, R. Waser, Ed., ed Weinheim, Germany: Wiley-VCH, 2013.
- ⁶² M. J. Marinella, "Emerging resistive switching memory technologies: Overview and current status," *2014 IEEE Intl Symp on Circuits and Systems (ISCAS)*, Melbourne VIC, 2014, pp. 830–833.
- ⁶³ D. M. Nminibapiel *et al.*, "Characteristics of resistive memory read fluctuations in endurance cycling," *El. Dev. Letters*, vol. 38, p. 326, 2017.
- ⁶⁴ D. Veksler *et al.*, "Switching variability factors in compliance-free metal oxide RRAM," *2019 IEEE International Reliability Physics Symposium (IRPS)*, 1-5, 2019.
- ⁶⁵ B. Govoreanu, *et al.*, "10x10nm² Hf/HfO_x crossbar resistive RAM with excellent performance, reliability and low-energy operation," *IEDM Tech. Dig.*, 2011, pp. 31.6.1–31.6.4.
- ⁶⁶ Z. Wei *et al.*, "Highly reliable TaO_x ReRAM and direct evidence of redox reaction mechanism," *IEDM*, 2008, pp. 1–4.
- ⁶⁷ M.-J. Lee, C. B. Lee, D. Lee, S. R. Lee, M. Chang, J. H. Hur, Y.-B. Kim, C.-J. Kim, D. H. Seo, S. Seo, U.-I. Chung, I.-K. Yoo, K. Kim, "A fast, high-endurance and scalable non-volatile memory device made from asymmetric Ta₂O₅-x/TaO₂-x bilayer structures," *Nature Mat.* July 2011.
- ⁶⁸ J. J. Yang, F. Miao, M. D. Pickett, D. A. A. Ohlberg, D. R. Stewart, C. N. Lau, *et al.*, "The mechanism of electroforming of metal oxide memristive switches," *Nanotechnology*, vol. 20, p. 215201, May 2009.
- ⁶⁹ G. Bersuker *et al.*, "Metal oxide RRAM switching mechanism based on conductive filament microscopic properties," *J. Appl. Phys.*, vol. 110, p. 125518, 2011.
- ⁷⁰ R. Waser, R. Dittmann, G. Staikov, and K. Szot, "Redox-Based Resistive Switching Memories - Nanoionic Mechanisms, Prospects, and Challenges," *Advanced Materials*, vol. 21, pp. 2632–2663, 2009.
- ⁷¹ J. J. Yang, D. B. Strukov, and D. R. Stewart, "Memristive devices for computing," *Nature Nanotech.*, vol. 8, pp. 13–24, 2013.
- ⁷² G. Bersuker, D. C. Gilmer, D. Veksler, "Metal-oxide resistive random access memory (RRAM) technology: Material and operation details and ramifications" in *Advances in Non-Volatile Memory and Storage Technology*, pp. 35-102, 2019
- ⁷³ F. Pan, S. Gao, C. Chen, C. Song, and F. Zeng, "Recent progress in resistive random access memories: Materials, switching mechanisms, and performance," *Materials Science and Engineering: R: Reports*, vol. 83, pp. 1–59, Sept. 2014.

- ⁷⁴ B. Govoreanu, G. S. Kar, Y. Y. Chen, V. Paraschiv, S. Kubicek, A. Fantini, *et al.*, "10x10nm²Hf/HfO_xcrossbar resistive RAM with excellent performance, reliability and low-energy operation," *Electron Devices Meeting (IEDM), 2011 IEEE International*, 2011, pp. 31.6.1–31.6.4.
- ⁷⁵ L. Kai-Shin, C. Ho, L. Ming-Taou, C. Min-Cheng, H. Cho-Lun, J. M. Lu, *et al.*, "Utilizing Sub-5 nm sidewall electrode technology for atomic-scale resistive memory fabrication," *VLSI Technology (VLSI-Technology): Digest of Technical Papers, 2014 Symposium on*, 2014, pp. 1–2.
- ⁷⁶ Chang, K.C., Zhang, R., Chang, T.C., Tsai, T.M., Chu, T.J., Chen, H.L., Shih, C.C., Pan, C.H., Su, Y.T., Wu, P.J. and Sze, S.M., , December, "High performance, excellent reliability multifunctional graphene oxide doped memristor achieved by self-protective compliance current structure," in *Electron Devices Meeting (IEDM), 2014 IEEE International*, 2014, (pp. 33-3), IEEE.
- ⁷⁷ C. T. Antonio, J. P. Strachan, M.-R. Gilbeto, and R. S. Williams, "Sub-nanosecond switching of a tantalum oxide memristor," *Nanotechnology*, vol. 22, p. 485203, 2011.
- ⁷⁸ B. J. Choi, A. C. Torrezan, J. P. Strachan, P. G. Kotula, A. J. Lohn, M. J. Marinella, R. S. Williams and J. Joshua Yang, "High-speed and low-energy nitride memristors" , *Advanced Functional Materials* 26, 5290 (2016)
- ⁷⁹ Lee, M.J., Lee, C.B., Lee, D., Lee, S.R., Chang, M., Hur, J.H., Kim, Y.B., Kim, C.J., Seo, D.H., Seo, S. and Chung, U.I., "A fast, high-endurance and scalable non-volatile memory device made from asymmetric Ta₂O₅-x/TaO₂-x bilayer structures," in *Nature Materials*, 10(8), 2011, p.625.
- ⁸⁰ S. Choi, J. Lee, S. Kim, and W. D. Lu, "Retention failure analysis of metal-oxide based resistive memory," *Applied Physics Letters*, vol. 105, p. 113510, 2014.
- ⁸¹ M. Wang, S. Cai, C. Pan, C. Wang, X. Lian, K. Xu, Y. Zhuo, J. Joshua Yang, P. Wang and F. Miao, "Robust memristors based on fully layered two-dimensional materials" , *Nature Electronics* 1, (2018). doi:10.1038/s41928-018-0021-4
- ⁸² S. Lee, J. Sohn, Z. Jiang, H.-Y. Chen, and H. S. Philip Wong, "Metal oxide-resistive memory using graphene-edge electrodes," *Nat Commun*, vol. 6, Sept. 2015.
- ⁸³ Kang, Chen-Fang, *et al.* "Self-formed conductive nanofilaments in (Bi, Mn) Ox for ultralow-power memory devices." *Nano Energy* 13 (2015): 283–290.
- ⁸⁴ T. Y. Liu, T. H. Yan, R. Scheuerlein, Y. Chen, J. K. Lee, G. Balakrishnan, *et al.*, "A 130.7 mm² 2-layer 32Gb ReRAM memory device in 24 nm technology," *Solid-State Circuits Conference Digest of Technical Papers (ISSCC), 2013 IEEE International*, 2013, pp. 210–211.
- ⁸⁵ J. Sung Hyun, T. Kumar, S. Narayanan, W. D. Lu, and H. Nazarian, "3D-stackable crossbar resistive memory based on Field Assisted Superlinear Threshold (FAST) selector," *Electron Devices Meeting (IEDM), 2014 IEEE International*, 2014, pp. 6.7.1-6.7.4.
- ⁸⁶ G. Bersuker *et al.*, "Toward reliable RRAM performance: macro- and micro-analysis of operation processes," *J. Comput. Electronics*, vol. 16, pp. 1085–1094, 2017.
- ⁸⁷ A. Sawa *et al.*, "Hysteretic current–voltage characteristics and resistance switching at a rectifying Ti/Pr_{0.7}Ca_{0.3}MnO₃ interface," *APL*, 85, p. 4073, 2004.
- ⁸⁸ Hyejung Choi *et al.*, "The Effect of Tunnel Barrier at Resistive Switching Device for Low Power Memory Applications," *IEEE International Memory Workshop*, 2011.
- ⁸⁹ R. Meyer *et al.*, "Oxide Dual-Layer Memory Element for Scalable Non-Volatile Cross-Point Memory Technology," *IEEE NVMTS*, 2008.
- ⁹⁰ C.J. Chevallier *et al.*, "A 0.13 μm 64 Mb multi-layered conductive metal-oxide memory," *IEEE Solid-State Circuits Conference Digest of Technical Papers (ISSCC)*, 2010.
- ⁹¹ H. Schroeder, V. V. Zhirnov, R. K. Cavin, and R. Waser, "Voltage-time dilemma of pure electronic mechanisms in resistive switching memory cells," *Journal of Applied Physics*, vol. 107, p. 54517, Mar. 2010.
- ⁹² X. Huang, H. Wu, B. Gao, D.C. Sekar, L. Dai, M. Kellam, G. Bronner, N. Deng, and H. Qian, "HfO₂/Al₂O₃ multilayer for RRAM arrays: a technique to improve tail-bit retention," *Nanotech.* 27, 395201 (2016).
- ⁹³ I. G. Baek, *et al.*, "Highly scalable nonvolatile resistive memory using simple binary oxide driven by asymmetric unipolar voltage pulses," *2004 IEDM Tech. Dig.*, 2004, pp. 587–590.
- ⁹⁴ L. Goux, *et al.*, "Field-driven ultrafast sub-ns programming in WAl₂O₃TiCuTe-based 1T1R CBRAM system," *VLSI Technology (VLSIT), 2012 Symposium on*, 2012, pp. 69–70.
- ⁹⁵ C. Cagli, *et al.*, "Experimental and theoretical study of electrode effects in HfO₂ based RRAM," *Electron Devices Meeting (IEDM), 2011 IEEE International*, 2011, pp. 28.7.1–28.7.4.
- ⁹⁶ C. Cagli, D. Ielmini, F. Nardi, and A. L. Lacaita, "Evidence for threshold switching in the set process of NiO-based RRAM and physical modeling for set, reset, retention and disturb prediction," *IEDM Tech. Dig.*, 2008, pp. 1–4.
- ⁹⁷ U. Russo, *et al.*, "Conductive-filament switching analysis and self-accelerated thermal dissolution model for reset in NiO-based RRAM," *IEDM Tech. Dig.*, 2007, pp. 775–778.
- ⁹⁸ L. Goux, *et al.*, "Coexistence of the bipolar and unipolar resistive-switching modes in NiO cells made by thermal oxidation of Ni layers," *Journal of Applied Physics*, vol. 107, p. 024512, Jan. 2010.
- ⁹⁹ D. S. Jeong, H. Schroeder, and R. Waser, "Coexistence of Bipolar and Unipolar Resistive Switching Behaviors in a Pt/TiO₂/Pt Stack," *Electrochemical and Solid-State Letters*, vol. 10, pp. G51-G53, Aug. 2007.
- ¹⁰⁰ L. Goux, *et al.*, "Roles and Effects of TiN and Pt Electrodes in Resistive-Switching HfO₂ Systems," *Electrochemical and Solid-State Letters*, vol. 14, pp. H244-H246, June 2011.
- ¹⁰¹ YY. Chen *et al.* "Insights into Ni-filament formation in unipolar-switching Ni/HfO₂/TiN resistive random access memory device," *Appl. Phys. Lett.* 100, 113513 (2012)
- ¹⁰² T. Yanagida, *et al.*, "Scaling Effect on Unipolar and Bipolar Resistive Switching of Metal Oxides," *Sci. Rep.*, vol. 3, p. 1657, Apr. 2013.

- ¹⁰³ X. A. Tran, *et al.*, "High performance unipolar $\text{AlO}_x\text{HfO}_x\text{Ni}$ based RRAM compatible with Si diodes for 3D application," *VLSI Technology (VLSIT), 2011 Symposium on*, 2011, pp. 44–45.
- ¹⁰⁴ Y.-H. Tseng, *et al.*, "High density and ultra-small cell size of Contact ReRAM (CR-RAM) in 90 nm CMOS logic technology and circuits," *IEDM Tech. Dig.*, 2009, pp. 1–4.
- ¹⁰⁵ Y.-H. Tseng, *et al.*, "Electron trapping effect on the switching behavior of contact RRAM devices through random telegraph noise analysis," *IEDM Tech. Dig.*, 2010, pp. 28.5.1–28.5.4.
- ¹⁰⁶ W. C. Shen, *et al.*, "High- κ metal gate contact RRAM (CRRAM) in pure 28 nm CMOS logic process," *IEDM Tech. Dig.*, 2012, pp. 31.6.1–31.6.4.
- ¹⁰⁷ M.-F. Chang, *et al.*, "A 0.5V 4Mb logic-process compatible embedded resistive RAM (ReRAM) in 65 nm CMOS using low-voltage current-mode sensing scheme with 45 ns random read time," *ISSCC Tech. Dig.*, 2012, pp. 434–436.
- ¹⁰⁸ W.-C. Chien, *et al.*, "A novel high performance WO_x ReRAM based on thermally-induced SET operation," *VLSI Technology*, 2013, pp. T100–T101.
- ¹⁰⁹ Kozicki, M. N., *et al.* "Information storage using nanoscale electrodeposition of metal in solid electrolytes." *Superlattices and microstructures*, vol. 34.3, pp. 459–465 2003.
- ¹¹⁰ K. Terabe, T. Hasegawa, T. Nakayama and M. Aono, "Quantized conductance atomic switch", *Nature* 433, 47–50 (2005).
- ¹¹¹ Valov, Ilia, *et al.* "Electrochemical metallization memories—fundamentals, applications, prospects." *Nanotechnology* 22.25 (2011): 254003.
- ¹¹² M. Tada, *et al.*, "Polymer Solid-electrolyte Switch Embedded on CMOS for Nonvolatile Crossbar Switch", *IEEE Transactions on Electron Devices*, volume 58, Issue 12, pp. 4398–4406, 2011.
- ¹¹³ Otsuka, Wataru, *et al.* "A 4mb conductive-bridge resistive memory with 2.3 gb/s read-throughput and 216mb/s program-throughput." *Solid-State Circuits Conference Digest of Technical Papers (ISSCC)*, 2011 IEEE
- ¹¹⁴ Gilbert, Nad, *et al.* "A 0.6 V 8 pJ/write Non-Volatile CBRAM Macro Embedded in a Body Sensor Node for Ultra Low Energy Applications." *VLSI Circuits (VLSIC), 2013 Symposium on. IEEE*, 2013.
- ¹¹⁵ S. Sills, *et al.*, "A Copper ReRAM Cell for Storage Class Memory Applications", 2014 Symposium on VLSI Technology Digest of Technical Papers, DOI: 10.1109/VLSIT.2014.6894368.
- ¹¹⁶ T. Sakamoto *et al.*, "Atom switch with Improved Cycle Endurance using Field Enhancement for Nonvolatile SoC", 2018 IEEE International Memory Workshop (IMW), doi: 10.1109/IMW.2018.8388781.
- ¹¹⁷ Gopalan, C., *et al.* "Demonstration of conductive bridging random access memory (CBRAM) in logic CMOS process." *Solid-State Electronics* 58.1 (2011): 54–61.
- ¹¹⁸ R. Fackenthal *et al.*, "19.7 A 16Gb ReRAM with 200MB/s write and 1GB/s read in 27nm technology," *2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC)*, San Francisco, CA, 2014, pp. 338–339.
- ¹¹⁹ J. Zahurak *et al.*, "Process integration of a 27 nm, 16Gb Cu ReRAM," *2014 IEEE International Electron Devices Meeting*, San Francisco, CA, 2014, pp. 6.2.1–6.2.4.
- ¹²⁰ Prall, Kirk, *et al.* "An Update on Emerging Memory: Progress to 2Xnm." *Memory Workshop (IMW)*, 2012 4th IEEE International. IEEE, 2012.
- ¹²¹ Sankaran, Kiroubanand, *et al.* "Modeling of Copper Diffusion in Amorphous Aluminum Oxide in CBRAM Memory Stack." *ECS Transactions* 45.3 (2012): 317–330.
- ¹²² J. R. Jameson *et al.*, "Towards Automotive Grade Embedded RRAM" 2018 48th European Solid-State Device Research Conference (ESSDERC) doi: 10.1109/ESSDERC.2018.8486890.
- ¹²³ Miyamura, Makoto, *et al.* "Programmable cell array using rewritable solid-electrolyte switch integrated in 90 nm CMOS." *Solid-State Circuits Conference Digest of Technical Papers (ISSCC)*, 2011 IEEE International. IEEE, 2011.
- ¹²⁴ Suri, M., *et al.* "CBRAM devices as binary synapses for low-power stochastic neuromorphic systems: Auditory (Cochlea) and visual (Retina) cognitive processing applications." *Electron Devices Meeting (IEDM)*, 2012 IEEE International. IEEE, 2012.
- ¹²⁵ Xu Bai, Makoto Miyamura, Ryusuke Nebashi, Ayuka Morioka, Naoki Banno, Koichiro Okamoto, Hideaki Numata, Hiromitsu Hada, Tadahiko Sugibayashi, Toshitsugu Sakamoto, and Munehiro Tada, "An atom-switch-based field-programmable gate array with optimized driving capability buffer," *Japanese Journal of Applied Physics* 58, SBBB04 (2019), pp. SBBB04-1-6, <https://doi.org/10.7567/1347-4065/aaf899>.
- ¹²⁶ <https://www.kenkai.jaxa.jp/eng/research/innovative/innovative.html>
- ¹²⁷ X. Bai *et al.*, "Via-Switch FPGA: 65-nm CMOS Implementation and Evaluation," in *IEEE Journal of Solid-State Circuits*, vol. 57, no. 7, pp. 2250–2262, July 2022, doi: 10.1109/JSSC.2021.3117260.1
- ¹²⁸ Masanori Hashimoto *et al.* 2022 *Jpn. J. Appl. Phys.* 61 SM0804.
- ¹²⁹ Choi, S., *et al.* "Resistance drift model for conductive-bridge (CB) RAM by filament surface relaxation." *Memory Workshop (IMW)*, 2012 4th IEEE International. IEEE, 2012.
- ¹³⁰ A Kanwal, S. Paul, M. Chhowalla, "Organic Memory Devices Using C60 and Insulator Polymer," *MRS Proceedings on Materials and Processes for Nonvolatile Memories*, 2005, 830, D7.2.1.
- ¹³¹ J.Y Ouyang, C.W Chu CW, C.R Szmanda, L.P Ma, and Y Yang, "Programmable polymer thin film and non-volatile memory device," *Nature Materials*, 3, 918 (2004)
- ¹³² S.-H. Liu ; W.-L. Yang ; C.-C. Wu ; T.-S. Chao ; M.-R. Ye ; Y.-Y. Su ; P.-Y. Wang ; M.-J. Tsai, "High-Performance Polyimide-Based ReRAM for Nonvolatile Memory Application " *IEEE Electron Dev. Lett.* 34, 123 – 125, 2013.
- ¹³³ W. Bai, R. Huang ; Y. Cai ; Y. Tang ; X. Zhang ; Y. Wang "Record Low-Power Organic RRAM With Sub-20-nA Reset Current," *IEEE Electron. Dev. Lett.* 34, 223–225, 2013.

- ¹³⁴ Y.-C. Lai, D.Y. Wang, I-S. Huang, Y.T. Chen, Y.-H. Hsu, T.-Y. Lin, H.-F. Meng, T.C. Chang, Y.J. Yang, C.C. Chen, F.-C. Hsu, Y.-F. Chen J. , “Low operation voltage macromolecular composite memory assisted by graphene nanoflakes,” *Mater. Chem. C* 1,552-559, 2013.
- ¹³⁵ J.J. Kim, B. Cho; K.S. Kim; T. Lee, G.Y. Jung, “Electrical Characterization of Unipolar Organic Resistive Memory Devices Scaled Down by a Direct Metal-Transfer Method,” *Adv. Mater.* 23, 2104-2107, 2011.
- ¹³⁶ S. Song, J. Jang, Y. Ji, S. Park, T.W. Kim, Y. Song, M.H. Yoon, H.C. Ko, G.Y. Jung, T. Lee, “Twistable nonvolatile organic resistive memory devices,” *Org. Electron.* 14, 2087-2092, 2013.
- ¹³⁷ Y. Chai, Y. Wu, K. Takei, H.Y. Chen, S.M. Yu, P.C.H. Chan, A. Javey, H.S.P. Wong, “Nanoscale Bipolar and Complementary Resistive Switching Memory Based on Amorphous Carbon,” *IEEE Trans. Electron. Dev.* 2011, 58, 3933-3939.
- ¹³⁸ C.-L. Tsai, F. Xiong, E. Pop, S. Moonsub, “Resistive Random Access Memory Enabled by Carbon Nanotube Crossbar Electrodes,” *ACS Nano* 7, 5360-5366, 2013.
- ¹³⁹ P. Siebeneicher, H. Kleemann, K. Leo, and B. Lüssem, “Non-volatile organic memory devices comprising SiO₂ and C₆₀ showing 10⁴ switching cycles,” *Appl. Phys. Lett.* 100, 193301 2012.
- ¹⁴⁰ Q. Chen, B. F. Bory, A. Kiazadeh, P. R. F. Rocha, H. L. Gomes, F. Verbakel, D. M. De Leeuw, S. C. J. Meskers, “Opto-electronic characterization of electron traps upon forming polymer oxide memory diodes,” *Appl. Phys. Lett.* 99, 083305, 2011.
- ¹⁴¹ Lin, W.-P., Liu, S.-J., Gong, T., Zhao, Q. and Huang, W., “Polymer-Based Resistive Memory Materials and Devices,” *Adv. Mater.*, 26: 570–606, 2014.
- ¹⁴² I. Salaoru, S. Alotaibi (a1), Z. Al Halafi (a1) and S Paul, “Creating Electrical Bistability Using Nano-bits – Application in 2-Terminal Memory Devices,” *MRS Advances*, 2, 195-208, 2017.
- ¹⁴³ Paul, S., “Realization of nonvolatile memory devices using small organic molecules and polymer,” *IEEE Transactions on Nanotechnology*, vol. 6, no. 2, pp. 191-195, 2007.
- ¹⁴⁴ K. Asadi, D.M. de Leeuw, B. de Boer, B. P.W.M. Blom, “Organic non-volatile memories from ferroelectric phase-separated blends,” *Nat. Mater.* 7, 547-550, 2008.
- ¹⁴⁵ N. Tsutsumi, X. Bai, and W. Sakai, “Towards nonvolatile memory devices based on ferroelectric polymers,” *AIP Advances* 2, 012154, 2012.
- ¹⁴⁶ K. Toprasertpong, M. Takenaka, and S. Takagi, “Memory window in ferroelectric field-effect transistors: Analytical approach,” *IEEE Trans. Electron Devices* 69, 7113-7119, 2022.
- ¹⁴⁷ J. Müller, P. Polakowski, S. Mueller, and T. Mikolajick, “Ferroelectric hafnium oxide based materials and devices: Assessment of current status and future prospects,” *ECS J. Solid State Sci. Technol. (ECS Journal of Solid State Science and Technology)* 4, N30-N35 (2015).
- ¹⁴⁸ T.S. Böske, J. Müller, D. Bräuhäus, U. Schröder, and U. Böttger, “Ferroelectricity in hafnium oxide: CMOS compatible ferroelectric field effect transistors,” *IEEE International Electron Devices Meeting (IEDM)*, Washington D.C., USA, 5-7 December (2011.), pp. 547–550.
- ¹⁴⁹ J. Müller, E. Yurchuk, T. Schlosser, J. Paul, R. Hoffmann, S. Müller, D. Martin, S. Slesazeck, P. Polakowski, J. Sundqvist, M. Czernohorsky, K. Seidel, P. Kucher, R. Boschke, M. Trentzsch, K. Gebauer, U. Schroder, and T. Mikolajick, “Ferroelectricity in HfO₂ enables nonvolatile data storage in 28 nm HKMG, in Symposium on VLSI Technology (VLSIT),” (2012.), pp. 25–26.
- ¹⁵⁰ K. Toprasertpong, M. Takenaka, and S. Takagi, “Direct observation of charge dynamics in FeFET by quasi-static split C-V and Hall techniques: Revealing FeFET operation,” *IEEE International Electron Devices Meeting (IEDM)*, San Francisco, CA, USA, 7-11 December 2019, pp. 570–573.
- ¹⁵¹ T.P. Ma and J.-P. Han, “Why is nonvolatile ferroelectric memory field-effect transistor still elusive?” *IEEE Electron Device Lett.* 23, 386 (2002).
- ¹⁵² S. Sakai, R. Ilangovan, and M. Takahashi, “Pt/SrBi₂Ta₂O₉/Hf-Al-O/Si field-effect-transistor with long retention using unsaturated ferroelectric polarization switching,” *Jap. J. Apl. Phys.* 43, 7876 (2004).
- ¹⁵³ J. Müller, S. Müller, P. Polakowski, and T. Mikolajick, “Ferroelectric hafnium oxide: a game changer to FRAM?,” *14th Non-Volatile Memory Technology Symposium (NVMTS)*, Jeju, South Korea (2014.).
- ¹⁵⁴ C.-H. Cheng and A. Chin, “Low-Leakage-Current DRAM-Like Memory Using a One-Transistor Ferroelectric MOSFET with a Hf-Based Gate Dielectric,” *IEEE Electron Device Lett.* 35, 138 (2014).
- ¹⁵⁵ J. Müller, T.S. Böske, S. Müller, E. Yurchuk, P. Polakowski, J. Paul, D. Martin, T. Schenk, K. Khullar, A. Kersch, W. Weinreich, S. Riedel, K. Seidel, A. Kumar, T.M. Arruda, S.V. Kalinin, T. Schlösser, R. Boschke, R. van Bentum, U. Schröder, and T. Mikolajick, “Ferroelectric hafnium oxide: A CMOS-compatible and highly scalable approach to future ferroelectric memories,” *IEEE International Electron Devices Meeting (IEDM)*, Washington (2013.), pp. 10.8.1 - 10.8.4.
- ¹⁵⁶ W. Zhang, M. Takahashi, and S. Sakai, “Electrical properties of CaxSr1-xBi₂Ta₂O₉ ferroelectric-gate field-effect transistors,” *Semicond. Sci. Technol.* 28, 85003 (2013).
- ¹⁵⁷ X. Zhang, M. Takahashi, K. Takeuchi, and S. Sakai, “64 kbit ferroelectric-gate-transistor-integrated NAND flash memory with 7.5 V program and long data retention,” *Jap. J. Apl. Phys.* 51, 04DD01 (2012).
- ¹⁵⁸ E. Yurchuk, S. Mueller, D. Martin, S. Slesazeck, U. Schroeder, T. Mikolajick, J. Müller, J. Paul, R. Hoffmann, J. Sundqvist, T. Schlosser, R. Boschke, R. van Bentum, and M. Trentzsch, “Origin of the endurance degradation in the novel HfO₂-based 1T ferroelectric non-volatile memories,” *IEEE International Reliability Physics Symposium (IRPS)* (2014.), pp. 2E.5.1 - 2E.5.5.
- ¹⁵⁹ H.-T. Lue, C.-J. Wu, and T.-Y. Tseng, “Device modeling of ferroelectric memory field-effect transistor (FeMFET),” *IEEE T. Electron Dev.* 49, 1790 (2002).
- ¹⁶⁰ P.D. Lomenzo, Q. Takmeel, C.M. Fancher, C. Zhou, N.G. Rudawski, S. Moghaddam, J.L. Jones, and T. Nishida, “Ferroelectric Si-Doped HfO₂ Device Properties on Highly Doped Germanium,” *Electron Device Letters*, IEEE 36, 766 (2015).

- ¹⁶¹ Z. Lin, M. Si, and P. D. Ye, "Ultra-Fast Operation of BEOL-Compatible Atomic-Layer-Deposited In_2O_3 Fe-FETs: Achieving Memory Performance Enhancement with Memory Window of 2.5 V and High Endurance $> 10^9$ Cycles without V_T Drift Penalty," Symposium on VLSI Technology, 12-17 June 2022, pp. 391–392.
- ¹⁶² E. D. Grimley, T. Schenk, T. Mikolajick, U. Schroeder, and J. M. LeBeau, "Atomic Structure of Domain and Interphase Boundaries in Ferroelectric HfO_2 . Advanced Materials Interfaces," 1701258 (2018). Doi: 10.1002/admi.201701258
- ¹⁶³ M. Pešić, F. P. G. Fengler, L. Larcher, A. Padovani, T. Schenk, E. D. Grimley, X. Sang, J. M. LeBeau, S. Slesazeck, U. Schroeder, and T. Mikolajick, "Physical Mechanisms behind the Field-Cycling Behavior of HfO_2 -Based Ferroelectric Capacitors," *Advanced Functional Materials* 26, 4601 (2016).
- ¹⁶⁴ H. Mulaosmanovic, J. Ocker, S. Müller, U. Schroeder, J. Müller, P. Polakowski, S. Flachowsky, R. van Bentum, T. Mikolajick, and S. Slesazeck, "Switching Kinetics in Nanoscale Hafnium Oxide Based Ferroelectric Field-Effect Transistors," *ACS applied materials & interfaces* 9, 3792 (2017).
- ¹⁶⁵ M. Trentzsch, S. Flachowsky, R. Richter, J. Paul, B. Reimer, D. Utess, S. Jansen, H. Mulaosmanovic, S. Müller, S. Slesazeck, J. A. Ocker. "A 28nm HKMG super low power embedded NVM technology based on ferroelectric FETs," in *Electron Devices Meeting (IEDM), 2016 IEEE International*, 2016 Dec 3 (pp. 11-5). IEEE.
- ¹⁶⁶ S. Dünkel S, M. Trentzsch, R. Richter, P. Moll, C. Fuchs, O. Gehring, M. Majer, S. Wittek, B. Müller, T. Melde, H. Mulaosmanovic, "A FeFET based super-low-power ultra-fast embedded NVM technology for 22nm FDSOI and beyond," in *Electron Devices Meeting (IEDM), 2017 IEEE International*, 2017 Dec 2 (pp. 19-7). IEEE.
- ¹⁶⁷ M. Pesic, S. Knebel, M. Hoffmann, C. Richter, T. Mikolajick, U. Schroeder, "How to make DRAM non-volatile? Anti-ferroelectrics: A new paradigm for universal memories," *Electron Devices Meeting (IEDM), 2016 IEEE International* 2016 Dec 3 (pp. 11-6). IEEE.
- ¹⁶⁸ V. Garcia and M. Bibes, "Ferroelectric tunnel junctions for information storage and processing," *Nature Communications* 5 (2014).
- ¹⁶⁹ H. Yamada, V. Garcia, S. Fusil, S. Boyn, M. Marinova, A. Gloter, S. Xavier, J. Grollier, E. Jacquet, C. Carrétéro, C. Deranlot, M. Bibes, and A. Barthélémy, "Giant Electroresistance of Super-tetragonal BiFeO_3 -Based Ferroelectric Tunnel Junctions," *ACS Nano* 7, 5385 (2013).
- ¹⁷⁰ Z. Wen, C. Li, D. Wu, A. Li, and N. Ming, "Ferroelectric-field-effect-enhanced electroresistance in metal/ferroelectric/semiconductor tunnel junctions," *Nature Materials* 12, 617 (2013).
- ¹⁷¹ C.H. Ahn, K.M. Rabe, and J.-M. Triscone, "Ferroelectricity at the nanoscale: local polarization in oxide thin films and heterostructures," *Science* 303, 488 (2004).
- ¹⁷² S. Boyn, S. Girod, V. Garcia, S. Fusil, S. Xavier, C. Deranlot, H. Yamada, C. Carrétéro, E. Jacquet, M. Bibes, A. Barthélémy, and J. Grollier, "High-performance ferroelectric memory based on fully patterned tunnel junctions," *Appl. Phys. Lett.* 104, 52909 (2014).
- ¹⁷³ A. Chanthbouala, A. Crassous, V. Garcia, K. Bouzehouane, S. Fusil, X. Moya, J. Allibe, B. Dlubak, J. Grollier, S. Xavier, C. Deranlot, A. Moshar, R. Proksch, N.D. Mathur, M. Bibes, and A. Barthélémy, "Solid-state memories based on ferroelectric tunnel junctions," *Nat Nano* 7, 101 (2012).
- ¹⁷⁴ X.S. Gao, J.M. Liu, K. Au, and J.Y. Dai, "Nanoscale ferroelectric tunnel junctions based on ultrathin BaTiO_3 film and Ag nanoelectrodes," *Applied Physics Letters* 101, 142905 (2012).
- ¹⁷⁵ A. Chen, "Emerging memory selector devices," *13th Non-Volatile Memory Technology Symposium (NVMTS)*, Singapore (2013.), pp. 1–5.
- ¹⁷⁶ Y. Kim, Y. Kim, H. Han, S. Jesse, S. Hyun, W. Lee, S.V. Kalinin, and J.K. Kim, "Towards the limit of ferroelectric nanostructures: switchable sub-10 nm nanoisland arrays," *J. Mater. Chem. C* 1, 5299 (2013).
- ¹⁷⁷ F.Y. Bruno, S. Boyn, V. Garcia, S. Fusil, S. Girod, C. Carrétéro, M. Marinova, A. Gloter, S. Xavier, C. Deranlot, M. Bibes, and A. Barthélémy, "Million-fold Resistance Change in Ferroelectric Tunnel Junctions Based on Nickelate Electrodes," *Advanced Electronic Materials* 2 (2016). DOI: 10.1002/aelm.201500245
- ¹⁷⁸ G. Sanchez-Santolino, J. Tornos, D. Hernandez-Martin, J. I. Beltran, C. Munuera, M. Cabero, A. Perez-Muñoz, J. Ricote, F. Mompean, M. Garcia-Hernandez, Z. Sefrioui, "Resonant electron tunneling assisted by charged domain walls in multiferroic tunnel junctions," *Nature Nanotechnology* 12, 655 (2017).
- ¹⁷⁹ A. Chanthbouala, V. Garcia, R.O. Cherifi, K. Bouzehouane, S. Fusil, X. Moya, S. Xavier, H. Yamada, C. Deranlot, N.D. Mathur, M. Bibes, A. Barthélémy, and J. Grollier, "A ferroelectric memristor," *Nature Materials* 11, 860 (2012).
- ¹⁸⁰ H. Kohlstedt, A. Petraru, K. Szot, A. Rüdiger, P. Meuffels, H. Haselier, R. Waser, and V. Nagarajan, "Method to distinguish ferroelectric from nonferroelectric origin in case of resistive switching in ferroelectric capacitors," *Applied Physics Letters* 92, 62907 (2008).
- ¹⁸¹ X. Tian, and A. Toriumi, "New opportunity of ferroelectric tunnel junction memory with ultrathin HfO_2 -based oxides," *Electron Devices Technology and Manufacturing Conference (EDTM), 2017 IEEE* (pp. 36-64). IEEE.
- ¹⁸² F. Ambriz-Vargas, G. Kolhatkar, M. Broyer, A. Hadj-Youssef, R. Nouar, A. Sarkissian, R. Thomas, C. Gomez-Yáñez, M. A. Gauthier, A. Ruediger, "A Complementary Metal Oxide Semiconductor Process-Compatible Ferroelectric Tunnel Junction," *ACS applied materials & interfaces* 9, 13262 (2017).
- ¹⁸³ S. Fujii, Y. Kamimuta, T. Ino, Y. Nakasaki, R. Takaishi, M. Saitoh, "First demonstration and performance improvement of ferroelectric HfO_2 -based resistive switch with low operation current and intrinsic diode property," in *VLSI Technology, 2016 IEEE Symposium on*, 2016 Jun 14 (pp. 1-2). IEEE.
- ¹⁸⁴ Rebooting the IT Revolution: A Call for Action. SIA-SRC Report (2015): <https://www.semiconductors.org/clientuploads/Resources/RITR%20WEB%20version%20FINAL.pdf>
- ¹⁸⁵ M. Hilbert and P. Lopez, "The World's Technological Capacity to Store, Communicate, and Compute Information," *Science* 332 (2011) 60

- ¹⁸⁶ G. M. Church, Y. Gao, K. Yuan, S. Kosuri, "Next-generation digital information storage in DNA," *Science* 337 (2012) 1628
- ¹⁸⁷ N. Goldman, P. Bertone, S. Chen, C. Dessimoz, E. M. LeProust, B. Sipo, E. Birney, "Towards practical, high-capacity, low-maintenance information storage in synthesized DNA," *Nature* 494 (2013) 77
- ¹⁸⁸ S. L. Shipman, J. Nivala, J. D. Macklis, G. M. Church, "CRISPR-Cas encoding of a digital movie into the genomes of a population of living bacteria," *Nature* 547 (2017) 345
- ¹⁸⁹ J. Bornholt, L. Ceze, R. Lopez, G. Seelig, D. M. Carmean, K. Strauss, "A DNA-Based Archival Storage System," *OPERATING SYSTEMS REVIEW* 50 (2017) 637-649
- ¹⁹⁰ J. Bornholt, R. Lopez, D. M. Carmean, L. Ceze, K. Strauss, "Toward a DNA-based archival storage system," *IEEE Micro* 37 (2017)
- ¹⁹¹ V. Zhirnov, R. M. Zadegan, G. S. Sandhu, G. M. Church, W. L. Hughes, "Nucleic acid memory," *Nature Materials* 15 (2016) 366
- ¹⁹² Y. Erlich and D. Zielinski, "DNA Fountain enables a robust and efficient storage architecture," *Science* 355 (2017) 950
- ¹⁹³ N. F. Mott, *Metal-Insulator Transitions*, 2nd ed. (Taylor & Francis, London, 1990)
- ¹⁹⁴ T. Oka and N. Nagaosa, "Interfaces of Correlated Electron Systems: Proposed Mechanism for Colossal Electroresistance," *Phys. Rev. Lett.* 95 (2005) 266403
- ¹⁹⁵ A. Asamitsu, Y. Tomioka, H. Kuwahara, and Y. Tokura, "Current switching of resistive states in magnetoresistive manganites," *Nature* 388 (1997) 50
- ¹⁹⁶ S. Q. Liu, N. J. Wu, and A. Ignatiev, "Electric-pulse-induced reversible resistance change effect in magnetoresistive films," *Appl. Phys. Lett.* 76 (2000) 2749
- ¹⁹⁷ A. Sawa, T. Fujii, M. Kawasaki, and Y. Tokura, "Hysteretic current-voltage characteristics and resistance switching at a rectifying Ti/Pr_{0.7}Ca_{0.3}MnO₃ interface," *Appl. Phys. Lett.* 85 (2004) 4073
- ¹⁹⁸ D. Ruzmetov, G. Gopalakrishnan, J. Deng, V. Narayanamurti, S. Ramanathan, "Electrical triggering of metal-insulator transition in nanoscale vanadium oxide junctions," *J. Appl. Phys.* 106 (2009) 083702
- ¹⁹⁹ Y. Zhou, X. Chen, C. Ko, Z. Yang, and S. Ramanathan, "Voltage-Triggered Ultrafast Phase Transition in Vanadium Dioxide Switches," *IEEE Electron Device Letters* 34 (2013) 220
- ²⁰⁰ M. Son, J. Lee, J. Park, J. Shin, G. Choi, S. Jung, W. Lee, S. Kim, S. Park, and H. Hwang, "Excellent Selector Characteristics of Nanoscale VO₂ for High-Density Bipolar ReRAM Applications," *IEEE Electron Device Letters* 32 (2011) 1579
- ²⁰¹ S. D. Ha, G. H. Aydogdu, and S. Ramanathan, "Metal-insulator transition and electrically driven memristive characteristics of SmNiO₃ thin films," *Appl. Phys. Lett.* 98 (2011) 012105
- ²⁰² K-H. Xue, C. A. Paz de Araujo, J. Celinska, C. McWilliams, "A non-filamentary model for unipolar switching transition metal oxide resistance random access memories," *J. Appl. Phys.* 109 (2011) 091602
- ²⁰³ C. R. McWilliams, J. Celinska, C. A. Paz de Araujo, K-H. Xue, "Device characterization of correlated electron random access memories," *J. Appl. Phys.* 109 (2011) 091608
- ²⁰⁴ F. Nakamura, M. Sakaki, Y. Yamanaka, S. Tamaru, T. Suzuki, and Y. Maeno, "Electric-field-induced metal maintained by current of the Mott insulator Ca₂RuO₄," *Scientific Reports* 3 (2013) 2536
- ²⁰⁵ M. D. Pickett and R. S. Williams, "Sub-100 fJ and sub-nanosecond thermally driven threshold switching in niobium oxide crosspoint nanodevices," *Nanotechnology* 23 (2012) 215202
- ²⁰⁶ M. D. Pickett, G. Medeiros-Ribeiro, and R. S. Williams, "A scalable neuristor built with Mott memristors," *Nature Materials* 12 (2013) 114
- ²⁰⁷ P. Stoliar, L. Cario, E. Janod, B. Corraze, C. Guillot-Deudon, S. Salmon-Bourmand, V. Guiot, J. Tranchant, and M. Rozenberg, "Universal Electric-Field-Driven Resistive Transition in Narrow-Gap Mott Insulators," *Advanced Materials* 25 (2013) 3222
- ²⁰⁸ V. Guiot, L. Cario, E. Janod, B. Corraze, V. Ta Phuoc, M. Rozenberg, P. Stoliar, T. Cren, and D. Roditchev, "Avalanche breakdown in GaTa₄Se₈-xTex narrow-gap Mott insulators," *Nature Communications* 4 (2013) 1722
- ²⁰⁹ P. Stoliar, M. Rozenberg, E. Janod, B. Corraze, J. Tranchant, and L. Cario, "Nonthermal and purely electronic resistive switching in a Mott memory," *Physical Review B* 90 (2014) 045146
- ²¹⁰ E. Janod, J. Tranchant, B. Corraze, M. Querré, P. Stoliar, M. Rozenberg, T. Cren, D. Roditchev, V. T. Phuoc, M.-P. Besland, and L. Cario, "Resistive Switching in Mott Insulators and Correlated Systems," *Advanced Functional Materials* (2015); DOI: 10.1002/adfm.201500823
- ²¹¹ H. Oike, F. Kagawa, N. Ogawa, A. Ueda, H. Mori, M. Kawasaki, and Y. Tokura, "Phase-change memory function of correlated electrons in organic conductors," *Physical Review B* 91 (2015) 041101(R)
- ²¹² Fan Zuo, Priyadarshini Panda, Michele Kotiuga, Jiarui Li, Mingu Kang, Claudio Mazzoli, Hua Zhou, Andi Barbour, Stuart Wilkins, Badri Narayanan, Mathew Cherukara, Zhen Zhang, Subramanian K. R. S. Sankaranarayanan, Riccardo Comin, Karin M. Rabe, Kaushik Roy & Shriram Ramanathan, "Habituation based synaptic plasticity and organismic learning in a quantum perovskite," *Nature Communications* 8, 240 (2017).
- ²¹³ M. Hu *et al.*, "Dot-product engine for neuromorphic computing: Programming 1T1M crossbar to accelerate matrix-vector multiplication," *2016 53rd ACM/EDAC/IEEE Design Automation Conference (DAC)*, 2016, pp. 1-6.
- ²¹⁴ B. Chakrabarti *et al.*, "A multiply-add engine with monolithically integrated 3D memristor crossbar/CMOS hybrid circuit," *Scientific Reports*, Article vol. 7, p. 42429, 02/14/online 2017.
- ²¹⁵ P. Yao, H. Wu, B. Gao, S.B. Eryilmaz, X. Huang, W. Zhang, Q. Zhang, N. Deng, L. Shi, H.S.P. Wong, and H. Qian, "Face classification using electronic synapses," *Nature Comm.* 8, 1-8 (2017).
- ²¹⁶ T. Tsuruoka, T. Hasegawa, K. Terabe and M. Aono "Conductance Quantization and synaptic behavior in a Ta₂O₃-based atomic switch", *Nanotechnology*, 23(43), 435705 (2012).
- ²¹⁷ X. Sheng, et al. "Low-Conductance and Multilevel CMOS-Integrated Nanoscale Oxide Memristors." *Advanced Electronic Materials* (2019): 1800876

- ²¹⁸ W. Wu, H. Wu, B. Gao, N. Deng, S. Yu, and H. Qian, "Improving Analog Switching in HfO_x-Based Resistive Memory With a Thermal Enhanced Layer," *IEEE Electron. Dev. Lett.* 38(8), 1019-1022 (2017).
- ²¹⁹ G. W. Burr, et al. "Recent progress in phase-change memory technology," *IEEE J. Emerg. Sel. Top. Circ. Sys.* 6(2), 146-162 (2016).
- ²²⁰ G. W. Burr, R. M. Shelby, C. di Nolfo, et al., "Experimental Demonstration and Tolerancing of a Large-Scale Neural Network (165,000 Synapses), Using Phase-Change Memory as the Synaptic Weight Element," *IEDM Technical Digest*, 29.5, 2014.
- ²²¹ W. Kim, R.L. Bruce, T. Masuda, et al., "Confined PCM-based Analog Synaptic Devices offering Low Resistance-drift and 1000 Programmable States for Deep Learning," *VLSI Technology Symposium*, 2019.
- ²²² S. La Barbera, D. R. B. Ly, G. Navarro, et al., "Narrow Heater Bottom Electrode-Based Phase Change Memory as a Bidirectional Artificial Synapse," *Advanced Electronic Materials*, vol 4, no. 9, 1800223, 2018.
- ²²³ C. Mackin, H. Tsai, S. Ambrogio, P. Narayanan, A. Chen, and G. W. Burr, "Weight Programming in DNN Analog Hardware Accelerators in the Presence of NVM Variability," *Advanced Electronic Materials* vol 5, no. 9, 1900026 (2019).
- ²²⁴ H. Tsai, S. Ambrogio, C. Mackin, et al., "Inference of Long-Short Term Memory networks at software-equivalent accuracy using 2.5M analog Phase Change Memory devices," *VLSI Technology Symposium*, 2019.
- ²²⁵ S. Kim et al., "A phase change memory cell with metallic surfactant layer as a resistance drift stabilizer," *Electron Devices Meeting (IEDM)*, 2013 *IEEE International*, 30.7.
- ²²⁶ W. W. Koelmans, et al., "Projected phase-change memory devices," *Nature communications*, (6) 8181 (2015).
- ²²⁷ I. Giannopoulos, A. Sebastian, M. Le Gallo, et al., "8-bit precision in-memory multiplication with projected phase-change memory." 2018 *IEEE International Electron Devices Meeting (IEDM)*, pp. 27-7. 2018.
- ²²⁸ S. Ambrogio, M. Gallot, K. Spoon, et al., "Reducing the Impact of Phase-Change Memory Conductance Drift on the Inference of large-scale Hardware Neural Networks," *Electron Devices Meeting, 2019 IEEE International*.
- ²²⁹ E. J. Fuller, S. T. Keene, A. Melianas, Z. Wang, S. Agarwal, Y. Li, Y. Tuchman, C. D. James, M. J. Marinella, J. J. Yang, A. Salleo, and A. A. Talin, "Parallel programming of an ionic floating-gate memory array for scalable neuromorphic computing," *Science* 364, 570-574 (2019).
- ²³⁰ S. Mallik, T. Tsuruoka, T. Tsuchiya, and K. Terabe, "Effects of Mg doping to a LiCO₂ channel on the synaptic plasticity of Li ion-gated transistors," *ACS Applied Materials & Interfaces* 15, 47184-47195 (2023).
- ²³¹ M. Onen, N. Emond, B. Wang, D. Zhang, F. M. Ross, J. Li, B. Yildiz, and J. A. del Alamo, "Nanosecond protonic programmable resistors for analog deep learning," *Science* 377, 539-543 (2022).
- ²³² S. Kim, T. Todorov, M. Onen, T. Gokmen, D. Bishop, P. Solomon, K. -T. Lee, M. Copel, D. B. Farmer, J. A. Ott, T. Ando, H. Miyazoe, V. Narayanan, and J. Rozen, "Metal-oxide based, CMOS-compatible ECRAM for deep learning accelerator." *IEEE International Electron Devices Meeting (IEDM)*, 2022, pp. 35.7.1-4.
- ²³³ H. Kang, H. W. Kim, E. R. Hong, and J. Woo, "Analog synaptic behavior of mobile ion source-limited electrochemical RAM using CuO_x oxide electrode for deep learning accelerator," *Appl. Phys. Lett.* 120, 122101 (2021).
- ²³⁴ S. Kim, T. Todorov, M. Onen, T. Gokmen, D. Bishop, P. Solomon, K. -T. Lee, M. Copel, D. B. Farmer, J. A. Ott, T. Ando, H. Miyazoe, V. Narayanan, and J. Rozen, "Metal-oxide based, CMOS-compatible ECRAM for deep learning accelerator." *IEEE International Electron Devices Meeting (IEDM)*, 2022, pp. 35.7.1-4.
- ²³⁵ H. Kang, H. W. Kim, E. R. Hong, and J. Woo, "Analog synaptic behavior of mobile ion source-limited electrochemical RAM using CuO_x oxide electrode for deep learning accelerator," *Appl. Phys. Lett.* 120, 122101 (2021).
- ²³⁶ D. Morris, D. Bromberg, J.-G. J. Zhu, and L. Pileggi, "mLogic: Ultra-low voltage non-volatile logic circuits using STT-MTJ devices," *Proceedings of the 49th Annual Design Automation Conference*, 2012, pp. 486-491.
- ²³⁷ C. Pan and A. Naeemi, "Non-Boolean computing benchmarking for beyond-CMOS devices based on cellular neural network," *Exploratory Solid-State Computational Devices and Circuits, IEEE Journal on*, vol. 2, pp. 36-43, 2016.
- ²³⁸ J. Hasler, C. Diorio, B. A. Minch, and C. A. Mead, "Single transistor learning synapses," Gerald Tesauero, David S. Touretzky, and Todd K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, MIT Press, Cambridge, MA, 1994, pp. 817-824.
- ²³⁹ A. Bandyopadhyay, G. Serrano, and P. Hasler, "Adaptive algorithm using hot-electron injection for programming analog computational memory elements within 0.2 percent of accuracy over 3.5 decades," *IEEE Journal of Solid-State Circuits*, vol. 41, no. 9, pp. 2107-2114, Sept. 2006.
- ²⁴⁰ S. Kim, J. Hasler, and S. George, "Integrated Floating-Gate Programming Environment for System-Level ICs," *IEEE Transactions on VLSI*, vol. 24, no. 6, 2016. pp. 2244-2252.
- ²⁴¹ Chakrabartty S and Cauwenberghs G 2007 Sub-microwatt analog VLSI trainable pattern classifier *IEEE J. Solid-State Circuits* 42 1169-1179
- ²⁴² Lu J, Young S, Arel I, and Holleman J 2015 A 1 TOPs/W analog deep machine-learning engine with floating-gate storage in 0.13 μ m CMOS *IEEE J. Solid-State Circuits* 50 270-281
- ²⁴³ Hasler J and Marr H 2013 Finding a roadmap to achieve large neuromorphic hardware systems *Frontiers Neurosci.* 7 118
- ²⁴⁴ Guo X, Merrikh Bayat F, Bavandpour M, Klachko M, Mahmoodi M R, Prezioso M, Likharev K K and Strukov D 2017 Fast, energy-efficient, robust, and reproducible mixed-signal neuromorphic classifier based on embedded NOR flash memory technology *IEEE Int. Electron Device Meeting (San Francisco, CA)* pp. 6.5.1-4

- ²⁴⁵ X. Guo, F. Merrikh Bayat, M. Prezioso, Y. Chen, B. Nguyen, N. Do, and D.B. Strukov, "Temperature-insensitive analog vector-by-matrix multiplier based on 55 nm NOR flash memory cells", in: Proc. CICC'17, Austin, TX, Apr.-May 2017, pp. 1-4.
- ²⁴⁶ Silicon Storage Technology, Inc. <http://www.sst.com/products-and-services/superflash-technology-products>
- ²⁴⁷ Fick L, Blaauw D, Sylvester D, Skrzyniarz S, Parikh M, and Fick D 2017 Analog in-memory subthreshold deep neural network accelerator *IEEE Custom Integrated Circuits Conf.* (Austin TX) pp. 1-4
- ²⁴⁸ Agarwal S, Garland D, Niroula J, Jacobs-Gedrim R B, Hsia A, Van Heukelom M S, Fuller E, Draper B and Marinella M 2019 Using floating gate memory to train ideal accuracy neural networks *IEEE J. Explor. Solid-State Computat. Devices Circuits* **5** 52-7
- ²⁴⁹ G.G. Marotta, A. Macerola, A. D'Alessandro, A. Torsi, C. Cerafogli, C. Lattaro, C. Musilli, D. Rivers, E. Sirizotti, and F. Paolini, *et al.*, "A 3bit/cell 32Gb NAND flash memory at 34nm with 6MB/s program throughput and with dynamic 2b/cell blocks configuration mode for a program throughput increase up to 13MB/s," *ISSCC*, 2010.
- ²⁵⁰ D. Lee, I. J. Chang, S.Y. Yoon, J. Jang, *et al.*, "A 64Gb 533Mb/s DDR Interface MLC NAND Flash in Sub-20 nm Technology," *ISSCC*, 2012, pp. 430-431.
- ²⁵¹ N. Shibata, K. Kanda, T. Hisada, K. Isobe1, *et al.*, "A 19nm 112.8mm² 64Gb Multi-Level Flash Memory with 400Mb/s/pin 1.8V Toggle Mode Interface," *ISSCC*, 2012, pp. 422-423.
- ²⁵² Y. Li, S. Lee, K. Oowada, H. Nguyen, *et al.*, "128Gb 3b/Cell NAND Flash Memory in 19nm Technology with 18MB/s Write Rate and 400Mb/s Toggle Mode," *ISSCC*, 2012, pp. 436-437.
- ²⁵³ C. Kim, D.H. Kim, W. Jeong, H.J. Kim, I.H. Park, H.W. Park, J. Lee, J. Park, Y.L. Ahn, J.Y. Lee, and S.B. Kim, "A 512Gb 3b/cell 64-stacked WL 3D V-NAND flash memory," in IEEE International Solid-State Circuits Conference (ISSCC), San Francisco, CA, pp. 202-203, 2017.
- ²⁵⁴ C.M. Compagnoni, A. Goda, A.S. Spinelli, P. Feeley, A.L. Lacaita, and A. Visconti, "Reviewing the evolution of the NAND flash technology," *Proc. IEEE*, vol. 105, no. 9, pp. 1609-1633, 2017.
- ²⁵⁵ Bavandpour M, Sahay S, Mahmoodi M R and Strukov D 2019 3D-aCortex: An ultra-compact energy-efficient neurocomputing platform based on commercial 3D-NAND flash memories *ArXiv (submitted)*
- ²⁵⁶ Bavandpour M, Sahay S, Mahmoodi M R and Strukov D 2019 3D-aCortex: An ultra-compact energy-efficient neurocomputing platform based on commercial 3D-NAND flash memories *ArXiv (submitted)*
- ²⁵⁷ S. Kim, T. Gokmen, H. M. Lee, and W. E. Haensch, "Analog CMOS-based resistive processing unit for deep neural network training," In *2017 IEEE 60th International Midwest Symposium on Circuits and Systems*, 422-425, 2017.
- ²⁵⁸ Y. Li, S. Kim, X. Sun, P. Solomon, T. Gokmen, H. Tsai, S. Koswatta, Z. Ren, R. Mo, C. C. Yeh, W. Haensch and E. Leobandung, "Capacitor-based Cross-point Array for Analog Neural Network with Record Symmetry and Linearity," *IEEE VLSI Systems Symposium*, 2018.
- ²⁵⁹ G. Cristiano, M. Giordano, S. Ambrogio, L. P. Romero, C. Cheng, P. Narayanan, *et al.*, "Perspective on Training Fully Connected Networks with Resistive Memories: Device Requirements for Multiple Conductances of Varying Significance," *J. Appl. Phys.*, vol. 124, no. 15, 151901, 2018.
- ²⁶⁰ R. Genov, G. Cauwenberghs, G. Mulliken, and F. Adil, "A 5.9mW 6.5GMACS CID/DRAM array processor," *Proceedings of the 28th European Solid-State Circuits Conference*, pp. 715-718, Sept 2002.
- ²⁶¹ R. Karakiewicz, R. Genov, and G. Cauwenberghs, "480-GMACS/mW Resonant Adiabatic Mixed-Signal Processor Array for Charge-Based Pattern Recognition," *IEEE Journal of Solid-State Circuits*, vol. 42, no. 11, pp. 2573-2584, Nov 2007.
- ²⁶² R. Karakiewicz, R. Genov, and G. Cauwenberghs, "1.1 TMACS/mW Fine-Grained Stochastic Resonant Charge-Recycling Array Processor," *IEEE Sensors Journal*, vol. 12, no. 4, pp. 785-792, April 2012.
- ²⁶³ S. Joshi, C. Kim, S. Ha, M. Y. Chi, and G. Cauwenberghs, "A 2pJ/MAC 14-b 8x8 Linear Transform Mixed-Signal Spatial Filter in 65 nm CMOS with 84 dB Interference Suppression," *IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC)*, February 2017.
- ²⁶⁴ E. H. Lee and S. S. Wong, "A 2.5 GHz 7.7 TOPS/W switched capacitor matrix multiplier with co-designed local memory in 40nm," *2016 IEEE International Solid-State Circuits Conference (ISSCC)*. IEEE, pp. 418-419, 2016.
- ²⁶⁵ J. Zhang, Z. Wang, and N. Verma, "A Matrix-Multiplying ADC Implementing a Machine-Learning Classifier Directly with Data Conversion," *IEEE Int. Solid-State Circ. Conf. (ISSCC)*, 2015.
- ²⁶⁶ E.H Lee and S.S. Wong, "A 2.5 GHz 7.7TOPS/W Switched-Capacitor Matrix Multiplier with Co-designed Local Memory in 40nm," *IEEE Int. Solid-State Circ. Conf. (ISSCC)*, 2016.
- ²⁶⁷ F. N. Buhler, A. E. Mendrela, Y. Lim, J. A. Fredenburg, and M. P. Flynn, "A 16-channel Noise-Shaping Machine Learning Analog-Digital Interface," *IEEE VLSI Systems Symposium*, 2016.
- ²⁶⁸ W. Guo and N. Sun, "A 12b-ENOB 61 μ W noise-shaping SAR ADC with a passive integrator," *42nd IEEE European Solid-State Circuits Conference (ESSCIRC 2016)*, 405-408, 2016.
- ²⁶⁹ V. V. Zhirnov, R. K. Cavin, S. Menzel, E. Linn, S. Schmelzer, D. Bräuhäus, C. Schindler and R. Waser, "Memory Devices: Energy-Space-Time Trade-offs," *Proc. IEEE* **98** (2010) 2185-2200
- ²⁷⁰ G. W. Burr, R. S. Shenoy, P. Narayanan, K. Virwani, A. Padilla B. Kurdi, and H. Hwang, "Selection devices for 3-D crosspoint memory," *Journal of Vacuum Science & Technology B*, 32(4), 040802 (2014).
- ²⁷¹ C. Kügeler, R. Rosezin, E. Linn, R. Bruchhaus, R. Waser, "Materials, technologies, and circuit concepts for nanocrossbar-based bipolar RRAM," *Appl. Phys. A* (2011) 791-809

- ²⁷² A. Chen, "Nonlinearity and Asymmetry for Device Selection in Cross-bar Memory Arrays," *IEEE Trans. Electron Dev.*, 62(9), 2857-2864, (2015)
- ²⁷³ A. Chen, "Memory selector devices and crossbar array design: a modeling-based assessment," *J Comput Electron* (2017) 16:1186-1200 DOI 10.1007/s10825-017-1059-7
- ²⁷⁴ The IRDS "Beyond-CMOS" Chapter 2021.
- ²⁷⁵ G. W. Burr, B. N. Kurdi, J. C. Scott, C. H. Lam, K. Gopalakrishnan, and R. S. Shenoy, "Overview of Candidate Device Technologies for Storage-Class Memory," *IBM J. Res. & Dev.* 52, No. 4/5, 449-464 (2008).
- ²⁷⁶ R. F. Freitas and W. W. Wilcke, "Storage-Class Memory: The Next Storage System Technology," *IBM J. Res. & Dev.* 52, No. 4/5, 439-447 (2008).
- ²⁷⁷ www.micron.com/about/innovations/3d-xpoint-technology
- ²⁷⁸ www.computerworld.com/article/2951869/computer-hardware/intel-and-micron-unveil-3d-xpoint-a-new-class-of-memory.html
- ²⁷⁹ <https://www.micron.com/products/advanced-solutions/3d-xpoint-technology>
- ²⁸⁰ <https://www.intel.com/content/www/us/en/architecture-and-technology/intel-micron-3d-xpoint-webcast.html>
- ²⁸¹ Kevin P.O'Brien, Carl H. Naylor, Chelsey Dorow, KirbyMaxey, Ashish Verma Penumatcha, Andrey Vyatskikh, Ting Zhong, Ande Kitamura, Sudarat Lee, Carly Rogan, Wouter Mortelmans, Mahmut Sami Kavrik, Rachel Steinhardt, Pratyush Buragohain, Sourav Dutta, Tristan Tronic, Scott Clendenning, Paul Fischer, Ernisse S. Putna, Marko Radosavljevic, Matt Metz & Uygur Avci. Process integration and future outlook of 2D transistors. *Nature Communications* volume 14, Article number: 6400 (2023)
- ²⁸² Kang, J., Tongay, S., Zhou, J., Li, J., Wu, J., "Band offsets and heterostructures of two-dimensional semiconductors," *Applied Physics Letters* 2013, 102 (1), 012111.
- ²⁸³ Z. Chen, Y.-M. Lin, M. J. Rooks, Ph. Avouris, "Graphene nano-ribbon electronics," *Physica E*, Vol. 40, p. 228 (2007)
- ²⁸⁴ Hai Li, Jumiat Wu, Zongyou Yin, and Hua Zhang. Preparation and Applications of Mechanically Exfoliated Single-Layer and Multilayer MoS₂ and WSe₂ Nanosheets. *Acc. Chem. Res.* 2014, 47, 4, 1067-1075.
- ²⁸⁵ Walth, M. et al. "Perspective of 2D Integrated Electronic Circuits: Scientific Pipe Dream or Disruptive Technology?" *Advanced Materials* 2201082 (2022) doi:10.1002/ADMA.202201082.
- ²⁸⁶ Radisavljevic, B., Kis, A., "Mobility engineering and a metal-insulator transition in monolayer MoS₂," *Nature Materials* 2013, 12, 815.
- ²⁸⁷ Li, S.-L., Wakabayashi, K., Xu, Y., Nakaharai, S., Komatsu, K., Li, W.-W., Lin, Y.-F., Aparecido-Ferreira, A., Tsukagoshi, K., "Thickness-Dependent Interfacial Coulomb Scattering in Atomically Thin Field-Effect Transistors," *Nano Letters* 2013, 13 (8), 3546-3552.
- ²⁸⁸ Ma, N. and Jena, D., "Charge Scattering and Mobility in Atomically Thin Semiconductors," *Phys. Rev. X*, 4, 011043 (2014)
- ²⁸⁹ C.-S. Pang, R. Zhou, X. Liu, P. Wu, T.Y.T. Hung, S. Guo, M.E. Zaghoul, S. Krylyuk, A.V. Davydov, J. Appenzeller, and Z. Chen, "Mobility Extraction in 2D Transition Metal Dichalcogenide Devices—Avoiding Contact Resistance Implied Overestimation", *Small*, 2100940 (2021). DOI: <https://doi.org/10.1002/sml.202100940>
- ²⁹⁰ Das, S. et al. Transistors based on two-dimensional materials for future integrated circuits. *Nature Electronics* 4, 786-799 (2021).
- ²⁹¹ Dorow, C. J. et al. Gate length scaling beyond Si: Mono-layer 2D Channel FETs Robust to Short Channel Effects. in 2022 International Electron Devices Meeting (IEDM) 7.5.1-7.5.4 (2022). doi:10.1109/IEDM45625.2022.10019524.
- ²⁹² Retamal, J. R. D., Periyangounder, D., Ke, J.-J., Tsai, M.-L. & He, J.-H. Charge carrier injection and transport engineering in two-dimensional transition metal dichalcogenides. *Chem. Sci.* 9, 7727-7745 (2018).
- ²⁹³ Wang, Y. & Chhowalla, M. Making clean electrical contacts on 2D transition metal dichalcogenides. *Nature Reviews Physics* 2021 4:2 4, 101-112 (2021).
- ²⁹⁴ Radisavljevic, B., Radenovic, A., Brivio, J., Giacometti, V., Kis, A., "Single-layer MoS₂ transistors," *Nature Nanotechnology* 2011, 6, 147.
- ²⁹⁵ Lee, Y.-H., Yu, L., Wang, H., Fang, W., Ling, X., Shi, Y., Lin, C.-T., Huang, J.-K., Chang, M.-T., Chang, C.-S., Dresselhaus, M., Palacios, T., Li, L.-J., Kong, J., "Synthesis and Transfer of Single-Layer Transition Metal Disulfides on Diverse Surfaces," *Nano Letters* 2013, 13 (4), 1852-1857.
- ²⁹⁶ Lee, Y.-H., Zhang, X.-Q., Zhang, W., Chang, M.-T., Lin, C.-T., Chang, K.-D., Yu, Y.-C., Wang, J. T.-W., Chang, C.-S., Li, L.-J., Lin, T.-W., "Synthesis of Large-Area MoS₂ Atomic Layers with Chemical Vapor Deposition," *Advanced Materials* 2012, 24 (17), 2320-2325.
- ²⁹⁷ Shaw, J. C., Zhou, H., Chen, Y., Weiss, N. O., Liu, Y., Huang, Y., Duan, X., "Chemical vapor deposition growth of monolayer MoSe₂ nanosheets," *Nano Research* 2014, 7 (4), 511-517.
- ²⁹⁸ Cong, C., Shang, J., Wu, X., Cao, B., Peimyoo, N., Qiu, C., Sun, L., Yu, T., "Synthesis and Optical Properties of Large-Area Single-Crystalline 2D Semiconductor WS₂ Monolayer from Chemical Vapor Deposition," *Advanced Optical Materials* 2014, 2 (2), 131-136.
- ²⁹⁹ Lin, Y.-C., Lu, N., Perea-Lopez, N., Li, J., Lin, Z., Peng, X., Lee, C. H., Sun, C., Calderin, L., Browning, P. N., Bresnehan, M. S., Kim, M. J., Mayer, T. S., Terrones, M., Robinson, J. A., "Direct Synthesis of van der Waals Solids," *ACS Nano* 2014, 8 (4), 3715-3723.
- ³⁰⁰ K. K. H. Smithe, et al. Low Variability in Synthetic Monolayer MoS₂ Devices. *ACS Nano*, 11, 8456-8463 (2017).
- ³⁰¹ Quentin Smets, Goutham Arutchelvan, Julien Jussot, Devin Verreck, Inge Asselberghs, Ankit Nalin Mehta, Abhinav Gaur, Dennis Lin, Salim El Kazzi, Benjamin Groven, Matty Caymax, Iuliana Radu. Ultra-scaled MOCVD MoS₂ MOSFETs with 42nm contact pitch and 250µA/µm drain current. In Proceedings of the 2019 IEEE International Electron Devices Meeting, 10.1109/IEDM19573.2019.8993650.
- ³⁰² Weisheng Li, Xiaoshu Gong, Zhihao Yu, Liang Ma, Wenjie Sun, Si Gao, Çağrı Köroğlu, Wenfeng Wang, Lei Liu, Taotao Li, Hongkai Ning, Dongxu Fan, Yifei Xu, Xuecou Tu, Tao Xu, Litao Sun, Wenhui Wang, Junpeng Lu, Zhenhua Ni, Jia Li, Xidong Duan, Peng Wang, Yuefeng Nie, Hao Qiu, Yi Shi, Eric Pop, Jinlan Wang, Xinran Wang. "Approaching the quantum limit in two-dimensional semiconductor contacts". *Nature* 613, 274-279 (2023).

- ³⁰³ J.D. Wood, G.P. Doidge, E.A. Carrion, J.C. Koepke, J.A. Kaitz, I. Datye, A. Behnam, J. Hewaparakrama, B. Aruin, Y. Chen, H. Dong, R.T. Haasch, J.W. Lyding, E. Pop, "Annealing Free, Clean Graphene Transfer using Alternative Polymer Scaffolds," *Nanotechnology* 26, 055302 (2015)
- ³⁰⁴ T. Li, W. Guo, L. Ma, W. Li, Z. Yu, Z. Han, S. Gao, L. Liu, D. Fan, Z. Wang, Y. Yang, W. Lin, Z. Luo, X. Chen, N. Dai, X. Tu, D. Pan, Y. Yao, P. Wang, Y. Nie, J. Wang, Y. Shi, X. Wang, "Epitaxial growth of wafer-scale molybdenum disulfide semiconductor single crystals on sapphire". *Nature Nanotechnology* 16, 1201–1207 (2021).
- ³⁰⁵ Maki Nakatani, Satoru Fukamachi, Pablo Solís-Fernández, Satoshi Honda, Kenji Kawahara, Yuta Tsuji, Yosuke Sumiya, Mai Kuroki, Kou Li, Qiunan Liu, Yung-Chang Lin, Aika Uchida, Shun Oyama, Hyun Goo Ji, Kenichi Okada, Kazu Suenaga, Yukio Kawano, Kazunari Yoshizawa, Atsushi Yasui & Hiroki Ago Ready-to-transfer two-dimensional materials using tunable adhesive force tapes. *Nature Electronics* 7, 119–130 (2024).
- ³⁰⁶ Steven Brems, Souvik Ghosh, Quentin Smets, Marie-Emmanuelle Boulon, Andries Boelen, Koen Kennes, Hung-Chieh Tsai, Francois Chancerel, Clement Merckling, Pieter-Jan Wyndaele, Jean-Francois de Marneffe, Tom Schram, Pawan Kumar, Stefanie Sergeant, Thomas Nuytten, Stefan De Gendt, Henry Medina Silva, Benjamin Groven, Pierre Morin, Gouri Sankar Kar, César Lockhart de la Rosa, Didit Yudistira, Joris Van Campenhout, Inge Asselberghs and Alain Phommahaxay. "Overview of scalable transfer approaches to enable epitaxial 2D material integration". 2023 International VLSI Symposium on Technology, Systems and Applications (VLSI-TSA/VLSI-DAT). DOI: 10.1109/VLSI-TSA/VLSI-DAT57221.2023.10134381.
- ³⁰⁷ Jiadi Zhu, Ji-Hoon Park, Steven A. Vitale, Wenjun Ge, Gang Seob Jung, Jiangtao Wang, Mohamed Mohamed, Tianyi Zhang, Maitreyi Ashok, Mantian Xue, Xudong Zheng, Zhien Wang, Jonas Hansryd, Anantha P. Chandrakasan, Jing Kong & Tomás Palacios. "Low-thermal-budget synthesis of monolayer molybdenum disulfide for silicon back-end-of-line integration on a 200 mm platform". *Nature Nanotechnology* 18, 456–463 (2023).
- ³⁰⁸ Radisavljevic, B., M. B. Whitwick, and A. Kis "Integrated circuits and logic operations based on single-layer MoS₂", *ACS Nano*, 5: 9934–8, (2011).
- ³⁰⁹ Wang, H., L. Yu, Y. H. Lee, Y. Shi, A. Hsu, M. L. Chin, L. J. Li, M. Dubey, J. Kong, and T. Palacios, "Integrated circuits based on bilayer MoS₂ transistors", *Nano Lett*, 12: 4674–80, (2012).
- ³¹⁰ Li, Na, Qinqin Wang, Cheng Shen, Zheng Wei, Hua Yu, Jing Zhao, Xiaobo Lu, Guole Wang, Congli He, Li Xie, Jianqi Zhu, Luoju Du, Rong Yang, Dongxia Shi, and Guangyu Zhang, "Large-scale flexible and transparent electronics based on monolayer molybdenum disulfide field-effect transistors", *Nature Electronics*, 3: 711–17, (2020).
- ³¹¹ Liu, H., A. T. Neal, Z. Zhu, Z. Luo, X. Xu, D. Tomanek, and P. D. Ye, "Phosphorene: an unexplored 2D semiconductor with a high hole mobility," *ACS Nano*, 8: 4033–41, (2014).
- ³¹² Sachid, A. B., M. Tosun, S. B. Desai, C. Y. Hsu, D. H. Lien, S. R. Madhupathy, Y. Z. Chen, M. Hettick, J. S. Kang, Y. Zeng, J. H. He, E. Y. Chang, Y. L. Chueh, A. Javey, and C. Hu, "Monolithic 3D CMOS Using Layered Semiconductors", *Adv Mater*, 28: 2547–54, (2016).
- ³¹³ C.-S. Pang, N. Thakuria, S. Gupta, Z. Chen, "First Demonstration of WSe₂ Based CMOS-SRAM", *IEEE International Electron Devices Meeting (IEDM) Proceeding*, 2018
- ³¹⁴ P. Wu and J. Appenzeller, "Towards CMOS like devices from two-dimensional channel materials," *Applied Physics Letters Materials* 7, 100701-1–100701-7 (2019).
- ³¹⁵ P. Wu, D. Reis, X.S. Hu and J. Appenzeller, "Two-Dimensional Transistors with Reconfigurable Polarities for Secure Circuits," *Nature Electronics* 4, 45–53 (2020).
- ³¹⁶ J. Appenzeller, F. Zhang, S. Das, and J. Knoch, "Transition Metal Dichalcogenide Schottky Barrier Transistors – A Device Analysis and Material Comparison," chapter 8, pp. 207–240 in *2D Materials for Nanoelectronics*, (Editors: M. Houssa, T. Dimoulas and A. Molle), Taylor and Francis (2016).
- ³¹⁷ P.-C. Shen, C. Su, Y. Lin, A.-S. Chou, C.-C. Cheng, J.-H. Park, M.-H. Chiu, A.-Y. Lu, H.-L. Tang, M.M. Tavakoli, G. Pitner, X. Ji, Z. Cai, N. Mao, J. Wang, V. Tung, J. Li, J. Bokor, A. Zettl, C.-I. Wu, T. Palacios, L.-J. Li, and J. Kong, "Ultralow contact resistance between semimetal and monolayer semiconductors," *Nature* 593, 211–217 (2021).
- ³¹⁸ D. Kiriya, M. Tosun, P. Zhao, J. S. Kang, and A. Javey, "Air-Stable Surface Charge Transfer Doping of MoS₂ by Benzyl Viologen," *J. Am. Chem. Soc.*, vol. 136, no. 22, pp. 7853–7856, Jun. 2014, doi: 10.1021/JA5033327.
- ³¹⁹ L. Cai, C. J. McClellan, A. L. Koh, H. Li, E. Yalon, E. Pop, and X. Zheng, "Rapid Flame Synthesis of Atomically Thin MoO₃ down to Monolayer Thickness for Effective Hole Doping of WSe₂," *Nano Lett.*, vol. 17, no. 6, pp. 3854–3861, Jun. 2017, doi: 10.1021/ACS.NANOLETT.7B01322.
- ³²⁰ C.-S. Pang, T.Y.T. Hung, A. Khosravi, R. Addou, Q. Wang, M.J. Kim, R.M. Wallace, Z. Chen, "Atomically Controlled Tunable Doping in High Performance WSe₂ Devices," *Advanced Electronic Materials*, 6, p.1901304 (2020)
- ³²¹ H. Fang, S. Chuang, T. C. Chang, K. Takei, T. Takahashi, and A. Javey, "High-Performance Single Layered WSe₂ p-FETs with Chemically Doped Contacts," *Nano Lett.*, vol. 12, no. 7, pp. 3788–3792, Jul. 2012, doi: 10.1021/NL301702R.
- ³²² C.-C. Chiang, H.-Y. Lan, C.-S. Pang, J. Appenzeller, Z. Chen, "Air-Stable P-doping in Record High-Performance Monolayer WSe₂ Devices," *Electron Device Letters*, 42 (2), 319–322
- ³²³ O'Brien, K. P. et al. Advancing 2D Monolayer CMOS Through Contact, Channel and Interface Engineering. in 2021 IEEE International Electron Devices Meeting (IEDM) 7.1.1–7.1.4 (2021). doi:10.1109/IEDM19574.2021.9720651.
- ³²⁴ Li, X. et al. One-dimensional semimetal contacts to two-dimensional semiconductors. *Nat Commun* 14, 111 (2023).
- ³²⁵ S. Das, J. Appenzeller, J., "Where Does the Current Flow in Two-Dimensional Layered Systems?" *Nano Letters* 2013, 13 (7), 3396–3402.

- ³²⁶ D. Jena and A. Konar, "Enhancement of Carrier Mobility in Semiconductor Nanostructures by Dielectric Engineering", *Physical Review Letters* 98, 136805 (2007).
- ³²⁷ F. Palumbo, C. Wen, S. Lombardo, S. Pazos, F. Aguirre, M. Eizenberg, F. Hui, M. Lanza, "A review on dielectric breakdown in thin dielectrics: silicon dioxide, high-k and layered dielectrics", *Advanced Functional Materials*, 30 (18), 1900657, 2019.
- ³²⁸ A. Pierret, D. Mele, H. Graef, J. Palomo, T. Taniguchi, K. Watanabe, Y. Li, B. Toury, C. Journet, P. Steyer, V. Garnier, A. Loiseau, J-M. Berroir, E. Bocquillon, G. Fève, C. Voisin, E. Baudin, M. Rostiche and B. Plaçais. "Dielectric permittivity, conductivity and breakdown field of hexagonal boron nitride". *Materials Research Express* 9 (2022) 065901.
- ³²⁹ W. Li et al. Uniform and ultrathin high- κ gate dielectrics for two-dimensional electronic devices. *Nat. Electron.* 2, 563–571 (2019).
- ³³⁰ T. Li et al. A native oxide high- κ gate dielectric for two-dimensional electronics. *Nat. Electron.* 3, 473–478 (2020).
- ³³¹ Y. Y. Illarionov et al. Ultrathin calcium fluoride insulators for two-dimensional field-effect transistors. *Nat. Electron.* 2, 230–235 (2019).
- ³³² Chen, T.-A. et al. Wafer-scale single-crystal hexagonal boron nitride monolayers on Cu (111). *Nature* 579, 219–223 (2020).
- ³³³ Illarionov, Y. Y. et al. Process implications on the stability and reliability of 300 mm FAB MoS₂ field-effect transistors. *npj 2D Mater Appl* 8, 1–7 (2024).
- ³³⁴ Stampfer, B. et al. Characterization of Single Defects in Ultrascaled MoS₂ Field-Effect Transistors. *ACS Nano* 12, 5368–5375 (2018).
- ³³⁵ Sarkar, D., X. Xie, W. Liu, W. Cao, J. Kang, Y. Gong, S. Kraemer, P. M. Ajayan, and K. Banerjee, "A subthermionic tunnel field-effect transistor with an atomically thin channel," *Nature*, 526: 91-5, (2015).
- ³³⁶ Qiu, C., F. Liu, L. Xu, B. Deng, M. Xiao, J. Si, L. Lin, Z. Zhang, J. Wang, H. Guo, H. Peng, and L. M. Peng, "Dirac-source field-effect transistors as energy-efficient, high-performance electronic switches," *Science*, 361: 387-92, (2018).
- ³³⁷ P. Wu and J. Appenzeller, "Reconfigurable Black Phosphorus Vertical Tunneling Field-Effect Transistor with Record High ON-Currents," *IEEE Electron Device Letters* 40, 981-984 (2019).
- ³³⁸ C.-S. Pang, C.-Y. Chen, T. Ameen, S. Zhang, H. Ilatikhameneh, R. Rahman, G. Klimeck, and Z. Chen, "WSe₂ Homojunction Devices: Electrostatically Configurable as Diodes, MOSFETs, and Tunnel FETs for Reconfigurable Computing," *Small*, 15, 2019, p.1902770
- ³³⁹ H. Lu and A. Seabaugh, "Tunnel Field-Effect Transistors: State-of-the-Art," *Electron Devices Society, IEEE Journal of the*, vol. 2, pp. 44-49, 2014.
- ³⁴⁰ S. Agarwal and E. Yablonovitch, "Designing a Low Voltage, High Current Tunneling Transistor," *CMOS and Beyond Logic Switches for Terascale Integrated Circuits*, T.-J. King-Liu and K. Kuhn, Eds., ed Cambridge University Press, 2015.
- ³⁴¹ A. C. Seabaugh and Z. Qin, "Low-Voltage Tunnel Transistors for Beyond CMOS Logic," *Proceedings of the IEEE*, vol. 98, pp. 2095-2110, 2010.
- ³⁴² S. H. Kim, H. Kam, C. Hu, and T.-J. K. Liu, "Germanium-Source Tunnel Field Effect Transistors with Record High I_{ON}/I_{OFF} ," presented at the *2009 Symposium on VLSI Technology*, Kyoto, Japan, 2009.
- ³⁴³ W. Y. Choi, B. G. Park, J. D. Lee, and T. J. K. Liu, "Tunneling Field-Effect Transistors (TFETs) With Subthreshold Swing (SS) Less Than 60 mV/dec," *IEEE Electron Device Letters*, vol. 28, pp. 743-745, Aug 2007.
- ³⁴⁴ K. Jeon, W.-Y. Loh, P. Patel, C. Y. Kang, O. Jungwoo, A. Bowonder, et al., "Si Tunnel Transistors With a Novel Silicided Source and 46mV/dec Swing," presented at the *2010 IEEE Symposium on VLSI Technology*, Honolulu, Hawaii, 2010.
- ³⁴⁵ T. Krishnamohan, K. Donghyun, S. Raghunathan, and K. Saraswat, "Double-Gate Strained-Ge Heterostructure Tunneling FET (TFET) with Record High Drive Currents and < 60 mV/dec Subthreshold Slope," presented at the *IEDM 2008. IEEE International Electron Devices Meeting*, San Francisco, CA., 2008.
- ³⁴⁶ U. E. Avci and I. A. Young, "Heterojunction TFET Scaling and resonant-TFET for steep subthreshold slope at sub-9nm gate-length," *Electron Devices Meeting (IEDM), 2013 IEEE International*, 2013, pp. 4.3.1–4.3.4.
- ³⁴⁷ E. Memisevic, E. Lind, M. Hellenbrand, J. Svensson and L. E. Wernersson, "Impact of Band-Tails on the Subthreshold Swing of III-V Tunnel Field-Effect Transistor," *IEEE Electron Device Letters*, vol. 38, no. 12, pp. 1661-1664, Dec. 2017.
- ³⁴⁸ E. Memisevic, J. Svensson, M. Hellenbrand, E. Lind, and L. E. Wernersson, "Vertical InAs/GaAsSb/GaSb tunneling field-effect transistor on Si with $S = 48$ mV/decade and $I_{on} = 10$ μ A/ μ m for $I_{off} = 1$ nA/ μ m at $V_{ds} = 0.3$ V," *2016 IEEE International Electron Devices Meeting (IEDM)*, 2016, p. 19.1.1–19.1.4.
- ³⁴⁹ D. Leonelli, A. Vandooren, R. Rooyackers, A. S. Verhulst, S. D. Gendt, M. M. Heyns, et al., "Performance Enhancement in Multi Gate Tunneling Field Effect Transistors by Scaling the Fin-Width," *Japanese Journal of Applied Physics*, vol. 49, p. 04DC10, 2010.
- ³⁵⁰ R. Gandhi, Z. Chen, N. Singh, K. Banerjee, and S. Lee, "Vertical Si-Nanowire-Type Tunneling FETs With Low Subthreshold Swing (<50mV/decade) at Room Temperature," *Electron Device Letters, IEEE*, vol. 32, pp. 437–439, 2011.
- ³⁵¹ R. Gandhi, C. Zhixian, N. Singh, K. Banerjee, and L. Sungjoo, "CMOS-Compatible Vertical-Silicon-Nanowire Gate-All-Around p-Type Tunneling FETs With <50 mV/decade Subthreshold Swing," *Electron Device Letters, IEEE*, vol. 32, pp. 1504–1506, 2011.
- ³⁵² H. Qianqian, H. Ru, Z. Zhan, Q. Yingxin, J. Wenzhe, W. Chunlei, et al., "A novel Si tunnel FET with 36mV/dec subthreshold slope based on junction depleted-modulation through striped gate configuration," *Electron Devices Meeting (IEDM), 2012 IEEE International*, 2012, pp. 8.5.1–8.5.4.
- ³⁵³ L. Knoll, Z. Qing-Tai, A. Nichau, S. Trellenkamp, S. Richter, A. Schafer, et al., "Inverters With Strained Si Nanowire Complementary Tunnel Field-Effect Transistors," *Electron Device Letters, IEEE*, vol. 34, pp. 813-815, 2013.
- ³⁵⁴ A. Villalon, C. Le Royer, M. Casse, D. Cooper, B. Previtali, C. Tabone, et al., "Strained tunnel FETs with record I_{on} : first demonstration of ETSOI TFETs with SiGe channel and RSD," *VLSI Technology (VLSIT), 2012 Symposium on*, 2012, pp. 49–50.

- ³⁵⁵ K. Sung Hwan, K. Hei, H. Chenming, and L. Tsu-Jae King, "Germanium-source tunnel field effect transistors with record high Ion/Ioff," *VLSI Technology, 2009 Symposium on*, 2009, pp. 178–179.
- ³⁵⁶ T. Krishnamohan, K. Donghyun, S. Raghunathan, and K. Saraswat, "Double-Gate Strained-Ge Heterostructure Tunneling FET (TFET) With record high drive currents and $\ll 60\text{mV/dec}$ subthreshold slope," *Electron Devices Meeting, 2008. IEDM 2008. IEEE International*, 2008, pp. 1–3.
- ³⁵⁷ B. Ganjipour, J. Wallentin, M. T. Borgström, L. Samuelson, and C. Thelander, "Tunnel Field-Effect Transistors Based on InP-GaAs Heterostructure Nanowires," *ACS Nano*, vol. 6, pp. 3109–3113, 2012/04/24 2012.
- ³⁵⁸ K. Tomioka, M. Yoshimura, and T. Fukui, "Steep-slope tunnel field-effect transistors using III-V nanowire/Si heterojunction," *VLSI Technology (VLSIT), 2012 Symposium on*, 2012, pp. 47–48.
- ³⁵⁹ W. G. Vandenberghe, A. S. Verhulst, B. Sorée, W. Magnus, G. Groeseneken, Q. Smets, *et al.*, "Figure of merit for and identification of sub-60 mV/decade devices," *Applied Physics Letters*, vol. 102, p. 013510, 2013.
- ³⁶⁰ J. Knoch, S. Mantl, and J. Appenzeller, "Impact of the dimensionality on the performance of tunneling FETs: Bulk versus one-dimensional devices," *Solid-State Electronics*, vol. 51, pp. 572–578, 4// 2007.
- ³⁶¹ S. Johnson and T. Tiedje, "Temperature dependence of the Urbach edge in GaAs," *Journal of Applied Physics*, vol. 78, pp. 5609–5613, 1995.
- ³⁶² S. Agarwal and E. Yablonovitch, "Band-Edge Steepness Obtained From Esaki/Backward Diode Current-Voltage Characteristics," *Electron Devices, IEEE Transactions on*, vol. 61, pp. 1488–1493, 2014.
- ³⁶³ U. E. Avci *et al.*, "Study of TFET non-ideality effects for determination of geometry and defect density requirements for sub-60mV/dec Ge TFET," *2015 IEEE International Electron Devices Meeting (IEDM)*, 2015, p. 34.5.1–34.5.4.
- ³⁶⁴ R. N. Sajjad, W. Chern, J. L. Hoyt, and D. A. Antoniadis, "Trap Assisted Tunneling and Its Effect on Subthreshold Swing of Tunnel FETs," *IEEE Trans. Electron Devices*, vol. 63, no. 11, pp. 4380–4387, Nov. 2016.
- ³⁶⁵ M. Pala and D. Esseni, "(Invited) Modeling the Influence of Interface Traps on the Transfer Characteristics of InAs Tunnel-FETs and MOSFETs," *Meet. Abstr.*, vol. MA2014-01, no. 36, pp. 1372–1372, Apr. 2014.
- ³⁶⁶ J. T. Teherani, S. Agarwal, W. Chern, P. M. Solomon, E. Yablonovitch, and D. A. Antoniadis, "Auger generation as an intrinsic limit to tunneling field-effect transistor performance," *J. Appl. Phys.*, vol. 120, no. 8, p. 084507, Aug. 2016.
- ³⁶⁷ C. Zhou, D. M. Newns, J. A. Misewich and P. C. Pattnaik, "A field effect transistor based on the Mott transition in a molecular layer," *Appl Phys Lett* 70 (5), 598–600 (1997)
- ³⁶⁸ D. M. Newns, J. A. Misewich, C. C. Tsuei, A. Gupta, B. A. Scott and A. Schrott, "Mott transition field effect transistor," *Appl Phys Lett* 73 (6), 780–782 (1998)
- ³⁶⁹ Y. Zhou and S. Ramanathan, "Correlated electron materials and field effect transistors for logic: a review," *Crit Rev Solid State Mater Sci*, 38(4), 286–317, 2013.
- ³⁷⁰ H. Akinaga, "Recent advances and future prospects in functional-oxide nanoelectronics: the emerging materials and novel functionalities that are accelerating semiconductor device research and development," *Jpn. J. Appl. Phys.*, 52(100001), 2013.
- ³⁷¹ Z. Yang, C. Ko and S. Ramanathan, "Oxide Electronics Utilizing Ultrafast Metal-Insulator Transitions," *Annual Review of Materials Research* 41, 8.1 (2011).
- ³⁷² A. Cavalleri, Cs. Toth, C. W. Siders, J. A. Squier, F. Raksi, P. Forget and J. C. Keiffer, "Femtosecond Structural Dynamics in VO₂ during an Ultrafast Solid-Solid Phase Transition," *Phy. Rev. Lett.* 87, 237401 (2001)
- ³⁷³ S. Hormoz and S. Ramanathan, "Limits on vanadium oxide Mott metal–insulator transition field-effect transistors," *Solid State Electron* 54 (6), 654–659 (2010)
- ³⁷⁴ H. Ji, Jiang Wei, and D. Natelson, "Modulation of the Electrical Properties of VO₂ Nanobeams Using an Ionic Liquid as a Gating Medium," *Nano Lett.*, 12, 2988–2992 (2012)
- ³⁷⁵ J. Jeong, N. Aetukuri, T. Graf, TD Schladt, MG Samant and S. S. Parkin, "Suppression of Metal-Insulator Transition in VO₂ by Electric Field-Induced Oxygen Vacancy Formation," *Science*, 339(6126), 1402–1405 (2013).
- ³⁷⁶ J. Shi, S. D. Ha, Y. Zhou, F. Schoofs and S. Ramanathan, "A correlated nickelate synaptic transistor," *Nature Communications*, 4, 2676 (2013)
- ³⁷⁷ S. D. Ha, J. Shi, Y. Meroz, L. Mahadevan and S. Ramanathan, "Neuromimetic Circuits with Synaptic Devices Based on Strongly Correlated Electron Systems," *Physical Review Applied*, 2, 064003 (2014),
- ³⁷⁸ M. D. Pickett, G. Medeiros-Ribeiro and R. Stanley Williams, "A scalable neuristor built with Mott memristors," *Nature Materials* 12, 114–117 (2013)
- ³⁷⁹ N. Shukla, A. V. Thathachary, A. Agrawal, H. Paik, A. Aziz, D. G. Schlom, S. K. Gupta, R. Engel-Herbert and S. Datta, "A steep-slope transistor based on abrupt electronic phase transition," *Nature Communications* 6, 7812 (2015)
- ³⁸⁰ Philipp Scheiderer, Matthias Schmitt, Judith Gabel, Michael Zapf, Martin Stübinger, Philipp Schütz, Lenart Dudy, Christoph Schlueter, Tien - Lin Lee, Michael Sing, Ralph Claessen, "Tailoring Materials for Mottronics: Excess Oxygen Doping of a Prototypical Mott Insulator", *Advanced Materials* 30, 1706708 (2018), <https://doi.org/10.1002/adma.201706708>
- ³⁸¹ T. Wei, T. Kanki, K. Fujiwara, M. Chikanari and H. Tanaka, "Electric field-induced transport modulation in VO₂ FETs with high-k oxide/organic parylene-C hybrid gate dielectric," *Appl. Phys. Lett.* 108 (2016) 053503.

- 382 T. Wei, T. Kanki, M. Chikanari, T. Uemura, T. Sekitani, and H. Tanaka, "Enhancement of electronic transport modulation in single crystalline VO₂ nanowire-based solid-state field-effect transistor," *Scientific Reports*, 7, 17215 (2017), <https://doi.org/10.1038/s41598-017-17468-x>
- 383 M. Yamamoto, R. Nouch, T. Kanki, A. N. Hattori, K. Watanabe, T. Taniguchi, K. Ueno and H. Tanaka, "Gate-Tunable Thermal Metal-Insulator Transition in VO₂ Monolithically Integrated into a WSe₂ Field-Effect Transistors", *ACS Appl. Mater. and Inter.* 11, 3224-3230 (2019), <https://doi.org/10.1021/acsami.8b18745>
- 384 A. N. Hattori, A. I. Osaka, K. Hattori, Y. Naitoh, H. Shima, H. Akinaga, H. Tanaka, " Investigation of Statistical Metal-Insulator Transition Properties of Electronic Domains in Spatially Confined VO₂ Nanostructure", *Crystals* 10, 631 (2020), <https://doi.org/10.3390/cryst10080631>
- 385 R. Rakshit, A. N. Hattori, Y. Naitoh, H. Shima, H. Akinaga, H. Tanaka, "Three-Dimensional Nanoconfinement Supports Verwey Transition in Fe₃O₄ Nanowire at 10 nm Length Scale", *Nano Lett.* 19, 5003-5010 (2019), <https://doi.org/10.1021/acs.nanolett.9b01222>
- 386 R. Yukawa, M. Kobayashi, T. Kanda, D. Shiga, K. Yoshimatsu, S. Ishibashi, M. Minohara, M. Kitamura, K. Horiba, A. F. Santander-Syro and H. Kumigashira, "Resonant tunneling driven metal-insulator transition in double quantum-well structures of strongly correlated oxide", *Nat. Commun.* 12, 7070 (2021). <https://doi.org/10.1038/s41467-021-27327-z>
- 387 Y. Zhou, X. Chen, Z. Yang, C. Mouli and S. Ramanathan, "Voltage-Triggered Ultrafast Phase Transition in Vanadium Dioxide Switches," *IEEE Electron Device Letters*, 34, 220 (2013)
- 388 J. Leroy, A. Crunteanu, A. Bessaudou, F. Cosset, C. Champeaux, and J. C. Orlianges, "High-speed metal-insulator transition in vanadium dioxide films induced by an electrical pulsed voltage over nano-gap electrodes," *Applied Physics Letters*, 100, 213507 (2012).
- 389 T. Yajima, T. Nishimura, T. Tanaka, K. Uchida and A. Toriumi, "Modulation of VO₂ Metal-Insulator Transition by Ferroelectric HfO₂ Gate Insulator", *Adv. Electron.Mater.*, 6, 1901356 (2020). <https://doi.org/10.1002/aelm.201901356>
- 390 J.del Valle, N. M. Vargas, R. Rocco, P. Salev, Y. Kalcheim, P. N. Lapa, C. Adda, M.-H. Lee, P. Y. Wang, L. Fratino, M. J. Rozenberg and I. K. Schuller, "Spatiotemporal characterization of the field-induced insulator-to-metal transition", *Science* 373, 907 (2021). <https://doi.org/10.1126/science.abd9088>
- 391 S. D. Ha, G. H. Aydogdu and S. Ramanathan, "Metal-insulator transition and electrically driven memristive characteristics of SmNiO₃ thin films," *Appl. Phys. Lett.*, 98, 012105 (2011)
- 392 W. L. Lim, E. J. Moon, J. W. Freeland, D. J. Meyers, M. Kareev, J. Chakhalian, and S. Urazhdin, "Field-effect diode based on electron-induced Mott transition in NdNiO₃," *Appl. Phys. Lett.* 101, 143111 (2012)
- 393 R. Scherwitzl, P. Zubko, I. G. Lezama, S. Ono, A. F. Morpurgo, G. Catalan and J.-M. Triscone, "Electric-Field Control of the Metal-Insulator Transition in Ultrathin NdNiO₃ Films," *Adv. Mater.* 22, 5517 (2010).
- 394 S. D. Ha, U. Vetter, J. Shi, and S. Ramanathan, "Electrostatic gating of metallic and insulating phases in SmNiO₃ ultrathin films," *Appl. Phys. Lett.* 102, 183102 (2013)
- 395 S. Bubel, A. J. Hauser, A. M. Glaudell, T. E. Mates, S. Stemmer and M. L. Chabiny, "The electrochemical impact on electrostatic modulation of the metal-insulator transition in nickelates," *Appl. Phys. Lett.* 106, 122102 (2015)
- 396 S. Bubel, A. J. Hauser, A. M. Glaudell, T. E. Mates, S. Stemmer and M. L. Chabiny, "The electrochemical impact on electrostatic modulation of the metal-insulator transition in nickelates," *Appl. Phys. Lett.* 106, 122102 (2015)
- 397 J. Shi, Y. Zhou and S. Ramanathan, "Colossal resistance switching and band gap modulation in a perovskite nickelate by electron doping," *Nature Communications*, 5, 4860 (2014)
- 398 C. Zhou, D. M. Newns, J. A. Misewich and P. C. Pattnaik, "A field effect transistor based on the Mott transition in a molecular layer," *Appl Phys Lett* 70 (5), 598–600 (1997)
- 399 D. M. Newns, J. A. Misewich, C. C. Tsuei, A. Gupta, B. A. Scott and A. Schrott, "Mott transition field effect transistor," *Appl Phys Lett* 73 (6), 780–782 (1998)
- 400 Y. Zhou and S. Ramanathan, "Correlated electron materials and field effect transistors for logic: a review," *Crit Rev Solid State Mater Sci*, 38(4), 286–317, 2013.
- 401 H. Akinaga, "Recent advances and future prospects in functional-oxide nanoelectronics: the emerging materials and novel functionalities that are accelerating semiconductor device research and development," *Jpn. J. Appl. Phys.*, 52(100001), 2013.
- 402 Z. Yang, C. Ko and S. Ramanathan, "Oxide Electronics Utilizing Ultrafast Metal-Insulator Transitions," *Annual Review of Materials Research* 41, 8.1 (2011).
- 403 A. Cavalleri, Cs. Toth, C. W. Siders, J. A. Squier, F. Raksi, P. Forget and J. C. Keiffer, "Femtosecond Structural Dynamics in VO₂ during an Ultrafast Solid-Solid Phase Transition," *Phy. Rev. Lett.* 87, 237401 (2001)
- 404 S. Hormoz and S. Ramanathan, "Limits on vanadium oxide Mott metal-insulator transition field-effect transistors," *Solid State Electron* 54 (6), 654–659 (2010)
- 405 H. Ji, Jiang Wei, and D. Natelson, "Modulation of the Electrical Properties of VO₂ Nanobeams Using an Ionic Liquid as a Gating Medium," *Nano Lett.*, 12, 2988–2992 (2012)
- 406 J. Jeong, N Aetukuri, T Graf, TD Schladt, MG Samant and S. S. Parkin, "Suppression of Metal-Insulator Transition in VO₂ by Electric Field-Induced Oxygen Vacancy Formation," *Science*, 339(6126), 1402–1405 (2013).
- 407 J. Shi, S. D. Ha, Y. Zhou, F. Schoofs and S. Ramanathan, "A correlated nickelate synaptic transistor," *Nature Communications*, 4, 2676 (2013)

- ⁴⁰⁸ S. D. Ha, J. Shi, Y. Meroz, L. Mahadevan and S. Ramanathan, "Neuromimetic Circuits with Synaptic Devices Based on Strongly Correlated Electron Systems," *Physical Review Applied*, 2, 064003 (2014),
- ⁴⁰⁹ M. D. Pickett, G. Medeiros-Ribeiro and R. Stanley Williams, "A scalable neuristor built with Mott memristors," *Nature Materials* 12, 114–117 (2013)
- ⁴¹⁰ N. Shukla, A. V. Thathachary, A. Agrawal, H. Paik, A. Aziz, D. G. Schlom, S. K. Gupta, R. Engel-Herbert and S. Datta, "A steep-slope transistor based on abrupt electronic phase transition," *Nature Communications* 6, 7812 (2015)
- ⁴¹¹ Philipp Scheiderer, Matthias Schmitt, Judith Gabel, Michael Zapf, Martin Stübinger, Philipp Schütz, Lenart Dudy, Christoph Schlueter, Tien - Lin Lee, Michael Sing, Ralph Claessen, "Tailoring Materials for Mottronics: Excess Oxygen Doping of a Prototypical Mott Insulator", *Advanced Materials* 30, 1706708 (2018), <https://doi.org/10.1002/adma.201706708>
- ⁴¹² T. Wei, T. Kanki, K. Fujiwara, M. Chikanari and H. Tanaka, "Electric field-induced transport modulation in VO₂ FETs with high-k oxide/organic parylene-C hybrid gate dielectric," *Appl. Phys. Lett.* 108 (2016) 053503.
- ⁴¹³ T. Wei, T. Kanki, M. Chikanari, T. Uemura, T. Sekitani, and H. Tanaka, "Enhancement of electronic transport modulation in single crystalline VO₂ nanowire-based solid-state field-effect transistor," *Scientific Reports*, 7, 17215 (2017), <https://doi.org/10.1038/s41598-017-17468-x>
- ⁴¹⁴ M. Yamamoto, R. Nouch, T. Kanki, A. N. Hattori, K. Watanabe, T. Taniguchi, K. Ueno and H. Tanaka, "Gate-Tunable Thermal Metal-Insulator Transition in VO₂ Monolithically Integrated into a WSe₂ Field-Effect Transistors", *ACS Appl. Mater. and Inter.* 11, 3224-3230 (2019), <https://doi.org/10.1021/acsami.8b18745>
- ⁴¹⁵ A. N. Hattori, A. I. Osaka, K. Hattori, Y. Naitoh, H. Shima, H. Akinaga, H. Tanaka, "Investigation of Statistical Metal-Insulator Transition Properties of Electronic Domains in Spatially Confined VO₂ Nanostructure", *Crystals* 10, 631 (2020), <https://doi.org/10.3390/cryst10080631>
- ⁴¹⁶ R. Rakshit, A. N. Hattori, Y. Naitoh, H. Shima, H. Akinaga, H. Tanaka, "Three-Dimensional Nanoconfinement Supports Verwey Transition in Fe₃O₄ Nanowire at 10 nm Length Scale", *Nano Lett.* 19, 5003-5010 (2019), <https://doi.org/10.1021/acs.nanolett.9b01222>
- ⁴¹⁷ R. Yukawa, M. Kobayashi, T. Kanda, D. Shiga, K. Yoshimatsu, S. Ishibashi, M. Minohara, M. Kitamura, K. Horiba, A. F. Santander-Syro and H. Kumigashira, "Resonant tunneling driven metal-insulator transition in double quantum-well structures of strongly correlated oxide", *Nat. Commun.* 12, 7070 (2021). <https://doi.org/10.1038/s41467-021-27327-z>
- ⁴¹⁸ Y. Zhou, X. Chen, Z. Yang, C. Mouli and S. Ramanathan, "Voltage-Triggered Ultrafast Phase Transition in Vanadium Dioxide Switches," *IEEE Electron Device Letters*, 34, 220 (2013)
- ⁴¹⁹ J. Leroy, A. Crunteanu, A. Bessaudou, F. Cosset, C. Champeaux, and J. C. Orlianges, "High-speed metal-insulator transition in vanadium dioxide films induced by an electrical pulsed voltage over nano-gap electrodes," *Applied Physics Letters*, 100, 213507 (2012).
- ⁴²⁰ T. Yajima, T. Nishimura, T. Tanaka, K. Uchida and A. Toriumi, "Modulation of VO₂ Metal-Insulator Transition by Ferroelectric HfO₂ Gate Insulator", *Adv. Electron. Mater.*, 6, 1901356 (2020). <https://doi.org/10.1002/aelm.201901356>
- ⁴²¹ J. del Valle, N. M. Vargas, R. Rocco, P. Salev, Y. Kalcheim, P. N. Lapa, C. Adda, M.-H. Lee, P. Y. Wang, L. Fratino, M. J. Rozenberg and I. K. Schuller, "Spatiotemporal characterization of the field-induced insulator-to-metal transition", *Science* 373, 907 (2021). <https://doi.org/10.1126/science.abd9088>
- ⁴²² S. D. Ha, G. H. Aydogdu and S. Ramanathan, "Metal-insulator transition and electrically driven memristive characteristics of SmNiO₃ thin films," *Appl. Phys. Lett.*, 98, 012105 (2011)
- ⁴²³ W. L. Lim, E. J. Moon, J. W. Freeland, D. J. Meyers, M. Kareev, J. Chakhalian, and S. Urazhdin, "Field-effect diode based on electron-induced Mott transition in NdNiO₃," *Appl. Phys. Lett.* 101, 143111 (2012)
- ⁴²⁴ R. Scherwitzl, P. Zubko, I. G. Lezama, S. Ono, A. F. Morpurgo, G. Catalan and J.-M. Triscone, "Electric-Field Control of the Metal-Insulator Transition in Ultrathin NdNiO₃ Films," *Adv. Mater.* 22, 5517 (2010).
- ⁴²⁵ S. D. Ha, U. Vetter, J. Shi, and S. Ramanathan, "Electrostatic gating of metallic and insulating phases in SmNiO₃ ultrathin films," *Appl. Phys. Lett.* 102, 183102 (2013)
- ⁴²⁶ S. Bubel, A. J. Hauser, A. M. Glauddell, T. E. Mates, S. Stemmer and M. L. Chabiny, "The electrochemical impact on electrostatic modulation of the metal-insulator transition in nickelates," *Appl. Phys. Lett.* 106, 122102 (2015)
- ⁴²⁷ S. Bubel, A. J. Hauser, A. M. Glauddell, T. E. Mates, S. Stemmer and M. L. Chabiny, "The electrochemical impact on electrostatic modulation of the metal-insulator transition in nickelates," *Appl. Phys. Lett.* 106, 122102 (2015)
- ⁴²⁸ J. Shi, Y. Zhou and S. Ramanathan, "Colossal resistance switching and band gap modulation in a perovskite nickelate by electron doping," *Nature Communications*, 5, 4860 (2014)
- ⁴²⁹ IRDS 2021 edition.
- ⁴³⁰ S. Datta and B. Das, "Electronic analog of the electro-optic modulator," *Appl. Phys. Lett.* vol. 56, pp.665–667, 1990.
- ⁴³¹ S. Sugahara and M. Tanaka, "A spin metal-oxide-semiconductor field-effect transistor using half-metallic-ferromagnet contacts for the source and drain," *Appl. Phys. Lett.* vol. 84, pp.2307–2309, 2004.
- ⁴³² S. Sugahara and J. Nitta, "Spin-Transistor Electronics: An Overview and Outlook," *Proc. IEEE*, vol. 98, pp. 2124–2154, 2010.
- ⁴³³ T. Tanamoto, H. Sugiyama, T. Inokuchi, T. Marukame, M. Ishikawa, K. Ikegami, and Y. Saito, "Scalability of spin field programmable gate array: A reconfigurable architecture based on spin metal-oxide-semiconductor field effect transistor," *J. Appl. Phys.* vol. 109, pp. 07C312/1-3, 2011.

- 434 S. Shuto, S. Yamamoto, and S. Sugahara, "Nonvolatile Static Random Access memory based on spin-transistor architecture," *J. Appl. Phys.* vol. 105, pp. 07C933/1-3, 2009.
- 435 S. Yamamoto, and S. Sugahara, "Nonvolatile Delay Flip-Flop Based on Spin-Transistor Architecture and Its Power-Gating Applications," *Jpn. J. Appl. Phys.* vol. 49, pp. 090204/1-3, 2010.
- 436 S. Sugahara, Y. Shuto, and S. Yamamoto, "Nonvolatile Logic Systems Based on CMOS/Spintronics Hybrid Technology: An Overview," *Magnetics Japan*, vol. 6, pp. 5–15, 2011.
- 437 Y. Shuto, S. Yamamoto, H. Sukegawa, Z. C. Wen, R. Nakane, S. Mitani, M. Tanaka, K. Inomata, and S. Sugahara, "Design and performance of pseudo-spin-MOSFETs using nano-CMOS devices," *2012 IEEE International Electron Devices Meeting*, pp. 29.6.1–29.6.4, 2012.
- 438 S. Yamamoto, Y. Shuto, and S. Sugahara, "Nonvolatile flip-flop based on pseudo-spin transistor architecture and its nonvolatile power-gating applications for low-power CMOS logic," *EPJ Appl. Phys.* vol. 63, pp. 14403, 2013.
- 439 D. Osintev, V. Sverdlov, Z. Stanojevic, A. Makarov, and S. Selberherr, "Temperature dependence of the transport properties of spin field-effect transistors built with InAs and Si channels," *Solid-state Electronics* vol. 71, pp. 25–29, 2012.
- 440 Y. Saito, T. Marukame, T. Inokuchi, M. Ishikawa, H. Sugiyama, and T. Tanamoto, "Spin injection, transport and read/write operation in spin-based MOSFET," *Thin Solid Films* vol. 519, pp. 8266 – 8273, 2011.
- 441 T. Sasaki, Y. Ando, M. Kameno, T. Tahara, H. Koike, T. Oikawa, T. Suzuki, and M. Shiraishi, "Spin transport in nondegenerate Si with a spin MOSFET structure at room temperature," *Phys. Rev. Appl.* Vol. 2, pp.034005/1-6, 2014.
- 442 T. Tahara, H. Koike, M. Kameno, T. Sasaki, Y. Ando, K. Tanaka, S. Miwa, Y. Suzuki, and M. Shiraishi, "Room-temperature operation of Si spin MOSFET with high on/off spin signal ratio," *Appl. Phys. Express* vol. 8, pp. 113004/1-3, 2015.
- 443 S. Sato, M. Ichihara, M. Tanaka, and R. Nakane "Electron spin and momentum lifetimes in two-dimensional Si accumulation channels: Demonstration of Schottky-barrier spin metal-oxide-semiconductor field-effect transistors at room temperature", *Phys. Rev. B* 99, pp. 165301/1-9, 2019.
- 444 I. Appelbaum, B. Huang, and D. J. Monsma, "Electronic measurement and control of spin transport in silicon," *Nature*, vol. 447, pp. 295–298, 2007.
- 445 X. Lou, C. Adelmann, S. A. Crooker, E. S. Garlid, J. Zhang, K. S. M. Reddy, S. D. Flexner, C. J. Palmström, and P. A. Crowell, "Electrical detection of spin transport in lateral ferromagnet-semiconductor devices", *Nat. Phys.* vol. 3, pp. 197-202, 2007.
- 446 O. M. J. van't Erve, A. T. Hanbicki, M. Holub, C. H. Li, C. Awo-Affouda, P. E. Thompson, and B. T. Jonker, "Electrical injection and detection of spin-polarized carriers in silicon in a lateral transport geometry", *Appl. Phys. Lett.* vol. 91, pp. 212109, 2007.
- 447 G. Schmidt, D. Ferrand, L. W. Molenkamp, A. T. Filip, and B. J. van Wees, "Fundamental obstacle for electrical spin injection from a ferromagnetic metal into a diffusive semiconductor," *Phys. Rev. B* vol. 62, pp. R4790–R4793, 2000.
- 448 E. I. Rashba, "Theory of electrical spin injection: Tunnel contacts as a solution of the conductivity mismatch problem," *Phys. Rev. B* vol. 62, pp. R16267–R16270, 2000.
- 449 A. Fert, and H. Jaffrès, "Conditions for efficient spin injection from a ferromagnetic metal into a semiconductor," *Phys. Rev. B* vol. 64, pp. 184420/1-9, 2001.
- 450 T. Tanamoto, H. Sugiyama, T. Inokuchi, M. Ishikawa, and Y. Saito, "Effects of interface resistance asymmetry on local and non-local magnetoresistance structures," *Jpn. J. Appl. Phys.* vol. 52, pp. 04CM03/1-4, 2013.
- 451 M. Ishikawa, T. Oka, Y. Fujita, H. Sugiyama, Y. Saito, and K. Hamaya, "Spin relaxation through lateral spin transport in heavily doped n-type silicon", *Phys. Rev. B* vol. 95, pp. 115302/1-6, 2017.
- 452 A. Spiessner, H. Saito, Y. Fujita, S. Yamada, K. Hamaya, S. Yuasa, and R. Jansen, "Giant Spin Accumulation in Silicon Nonlocal Spin-Transport Devices", *Phys. Rev. Appl.* vol. 8, pp. 064023/1-10, 2017.
- 453 R. Jansen, A. Spiessner, H. Saito, Y. Fujita, S. Yamada, K. Hamaya, and S. Yuasa, "Nonlinear Electrical Spin Conversion in a Biased Ferromagnetic Tunnel Contact", *Phys. Rev. Appl.* vol.10, pp. 064050/1-13, 2018.
- 454 A. Spiessner, H. Saito, S. Yuasa, and R. Jansen, "Tunnel spin polarization of Fe/MgO/Si contacts reaching 90% with increasing MgO thickness", *Phys. Rev. B* vol. 99, pp. 224427/1-11, 2019.
- 455 T. Akiho, J. Shan, H. X. Liu, K. Matsuda, M. Yamamoto, and T. Uemura, "Electrical injection of spin-polarized electrons and electrical detection of dynamic nuclear polarization using a Heusler alloy spin source", *Phys. Rev. B* vol. 87, pp. 235205/1-7, 2013.
- 456 T. Saito, N. Tezuka, M. Matsuura, and S. Sugimoto, "Spin injection, transport, and detection at room temperature in a lateral spin transport device with Co₂FeAl_{0.5}Si_{0.5}/n-GaAs Schottky tunnel junctions," *Appl. Phys. Express*, vol. 6, pp. 103006/1-4, 2013.
- 457 K. Kasahara, Y. Fujita, S. Yamada, K. Sawano, M. Miyao, and K. Hamaya, "Greatly enhanced generation efficiency of pure spin currents in Ge using Heusler compound Co₂FeSi electrodes," *Appl. Phys. Express*, vol. 7, pp. 033002/1-4, 2014.
- 458 T. A. Peterson, S. J. Patel, C. C. Geppert, K. D. Christie, A. Rath, D. Pennachio, M. E. Flatté, P. M. Voyles, C. J. Palmström, and P. A. Crowell, "Spin injection and detection up to room temperature in Heusler alloy/n-GaAs spin valves", *Phys. Rev. B* vol. 94, pp. 235309/1-12, 2016.
- 459 Y. Fujita, M. Yamada, M. Tsukahara, T. Oka, S. Yamada, T. Kanashima, K. Sawano, and K. Hamaya, "Spin Transport and Relaxation up to 250 K in Heavily Doped n-Type Ge Detected Using Co₂FeAl_{0.5}Si_{0.5} Electrodes", *Phys. Rev. Appl.* vol. 8, pp. 014007/1-9 2017.
- 460 M. Yamada, Y. Fujita, S. Yamada, K. Sawano, and K. Hamaya, "Control of electrical properties in Heusler-alloy/Ge Schottky tunnel contacts by using phosphorous δ-doping with Si-layer insertion", *Mat. Sci. Semicon. Proc.* vol. 70, pp. 83-85, 2017.

- 461 M. Yamada, M. Tsukahara, Y. Fujita, T. Naito, S. Yamada, K. Sawano, and K. Hamaya, "Room-temperature spin transport in *n*-Ge probed by four-terminal nonlocal measurements", *Appl. Phys. Express* vol. 10, pp. 093001/1-4, 2017.
- 462 M. Tsukahara, M. Yamada, T. Naito, S. Yamada, K. Sawano, V. K. Lazarov, and K. Hamaya, "Room-temperature local magnetoresistance effect in *n*-Ge devices with low-resistive Schottky-tunnel contacts", *Appl. Phys. Express* vol. 12, pp. 033002/1-4, 2019.
- 463 M. Yamada, F. Kuroda, M. Tsukahara, S. Yamada, T. Fukushima, K. Sawano, T. Oguchi, and K. Hamaya, "Spin injection through the energy-band symmetry matching with high spin polarization in atomically controlled ferromagnet/ferromagnet/semiconductor structures", *NPG Asia Mater.* vol. 12, pp. 47/1-9 (2020).
- 464 K. Kudo, M. Yamada, S. Honda, Y. Wagatsuma, S. Yamada, K. Sawano, and K. Hamaya, "Room-temperature two-terminal magnetoresistance ratio reaching 0.1 % in semiconductor-based lateral devices with *L*₂₁-ordered Co₂MnSi", *Appl. Phys. Lett.* vol. 118, pp. 162404/1-6 2021.
- 465 T. Naito, M. Yamada, S. Yamada, K. Sawano, and K. Hamaya, "Suppression of Donor-Driven Spin Relaxation in Strained Si_{0.1}Ge_{0.9}", *Phys. Rev. Appl.* vol. 13, 054025/1-8 2020.
- 466 K. Yamamoto, T. Matsuo, M. Yamada, Y. Wagatsuma, K. Sawano, and K. Hamaya, "Electrical properties of a low-temperature fabricated Ge-based top-gate MOSFET structure with epitaxial ferromagnetic Heusler-alloy Schottky-tunnel source and drain, *Mat. Sci. Semicon. Proc.* vol.167, 107763/1-7 2023.
- 467 Y. Shuto, R. Nakane, W. Wang, H. Sukegawa, S. Yamamoto, M. Tanaka, K. Inomata, and S. Sugahara, "A New Spin-Functional Metal–Oxide–Semiconductor Field-Effect Transistor Based on Magnetic Tunnel Junction Technology: Pseudo-Spin-MOSFET," *Appl. Phys. Express* vol. 3, pp. 013003/1-3, 2010.
- 468 R. Nakane, Y. Shuto, H. Sukegawa, Z. C. Wen, S. Yamamoto, S. Mitani, M. Tanaka, K. Inomata, and S. Sugahara, "Fabrication of pseudo-spin-MOSFETs using a multi-project wafer CMOS chip," *Solid-State Elec.*, vol. 102, pp. 52–58, 2014.
- 469 H. C. Koo, J. H. Kwon, J. Eom, J. Chang, S. H. Han, and M. Johnson, "Control of Spin Precession in a Spin-Injected Field Effect Transistor", *Science* vol. 325, pp. 1515–1518, 2009.
- 470 P. Chuang, S.-C. Ho, L.W. Smith, F. Sfigakis, M. Pepper, C.-H. Chen, J.-C. Fan, J. P. Griffiths, I. Farrer, H. E. Beere, G. A. C. Jones, D. A. Ritchie, and T.-M. Chen, "All-electric all-semiconductor spin field-effect transistors", *Nat. Nanotechnol.* vol. 10, pp. 35-39, 2015.
- 471 F. Eberle, D. Schuh, B. Grunewald, D. Bougeard, D. Weiss, and M. Ciorga, "Controlled Rotation of Electrically Injected Spins in a Nonballistic Spin-Field-Effect Transistor", *Nano Lett.* vol. 23, pp. 4815-4821, 2023.
- 472 S. Lee, H. Koike, M. Goto, S. Miwa, Y. Suzuki, N. Yamashita, R. Ohshima, E. Shigematsu, Y. Ando, and M. Shiraishi, "Synthetic Rashba spin–orbit system using a silicon metal-oxide semiconductor", *Nat. Mater.* vol. 20, pp. 1228-1233, 2021.
- 473 K. He, B. Barut, S. Yin, M. D. Randle, R. Dixit, N. Arabchigavkani, J. Nathawat, A. Mahmood, W. Echtenkamp, C. Binek, P. A. Dowben, and J. P. Bird, "Graphene on Chromia: A System for Beyond-Room-Temperature Spintronics", *Adv. Mater.* **34**, 2105023 (2022); doi: 10.1002/adma.202105023
- 474 P. A. Dowben, C. Binek, K. Zhang, L. Wang, W.-N. Mei, J. P. Bird, U. Singiseti, X. Hong, K. L. Wang, D. Nikonov, "Towards a Strong Spin-Orbit Coupling Magnetoelectric Transistor", *IEEE Journal of Exploratory Solid-State Computational Devices and Circuits* **4**, 1-9 (2018); Online ISSN: 2329-9231; doi: 10.1109/JXCDC.2018.2809640
- 475 S. Cao, Z. Xiao, C.-P. Kwan, K. Zhang, J. P. Bird, L. Wang, W.-N. Mei, J. Hong, and P. A. Dowben, "Moving towards the Magnetoelectric Graphene Transistor", *Appl. Phys. Lett.* **111**, 182402 (2017); doi: 10.1063/1.4999643
- 476 N. Sharma, J. P. Bird, C. Binek, P. A. Dowben, D. Nikonov, and A. Marshall, "Evolving Magneto-electric Device Technologies", *Semicond. Sci. and Technol.* **35**, 073001 (2020); doi: 10.1088/1361-6641/ab8438
- 477 D. E. Nikonov, C. Binek, X. Hong, J. P. Bird, K. L. Wang, P. A. Dowben, "Magneto-Electric Logic Devices Using Semiconductor Channel with Large Spin-Orbit Coupling", U.S. Patent 10,361,292 issued July 23, 2019
- 478 C. Pan, A. Naeemi, "An expanded benchmarking of beyond-CMOS devices based on boolean and neuromorphic representative circuits", *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits* **3**, 101-110 (2018); doi: 10.1109/JXCDC.2018.2793536
- 479 N. Sharma, C. Binek, A. Marshall, J. P. Bird, P. A. Dowben and D. Nikonov, "Compact Modeling and Design of Magneto-Electric Transistor Devices and Circuits," *2018 31st IEEE International System-on-Chip Conference (SOCC)*, Arlington, VA, USA, *IEEE Explore* 146-151 (2018); doi: 10.1109/SOCC.2018.8618494
- 480 C. Pan, A. Naeemi, "Complementary Logic Implementation for Antiferromagnet Field-Effect Transistors", *IEEE J. Exploratory Solid-State Computational Devices and Circuits* **4**, 69-75 (2018); doi: 10.1109/JXCDC.2018.2878635
- 481 N. S. Gaul, A. Marshall, P. A. Dowben, D. E. Nikonov, "Circuits Based on Magnetoelectric Transistor Devices", U.S. Patent 11,349,480 B2, issued May 31, 2022
- 482 N. S. Gaul, A. Marshall, P. A. Dowben, D. E. Nikonov, "Magnetoelectric Inverter", U.S. Patent application 11,658,663 B2, issued May 23, 2023
- 483 N. S. Gaul, A. Marshall, P. A. Dowben, D. E. Nikonov, "Magnetoelectric Xnor Logic Gate Device", U.S. Patent 11757449 B2, issued Sept. 12, 2023
- 484 N. S. Gaul, A. Marshall, P. A. Dowben, D. E. Nikonov, "Magnetoelectric Majority Gate Device", U.S. Patent 11764786 B2, issued Sept. 19, 2023
- 485 N. S. Gaul, A. Marshall, P. A. Dowben, D. E. Nikonov, "Circuits Based on Magnetoelectric Transistor Devices", U.S. Patent application 16581691, publication date: March 26, 2020, filed September 24, 2019

- ⁴⁸⁶ P. A. Dowben, D. E. Nikonov, A. Marshall and C. Binek, "Magneto-electric antiferromagnetic spin-orbit logic devices", *Appl. Phys. Lett.* **116**, 080502 (2020); doi: 10.1063/1.5141371
- ⁴⁸⁷ A. Marshall and P. A. Dowben, "Magneto-electric Transistor Devices and Circuits with Steering Logic", 2020 IEEE 14th Dallas Circuits and Systems Conference (DCAS), *IEEE Explore* (2021) pp 1-4, doi: 10.1109/DCAS51144.2020.9330662
- ⁴⁸⁸ A. Marshall and P. A. Dowben, "Magneto-electric Transistor Devices and Circuits with Steering Logic", in *Advances in Semiconductor Technologies: Selected Topics Beyond Conventional CMOS*, edited by An Chen, John Wiley & Sons, Inc. (Hoboken, New Jersey) (2022), pp. 61-77; ISBN 9781119869597 (adobe pdf) | ISBN 9781119869603 (epub)
- ⁴⁸⁹ S. Angizi, N. Khoshavi, A. Marshall, P. Dowben, and D. Fan, "MeF-RAM: A New Non-Volatile Cache Memory Based on Magneto-Electric FET", *ACM Transactions on Automation of Electronic Systems (TODAES)* **27**, 18 (2021); doi: 10.1145/3484222
- ⁴⁹⁰ S. Angizi, D. Fan, A. Marshall, P. A. Dowben, "Non-Volatile Based Memory Architectures using Magneto-Electric FETs", in *Advances in Semiconductor Technologies: Selected Topics Beyond Conventional CMOS*, edited by An Chen, John Wiley & Sons, Inc. (Hoboken, New Jersey) (2022), pp. 79-92; ISBN 9781119869597 (adobe pdf) | ISBN 9781119869603 (epub)
- ⁴⁹¹ X. He, Y. Wang, N. Wu, A. N. Caruso, E. Vescovo, K. D. Belashchenko, P. A. Dowben and C. Binek, "Robust isothermal electric control of exchange bias at room temperature", *Nat. Mater.* **9**, 579-585 (2010); <https://doi.org/10.1038/nmat2785>
- ⁴⁹² N. Wu, X. He, A. Wysocki, U. Lanke, T. Komesu, K. D. Belashchenko, C. Binek, P. A. Dowben, "Imaging and control of surface magnetization domains in a magnetoelectric antiferromagnet", *Phys. Rev. Lett.* **106**, 087202 (2011); <https://doi.org/10.1103/PhysRevLett.106.087202>
- ⁴⁹³ S. Cao, X. Zhang, N. Wu, A. T. N'Diaye, G. Chen, A. K. Schmid, X. Chen, W. Echtenkamp, A. Enders, C. Binek, P. A. Dowben, "Spin Polarization Asymmetry at the Surface of Chromia", *New J. Phys.* **16**, 073021 (2014); doi:10.1088/1367-2630/16/7/073021
- ⁴⁹⁴ K. D. Belashchenko, "Equilibrium Magnetization at the Boundary of a Magnetoelectric Antiferromagnet", *Phys. Rev. Lett.* **105**, 147204 (2010); doi: 10.1103/PhysRevLett.105.147204
- ⁴⁹⁵ M. Street, W. Echtenkamp, T. Komesu, S. Cao, P. A. Dowben, and C. Binek, "Increasing the Néel Temperature of Magnetoelectric Chromia for Voltage-Controlled Spintronics", *Appl. Phys. Lett.* **104**, 222402 (2014); doi: 10.1063/1.4880938
- ⁴⁹⁶ A. Mahmood, W. Echtenkamp, M. Street, J.-L. Wang, S. Cao, T. Komesu, P. A. Dowben, P. Buragohain, H. Lu, A. Gruverman, A. Parthasarathy, S. Rakheja, C. Binek, "Voltage controlled Néel vector rotation in zero magnetic field at CMOS-compatible temperatures", *Nat. Commun.* **12**, 1674 (2021); doi: 10.21203/rs.3.rs-38435/v1
- ⁴⁹⁷ A. Mahmood, J. Weaver, S. Q. A. Shah, W. Echtenkamp, Je. W. Lynn, P. A. Dowben, C. Binek, "Post deposition interfacial Néel temperature tuning in magnetoelectric B:Cr₂O₃", *Adv. Phys. Res.* **3**, 2300061 (2023); doi: 10.1002/apxr.202300061
- ⁴⁹⁸ E. Parsonnet, Y.-L. Huang, T. Gosavi, A. Qualls, D. Nikonov, C.-C. Lin, I. Young, J. Bokor, L. W. Martin, and R. Ramesh, "Toward Intrinsic Ferroelectric Switching in Multiferroic BiFeO₃", *Phys. Rev. Lett.* **125**, 067601 (2020)
- ⁴⁹⁹ G. Hao, A. Mosey, X. Jiang, A. J. Yost, K. R. Sapkota, G. T. Wang, X. Zhang, J. Zhang, A. T. N'Diaye, R. Cheng, X. Xu, P. A. Dowben, "Nonvolatile Voltage Controlled Molecular Spin State Switching", *Appl. Phys. Lett.* **114**, 032901 (2019); doi: 10.1063/1.5054909
- ⁵⁰⁰ A. Mosey, A. S. Dale, G. Hao, A. N'Diaye, P. A. Dowben, R. Cheng, "Quantitative Study of Energy Change in Voltage-controlled Spin Crossover Molecules for Molecular Spintronics", *J. Phys. Chem. Lett.* **11**, 8231-8237 (2020); doi: 10.1021/acs.jpclett.0c02209; 'Correction to "Quantitative Study of the Energy Changes in Voltage-Controlled Spin Crossover Molecular Thin Films"' A. Mosey, A. S. Dale, G. Hao, A. N'Diaye, P. A. Dowben, and R. Cheng. *J. Phys. Chem. Lett.* **12**, 2463 (2021); doi: 10.1021/acs.jpclett.1c00608
- ⁵⁰¹ T. K. Ekanayaka, G. Hao, A. Mosey, A. S. Dale, X. Jiang, A. J. Yost, K. R. Sapkota, G. T. Wang, J. Zhang, A. T. N'Diaye, A. Marshall, R. Cheng, A. Naeemi, X. Xu, and P. A. Dowben, "Nonvolatile voltage controlled molecular spin state switching for memory applications", *Magnetochemistry* **7**, 37 (2021); doi: 10.3390/magnetochemistry7030037
- ⁵⁰² J. A. Currivan, M. A. Baldo & C. A. Ross. (12) Patent Application Publication (10) Pub . No .: US 2013 / 0181158 A1. *US Pat.* **1**, 12-15 (2013).
- ⁵⁰³ J. A. Currivan, Y. Jang, M. D. Mascaro, M. A. Baldo & C. A. Ross. Low Energy Magnetic Domain Wall Logic in Short, Narrow, Ferromagnetic Wires. *IEEE Magn. Lett.* **3**, 3000104 (2012)
- ⁵⁰⁴ J. A. Currivan-Incorvia, S. Siddiqui, S. Dutta, E. R. Everts, J. Zhang, D. Bono, C. A. Ross & M. A. Baldo. Logic circuit prototypes for three-terminal magnetic tunnel junctions with mobile domain walls. *Nat. Commun.* **7**, 10275 (2016).
- ⁵⁰⁵ M. Alamdar, L. J. Chang, K. Jarvis, P. Kotula, C. Cui, R. Gearba-Dolocan, Y. Liu, E. Antunano, J. E. Manuel, G. Vizkelethy, L. Xue, R. Jacobs-Gedrim, C. H. Bennett, T. P. Xiao, D. Hughart, E. Bielejec, M. J. Marinella, & J. A. C. Incorvia. Irradiation Effects on Perpendicular Anisotropy Spin Orbit Torque Magnetic Tunnel Junctions. *IEEE Transactions on Nuclear Science*, **68**(5), 665-670 (2021). <https://doi.org/10.1109/TNS.2021.3066070>
- ⁵⁰⁶ C. H. Bennett, T. P. Xiao, T. Leonard, X. Zhan, R. Gearba-Dolocan, J. Young, G. Vizkelethy, E. Bielejec, D. Hughart, M. J. Marinella, & J. A. C. Incorvia. *Radiation Response of Domain-Wall Magnetic Tunnel Junction Logic Devices*. IEEE Transactions on Nuclear Science, 1-1 (2023). <https://doi.org/10.1109/TNS.2023.3333819>
- ⁵⁰⁷ R. Blasing, A. A. Khan, P. C. Filippou, C. Garg, F. Hameed, J. Castrillon & S. S. P. Parkin. Magnetic Racetrack Memory: From Physics to the Cusp of Applications within a Decade. *Proceedings of the IEEE* **108**, 1303-1321 (2020).
- ⁵⁰⁸ D. Morris, D. Bromberg, J.-G. J.-. G. J. Zhu & L. Pileggi. mLogic: Ultra-low voltage non-volatile logic circuits using STT-MTJ devices. *Proc. 49th Annu. Des. Autom. Conf.* 486-491 (2012).
- ⁵⁰⁹ V. Sokalski, D. M. Bromberg, D. Morris, M. T. Moneck, E. Yang, L. Pileggi & J.-G.-. G. Zhu. Naturally Oxidized FeCo as a Magnetic Coupling Layer for Electrically Isolated Read/Write Paths in mLogic. *IEEE Trans. Magn.* **49**, 4351-4354 (2013).

- ⁵¹⁰ D. M. Bromberg, M. T. Moneck, V. M. Sokalski, J. Zhu, L. Pileggi & J. G. Zhu. Experimental demonstration of four-terminal magnetic logic device with separate read- and write-paths. *Tech. Dig. - Int. Electron Devices Meet. IEDM 2015-Febru*, 33.1.1-33.1.4 (2015).
- ⁵¹¹ D. M. Bromberg, J. Zhu, L. Pileggi, V. Sokalski, M. Moneck & P. C. Richardson. (12) United States Patent.
- ⁵¹² J. G. J. Zhu, D. M. Bromberg, M. Moneck, V. Sokalski & L. Pileggi. MLogic: All spin logic device and circuits. *2015 4th Berkeley Symp. Energy Effic. Electron. Syst. E3S 2015 - Proc.* 1–2 (2015). doi:10.1109/E3S.2015.7336781
- ⁵¹³ N. Zogbi, S. Liu, C. H. Bennett, S. Agarwal, M. J. Marinella, J. A. C. Incorvia & T. P. Xiao. Massively Parallel Matrix Multiplication Using Voltage Controlled Magnetic Anisotropy Domain Wall Logic. *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits* **9**, 65–73 (2023).
- ⁵¹⁴ D. E. Nikonov & I. A. Young. Overview of beyond-CMOS devices and a uniform methodology for their benchmarking. *Proc. IEEE* **101**, 2498–2533 (2013).
- ⁵¹⁵ D. E. Nikonov & I. A. Young. Benchmarking of Beyond-CMOS Exploratory Devices for Logic Integrated Circuits. *IEEE J. Explor. Solid-State Comput. Devices Circuits* **1**, 3–11 (2015).
- ⁵¹⁶ D. E. Nikonov & I. A. Young. Uniform methodology for benchmarking beyond-CMOS logic devices. *Electron Devices Meet. (IEDM), 2012 IEEE Int.* 24–25 (2012).
- ⁵¹⁷ X. Hu, A. Timm, W. H. Brigner, J. A. C. Incorvia & J. S. Friedman. SPICE-only model for spin-transfer torque domain-wall MTJ logic. *IEEE Trans. Electron Devices* **66**, 2817–2821 (2019).
- ⁵¹⁸ Y. Kang, J. Bokor & V. Stojanović. Design Requirements for a Spintronic MTJ Logic Device for Pipelined Logic Applications. *IEEE Trans. Electron Devices* **63**, 1754–1761 (2016).
- ⁵¹⁹ T. P. Xiao, M. J. Marinella, C. H. Bennett, X. Hu, B. Feinberg, R. Jacobs-Gedrim, S. Agarwal, J. Brunhaver, J. S. Friedman & J. A. C. Incorvia. Energy and Performance Benchmarking of a Domain Wall-Magnetic Tunnel Junction Multibit Adder. *IEEE J. Explor. Solid-State Comput. Devices Circuits* **PP**, 1 (2019).
- ⁵²⁰ M. Alamdar, T. Leonard, C. Cui, B. P. Rimal, L. Xue, O. G. Akinola, T. P. Xiao, J. S. Friedman, C. H. Bennett, M. J. Marinella, & J. A. C. Incorvia. Domain wall-magnetic tunnel junction spin-orbit torque devices and circuits for in-memory computing. *Applied Physics Letters* **118**, 112401 (2021).
- ⁵²¹ T. Leonard, S. Liu, M. Alamdar, C. Cui, O. G. Akinola, L. Xue, T. P. Xiao, J. S. Friedman, M. J. Marinella, C. H. Bennett & J. A. C. Incorvia. Shape-Dependent Multi-Weight Magnetic Artificial Synapses for Neuromorphic Computing. *Adv Electron Mater* **2200563**, 1–11 (2022).
- ⁵²² T. Leonard, S. Liu, H. Jin & J. A. C. Incorvia. Stochastic domain wall-magnetic tunnel junction artificial neurons for noise-resilient spiking neural networks. *Appl Phys Lett* **122**, (2023).
- ⁵²³ E. Raymenants, O. Bultynck, D. Wan, T. Devolder, K. Garello, L. Souriau, A. Thiam, D. Tsvetanova, Y. Canvel, D. E. Nikonov, I. A. Young, M. Heyns, B. Soree, I. Asselberghs, I. Radu, S. Couet & V. D. Nguyen. Nanoscale domain wall devices with magnetic tunnel junction read and write. *Nat. Electron.* **4**, 392–398 (2021).
- ⁵²⁴ E. Raymenants, D. Wan, S. Couet, L. Souriau, A. Thiam, Di. Tsvetanova, Y. Canvel, K. Garello, G. S. Kar, M. Heyns, I. Asselberghs, D. E. Nikonov, I. A. Young, S. Pizzini, I. Radu & V. Dai Nguyen. All-Electrical Control of Scaled Spin Logic Devices Based on Domain Wall Motion. *IEEE Trans. Electron Devices* **68**, 2116–2122 (2021).
- ⁵²⁵ C. Garg, S. H. Yang, L. Thompson, T. Topuria, A. Capua, B. Hughes, T. Phung, P. C. Filippou & S. S. P. Parkin. Efficient chiral-domain-wall motion driven by spin-orbit torque in metastable platinum films. *Phys. Rev. Appl.* **14**, 1 (2020).
- ⁵²⁶ K. Cai, Z. Zhu, J. M. Lee, R. Mishra, L. Ren, S. D. Pollard, P. He, G. Liang, K. L. Teo & H. Yang. Ultrafast and energy-efficient spin-orbit torque switching in compensated ferrimagnets. *Nat. Electron.* **3**, 37–42 (2020).
- ⁵²⁷ A. Sengupta, Y. Shim & K. Roy. Proposal for an all-spin artificial neural network: Emulating neural and synaptic functionalities through domain wall motion in ferromagnets. *IEEE Trans. Biomed. Circuits Syst.* **10**, 1152–1160 (2016).
- ⁵²⁸ S. Dutta, S. A. Siddiqui, F. Buttner, L. Liu, C. A. Ross & M. A. Baldo. A logic-in-memory design with 3-Terminal magnetic tunnel junction function evaluators for convolutional neural networks. *Proc. IEEE/ACM Int. Symp. Nanoscale Archit. NANOARCH 2017* 83–88 (2017). doi:10.1109/NANOARCH.2017.8053724
- ⁵²⁹ D. Blaauw & Q. Dong. Binary-threshold, Rectified-linear and Recurrent Neural Networks Built with Spintronic Devices. *SPIN Neuron DAC (1).pdf* (2016).
- ⁵³⁰ R. Zand, K. Y. Camsari, S. D. Pyle, I. Ahmed, C. H. Kim & R. F. DeMara. Low-energy deep belief networks using intrinsic sigmoidal spintronic-based probabilistic neurons. in *Proceedings of the 2018 on Great Lakes Symposium on VLSI - GLSVLSI '18* 15–20 (ACM Press, 2018). doi:10.1145/3194554.3194558
- ⁵³¹ G. Srinivasan, A. Sengupta & K. Roy. Magnetic Tunnel Junction Based Long-Term Short-Term Stochastic Synapse for a Spiking Neural Network with On-Chip STDP Learning. *Sci Rep* **6**, 29545 (2016).
- ⁵³² O. Akinola, X. Hu, C. H. Bennett, M. Marinella, J. S. Friedman & J. A. C. Incorvia. Three-terminal magnetic tunnel junction synapse circuits showing spike-timing-dependent plasticity. *J. Phys. D. Appl. Phys.* **52**, 49LT01 (2019).
- ⁵³³ D. Bhowmik, U. Saxena, A. Dankar, A. Verma, D. Kaushik, S. Chatterjee & U. Singh. On-chip learning for domain wall synapse based Fully Connected Neural Network. *J. Magn. Mater.* **489**, 165434 (2019).
- ⁵³⁴ N. Hassan, X. Hu, L. Jiang-Wei, W. H. Brigner, O. G. Akinola, F. Garcia-Sanchez, M. Pasquale, C. H. Bennett, J. A. C. Incorvia & J. S. Friedman. Magnetic domain wall neuron with lateral inhibition. *J. Appl. Phys.* **124**, (2018).

- ⁵³⁵ W. H. Brigner, X. Hu, N. Hassan, C. H. Bennett, J. A. C. Incorvia, F. Garcia-Sanchez & J. S. Friedman. Graded-Anisotropy-Induced Magnetic Domain Wall Drift for an Artificial Spintronic Leaky Integrate-and-Fire Neuron. *IEEE J. Explor. Solid-State Comput. Devices Circuits* **5**, 1–1 (2019).
- ⁵³⁶ H. Vakili, W. Zhou, C. T. Ma, S. J. Poon, M. G. Morshed, M. N. Sakib, S. Ganguly, M. Stan, T. Q. Hartnett, P. Balachandran, J. W. Xu, Y. Quessab, A. D. Kent, K. Litzius, G. S. D. Beach & A. W. Ghosh. Skyrmionics-Computing and memory technologies based on topological excitations in magnets. *J. Appl. Phys.* **130**, (2021).
- ⁵³⁷ T. Leonard, S. Liu, M. Alamdar, C. Cui, O. G. Akinola, L. Xue, T. P. Xiao, J. S. Friedman, M. J. Marinella, C. H. Bennett & J. A. C. Incorvia. Shape-Dependent Multi-Weight Magnetic Artificial Synapses for Neuromorphic Computing. *Under review*. Arxiv Prepr. arXiv: 2111.11516 (2021).
- ⁵³⁸ C. Cui, O. G. Akinola, N. Hassan, C. H. Bennett, M. J. Marinella, J. S. Friedman & J. A. C. Incorvia. Maximized lateral inhibition in paired magnetic domain wall racetracks for neuromorphic computing. *Nanotechnology* **31**, (2020).
- ⁵³⁹ W. H. Brigner, J. S. Friedman, N. Hassan, L. Jiang-Wei, X. Hu, D. Saha, C. H. Bennett, M. J. Marinella, J. A. C. Incorvia & F. Garcia-Sanchez. Shape-Based Magnetic Domain Wall Drift for an Artificial Spintronic Leaky Integrate-and-Fire Neuron. *IEEE Trans. Electron Devices* **66**, 4970–4975 (2019).
- ⁵⁴⁰ S. Liu, C. H. Bennett, J. S. Friedman, M. J. Marinella, D. Paydarfar & J. A. Incorvia. Controllable reset behavior in domain wall-magnetic tunnel junction artificial neurons for task-adaptable computation. *IEEE Magn. Lett.* **11**, 1–5 (2021).
- ⁵⁴¹ S. Liu, T. P. Xiao, C. Cui, J. A. C. Incorvia, C. H. Bennett & J. Matthew. A domain wall-magnetic tunnel junction artificial synapse with notched geometry for accurate and efficient training of deep neural networks. *Appl. Phys. Lett.* **202405**, 1–6 (2021).
- ⁵⁴² S. Fukami, N. Ishiwata, N. Kasai, M. Yamanouchi, H. Sato, S. Ikeda & H. Ohno. Scalability prospect of three-terminal magnetic domain-wall motion device. *IEEE Trans. Magn.* **48**, 2152–2157 (2012).
- ⁵⁴³ A. Chumak et al., *IEEE Transactions on Magnetism*, **58**, 0800172 (2022).
- ⁵⁴⁴ A. Khitun and K. Wang, *Superlattices & Microstructures* **38**, 184 (2005).
- ⁵⁴⁵ A. Khitun, M. Bao and K. L. Wang, *IEEE Transactions on Magnetism* **44**, 2141 (2008).
- ⁵⁴⁶ M. P. Kostylev, A. A. Serga, T. Schneider, B. Leven, and B. Hillebrands, *Applied Physics Letters* **87**, 153501 (2005).
- ⁵⁴⁷ K.-S. Lee and S.-K. Kim, *Journal of Applied Physics* **104**, 053909 (2008).
- ⁵⁴⁸ G. Csaba, A. Papp, and W. Porod, *Physics Letters A* **381**, 1471 (2017).
- ⁵⁴⁹ R. Verba, V. Tiberkevich, I. Krivorotov, and A. Slavin, *Physical Review Applied* **1**, 044006 (2014).
- ⁵⁵⁰ Y.-J. Chen et al., *Nano Letters* **17**, 572 (2017).
- ⁵⁵¹ S. Dutta et al., *Scientific Reports* **5**, 09861 (2015).
- ⁵⁵² K. Vogt et al., *Nature Communications* **5**, 1 (2014).
- ⁵⁵³ B. Divinskiy et al., *Advanced Materials* **30**, 1802837 (2018).
- ⁵⁵⁴ L. J. Cornelissen, J. Liu, B. J. van Wees, and R. A. Duine, *Physical Review Letters* **120**, 097702 (2018).
- ⁵⁵⁵ J. Han, P. Zhang, J. T. Hou, A. A. Siddiqui and L. Liu, *Science* **366**, 1121 (2019).
- ⁵⁵⁶ R. Verba et al., *Physical Review Applied* **11**, 054040 (2019).
- ⁵⁵⁷ K. Baumgaertl and Dirk Grundler, *Nature Communications* **14**, 1490 (2023).
- ⁵⁵⁸ A. Khitun et al., *Journal of Nanoelectronics and Optoelectronics* **3**, 24 (2008).
- ⁵⁵⁹ A. Khitun and K. L. Wang, *Journal of Applied Physics* **110**, 034306 (2011).
- ⁵⁶⁰ S. Klingler et al., *Applied Physics Letters* **105**, 152410 (2014).
- ⁵⁶¹ A. Khitun, *Journal of Applied Physics* **113**, 164503 (2013).
- ⁵⁶² A. Khitun, *Journal of Applied Physics* **111**, 054307 (2012).
- ⁵⁶³ K. Wagner et al., *Nature Nanotechnology* **11**, 432 (2016).
- ⁵⁶⁴ Y. Wu et al., *Journal of Nanoelectronics and Optoelectronics* **4**, 394 (2009).
- ⁵⁶⁵ S. Cherepov et al., *Applied Physics Letters* **104**, 082403 (2014).
- ⁵⁶⁶ F. Gertz, A. Kozhevnikov, Y. Filimonov and A. Khitun, *Magnetism, IEEE Transactions on Magnetism* **51**, 1 (2014).
- ⁵⁶⁷ A. Kozhevnikov et al., *Applied Physics Letters* **106**, 142409 (2015).
- ⁵⁶⁸ A. Litvinenko et al., *Communications Physics* **6**, 227 (2023).
- ⁵⁶⁹ Y. Wang et al., *Science* **366**, 1125 (2019).
- ⁵⁷⁰ R. Salikhov et al., *Nature Physics* **19**, 529 (2023).
- ⁵⁷¹ C. Liu et al., *Nature Communications* **9**, 738 (2018).
- ⁵⁷² A. Khitun, D. E. Nikonov, and K. L. Wang, *Journal of Applied Physics* **106**, 123909 (2009).
- ⁵⁷³ R. Verba, V. Tiberkevich, I. Krivorotov, and A. Slavin, *Physical Review Applied* **1**, 044006 (2014).
- ⁵⁷⁴ B. Divinskiy et al., *Advanced Materials* **30**, 1802837 (2018).
- ⁵⁷⁵ P. Graczyk and M. Krawczyk, *Scientific Reports* **11**, 15692 (2021).
- ⁵⁷⁶ R. Lebrun et al., *Nature* **561**, 222 (2018).
- ⁵⁷⁷ J. Han et al., *Nature Nanotechnology* **15**, 563 (2020).
- ⁵⁷⁸ C. Jia, M. Chen, A. F. Schäffer, and J. Berakdar, *NPJ Computational Materials* **7**, 101 (2021).

- 579 J. Li et al., *Nature* **578**, 70 (2020).
- 580 P. Vaidya et al., *Science* **368**, 160 (2020).
- 581 Y.-Y. Jiang et al., *Physical Review B* **108**, 174439 (2023).
- 582 D. A. Connelly et al., *Scientific Reports* **11**, 18378 (2021).
- 583 C. Gong et al., *Nature* **546**, 265 (2017).
- 584 F. Schultz et al., *Nano Letters* **23**, 10126 (2023).
- 585 D. MacNeill, J. T. Hou, D. R. Klein, P. Zhang, P. Jarillo-Herrero, and L. Liu, *Physical Review Letters* **123**, 047204 (2019).
- 586 G. Chen et al., *Nature Communications* **12**, 6279 (2021).
- 587 K. Akarvardar et al., "Design considerations for complementary nanoelectromechanical logic gates," *IEEE International Electron Devices Meeting Technical Digest*, pp. 299–302, 2007.
- 588 F. Chen et al., "Integrated circuit design with NEM relays," *IEEE/ACM International Conference on Computer-Aided Design*, pp. 750–757, 2008.
- 589 H. Kam et al., "Design and reliability of a micro-relay technology for zero-standby-power digital logic applications," *IEEE International Electron Devices Meeting Technical Digest*, pp. 809–811, 2009.
- 590 Y.-B. Lee et al., "Sub-10 fJ/bit radiation-hard nanoelectromechanical non-volatile memory," *Nature Communications*, vol. 14, no. 1. Springer Science and Business Media LLC, Jan. 28, 2023. doi: 10.1038/s41467-023-36076-0.
- 591 U. Sikder et al., "Back-End-of-Line Nano-Electro-Mechanical Switches for Reconfigurable Interconnects," *IEEE Electron Device Letters*, vol. 41, no. 4, pp. 625–628, April 2020.
- 592 S. Natarajan et al., "A 14nm logic technology featuring 2nd-generation FinFET, air-gapped interconnects, self-aligned double patterning and a 0.0588 μm^2 SRAM cell size," *IEEE International Electron Devices Meeting Technical Digest*, pp. 3.7.1–3.7.3, 2014.
- 593 N. Xu et al., "Hybrid CMOS/BEOL-NEMS technology for ultra-low-power IC applications," *IEEE International Electron Devices Meeting Technical Digest*, pp. 28.8.1–28.8.4, 2014.
- 594 H. Fariborzi et al., "Analysis and demonstration of MEM-relay power gating," *Custom Integrated Circuits Conference*, 2010.
- 595 C. Chen et al., "Nano-electro-mechanical relays for FPGA routing: experimental demonstration and a design technique," *Design, Automation & Test in Europe Conference & Exhibition*, 2012.
- 596 J. S. Lee, J. Yoon and W. Y. Choi, "In-Memory Nearest Neighbor Search With Nanoelectromechanical Ternary Content-Addressable Memory," in *IEEE Electron Device Letters*, vol. 43, no. 1, pp. 154–157, Jan. 2022.
- 597 K. Kato, V. Stojanović and T.-J. K. Liu, "Embedded nano-electro-mechanical memory for energy-efficient reconfigurable logic," *IEEE Electron Device Letters*, vol. 37, no. 12, pp. 1563–1565, 2016.
- 598 J. O. Lee et al., "A sub-1-volt nanoelectromechanical switching device," *Nature Nanotechnology*, Vol. 8, pp. 36–40, 2013.
- 599 Feng, X. L., et al. "Low voltage nanoelectromechanical switches based on silicon carbide nanowires." *Nano Letters*, vol.10, no. 8, pp. 2891–2896, 2010.
- 600 T. He et al., "Silicon carbide (SiC) nanoelectromechanical switches and logic gates with long cycles and robust performance in ambient air and at high temperature," 2013 IEEE International Electron Devices Meeting, Washington, DC, 2013, pp. 4.6.1–4.6.4.
- 601 U. Zaghloul and G. Piazza, "Sub-1-volt Piezoelectric Nanoelectromechanical Relays With Millivolt Switching Capability," in *IEEE Electron Device Letters*, vol. 35, no. 6, pp. 669–671, June 2014.
- 602 R. Nathanael et al., "4-terminal relay technology for complementary logic," *IEEE International Electron Devices Meeting Technical Digest*, pp. 223–226, 2009.
- 603 Z. A. Ye et al., "Body-Biased Multiple-Gate Micro-Electro-Mechanical Relays," in *IEEE Electron Device Letters*, vol. 42, no. 3, pp. 402–405, March 2021.
- 604 B. Smith, M. A. Mamun, B. Horstmann, Ü. Özgür and V. Avrutin, "Multi-Gate In-Plane Actuated NEMS Relays for Effective Complementary Logic Gate Designs," in *Journal of Microelectromechanical Systems*, vol. 32, no. 6, pp. 604–611, Dec. 2023.
- 605 T.-J. K. Liu et al., "Prospects for MEM_relay logic switch technology," *IEEE International Electron Devices Meeting Technical Digest*, pp. 424–427, 2010.
- 606 H. Kam et al., "Design, optimization and scaling of MEM relays for ultra-low-power digital logic," *IEEE Transactions on Electron Devices*, Vol. 58, No. 1, pp. 236–250, 2011.
- 607 C. Qian et al., "Energy-delay performance optimization of NEM logic relay," *IEEE International Electron Devices Meeting Technical Digest*, pp. 18.1.1–18.1.4, 2015.
- 608 F. Chen et al., "Demonstration of integrated micro-electro-mechanical (MEM) switch circuits for VLSI applications," *International Solid State Circuits Conference*, pp. 150–151, 2010.
- 609 H. Fariborzi et al., "Design and demonstration of micro-electro-mechanical relay multipliers," *IEEE Asian Solid-State Circuits Conference*, 2011.
- 610 M. Spencer et al., "Demonstration of integrated micro-electro-mechanical relay circuits for VLSI applications," *IEEE Journal of Solid-State Circuits*, Vol. 46, No. 1, pp. 308–320, 2011.
- 611 L. Hutin et al., "Electromechanical diode cell scaling for high-density nonvolatile memory," *IEEE Transactions on Electron Devices*, Vol. 61, pp. 1382–1387, 2014.

- ⁶¹² H. S. Kwon et al., "Monolithic three-dimensional 65-nm CMOS-nanoelectromechanical reconfigurable logic for sub-1.2V operation," *IEEE Electron Device Letters*, vol. 38, no. 9, pp. 1317–1320, 2017.
- ⁶¹³ T.-J. K. Liu et al., "There's plenty of room at the top," in 2017 IEEE 30th International Conference on Micro Electro Mechanical Systems (MEMS). IEEE, 2017, pp. 1–4.
- ⁶¹⁴ U. Sikder, et al., "Vertical NV-NEM switches in CMOS Back-End-of-Line: first experimental demonstration and array programming scheme," 2020 IEEE International Electron Devices Meeting (IEDM), 2020, pp. 21.2.1-21.2.4.
- ⁶¹⁵ U. Sikder et al., "Toward Monolithically Integrated Hybrid CMOS-NEM Circuits," in *IEEE Transactions on Electron Devices*, vol. 68, no. 12, pp. 6430–6436, Dec. 2021.
- ⁶¹⁶ H. Kim, M. Cho, S. Lee, H. S. Kwon, W. Y. Choi, and Y. Kim, "Content-Addressable Memory System Using a Nanoelectromechanical Memory Switch," *Electronics*, vol. 11, no. 3. MDPI AG, p. 481, Feb. 07, 2022.
- ⁶¹⁷ T.-S. Kim, Y.-B. Lee, S.-H. Kim, S.-Y. Lee, S.-J. Lee, and J.-B. Yoon, "Monolithically 3-D Integrated Nanoelectromechanical (NEM) Configuration Memory for CMOS-NEM Hybrid Demultiplexer," *IEEE Electron Device Letters*, vol. 44, no. 12. Institute of Electrical and Electronics Engineers (IEEE), pp. 2055–2058, Dec. 2023.
- ⁶¹⁸ Y. Chen et al., "Reliability of MEM relays for zero leakage logic," *SPIE Conference Proceedings* Vol. 8614, 2013.
- ⁶¹⁹ E. Jo, Y. Kang, S. Sim, H. Lee, and J. Kim, "High-Temperature-Operable Electromechanical Computing Units Enabled by Aligned Carbon Nanotube Arrays," *ACS Nano*, vol. 17, no. 14. American Chemical Society (ACS), pp. 13310–13318, Jul. 07, 2023.
- ⁶²⁰ B. Saha et al., "Reducing adhesion energy of micro-relay electrodes by ion beam synthesized oxide nanolayers," in *APL Materials*, vol. 5, no. 3, pp. 036103, 2017.
- ⁶²¹ Z. A. Ye et al., "Demonstration of 50-mV Digital Integrated Circuits with Microelectromechanical Relays," 2018 IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, 2018, pp. 4.1.1-4.1.4.
- ⁶²² X. Hu et al., "Ultra-Low-Voltage Operation of MEM Relays for Cryogenic Logic Applications," 2019 IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, 2019, pp. 34.2.1-34.2.4.
- ⁶²³ X. Hu et al., "Study of DC-driven MEM Relay Oscillators for Implementation of Ising Machines," 2021 IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, 2022, pp. 23.6.1-23.6.4.
- ⁶²⁴ Chiu, C.-K., et al., *Classification of topological quantum matter with symmetries*. *Reviews of Modern Physics*, 2016. **88**(3).
- ⁶²⁵ Hasan, M.Z. and C.L. Kane, *Colloquium: Topological insulators*. *Reviews of Modern Physics*, 2010. **82**(4): p. 3045-3067.
- ⁶²⁶ Ren, Y., Z. Qiao, and Q. Niu, *Topological phases in two-dimensional materials: a review*. *Rep Prog Phys*, 2016. **79**(6): p. 066501.
- ⁶²⁷ Weng, H., et al., Quantum anomalous Hall effect and related topological electronic states. *Advances in Physics*, 2015. **64**(3): p. 227-282.
- ⁶²⁸ Fei, Z., et al., *Edge conduction in monolayer WTe₂*. *Nature Physics*, 2017. **13**(7): p. 677-682.
- ⁶²⁹ König, M., et al., Quantum Spin Hall Insulator State in HgTe Quantum Wells. *Science*, 2007. **318**: p. 766
- ⁶³⁰ Liu, C., et al., Helical Edge Transport in Millimeter-Scale Thin Films of Na₃Bi. *Nano Lett*, 2020. **20**(9): p. 6306-6312.
- ⁶³¹ Tang, F., et al., Comprehensive search for topological materials using symmetry indicators. *Nature*, 2019. **566**(7745): p. 486-489.
- ⁶³² Vergniory, M.G., et al., *A complete catalogue of high-quality topological materials*. *Nature*, 2019. **566**(7745): p. 480-485.
- ⁶³³ Zhang, T., et al., *Catalogue of topological electronic materials*. *Nature*, 2019. **566**(7745): p. 475-479.
- ⁶³⁴ Collins, J.L., et al., Electric-field-tuned topological phase transition in ultrathin Na₃Bi. *Nature*, 2018. **564**(7736): p. 390-394.
- ⁶³⁵ F. Reis, G.L., L. Dudy, M. Bauernfeind, S. Glass, W. Hanke, R. Thomale, J. Schäfer, R. Claessen, *Bismuthene on a SiC substrate: A candidate for a high-temperature quantum spin Hall material*. *Science*, 2017. **357**: p. 287-290.
- ⁶³⁶ Ezawa, M., Quantized conductance and field-effect topological quantum transistor in silicene nanoribbons. *Applied Physics Letters*, 2013. **102**(17).
- ⁶³⁷ Kane, C.L. and E.J. Mele, *Z₂ topological order and the quantum spin Hall effect*. *Phys Rev Lett*, 2005. **95**(14): p. 146802.
- ⁶³⁸ Liu, J., et al., Spin-filtered edge states with an electrically tunable gap in a two-dimensional topological crystalline insulator. *Nat Mater*, 2014. **13**(2): p. 178-83.
- ⁶³⁹ Liu, Q., et al., Switching a normal insulator into a topological insulator via electric field with application to phosphorene. *Nano Lett*, 2015. **15**(2): p. 1222-8.
- ⁶⁴⁰ Molle, A., et al., *Buckled two-dimensional Xene sheets*. *Nat Mater*, 2017. **16**(2): p. 163-169.
- ⁶⁴¹ Ni, Z., et al., *Tunable bandgap in silicene and germanene*. *Nano Lett*, 2012. **12**(1): p. 113-8.
- ⁶⁴² Pan, H., et al., Electric control of topological phase transitions in Dirac semimetal thin films. *Sci Rep*, 2015. **5**: p. 14639.
- ⁶⁴³ Qian, X., et al., Quantum spin Hall effect in two-dimensional transition metal dichalcogenides. *Science*, 2014. **346**: p. 1344-1347.
- ⁶⁴⁴ Vandenbergh, W.G. and M.V. Fischetti, Imperfect two-dimensional topological insulator field-effect transistors. *Nat Commun*, 2017. **8**: p. 14184.
- ⁶⁴⁵ Wray, L.A., *Topological transistor*. *Nature Physics*, 2012. **8**(10): p. 705-706.
- ⁶⁴⁶ Zhang, Z., et al., Magnetic quantum phase transition in Cr-doped Bi₂(SexTe_{1-x})₃ driven by the Stark effect. *Nat Nanotechnol*, 2017. **12**(10): p. 953-957.
- ⁶⁴⁷ Nadeem, M., et al., Overcoming Boltzmann's Tyranny in a Transistor via the Topological Quantum Field Effect. *Nano Lett*, 2021. **21**(7): p. 3155-3161.

- ⁶⁴⁸ Fuhrer, M.S., et al., Proposal for a Negative Capacitance Topological Quantum Field-Effect Transistor, in 67th Annual IEEE International Electron Devices Meeting (IEDM). 2021: San Francisco.
- ⁶⁴⁹ Bampoulis, P., et al., Quantum Spin Hall States and Topological Phase Transition in Germanene. *Physical Review Letters*, 2023. **130**(19): p. 196401.
- ⁶⁵⁰ Xu, Y., et al., *Quantized Field-Effect Tunneling between Topological Edge or Interface States*. *Phys Rev Lett*, 2019. **123**(20): p. 206801.
- ⁶⁵¹ Banerjee, S., et al., Robust Subthermionic Topological Transistor Action via Antiferromagnetic Exchange. *Physical Review Applied* (2022). **18**(5): p. 054088.
- ⁶⁵² Salahuddin, S. and S. Datta, Use of Negative Capacitance to Provide Voltage Amplification for Low Power Nanoscale Devices. *Nano Letters*, 2008. **8**: p. 405.
- ⁶⁵³ Cao, W. and K. Banerjee, *Is negative capacitance FET a steep-slope logic switch?* *Nat Commun*, 2020. **11**(1): p. 196.
- ⁶⁵⁴ Nikonov, D.E. and I.A. Young, *Benchmarking of Beyond-CMOS Exploratory Devices for Logic Integrated Circuits*. *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, 2015. **1**: p. 3-11.
- ⁶⁵⁵ Kadykov, A.M., et al., *Terahertz detection of magnetic field-driven topological phase transition in HgTe-based transistors*. *Applied Physics Letters*, 2015. **107**(15): p. 152101.
- ⁶⁵⁶ Ezawa, M., Topological Switch between Second-Order Topological Insulators and Topological Crystalline Insulators. *Phys Rev Lett*, 2018. **121**(11): p. 116801.
- ⁶⁵⁷ Guan, S., et al., *Artificial gravity field, astrophysical analogues, and topological phase transitions in strained topological semimetals*. *npj Quantum Materials*, 2017. **2**(1): p. 23.
- ⁶⁵⁸ Krishtopenko, S.S., et al., *Pressure- and temperature-driven phase transitions in HgTe quantum wells*. *Physical Review B*, 2016. **94**(24): p. 245402.
- ⁶⁵⁹ Liang, C., et al., Polarization-Driven Topological-Insulator Transition for Piezotronic Field-Effect Transistors with Subthreshold Swing of 5 mV/decade. *Physical Review Applied*, 2023. **20**(1): p. 014061.
- ⁶⁶⁰ Lindner, N.H., G. Refael, and V. Galitski, *Floquet topological insulator in semiconductor quantum wells*. *Nature Physics*, 2011. **7**(6): p. 490-495.
- ⁶⁶¹ Zheng, J., et al., *All-Optically Controlled Topological Transistor Based on Xenon*. *Physical Review Applied*, 2020. **14**(3): p. 034027.
- ⁶⁶² Chang, C.-Z., et al., Experimental Observation of the Quantum Anomalous Hall Effect in a Magnetic Topological Insulator. *Science* 2013. **340**: p. 167-170
- ⁶⁶³ Bhattacharyya, S., et al., Recent Progress in Proximity Coupling of Magnetism to Topological Insulators. *Adv Mater*, 2021. **33**(33): p. e2007795.
- ⁶⁶⁴ Serlin, M., et al., Intrinsic quantized anomalous Hall effect in a moiré heterostructure. *Science*, 2019. **367**(6480): p. 900-903.
- ⁶⁶⁵ Li, J. and Wu, R., Electrically Tunable Topological Phase Transition in van der Waals Heterostructures. *Nano Letters*, 2023. **23**(6): pp. 2173–2178.
- ⁶⁶⁶ You, J.Y., Dong, X.J., Gu, B. and Su, G, Electric field induced topological phase transition and large enhancements of spin-orbit coupling and Curie temperature in two-dimensional ferromagnetic semiconductors. *Physical Review B*, 2021. **103**(10): p. 104403.
- ⁶⁶⁷ Ju, L., et al., Topological valley transport at bilayer graphene domain walls. *Nature*, 2015. **520**(7549): p. 650-5.
- ⁶⁶⁸ Li, J., et al., Gate-controlled topological conducting channels in bilayer graphene. *Nat Nanotechnol*, 2016. **11**(12): p. 1060-1065.
- ⁶⁶⁹ Li, J., et al., *A valley valve and electron beam splitter*. *Science* 2018. **362**: p. 1149-1152.
- ⁶⁷⁰ D. Niu, Y. Xiao, and Y. Xie, "Low power memristor-based ReRAM design with error correcting code," in *2012 17th Asia and South Pacific Design Automation Conference (ASP-DAC)*, pp. 79–84, 2012.
- ⁶⁷¹ L. Bao et al., "Hybrid-Domain In-Memory Polynomial Acceleration based on 40nm RRAM Multi-Core Chip for Machine Vision Calibration," *2023 International Electron Devices Meeting (IEDM)*, San Francisco, CA, USA, 2023, pp. 1-4.
- ⁶⁷² S. Barve et al., "Power-Efficient Clustering Using Programmable VT FETs in Neuromorphic Architectures," *2023 International Electron Devices Meeting (IEDM)*, San Francisco, CA, USA, 2023, pp. 1-4
- ⁶⁷³ H. Lim, V. Kornijcuk, J. Y. Seok, S. K. Kim, I. Kim, C. S. Hwang, et al., "Reliability of neuronal information conveyed by unreliable neuristor-based leaky integrate-and-fire neurons: a model study," *Scientific Reports*, vol. 5, p. 9776, May 13, 2015.
- ⁶⁷⁴ T. Tuma, A. Pantazi, M. Le Gallo, A. Sebastian, and E. Eleftheriou, "Stochastic phase-change neurons," *Nature Nanotechnology*, vol. 11, pp. 693–9, Aug 2016.
- ⁶⁷⁵ H. Lim, H. W. Ahn, V. Kornijcuk, G. Kim, J. Y. Seok, I. Kim, et al., "Relaxation oscillator-realized artificial electronic neurons, their responses, and noise," *Nanoscale*, vol. 8, pp. 9629–40, May 14, 2016.
- ⁶⁷⁶ Boahen, K. Dendrocentric learning for synthetic intelligence. *Nature* **612**, 43–50 (2022).
- ⁶⁷⁷ J. . -Y. Chen et al., "Multi-gate FeFET Discriminates Spatiotemporal Pulse Sequences for Dendrocentric Learning," *2023 International Electron Devices Meeting (IEDM)*, San Francisco, CA, USA, 2023, pp. 1-4
- ⁶⁷⁸ T. P. Xiao, C. H. Bennett, B. Feinberg, S. Agarwal, and M. J. Marinella, "Analog architectures for neural network acceleration based on non-volatile memory," *Applied Physics Reviews*, vol. 7, 031301, 2020.
- ⁶⁷⁹ R. Genov and G. Cauwenberghs, "Charge-mode parallel architecture for vector-matrix multiplication," *IEEE Transactions on Circuits and Systems II: Analog and Digital Signal Processing*, vol. 48, no. 10, pp. 930–936, 2001.

- ⁶⁸⁰ A. Coates, B. Huval, T. Wang, D. J. Wu, A. Y. Ng, and B. Catanzaro, "Deep learning with COTS HPC systems," *Proceedings of the 30th International Conference on Machine Learning (ICML)*, Atlanta, Georgia, USA, 2013.
- ⁶⁸¹ T.-H. Yang, H.-Y. Cheng, C.-L. Yang, I.-C. Tseng, H.-W. Hu, H.-S. Chang, and H.-P. Li, "Sparse ReRAM Engine: Joint Exploration of Activation and Weight Sparsity in Compressed Neural Networks," *ACM/IEEE International Symposium on Computer Architecture (ISCA)*, Phoenix, AZ, USA, 2019.
- ⁶⁸² V. Bohossian, P. Hasler, and J. Bruck, "Programmable neural logic," *IEEE Transactions on Components, Packaging, and Manufacturing Technology, Part B: Advanced Packaging*, vol. 21, no. 4, pp. 346–351, Nov. 1998.
- ⁶⁸³ R. Robucci, J. Gray, Leung Kin Chiu, J. Romberg, and P. Hasler, "Compressive Sensing on a CMOS Separable-Transform Image Sensor," *IEEE proceedings*, vol. 98, no. 1, 2010, pp. 1089–1101.
- ⁶⁸⁴ S. Koziol, D. Lenz, S. Hilsenbeck, S. Chopra, P. Hasler, and A. Howard, "Using floating-gate based programmable analog arrays for real-time control of a game-playing robot," *Systems, Man, and Cybernetics (SMC), 2011 IEEE International Conference on*, 2011, pp. 3566–3571.
- ⁶⁸⁵ P. Hasler and J. Dugger, "An analog floating-gate node for supervised learning," *IEEE Transactions on Circuits and Systems I*, vol. 52, no. 5, pp. 834–845, May 2005.
- ⁶⁸⁶ S. Suh, A. Basu, C. Schlottmann, P. Hasler, and J. Barry, "Low-power discrete Fourier transform for OFDM: A programmable analog approach," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 58, no. 2, pp. 290–298, 2011.
- ⁶⁸⁷ S. Ramakrishnan and J. Hasler, "Vector-matrix multiply and winner-take-all as an analog classifier," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 22, no. 2, pp. 353–361, 2014.
- ⁶⁸⁸ S. Agarwal *et al.*, "Energy Scaling Advantages of Resistive Memory Crossbar Based Computation and its Application to Sparse Coding," *Frontiers in Neuroscience*, vol. 9, p. 484, 2016, Art. no. 484.
- ⁶⁸⁹ P. Narayanan *et al.*, "Toward on-chip acceleration of the backpropagation algorithm using nonvolatile memory," *IBM Journal of Research and Development*, vol. 61, no. 4/5, pp. 11:1–11:7, 2017.
- ⁶⁹⁰ M. J. Marinella *et al.*, "Multiscale Co-Design Analysis of Energy, Latency, Area, and Accuracy of a ReRAM Analog Neural Training Accelerator," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 8, no. 1, pp. 86–101, 2018.
- ⁶⁹¹ T. P. Xiao *et al.*, "Analysis and mitigation of parasitic resistance effects for analog in-memory neural network acceleration," *Semiconductor Science and Technology*, vol. 36, no. 11, pp. 114004, 2021.
- ⁶⁹² G. W. Burr *et al.*, "Access Devices for 3D Crosspoint Memory," *Journal of Vacuum Science and Technology B*, vol. 32, no. 4, p. 040802, 2014.
- ⁶⁹³ M. Bavandpour, S. Sahay, M. R. Mahmoodi, and D. Strukov, "Efficient Mixed-Signal Neurocomputing via Successive Integration and Rescaling," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 2019.
- ⁶⁹⁴ V. Reddi *et al.*, "MLPerf Inference Benchmark," *ACM/IEEE International Symposium on Computer Architecture (ISCA)*, Valencia, 2020.
- ⁶⁹⁵ T. P. Xiao *et al.*, "On the Accuracy of Analog Neural Network Inference Accelerators," *arXiv preprint*, arXiv:2109.01262, 2021.
- ⁶⁹⁶ S. Han, H. Mao, and W. J. Dally, "Deep Compression: Compressing Deep Neural Networks with Pruning, Trained Quantization and Huffman Coding," *arXiv preprint*, arXiv:1510.00149, 2016.
- ⁶⁹⁷ T. P. Xiao *et al.*, "An Accurate, Error-Tolerant, and Energy-Efficient Neural Network Inference Engine Based on SONOS Analog Memory," *IEEE Transactions on Circuits and Systems I: Regular Papers*, pp. 1–14, 2022.
- ⁶⁹⁸ M. N. Bojnordi and E. Ipek, "Memristive Boltzmann machine: A hardware accelerator for combinatorial optimization and deep learning," *2016 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, Barcelona, 2016, pp. 1–13.
- ⁶⁹⁹ A. Shafiee *et al.*, "ISAAC: A Convolutional Neural Network Accelerator with In-Situ Analog Arithmetic in Crossbars," *2016 ACM/IEEE 43rd Annual International Symposium on Computer Architecture (ISCA)*, Seoul, 2016, pp. 14–26.
- ⁷⁰⁰ P. Chi *et al.*, "PRIME: A Novel Processing-in-Memory Architecture for Neural Network Computation in ReRAM-Based Main Memory," *2016 ACM/IEEE 43rd Annual International Symposium on Computer Architecture (ISCA)*, Seoul, 2016, pp. 27–39.
- ⁷⁰¹ L. Song, Y. Zhuo, X. Qian, H. Li and Y. Chen, "GraphR: Accelerating Graph Processing Using ReRAM," *2018 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, Vienna, 2018, pp. 531–543.
- ⁷⁰² B. Feinberg, U. K. R. Vengalam, N. Whitehair, S. Wang and E. Ipek, "Enabling Scientific Computing on Memristive Accelerators," *2018 ACM/IEEE 45th Annual International Symposium on Computer Architecture (ISCA)*, Los Angeles, CA, 2018, pp. 367–382.
- ⁷⁰³ W. Wan, R. Kubendran, C. Schafer, S. B. Eryilmaz, W. Zhang, D. Wu, S. Deiss, P. Raina, H. Qian, B. Gao, S. Joshi, H. Wu, H.-S. P. Wong, and G. Cauwenberghs, "A compute-in-memory chip based on resistive random-access memory," *Nature*, vol. 608, pp. 504–512, 2022.
- ⁷⁰⁴ B. E. Boser, E. Sackinger, J. Bromley, Y. Le Cun, and L. D. Jackel, "An analog neural network processor with programmable topology," *IEEE Journal of Solid-State Circuits*, vol. 26, no. 12, pp. 2017–2025, 1991.
- ⁷⁰⁵ F. J. Kub, K. K. Moon, I. A. Mack, and F. M. Long, "Programmable analog vector-matrix multipliers," *IEEE Journal of Solid-State Circuits*, vol. 25, no. 1, pp. 207–214, 1990.
- ⁷⁰⁶ S. Kim, T. Gokmen, H.-M. Lee, and W. F. Haensch, "Analog CMOS-based resistive processing unit for deep neural network training," *IEEE International Midwest Symposium on Circuits and Systems (MWSCAS)*, 2017.
- ⁷⁰⁷ Rishabh Sehgal and Jaydeep P. Kulkarni, "Trends in Analog and Digital Intensive Compute-in-SRAM Designs," in *2021 IEEE 3rd International Conference on Artificial Intelligence Circuits and Systems (AICAS)*, 2021, 1–4, <https://doi.org/10.1109/AICAS51828.2021.9458576>.

- ⁷⁰⁸ Mingu Kang, Sujan K. Gonugondla, and Naresh R. Shanbhag, "A 19.4 nJ/Decision 364K Decisions/s in-Memory Random Forest Classifier in 6T SRAM Array," in *ESSCIRC 2017 - 43rd IEEE European Solid State Circuits Conference* (ESSCIRC 2017 - 43rd IEEE European Solid-State Circuits Conference, Leuven: IEEE, 2017), 263–66, <https://doi.org/10.1109/ESSCIRC.2017.8094576>.
- ⁷⁰⁹ Rishabh Sehgal et al., "Compute-MLROM: Compute-in-Multi Level Read Only Memory for Energy Efficient Edge AI Inference Engines," in *ESSCIRC 2023- IEEE 49th European Solid State Circuits Conference (ESSCIRC)*, 2023, 37–40, <https://doi.org/10.1109/ESSCIRC59616.2023.10268726>.
- ⁷¹⁰ Naresh R. Shanbhag and Saion K. Roy, "Benchmarking In-Memory Computing Architectures," *IEEE Open Journal of the Solid-State Circuits Society* 2 (2022): 288–300, <https://doi.org/10.1109/OJSSCS.2022.3210152>.
- ⁷¹¹ Pouya Houshmand, Jiacong Sun, and Marian Verhelst, "Benchmarking and Modeling of Analog and Digital SRAM In-Memory Computing Architectures" (arXiv, May 25, 2023), <https://doi.org/10.48550/arXiv.2305.18335>.
- ⁷¹² Mustafa Ali et al., "IMAC: In-Memory Multi-Bit Multiplication and ACcumulation in 6T SRAM Array," *IEEE Transactions on Circuits and Systems I: Regular Papers* 67, no. 8 (August 2020): 2521–31, <https://doi.org/10.1109/TCSI.2020.2981901>; Jian-Wei Su et al., "16.3 A 28nm 384kb 6T-SRAM Computation-in-Memory Macro with 8b Precision for AI Edge Chips," in *2021 IEEE International Solid-State Circuits Conference (ISSCC)*, vol. 64, 2021, 250–52, <https://doi.org/10.1109/ISSCC42613.2021.9365984>.
- ⁷¹³ Rishabh Sehgal et al., "A Bit-Serial, Compute-in-SRAM Design Featuring Hybrid-Integrating ADCs and Input Dependent Binary Scaled Precharge Eliminating DACs for Energy-Efficient DNN Inference," *IEEE Journal of Solid-State Circuits*, 2023, 1–16, <https://doi.org/10.1109/JSSC.2023.3235210>.
- ⁷¹⁴ Avishek Biswas and Anantha P. Chandrakasan, "CONV-SRAM: An Energy-Efficient SRAM With In-Memory Dot-Product Computation for Low-Power Convolutional Neural Networks," *IEEE Journal of Solid-State Circuits* 54, no. 1 (January 2019): 217–30, <https://doi.org/10.1109/JSSC.2018.2880918>.
- ⁷¹⁵ Xin Si et al., "A Local Computing Cell and 6T SRAM-Based Computing-in-Memory Macro With 8-b MAC Operation for Edge AI Chips," *IEEE Journal of Solid-State Circuits* 56, no. 9 (September 2021): 2817–31, <https://doi.org/10.1109/JSSC.2021.3073254>.
- ⁷¹⁶ Jintao Zhang, Zhuo Wang, and Naveen Verma, "In-Memory Computation of a Machine-Learning Classifier in a Standard 6T SRAM Array," *IEEE Journal of Solid-State Circuits* 52, no. 4 (April 2017): 915–24, <https://doi.org/10.1109/JSSC.2016.2642198>.
- ⁷¹⁷ Mingu Kang, Sujan K. Gonugondla, and Naresh R. Shanbhag, "Deep In-Memory Architectures in SRAM: An Analog Approach to Approximate Computing," *Proceedings of the IEEE* 108, no. 12 (December 2020): 2251–75, <https://doi.org/10.1109/JPROC.2020.3034117>.
- ⁷¹⁸ Xin Si et al., "A Twin-8T SRAM Computation-in-Memory Unit-Macro for Multibit CNN-Based AI Edge Processors," *IEEE Journal of Solid-State Circuits* 55, no. 1 (January 2020): 189–202, <https://doi.org/10.1109/JSSC.2019.2952773>.
- ⁷¹⁹ Sehgal et al., "A Bit-Serial, Compute-in-SRAM Design Featuring Hybrid-Integrating ADCs and Input Dependent Binary Scaled Precharge Eliminating DACs for Energy-Efficient DNN Inference."
- ⁷²⁰ Amogh Agrawal et al., "CASH-RAM: Enabling In-Memory Computations for Edge Inference Using Charge Accumulation and Sharing in Standard 8T-SRAM Arrays," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems* 10, no. 3 (September 2020): 295–305, <https://doi.org/10.1109/JETCAS.2020.3014250>.
- ⁷²¹ Shanbhag and Roy, "Benchmarking In-Memory Computing Architectures."
- ⁷²² Yu-Der Chih et al., "16.4 An 89TOPS/W and 16.3TOPS/Mm² All-Digital SRAM-Based Full-Precision Compute-In Memory Macro in 22nm for Machine-Learning Edge Applications," in *2021 IEEE International Solid-State Circuits Conference (ISSCC)*, vol. 64, 2021, 252–54, <https://doi.org/10.1109/ISSCC42613.2021.9365766>.
- ⁷²³ Chih et al.
- ⁷²⁴ Je-Min Hung et al., "An 8-Mb DC-Current-Free Binary-to-8b Precision ReRAM Nonvolatile Computing-in-Memory Macro Using Time-Space-Readout with 1286.4-21.6TOPS/W for Edge-AI Devices," in *2022 IEEE International Solid-State Circuits Conference (ISSCC)*, vol. 65, 2022, 1–3, <https://doi.org/10.1109/ISSCC42614.2022.9731715>.
- ⁷²⁵ Shanshan Xie et al., "Gain-Cell CIM: Leakage and Bitline Swing Aware 2T1C Gain-Cell eDRAM Compute in Memory Design with Bitline Precharge DACs and Compact Schmitt Trigger ADCs," in *2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, 2022, 112–13, <https://doi.org/10.1109/VLSITechnologyandCirc46769.2022.9830338>.
- ⁷²⁶ Tanvi Sharma et al., "WWW: What, When, Where to Compute-in-Memory" (arXiv, June 20, 2024), <https://doi.org/10.48550/arXiv.2312.15896>.
- ⁷²⁷ D. Soudry, D. D. Castro, A. Gal, A. Kolodny, and S. Kvatinsky, "Memristor-Based Multilayer Neural Networks With Online Gradient Descent Training," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 26, no. 10, 2015.
- ⁷²⁸ D. Kadedotad et al., "Parallel Architecture With Resistive Crosspoint Array for Dictionary Learning Acceleration," *Emerging and Selected Topics in Circuits and Systems, IEEE Journal on*, vol. 5, no. 2, pp. 194–204, 2015.
- ⁷²⁹ I. Gao et al., "Fully parallel write/read in resistive synaptic array for accelerating on-chip learning," *Nanotechnology*, vol. 26, 455204, 2015.
- ⁷³⁰ G. W. Burr et al., "Experimental Demonstration and Tolerancing of a Large-Scale Neural Network (165 000 Synapses) Using Phase-Change Memory as the Synaptic Weight Element," *Electron Devices, IEEE Transactions on*, vol. 62, no. 11, pp. 3498–3507, 2015.
- ⁷³¹ P. Y. Chen, L. Gao, and S. Yu, "Design of Resistive Synaptic Array for Implementing On-Chip Sparse Learning," *IEEE Transactions on Multi-Scale Computing Systems*, vol. 2, no. 4, pp. 257–264, 2016.
- ⁷³² S. Agarwal et al., "Resistive memory device requirements for a neural algorithm accelerator," *International Joint Conference on Neural Networks (IJCNN)*, Vancouver, BC, Canada, 2016.

- ⁷³³ B. D. Hoskins, et al., "Streaming Batch Eigenupdates for Hardware Neural Networks," *Frontiers in Neuroscience*, vol. 13, no. 793, 2019.
- ⁷³⁴ I. Richter et al., "Memristive Accelerator for Extreme Scale Linear Solvers," *40th Annual GOMACTech Conference*, St. Louis, MO, 2015.
- ⁷³⁵ Sun, Z., Pedretti, G., Ambrosi, E., Bricalli, A., Wang, W., Ielmini, D. Solving matrix equations in one step with crosspoint resistive arrays. *Proc. Natl. Acad. Sci. (PNAS)* 116(10), 4123-4128 (2019).
- ⁷³⁶ Sun, Z., Pedretti, G., Ielmini, D. Fast solution of linear systems with analog resistive switching memory (RRAM). 2019 International Conference on Rebooting Computing (ICRC) 2019.
- ⁷³⁷ Sun, Z., Pedretti, G., Bricalli, A., Ielmini, D. One-step regression and classification with crosspoint resistive memory arrays. *Sci. Adv.* (2019). Accepted for publication
- ⁷³⁸ S. R. Nandakumar, M. Le Gallo, I. Boybat, B. Rajendran, A. Sebastian, and E. Eleftheriou, "Mixed-precision architecture based on computational memory for training deep neural networks." *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2018.
- ⁷³⁹ B. Feinberg et al., "An Analog Preconditioner for Solving Linear Systems," *2021 IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, Seoul, Korea (South), 2021, pp. 761-774, doi: 10.1109/HPCA51647.2021.00069.
- ⁷⁴⁰ Xiaobo Sharon Hu, Michael T. Niemier, Arman Kazemi, Ann Franchesca B. Laguna, Kai Ni, Ramin Rajaei, Mohammad Mehdi Sharifi, Xunzhao Yin, "In-Memory Computing with Associative Memories: A Cross-Layer Perspective," *IEEE International Electron Devices Meeting (IEDM)*, 2021.
- ⁷⁴¹ H. Amrouch et al., "Cross-layer Design for Computing-in-Memory: From Devices, Circuits, to Architectures and Applications," *2021 26th Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2021, pp. 132-139.
- ⁷⁴² Ni, Kai, et al. "Ferroelectric ternary content-addressable memory for one-shot learning." *Nature Electronics* 2.11 (2019): 521-529.
- ⁷⁴³ Kazemi, Arman, et al. "In-memory nearest neighbor search with 1T1R multi-bit content-addressable memories." *2021 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 2021.
- ⁷⁴⁴ Kazemi, Arman, et al. "A flash-based multi-bit content-addressable memory with euclidean squared distance." *2021 IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*. IEEE, 2021.
- ⁷⁴⁵ Yin, Xunzhao, et al. "FeCAM: A universal compact digital and analog content addressable memory using ferroelectric." *IEEE Transactions on Electron Devices* 67.7 (2020): 2785-2792.
- ⁷⁴⁶ Rajaei, Ramin, et al. "Compact single-phase-search multistate content-addressable memory design using one FeFET/cell." *IEEE Transactions on Electron Devices* 68.1 (2020): 109-117.
- ⁷⁴⁷ Li, Chao, et al. "A scalable design of multi-bit ferroelectric content addressable memory for data-centric computing." *2020 IEEE International Electron Devices Meeting (IEDM)*. IEEE, 2020.
- ⁷⁴⁸ Li, C., Graves, C. E., Sheng, X., Miller, D., Foltin, M., Pedretti, G., & Strachan, J. P. (2020). Analog content-addressable memories with memristors. *Nature communications*, 11(1), 1-8.
- ⁷⁴⁹ Li, Jing, Robert Montoye, Masatoshi Ishii, Kevin Stawiasz, Takeshi Nishida, Kim Maloney, Gary Ditlow et al. "1Mb 0.41 μm^2 2T-2R cell nonvolatile TCAM with two-bit encoding and clocked self-referenced sensing." In *2013 Symposium on VLSI Technology*, pp. C104-C105. IEEE, 2013.
- ⁷⁵⁰ Yang, H. Z., et al. "A Novel High-Density and Low-Power Ternary Content Addressable Memory Design Based on 3D NAND Flash." *2020 IEEE Silicon Nanoelectronics Workshop (SNW)*. IEEE, 2020.
- ⁷⁵¹ Wang, Fei, et al. "Implementation of data search in multi-level NAND flash memory by complementary storage scheme." *IEEE Electron Device Letters* 41.8 (2020): 1189-1192.
- ⁷⁵² Matsunaga, Shoun, et al. "Design of a nine-transistor/two-magnetic-tunnel-junction-cell-based low-energy nonvolatile ternary content-addressable memory." *Japanese Journal of Applied Physics* 51.2S (2012): 02BM06.
- ⁷⁵³ C. Wang, D. Zhang, L. Zeng, E. Deng, J. Chen and W. Zhao, "A Novel MTJ-Based Non-Volatile Ternary Content-Addressable Memory for High-Speed, Low-Power, and High-Reliable Search Operation," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 66, no. 4, pp. 1454-1464, April 2019, doi: 10.1109/TCSI.2018.2885343.
- ⁷⁵⁴ Guo, Qing, et al. "A resistive TCAM accelerator for data-intensive computing." *Proceedings of the 44th Annual IEEE/ACM International Symposium on Microarchitecture*. ACM, 2011.
- ⁷⁵⁵ L. Yavits, S. Kvatinisky, A. Morad, and R. Ginosar, "Resistive Associative Processor," *IEEE Computer Architecture Letters*, vol. 14, no. 2, pp. 148-151, 2015.
- ⁷⁵⁶ Shinde, Rajendra, et al. "Similarity search and locality sensitive hashing using ternary content addressable memories." *Proceedings of the 2010 ACM SIGMOD International Conference on Management of data*. ACM, 2010.
- ⁷⁵⁷ M. Imani, D. Peroni, A. Rahimi, and T. Rosing, "Resistive CAM Acceleration for Tunable Approximate Computing," *IEEE Transactions on Emerging Topics in Computing*, vol. PP, no. 99, p. 1, 2016.
- ⁷⁵⁸ Ly, Denys RB, et al. "In-depth Characterization of Resistive Memory-Based Ternary Content Addressable Memories." *2018 IEEE International Electron Devices Meeting (IEDM)*. IEEE, 2018
- ⁷⁵⁹ C. Yakopcic, V. Bontupalli, R. Hasan, D. Mountain, and T. M. Taha, "Self-biasing memristor crossbar used for string matching and ternary content-addressable memory implementation," *Electronics Letters*, vol. 53, no. 7, pp. 463-465, 2017.
- ⁷⁶⁰ Graves, Catherine E., et al. "Regular Expression Matching with Memristor TCAMs." *2018 IEEE International Conference on Rebooting Computing (ICRC)*. IEEE, 2018.

- ⁷⁶¹ Zhi-Hua Zhou and Ji Feng, Deep Forest: Towards An Alternative to Deep Neural Networks, Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence, IJCAI-17, p 3553–3559, 2017.
- ⁷⁶² IMCRYPTO: An In-Memory Computing Fabric for AES Encryption and Decryption, Dayane Reis and Haoran Geng and Michael Niemier and Xiaobo Sharon Hu, 2021, 2112.02231, arXiv.
- ⁷⁶³ C. Schlottmann and P. Hasler, “A highly dense, low power, programmable analog vector-matrix multiplier: The FPAA implementation,” *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 1, no. 3, pp. 403–411, 2011.
- ⁷⁶⁴ C. Schlottmann, S. Shapero, S. Nease, and P. Hasler, “A digitally enhanced dynamically reconfigurable analog platform for low-power signal processing,” *IEEE Journal of Solid-State Circuits*, vol. 47, no. 9, pp. 2174–2184, 2012.
- ⁷⁶⁵ S. George, S. Kim, S. Shah, J. Hasler, M. Collins, F. Adil, R. Wunderlich, S. Nease, and S. Ramakrishnan, “A Programmable and Configurable Mixed-Mode FPAA SOC,” *IEEE Transactions on Very Large Scale Integration Systems*, vol. 24, no. 6, 2016. pp. 2253–2261.
- ⁷⁶⁶ J. Hasler, “Large-Scale Field-Programmable Analog Arrays,” IEEE Proceedings, Dec 2019
- ⁷⁶⁷ C. M. Twigg, J. D. Gray, and P. Hasler, “Programmable floating gate FPAA switches are not dead weight,” *IEEE International Symposium on Circuits and Systems*, May 2007, pp. 169–172.
- ⁷⁶⁸ S. Shapero and P. Hasler, “Mismatch characterization and calibration for accurate and automated analog design,” *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 60, no. 3, pp. 548–556, 2013.
- ⁷⁶⁹ S. Kim, S. Shah, and J. Hasler, “Calibration of Floating-Gate SoC FPAA System,” IEEE Transactions on VLSI, 2017.
- ⁷⁷⁰ S. Shah, H. Toreyin, J. Hasler, and A. Natarajan, “Models and Techniques For Temperature Robust Systems On A Reconfigurable Platform,” *Journal of Low Power Electronics Applications*, vol. 7, no. 21, August 2017. pp. 1–14.
- ⁷⁷¹ J. Hasler, C. Scholtmann, and S. Kozioł, “FPAA chips and tools as the center of an design-based analog systems education,” *IEEE Microelectronic Systems Education (MSE)*, 2011, pp. 47–51.
- ⁷⁷² M. Collins, J. Hasler, and S. George, “Analog systems education: An integrated toolset and FPAA SoC boards,” *Microelectronics Systems Education (MSE)*, 2015, pp. 32–35.
- ⁷⁷³ J. Hasler and H. Wang, “A Fine-Grain FPAA fabric for RF+Baseband,” *GOMAC*, March 2015.
- ⁷⁷⁴ J. Hasler, S. Kim, and F. Adil, “Scaling Floating-Gate Devices Predicting Behavior for Programmable and Configurable Circuits and Systems,” *Journal of Low Power Electronics Applications*, vol. 6, no. 13, 2016, pp. 1–19.
- ⁷⁷⁵ P. Blouw, X. Choo, E. Hunsberger, and C. Eliasmith, “Benchmarking keyword spotting efficiency on neuromorphic hardware,” Apr. 2019. arXiv:1812.01739. [Online]. Available: <https://arxiv.org/abs/1812.01739>
- ⁷⁷⁶ W. A. Wulf and S. A. McKee, “Hitting the memory wall: implications of the obvious”, *SIGARCH Computer Architecture News*, vol 23, no. 1, 1995.
- ⁷⁷⁷ V. Sze, Y.-H. Chen, T.-J. Yang, and J. S. Emer, “Efficient Processing of Deep Neural Networks: A Tutorial and Survey,” *Proceedings of the IEEE*, vol. 105, no. 12, pp. 2295–2329, 2017.
- ⁷⁷⁸ X. Guo et al, “Fast, energy-efficient, robust, and reproducible mixed-signal neuromorphic classifier based on embedded NOR flash memory technology,” *International Electron Devices Meeting (IEDM)*, pp. 6.5.1-6.5.4, 2017.
- ⁷⁷⁹ P. Yao, H. Wu, B. Gao, J. Tang, Q. Zhang, W. Zhang, J. J. Yang, and H. Qian, “Fully hardware-implemented memristor convolutional neural network,” *Nature*, vol. 577, no. 7792, pp. 641–646, 2020.
- ⁷⁸⁰ A. Ankit, et al., “PUMA: A programmable ultra-efficient memristor-based accelerator for machine learning inference,” *International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, pp. 715–731, 2019.
- ⁷⁸¹ M. Hu, et al., “Dot-product engine for neuromorphic computing: Programming 1T1M crossbar to accelerate matrix-vector multiplication,” *Design Automation Conference (DAC)*, pp. 1–6, 2016.
- ⁷⁸² V. Joshi, et al., “Accurate deep neural network inference using computational phase-change memory,” *Nature Communications*, vol. 11, no. 2473, 2020.
- ⁷⁸³ A. Nag, et al., “Newton: Gravitating towards the physical limits of crossbar acceleration,” *IEEE Micro*, vol. 38, no. 5, pp. 41–49, 2018.
- ⁷⁸⁴ L. Fick, S. Skrzyniarz, M. Parikh, M. B. Henry, and D. Fick, “Analog matrix processor for edge AI real-time video analytics,” *IEEE International Solid-State Circuits Conference (ISSCC)*, vol. 65, pp. 260–262, 2022.
- ⁷⁸⁵ W.-S. Khwa, Y.-C. Chiu, C.-J. Jhang, S.-P. Huang, C.-Y. Lee, T.-H. Wen, F.-C. Chang, S.-M. Yu, T.-Y. Lee, and M.-F. Chang, “A 40-nm, 2M-cell, 8b-precision, hybrid SLC-MLC PCM Computing-in-Memory Macro with 20.5–65.0 TOPS/W for Tiny-AI Edge Devices,” *IEEE International Solid-State Circuits Conference (ISSCC)*, vol. 65, pp. 1–3, 2022.
- ⁷⁸⁶ S. Ambrogio et al., “An analog-AI chip for energy-efficient speech recognition and transcription,” *Nature*, vol. 620, no. 7975, pp. 768–775, 2023.
- ⁷⁸⁷ Z. He, J. Lin, R. Ewetz, J.-S. Yuan, and D. Fan, “Noise injection adaption: End-to-end ReRAM crossbar non-ideal effect adaption for neural network mapping,” *Design Automation Conference (DAC)*, pp. 57:1–57:6, 2019.
- ⁷⁸⁸ S. Jain and A. Raghunathan, “CxDNN: Hardware-software compensation methods for deep neural networks on resistive crossbar systems,” *ACM Transactions on Embedded Computer Systems*, vol. 18, no. 6, 2019.
- ⁷⁸⁹ M. Klachko, M. R. Mahmoodi, and D. Strukov, “Improving noise tolerance of mixed-signal neural networks,” *International Joint Conference on Neural Networks (IJCNN)*, pp. 1–8, 2019.

- ⁷⁹⁰ Y. Long, X. She, and S. Mukhopadhyay, "Design of reliable DNN accelerator with unreliable ReRAM," *Design, Automation Test in European Conference Exhibition (DATE)*, pp. 1769-1774, 2019.
- ⁷⁹¹ Y. Jeong, M. A. Zidan, and W. D. Lu, "Parasitic effect analysis in memristor-array-based neuromorphic systems," *IEEE Transactions on Nanotechnology*, vol. 17, no. 1, pp. 184-193, 2018.
- ⁷⁹² F. Zhang and M. Hu, "Mitigate parasitic resistance in resistive crossbar-based convolutional neural networks," *ACM Journal on Emerging Technologies in Computing Systems*, vol. 16, no. 3, 2020.
- ⁷⁹³ S. Ambrogio, et al., "Reducing the Impact of Phase-Change Memory Conductance Drift on the Inference of Large-Scale Hardware Neural Networks," *IEEE International Electron Devices Meeting (IEDM)*, 2019.
- ⁷⁹⁴ J. Chen and X. Ran, "Deep learning with edge computing: A review," *Proceedings of the IEEE*, vol. 107, no. 8, pp. 1655-1674, 2019.
- ⁷⁹⁵ X. Xu, Y. Ding, S. X. Hu, M. Niemier, J. Cong, Y. Hu, and Y. Shi, "Scaling for edge inference of deep neural networks," *Nature Electronics*, vol. 1, no. 4, pp. 216-222, 2018.
- ⁷⁹⁶ A. Sebastian, M. Le Gallo, R. Khaddam-Aljameh, and E. Eleftheriou, "Memory devices and applications for in-memory computing," *Nature Nanotechnology*, vol. 15, no. 7, pp. 529-544, 2020.
- ⁷⁹⁷ M.-K. Song et al., "Recent Advances and Future Prospects for Memristive materials, Devices, and Systems," *ACS Nano*, vol. 17, no. 13, pp. 11994-12039, 2023.
- ⁷⁹⁸ M. Le Gallo et al., "A 64-core mixed-signal in-memory compute chip based on phase-change memory for deep neural network inference," *Nature Electronics*, vol. 6, no. 9, pp. 680-693, 2023.
- ⁷⁹⁹ V. Agrawal, T. P. Xiao, C. H. Bennett, B. Feinberg, S. Shetty, K. Ramkumar, H. Medu, K. Thekkekara, R. Chettuvetty, S. Leshner, Z. Luzada, L. Hinh, T. Phan, M. J. Marinella, and S. Agarwal, "Subthreshold operation of SONOS analog memory to enable accurate low-power neural network inference," *IEEE International Electron Devices Meeting (IEDM)*, 21.7.1-21.7.4, 2022.
- ⁸⁰⁰ W. Song et al., "Programming memristor arrays with arbitrarily high precision for analog computing," *Science*, vol. 383, no. 6685, pp. 903-910, 2024.
- ⁸⁰¹ Y. Huang, T. Ando, A. Sebastian, M.-F. Chang, J. J. Yang, and Q. Xia, "Memristor-based hardware accelerators for artificial intelligence," *Nature Reviews Electrical Engineering*, vol. 1, no. 5, pp. 286-299, 2024.
- ⁸⁰² M. D. Pickett, G. Medeiros-Ribeiro, and R. S. Williams, "A scalable neuristor built with Mott memristors," *Nature materials*, vol. 12, no. 2, pp. 114-117, 2013.
- ⁸⁰³ Locatelli, N., Cros, V. & Grollier, J. Spin-torque building blocks. *Nature Mater* **13**, 11–20 (2014).
- ⁸⁰⁴ Tuma, T., Pantazi, A., Le Gallo, M. et al. Stochastic phase-change neurons. *Nature Nanotech* **11**, 693–699 (2016).
- ⁸⁰⁵ J. M. Shainline, S. M. Buckley, A. N. McCaughan, J. T. Chiles, A. J. Salim, M. Castellanos-Beltran, C. A. Donnelly, M. L. Schneider, R. P. Mirin, and S. W. Nam, "Superconducting optoelectronic loop neurons," *Journal of Applied Physics*, vol. 126, no. 4, p. 044902, 2019
- ⁸⁰⁶ A. R. Young, M. E. Dean, J. S. Plank and G. S. Rose, "A Review of Spiking Neuromorphic Hardware Communication Systems," in *IEEE Access*, vol. 7, pp. 135606-135620, 2019, doi: 10.1109/ACCESS.2019.2941772.
- ⁸⁰⁷ S. R. Nandakumar, et al., "Mixed-precision deep learning based on computational memory," *Frontiers in Neuroscience*, vol. 14, no. 406, 2020.
- ⁸⁰⁸ C. H. Bennett, D. Garland, R. B. Jacobs-Gedrim, S. Agarwal, and M. J. Marinella, "Wafer-Scale TaO_x device variability and implications for neuromorphic computing applications," *IEEE International Reliability Physics Symposium (IRPS)*, pp. 1-4, 2019.
- ⁸⁰⁹ X. Sun and S. Yu, "Impact of Non-Ideal Characteristics of Resistive Synaptic Devices on Implementing Convolutional Neural Networks," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 9, no. 3, pp. 570-579, 2019.
- ⁸¹⁰ R. B. Jacobs-Gedrim et al., "Impact of Linearity and Write Noise of Analog Resistive Memory Devices in a Neural Algorithm Accelerator," presented at the *IEEE International Conference on Rebooting Computing (ICRC)* Washington, DC, November 2017.
- ⁸¹¹ Murat Onen et al., Nanosecond protonic programmable resistors for analog deep learning. *Science* **377**,539-543(2022).
- ⁸¹² S. Agarwal et al., "Achieving ideal accuracies in analog neuromorphic computing using periodic carry," *VLSI Technology, 2017 Symposium on*, 2017, pp. T174–T175: IEEE.
- ⁸¹³ I. Boybat et al., "Improved Deep Neural Network Hardware-Accelerators Based on Non-Volatile-Memory: The Local Gains Technique," *2017 IEEE International Conference on Rebooting Computing (ICRC)*, 2017, pp. 1–8.
- ⁸¹⁴ Gokmen, Tayfun, and Wilfried Haensch. "Algorithm for Training Neural Networks on Resistive Device Arrays." *arXiv preprint arXiv:1909.07908* (2019).
- ⁸¹⁵ S. Ambrogio, P. Narayanan, H. Tsai, R. M. Shelby, I. Boybat, C. di Nolfo, S. Sidler, et al. "Equivalent-Accuracy Accelerated Neural-Network Training Using Analogue Memory," *Nature*, vol. 558, no. 7708, pp. 60–67, 2018.
- ⁸¹⁶ S. Agarwal et al. (2017). *CrossSim*. Available: <http://cross-sim.sandia.gov>.
- ⁸¹⁷ T. Gokmen and Y. Vlasov, "Acceleration of deep neural network training with resistive cross-point devices," *arXiv preprint arXiv:1603.07341*, 2016.
- ⁸¹⁸ Zenke, F., and E. O. Neftci. "Brain-Inspired Learning on Neuromorphic Substrates." *Proceedings of the IEEE*, 2021, 1–16. <https://doi.org/10.1109/JPROC.2020.3045625>.
- ⁸¹⁹ Richards, Blake A, Timothy P Lillicrap, Philippe Beaudoin, Yoshua Bengio, Rafal Bogacz, Amelia Christensen, Claudia Clopath, et al. "A Deep Learning Framework for Neuroscience." *Nature Neuroscience* **22**, no. 11 (2019): 1761–70.

- 820 Gerstner, Wulfram, Marco Lehmann, Vasiliki Liakoni, Dane Corneil, and Johanni Brea. "Eligibility Traces and Plasticity on Behavioral Time Scales: Experimental Support of Neohebbian Three-Factor Learning Rules." *Frontiers in Neural Circuits* 12 (2018): 53.
- 821 Bellec, Guillaume, Franz Scherr, Anand Subramoney, Elias Hajek, Darjan Salaj, Robert Legenstein, and Wolfgang Maass. "A Solution to the Learning Dilemma for Recurrent Networks of Spiking Neurons." *Nature Communications* 11, no. 1 (2020): 1–15.
- 822 Yik, Jason, et al. "The neurobench framework for benchmarking neuromorphic computing algorithms and systems." *Nature Communications* 16.1 (2025): 1545.
- 823 Lillicrap, Timothy P, Adam Santoro, Luke Marris, Colin J Akerman, and Geoffrey Hinton. "Backpropagation and the Brain." *Nature Reviews Neuroscience*, 2020, 1–12.
- 824 Illing, Bernd, Wulfram Gerstner, and Guillaume Bellec. "Towards Truly Local Gradients with CLAPP: Contrastive, Local And Predictive Plasticity." *ArXiv Preprint ArXiv:2010.08262*, 2020.
- 825 Whittington, James C. R., and Rafal Bogacz. "Theories of Error Back-Propagation in the Brain." *Trends in Cognitive Sciences* 23, no. 3 (2019): 235–50. <https://doi.org/10.1016/j.tics.2018.12.005>.
- 826 Halvagal, Manu Srinath, and Friedemann Zenke. "The combination of Hebbian and predictive plasticity learns invariant object representations in deep sensory networks." *Nature Neuroscience* 26.11 (2023): 1906–1915.
- 827 Wybo, Willem AM, et al. "NMDA-driven dendritic modulation enables multitask representation learning in hierarchical sensory processing pathways." *Proceedings of the National Academy of Sciences* 120.32 (2023)
- 828 Ellenberger, Benjamin, et al. "Backpropagation through space, time, and the brain." *arXiv preprint arXiv:2403.16933* (2024).
- 829 Laskin, Michael, Luke Metz, Seth Nabarrao, Mark Saroufim, Badreddine Nouné, Carlo Luschi, Jascha Sohl-Dickstein, and Pieter Abbeel. "Parallel Training of Deep Networks with Local Updates." *ArXiv Preprint ArXiv:2012.03837*, 2020.
- 830 Dalgaty, Thomas, et al. "Mosaic: in-memory computing and routing for small-world spike-based neuromorphic systems." *Nature Communications* 15.1 (2024): 142.
- 831 Stewart, Kenneth M., and Emre O. Neftci. "Meta-learning spiking neural networks with surrogate gradient descent." *Neuromorphic Computing and Engineering* 2.4 (2022): 044002.
- 832 Yu, Zhenming, Nathan Leroux, and Emre Neftci. "Training-to-Learn with Memristive Devices." 2022 International Electron Devices Meeting (IEDM). IEEE, 2022.
- 833 Hinton, Geoffrey. "The forward-forward algorithm: Some preliminary investigations." *arXiv preprint arXiv:2212.13345* 2.3 (2022): 5.
- 834 Ali Momeni et al., Backpropagation-free training of deep physical neural networks. *Science* 382, 1297–1303 (2023).
- 835 P. Kanerva, "Hyperdimensional computing: An introduction to computing in distributed representation with high-dimensional random vectors," *Cognitive computation*, vol. 1, no. 2, pp. 139–159, 2009.
- 836 E. P. Frady, D. Kleyko, C. J. Kymn, B. A. Olshausen, and F. T. Sommer, "Computing on Functions Using Randomized Vector Representations," *arXiv preprint arXiv:2109.03429*, 2021.
- 837 Justin C. Wong. "Negative capacitance and hyperdimensional computing for unconventional low-power computing." PhD thesis, EECS Department, University of California, Berkeley, Dec 2018.
- 838 G. E. Hinton, "Mapping part-whole hierarchies into connectionist networks," *Artificial Intelligence*, vol. 46, no. 1-2, pp. 47–75, 1990.
- 839 T. Plate, "Holographic Reduced Representations: Convolution Algebra for Compositional Distributed Representations," in *IJCAI*, 1991: Citeseer, pp. 30–35.
- 840 Z. Zou, Y. Kim, F. Imani, H. Alimohamadi, R. Cammarota, and M. Imani, "Scalable edge-based hyperdimensional learning system with brain-like neural adaptation," in *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, 2021, pp. 1–15.
- 841 A. Mitrokhin, P. Sutor, C. Fermüller, and Y. Aloimonos, "Learning sensorimotor control with neuromorphic sensors: Toward hyperdimensional active perception," *Science Robotics*, vol. 4, no. 30, 2019.
- 842 A. Hernandez-Cane, N. Matsumoto, E. Ping, and M. Imani, "Onlinehd: Robust, efficient, and single-pass online learning using hyperdimensional system," in *2021 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, 2021: IEEE, pp. 56–61.
- 843 G. Karunaratne et al., "Robust high-dimensional memory-augmented neural networks," *Nature communications*, vol. 12, no. 1, pp. 1–12, 2021.
- 844 D. Kleyko et al., "Vector Symbolic Architectures as a Computing Framework for Nanoscale Hardware," *arXiv preprint arXiv:2106.05268*, 2021.
- 845 M. Imani, A. Rahimi, D. Kong, T. Rosing, and J. M. Rabaey, "Exploring hyperdimensional associative memory," in *2017 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, 2017: IEEE, pp. 445–456.
- 846 G. Karunaratne, M. Le Gallo, G. Cherubini, L. Benini, A. Rahimi, and A. Sebastian, "In-memory hyperdimensional computing," *Nature Electronics*, vol. 3, no. 6, pp. 327–337, 2020.
- 847 H. Li et al., "Hyperdimensional computing with 3D VRRAM in-memory kernels: Device-architecture co-design for energy-efficient, error-resilient language recognition," in *2016 IEEE International Electron Devices Meeting (IEDM)*, 2016: IEEE, pp. 16.1. 1–16.1. 4.
- 848 P. Poduval, Z. Zou, H. Najafi, H. Homayoun, and M. Imani, "Stochd: Stochastic hyperdimensional system for efficient and robust learning from raw data," in *2021 58th ACM/IEEE Design Automation Conference (DAC)*, 2021: IEEE, pp. 1195–1200.
- 849 A. Rahimi, P. Kanerva, and J. M. Rabaey, "A robust and energy-efficient classifier using brain-inspired hyperdimensional computing," in *Proceedings of the 2016 International Symposium on Low Power Electronics and Design*, 2016, pp. 64–69.

- ⁸⁵⁰ A. Hernández-Cano, R. Cammarota, and M. Imani, "Prid: Model inversion privacy attacks in hyperdimensional learning systems," in *2021 58th ACM/IEEE Design Automation Conference (DAC)*, 2021: IEEE, pp. 553-558.
- ⁸⁵¹ M. Imani *et al.*, "A framework for collaborative learning in secure high-dimensional space," in *2019 IEEE 12th International Conference on Cloud Computing (CLOUD)*, 2019: IEEE, pp. 435-446.
- ⁸⁵² R. Thapa, D. Ma, and X. Jiao, "HDXplore: Automated Blackbox Testing of Brain-Inspired Hyperdimensional Computing," *arXiv preprint arXiv:2105.12770*, 2021.
- ⁸⁵³ F. Yang and S. Ren, "Adversarial Attacks on Brain-Inspired Hyperdimensional Computing-Based Classifiers," *arXiv preprint arXiv:2006.05594*, 2020.
- ⁸⁵⁴ S. Aaronson, *Quantum Computing since Democritus*, Cambridge University Press, 2013.
- ⁸⁵⁵ T. P. Xiao, "Optoelectronics for refrigeration and analog circuits for combinatorial optimization," Ph.D. dissertation, Chapter 6, University of California, Berkeley, 2019.
- ⁸⁵⁶ T. Wang and J. Roychowdhury, "Oscillator-based Ising machine," *arXiv e-prints*, 2017. *arXiv*: 1709.08102
- ⁸⁵⁷ T. Wang, L. Wu, and J. Roychowdhury, "New computational results and hardware prototypes for oscillator-based Ising machines," *Proceedings of the 56th Annual Design Automation Conference*, 2019.
- ⁸⁵⁸ J. Chou, S. Bramhavar, S. Ghosh, and W. Herzog, "Analog coupled oscillator based weighted Ising machine," *arXiv e-prints*, 2019. *arXiv*: 1906.06312
- ⁸⁵⁹ A. Lucas, "Ising formulations of many NP problems," *Interdisciplinary Physics*, vol. 2, p. 5, 2014.
- ⁸⁶⁰ Z. Wang, A. Marandi, K. Wen, R. L. Byer, and Y. Yamamoto, "Coherent Ising machine based on degenerate optical parametric oscillators," *Phys. Rev. A*, vol. 88, p. 063853, 2013.
- ⁸⁶¹ T. Inagaki *et al.*, "A coherent Ising machine for 2000-node optimization problems," *Science*, vol. 354, no. 6312, pp. 603-606, 2016.
- ⁸⁶² P. L. McMahon, A. Marandi, Y. Haribara, R. Hamerly, C. Langrock, S. Tamate, T. Inagaki, H. Takesue, S. Utsunomiya, K. Aihara, R. L. Byer, M. M. Fejer, H. Mabuchi, and Y. Yamamoto, "A fully-programmable 100-spin coherent Ising machine with all-to-all connections," *Science*, p. aah5178, Oct. 2016.
- ⁸⁶³ I. Mahboob, H. Okamoto, and H. Yamaguchi, "An electromechanical Ising Hamiltonian," *Science Advances*, vol. 2, no. 6, 2016.
- ⁸⁶⁴ S. K. Vadlamani, T. P. Xiao, E. Yablonovitch, "Physics successfully implements Lagrange multiplier optimization," *Proceedings of the National Academy of Sciences*, vol. 117, no. 43, pp. 26639-26650, 2020.
- ⁸⁶⁵ Lo, H., Moy, W., Yu, H. *et al.* An Ising solver chip based on coupled ring oscillators with a 48-node all-to-all connected array architecture. *Nat Electron* **6**, 771-778 (2023).
- ⁸⁶⁶ S. K. Vadlamani, "Sharp Switching in Tunnel Transistors and Physics-based Machines for Optimization," Ph. D. dissertation, Chapter 5, University of California, Berkeley, 2021.
- ⁸⁶⁷ T. Leleu, Y. Yamamoto, P. L. McMahon, K. Aihara, "Destabilization of local minima in analog spin systems by correction of amplitude heterogeneity," *Physical Review Letters*, vol. 122, no. 4, p. 040607, 2019.
- ⁸⁶⁸ E. Goto, "New Parametron circuit element using nonlinear reactance," *KDD Kenkyu Shiryo*, 1954.
- ⁸⁶⁹ J. von Neumann, "Non-linear Capacitance or Inductance Switching, Amplifying, and Memory Organs," U. S. Patent No. 2,815,488, Dec. 1957.
- ⁸⁷⁰ E. Goto, "The parametron, a digital computing element which utilizes parametric oscillation," *Proceedings of the IRE*, vol. 47, no. 8, pp. 1304-1316, 1959.
- ⁸⁷¹ R. L. Wightington, "A New Concept in Computing," in *Proceedings of the IRE*, vol. 47, no. 4, pp. 516-523, April 1959, [doi:10.1109/JRPROC.1959.287311](https://doi.org/10.1109/JRPROC.1959.287311).
- ⁸⁷² M. Ercsey-Ravasz and Z. Toroczkai, "Optimization hardness as transient chaos in an analog approach to constraint satisfaction," *Nature Physics*, vol. 7, no. 12, pp. 966-970, Dec. 2011.
- ⁸⁷³ F. L. Traversa, C. Ramella, F. Bonani, and M. Di Ventra, "Memcomputing NP-complete problems in polynomial time using polynomial resources and collective states," *Science Advances*, vol. 1, no. 6, pp. e1500031-e1500031, Jul. 2015.
- ⁸⁷⁴ V. Elser, I. Rankenburg, and P. Thibault, "Searching with iterated maps," *Proceedings of the National Academy of Sciences*, vol. 104, no. 2, pp. 418-423, Jan. 2007.
- ⁸⁷⁵ G. Appa, L. S. Pitsoulis, and H. P. Williams, Eds., *Handbook on modelling for discrete optimization*. New York: Springer, 2006.
- ⁸⁷⁶ Soos, M., Nohl, K., and Castelluccia, C. "Extending SAT solvers to cryptographic problems." International Conference on Theory and Applications of Satisfiability Testing. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009.
- ⁸⁷⁷ Prasad, M. R., Biere, A., and Gupta, A. "A survey of recent advances in SAT-based formal verification." International Journal on Software Tools for Technology Transfer **7** (2005): 156-173
- ⁸⁷⁸ Davis, M., and Putnam, H.. "A computing procedure for quantification theory." Journal of the ACM (JACM) **7.3** (1960): 201-215.
- ⁸⁷⁹ Schoning, T. "A probabilistic algorithm for k-SAT and constraint satisfaction problems." 40th Annual Symposium on Foundations of Computer Science (Cat. No. 99CB37039). IEEE, 1999
- ⁸⁸⁰ Park, S., Nam, J. W., and Gupta, S. K. "HW-BCP: A Custom Hardware Accelerator for SAT Suitable for Single Chip Implementation for Large Benchmarks." Proceedings of the 26th Asia and South Pacific Design Automation Conference. 2021.
- ⁸⁸¹ Pedretti, G., *et al.* "Differentiable content addressable memory with memristors." Advanced electronic materials **8.8** (2022): 2101198.

- 882 Pedretti, G., et al. "Zeroth and higher-order logic with content addressable memories." 2023 International Electron Devices Meeting (IEDM). IEEE, 2023.
- 883 Mohseni, N., McMahon, P. L., and Byrnes, T. "Ising machines as hardware solvers of combinatorial optimization problems." *Nature Reviews Physics* 4.6 (2022): 363-379
- 884 Cai, F., et al. "Power-efficient combinatorial optimization using intrinsic noise in memristor Hopfield neural networks." *Nature Electronics* 3.7 (2020): 409-418
- 885 Fleury, A. B. K. F. M., and Heisinger, M. "Cadical, kissat, paracooba, plingeling and treengeling entering the sat competition 2020." SAT COMPETITION 2020 (2020): 50.
- 886 J. J. Hopfield, "Neural networks and physical systems with emergent collective computational abilities," *Proceedings of the National Academy of Sciences*, vol. 79, no. 8, pp. 2554–2558, Apr. 1982.
- 887 F. C. Hoppensteadt and E. M. Izhikevich, "Oscillatory neurocomputers with dynamic connectivity," *Physical Review Letters*, vol. 82, no. 14, p. 2983, 1999.
- 888 S. Levitan, Y. Fang, D. Dash, T. Shibata, D. Nikonov, and G. Bourianoff, "Non-Boolean associative architectures based on nano-oscillators," *2012 13th International Workshop on Cellular Nanoscale Networks and Their Applications (CNNA)*, 2012, pp. 1–6.
- 889 G. Csaba, M. Pufall, D. Nikonov, G. Bourianoff, A. Horvath, T. Roska, and W. Porod, "Spin torque oscillator models for applications in associative memories," *2012 13th International Workshop on Cellular Nanoscale Networks and Their Applications (CNNA)*, 2012, pp. 1–2.
- 890 M. Namba and Z. Zhang, "Cellular neural network for associative memory and its application to Braille image recognition," in *Neural Networks, 2006. IJCNN'06. International Joint Conference on*, 2006, pp. 2409–2414.
- 891 D. Liu and A. N. Michel, "Cellular neural networks for associative memories," *Circuits and Systems II: Analog and Digital Signal Processing, IEEE Transactions on*, vol. 40, pp. 119–121, 1993.
- 892 P. Szolgay, I. Szatmari, and K. Laszlo, "A fast fixed point learning method to implement associative memory on CNNs," *Circuits and Systems I: Fundamental Theory and Applications, IEEE Transactions on*, vol. 44, pp. 362–366, 1997.
- 893 D. J. Amit, H. Gutfreund, and H. Sompolinsky, "Storing infinite numbers of patterns in a spin-glass model of neural networks" 1985 *Phys. Rev. Lett.* 55:1530. 10.1103/PhysRevLett.55.1530
- 894 Ulmann, Bernd. Analog Computing, München: Oldenbourg Wissenschaftsverlag, 2013. <https://doi.org/10.1524/9783486755183>
- 895 G. E. R. Cowan, R. C. Melville and Y. P. Tsividis, "A VLSI analog computer/digital computer accelerator," in *IEEE Journal of Solid-State Circuits*, vol. 41, no. 1, pp. 42-53, Jan. 2006, doi: 10.1109/JSSC.2005.858618.
- 896 A. Basu et al., "A Floating-Gate-Based Field-Programmable Analog Array," in *IEEE Journal of Solid-State Circuits*, vol. 45, no. 9, pp. 1781-1794, Sept. 2010, doi: 10.1109/JSSC.2010.2056832.
- 897 N. Guo et al., "Energy-Efficient Hybrid Analog/Digital Approximate Computation in Continuous Time," in *IEEE Journal of Solid-State Circuits*, vol. 51, no. 7, pp. 1514-1524, July 2016, doi: 10.1109/JSSC.2016.2543729.
- 898 S. George et al., "A Programmable and Configurable Mixed-Mode FPAA SoC," in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 24, no. 6, pp. 2253-2261, June 2016, doi: 10.1109/TVLSI.2015.2504119.
- 899 G. E. R. Cowan, R. C. Melville and Y. P. Tsividis, "A VLSI analog computer/math co-processor for a digital computer," *ISSCC. 2005 IEEE International Digest of Technical Papers. Solid-State Circuits Conference*, 2005., 2005, pp. 82-586 Vol. 1, doi: 10.1109/ISSCC.2005.1493879.
- 900 N. Guo et al., "Continuous-time hybrid computation with programmable nonlinearities," *ESSCIRC Conference 2015 - 41st European Solid-State Circuits Conference (ESSCIRC)*, 2015, pp. 279-282, doi: 10.1109/ESSCIRC.2015.7313881.
- 901 Y. Huang, N. Guo, S. Sethumadhavan, M. Seok and Y. Tsividis, "A Case Study in Analog Co-Processing for Solving Stochastic Differential Equations," 2018 *IEEE 23rd International Conference on Digital Signal Processing (DSP)*, 2018, pp. 1-5, doi: 10.1109/ICDSP.2018.8631831.
- 902 J. Hasler, "Starting Framework for Analog Numerical Analysis," *Journal of Low Power Electronics Applications*, vol. 7, no. 17, June 2017.
- 903 N. Udayanga, S. I. Hariharan, S. Mandal, L. Belostotski, L. T. Bruton and A. Madanayake, "Continuous-Time Algorithms for Solving Maxwell's Equations Using Analog Circuits," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 66, no. 10, pp. 3941-3954, Oct. 2019, doi: 10.1109/TCSI.2019.2915200.
- 904 N. Udayanga et al., "A Radio Frequency Analog Computer for Computational Electromagnetics," in *IEEE Journal of Solid-State Circuits*, vol. 56, no. 2, pp. 440-454, Feb. 2021, doi: 10.1109/JSSC.2020.3018432.
- 905 Huang, Yipeng. Hybrid Analog-Digital Co-Processing for Scientific Computation. Columbia University, 2018.
- 906 Y. Huang, N. Guo, M. Seok, Y. Tsividis and S. Sethumadhavan, "Analog Computing in a Modern Context: A Linear Algebra Accelerator Case Study," in *IEEE Micro*, vol. 37, no. 3, pp. 30-38, 2017, doi: 10.1109/MM.2017.55.
- 907 Zhong Sun, Giacomo Pedretti, Elia Ambrosi, Alessandro Bricalli, Wei Wang, Daniele Ielmini. Solving matrix equations in one step with cross-point resistive arrays. *Proceedings of the National Academy of Sciences* Mar 2019, 116 (10) 4123-4128; DOI: 10.1073/pnas.1815682116
- 908 Y. Huang, N. Guo, M. Seok, Y. Tsividis, K. Mandli and S. Sethumadhavan, "Hybrid Analog-Digital Solution of Nonlinear Partial Differential Equations," 2017 50th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO), 2017, pp. 665-678.
- 909 M. Lukoševičius and H. Jaeger, 'Reservoir computing approaches to recurrent neural network training', *Computer science review*, vol. 3, no. 3, pp. 127–149, 2009

- ⁹¹⁰ D. J. Gauthier, E. Boltt, A. Griffith, and W. A. S. Barbosa, 'Next generation reservoir computing', *Nature Communications*, vol. 12, p. 5564, 2021.
- ⁹¹¹ M. Yan, C. Huang, P. Bienstman, P. Tino, W. Lin, and J. Sun, 'Emerging opportunities and challenges for the future of reservoir computing', *Nature Communications*, vol. 15, p. 2056, 2024.
- ⁹¹² P. Mujal et al., 'Opportunities in quantum reservoir computing and extreme learning machines', *Advanced Quantum Technologies*, vol. 4, no. 8, p. 2100027, 2021.
- ⁹¹³ S. Ghosh, K. Nakajima, T. Krisnanda, K. Fujii, and T. C. H. Liew, 'Quantum neuromorphic computing with reservoir computing networks', *Advanced Quantum Technologies*, vol. 4, no. 9, p. 2100053, 2021.
- ⁹¹⁴ C. Gallicchio, A. Micheli, and L. Pedrelli, "Deep reservoir computing: A critical experimental analysis," *Neurocomputing*, vol. 268, pp. 87–99, 2017.
- ⁹¹⁵ A. Lupo, E. Picco, M. Zajnulina, and S. Massar, 'Deep photonic reservoir computer based on frequency multiplexing with fully analog connection between layers', *Optica*, vol. 10, no. 11, pp. 1478–1485, 2023.
- ⁹¹⁶ G. V. der Sande, D. Brunner, and M. C. Soriano, "Advances in photonic reservoir computing," *Nanophotonics*, vol. 6, no. 3, pp. 561–576, 2017.
- ⁹¹⁷ A. Lugnan et al., 'Photonic neuromorphic information processing and reservoir computing', *APL Photonics*, vol. 5, no. 2, p. 020901, 2020.
- ⁹¹⁸ C. Du, F. Cai, M. A. Zidan, W. Ma, S. H. Lee, and W. D. Lu, 'Reservoir computing using dynamic memristors for temporal information processing', *Nature communications*, vol. 8, p. 2204, 2017.
- ⁹¹⁹ Y. Zhong, J. Tang, X. Li, B. Gao, H. Qian, and H. Wu, 'Dynamic memristor-based reservoir computing for high-efficiency temporal signal processing', *Nature Communications*, vol. 12, p. 408, 2021.
- ⁹²⁰ J. Torreon et al., "Neuromorphic computing with nanoscale spintronic oscillators," *Nature*, vol. 547, no. 7664, pp. 428–431, 2017.
- ⁹²¹ D. A. Allwood et al., 'A perspective on physical reservoir computing with nanomagnetic devices', *Applied Physics Letters*, vol. 122, no. 4, p. 040501, 2023.
- ⁹²² G. Dion, S. Mejaouri, and J. Sylvestre, "Reservoir computing with a single delay-coupled non-linear mechanical oscillator," *Journal of Applied Physics*, vol. 124, no. 15, p. 152132, 2018.
- ⁹²³ S. Abreu et al., "A Photonics Perspective on Computing with Physical Substrates", *Reviews in Physics*, vol. 12, 2024.
- ⁹²⁴ M. Mitzenmacher and E. Upfal, *Probability and Computing: Randomization and Probabilistic Techniques in Algorithms and Data Analysis*, 2nd ed., Cambridge University Press, 2017.
- ⁹²⁵ Buluc, Aydin, et al. "Randomized algorithms for scientific computing (RASC)." *arXiv preprint arXiv:2104.11079* (2021).
- ⁹²⁶ A. A. Faisal, L. P. Selen, and D. M. Wolpert, "Noise in the nervous system," *Nature reviews. Neuroscience*, vol. 9, p. 292, 2008.
- ⁹²⁷ M. D. McDonnell and L. M. Ward, "The benefits of noise in neural systems: bridging theory and experiment," *Nature reviews. Neuroscience*, vol. 12, p. 415, 2011.
- ⁹²⁸ W. Maass, "Noise as a resource for computation and learning in networks of spiking neurons," *Proceedings of the IEEE*, vol. 102, pp. 860–880, 2014.
- ⁹²⁹ M. Al-Shedivat, R. Naous, G. Cauwenberghs, and K. N. Salama, "Memristors empower spiking neurons with stochasticity," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 5, pp. 242–253, 2015.
- ⁹³⁰ Suzuki, Hideyuki, et al. "Chaotic boltzmann machines." *Scientific reports* 3.1 (2013): 1–5.
- ⁹³¹ S. Chowdhury et al., "A Full-Stack View of Probabilistic Computing With p-Bits: Devices, Architectures, and Algorithms," in *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, vol. 9, no. 1, pp. 1–11, June 2023.
- ⁹³² Kaiser, Jan, and Supriyo Datta. "Probabilistic computing with p-bits." *Applied Physics Letters* 119.15 (2021): 150503.
- ⁹³³ Joseph, Anosh. "Markov Chain Monte Carlo Methods in Quantum Field Theories: A Modern Primer." *arXiv preprint arXiv:1912.10997* (2019).
- ⁹³⁴ Mohseni, N., McMahon, P. L., & Byrnes, T. (2022). Ising machines as hardware solvers of combinatorial optimization problems. *Nature Reviews Physics*, 4(6), 363–379.
- ⁹³⁵ H. Goto, K. Tatsumura, and A. R. Dixon, "Combinatorial optimization by simulating adiabatic bifurcations in nonlinear Hamiltonian systems," *Sci. Adv.*, vol. 5, no. 4, 2019 Apr 1, Art no. eaav2372.
- ⁹³⁶ Fabio L Traversa and Massimiliano Di Ventra. Polynomial-time solution of prime factorization and np-complete problems with digital memcomputing machines. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 27(2):023107, 2017.
- ⁹³⁷ Parihar, A., Shukla, N., Jerry, M., Datta, S., & Raychowdhury, A. (2017). Vertex coloring of graphs via phase dynamics of coupled oscillatory networks. *Scientific reports*, 7(1), 911.
- ⁹³⁸ Ahmed, I., Chiu, P. W., Moy, W., & Kim, C. H. (2021). A probabilistic compute fabric based on coupled ring oscillators for solving combinatorial optimization problems. *IEEE Journal of Solid-State Circuits*, 56(9), 2870–2880.
- ⁹³⁹ Yamamoto, Y., Leleu, T., Ganguli, S., & Mabuchi, H. (2020). Coherent Ising machines—Quantum optics and neural network Perspectives. *Applied Physics Letters*, 117(16).
- ⁹⁴⁰ Yamaoka, Masanao, et al. "A 20k-spin Ising chip to solve combinatorial optimization problems with CMOS annealing." *IEEE Journal of Solid-State Circuits* 51.1 (2015): 303–309.
- ⁹⁴¹ Aramon, Maliheh, et al. "Physics-inspired optimization for quadratic unconstrained problems using a digital annealer." *Frontiers in Physics* 7 (2019): 48.

- ⁹⁴² Patel, Saavan, et al. "Ising model optimization problems on a FPGA accelerated restricted Boltzmann machine." *arXiv preprint arXiv:2008.04436* (2020).
- ⁹⁴³ Yamamoto, Kasho, et al. "STATICA: A 512-Spin 0.25 M-Weight Annealing Processor With an All-Spin-Updates-at-Once Architecture for Combinatorial Optimization With Complete Spin-Spin Interactions." *IEEE Journal of Solid-State Circuits* 56.1 (2020): 165-178.
- ⁹⁴⁴ Su, Yuqi, et al. "A 252 Spins Scalable CMOS Ising Chip Featuring Sparse and Reconfigurable Spin Interconnects for Combinatorial Optimization Problems." *2021 IEEE Custom Integrated Circuits Conference (CICC)*. IEEE, 2021.
- ⁹⁴⁵ Lucas, Andrew. "Ising formulations of many NP problems." *Frontiers in physics* 2 (2014)
- ⁹⁴⁶ Behin-Aein, Behtash, Vinh Diep, and Supriyo Datta. "A building block for hardware belief networks." *Scientific reports* 6 (2016): 29893.
- ⁹⁴⁷ Camsari, Kerem Yunus, et al. "Stochastic p-bits for invertible logic." *Physical Review X* 7.3 (2017): 031014.
- ⁹⁴⁸ Camsari, Kerem Yunus, Sayeef Salahuddin, and Supriyo Datta. "Implementing p-bits with embedded MTJ." *IEEE Electron Device Letters* 38.12 (2017): 1767-1770.
- ⁹⁴⁹ Zand, Ramtin, et al. "Composable probabilistic inference networks using MRAM-based stochastic neurons." *ACM Journal on Emerging Technologies in Computing Systems (JETC)* 15.2 (2019): 17.
- ⁹⁵⁰ N. A. Aadit, A. Grimaldi, M. Carpentieri, L. Theogarajan, J. M. Martinis, G. Finocchio, and K. Y. Camsari, "Massively parallel probabilistic computing with sparse Ising machines", *Nature Electronics* 5, 460 (2022)
- ⁹⁵¹ Sutton, Brian, et al. "Intrinsic optimization using stochastic nanomagnets." *Scientific reports* 7 (2017): 44370.
- ⁹⁵² Smithson, Sean C., et al. "Efficient CMOS invertible logic using stochastic computing." *IEEE Transactions on Circuits and Systems I: Regular Papers* 66.6 (2019): 2263-2274.
- ⁹⁵³ Onizawa, Naoya, et al. "In-Hardware Training Chip Based on CMOS Invertible Logic for Machine Learning." *IEEE Transactions on Circuits and Systems I: Regular Papers* (2019).
- ⁹⁵⁴ Sutton, Brian, et al. "Autonomous Probabilistic Coprocessing with Petaflips per Second." *arXiv preprint arXiv:1907.09664* (2019).
- ⁹⁵⁵ J. J. Hopfield and D. W. Tank, "Neural computation of decisions in optimization problems," *Biological Cybernetics*, vol. 52, no. 3, pp. 141–152, Jul. 1985.
- ⁹⁵⁶ S. Kumar, J. P. Strachan and R. S. Williams, "Chaotic dynamics in nanoscale NbO₂ Mott memristors for analogue computing," *Nature*, 548, 318 (2017)
- ⁹⁵⁷ A. Parihar, N. Shukla, M. Jerry, S. Datta, and A. Raychowdhury, "Vertex coloring of graphs via phase dynamics of coupled oscillatory networks," *Scientific Reports*, vol. 7, no. 1, p. 911, Apr. 2017.
- ⁹⁵⁸ H. Mostafa, L. K. Müller, and G. Indiveri, "An event-based architecture for solving constraint satisfaction problems," *Nature Communications*, vol. 6, p. 8941, Dec. 2015.
- ⁹⁵⁹ Inagaki, T., Haribara, Y., Igarashi, K., Sonobe, T., Tamate, S., Honjo, T., Marandi, A., McMahon, P.L., Umeki, T., Enbutsu, K., Tadanaga, O. A coherent Ising machine for 2000-node optimization problems. *Science*. 2016 Nov 4;354(6312):603-6.
- ⁹⁶⁰ Dutta, S., Khanna, A., Assoa, A.S., et al. An Ising Hamiltonian solver based on coupled stochastic phase-transition nano-oscillators. *Nature Electronics*. 2021 Jul;4(7):502-512.
- ⁹⁶¹ Moy, W., Ahmed, I., Chiu, P.W., Moy, J., Sapatnekar, S.S., Kim, C.H. A 1,968-node coupled ring oscillator circuit for combinatorial optimization problem solving. *Nature Electronics*. 2022 May;5(5):310-7.
- ⁹⁶² Tatsumura, K., Yamasaki, M., Goto, H. Scaling out Ising machines using a multi-chip architecture for simulated bifurcation. *Nature Electronics*. 2021 Mar;4(3):208-217.
- ⁹⁶³ Kiraly, E. J. Knol, W. M. J. van Weerdenburg, H. J. Kappen, and A. A. Khajetoorians, "An atomic Boltzmann machine capable of self-adaption," *Nature Nanotechnology*, vol. 16, no. 4, pp. 414--420, 2021.
- ⁹⁶⁴ S. Niazi, S. Chowdhury, N. A. Aadit, M. Mohseni, Y. Qin, and K. Y. Camsari, "Training deep Boltzmann networks with sparse Ising machines," accepted in *Nature Electronics*, 2024 (in press).
- ⁹⁶⁵ Faria, R., Camsari, K. Y., & Datta, S. (2018). Implementing Bayesian networks with embedded stochastic MRAM. *AIP Advances*, 8(4). AIP Publishing.
- ⁹⁶⁶ J. Kaiser, W. A. Borders, K. Y. Camsari, S. Fukami, H. Ohno, and S. Datta, "Hardware-aware in situ learning based on stochastic magnetic tunnel junctions," *Physical Review Applied*, vol. 17, no. 1, p. 014016, 2022.
- ⁹⁶⁷ Hirtzlin, B. Penkovsky, M. Bocquet, J.-O. Klein, J.-M. Portal, and D. Querlioz, "Stochastic computing for hardware implementation of binarized neural networks", *IEEE Access* 7, 76394 (2019).
- ⁹⁶⁸ T. Dalgaty, N. Castellani, C. Turck, K.-E. Harabi, D. Querlioz, and E. Vianello, "In situ learning using intrinsic memristor variability via Markov chain Monte Carlo sampling," *Nature Electronics*, vol. 4, no. 2, pp. 151-161, 2021.
- ⁹⁶⁹ J. Laydevant, D. Marković, and J. Grollier, "Training an ising machine with equilibrium propagation," *Nature Communications*, vol. 15, no. 1, pp. 3671, 2024.
- ⁹⁷⁰ Camsari, Kerem Y., Brian M. Sutton, and Supriyo Datta. "p-bits for probabilistic spin logic." *Applied Physics Reviews* 6.1 (2019): 011305.
- ⁹⁷¹ K. Y. Camsari, S. Chowdhury, and S. Datta, "Scalable Emulation of Sign-Problem-Free Hamiltonians with Room-Temperature p-bits," *Physical Review Applied*, vol. 12, no. 3, p. 34061, Sep. 2019, doi: 10.1103/physrevapplied.12.034061.
- ⁹⁷² S. Chowdhury, S. Datta, and K. Y. Camsari, "A Probabilistic Approach to Quantum Inspired Algorithms," in 2019 IEEE International Electron Devices Meeting (IEDM), IEEE, Dec. 2019. doi: 10.1109/iedm19573.2019.8993655.

- ⁹⁷³ S. Chowdhury, K. Y. Çamsari, and S. Datta, "Emulating Quantum Circuits with Generalized Ising Machines," *IEEE Access*, vol. 11, pp. 116944–116955, 2023, doi: 10.1109/access.2023.3323847.
- ⁹⁷⁴ T. Okuyama, M. Hayashi, and M. Yamaoka, "An Ising Computer Based on Simulated Quantum Annealing by Path Integral Monte Carlo Method," in 2017 IEEE International Conference on Rebooting Computing (ICRC), IEEE, Nov. 2017. doi: 10.1109/icrc.2017.8123652.
- ⁹⁷⁵ H. M. Waidyasooriya and M. Hariyama, "Highly-Parallel FPGA Accelerator for Simulated Quantum Annealing," *IEEE Transactions on Emerging Topics in Computing*, vol. 9, no. 4, pp. 2019–2029, Oct. 2021, doi: 10.1109/tetc.2019.2957177
- ⁹⁷⁶ A. Grimaldi et al., "Experimental evaluation of simulated quantum annealing with MTJ-augmented p-bits," in 2022 International Electron Devices Meeting (IEDM), IEEE, Dec. 2022. doi: 10.1109/iedm45625.2022.10019530.
- ⁹⁷⁷ S. Chowdhury, K. Y. Camsari, and S. Datta, "Accelerated quantum Monte Carlo with probabilistic computers," *Communications Physics*, vol. 6, no. 1, Apr. 2023, doi: 10.1038/s42005-023-01202-3.
- ⁹⁷⁸ S. Chowdhury and K. Y. Camsari, "Machine Learning Quantum Systems with Magnetic p-bits," in 2023 IEEE International Magnetic Conference - Short Papers (INTERMAG Short Papers), IEEE, May 2023. doi: 10.1109/intermagshortpapers58606.2023.10228205.
- ⁹⁷⁹ Burri, Samuel, et al. "SPADs for quantum random number generators and beyond." *2014 19th Asia and South Pacific Design Automation Conference (ASP-DAC)*. IEEE, 2014.
- ⁹⁸⁰ Vodenicarevic, Damir, et al. "Low-energy truly random number generation with superparamagnetic tunnel junctions for unconventional computing." *Physical Review Applied* 8.5 (2017): 054045.
- ⁹⁸¹ Valle, Javier del, et al. "Generation of Tunable Stochastic Sequences Using the Insulator–Metal Transition." *Nano Letters* (2022).
- ⁹⁸² Patel, Ravi G., T. Patrick Xiao, Sapan Agarwal, and Christopher Bennett. "Analog Bayesian neural networks are insensitive to the shape of the weight distribution." *arXiv preprint arXiv:2501.05564* (2025).
- ⁹⁸³ S. Oh et al., "Bayesian Neural Network Implemented by Dynamically Programmable Noise in Vanadium Oxide," *2023 International Electron Devices Meeting (IEDM)*, San Francisco, CA, USA, 2023, pp. 1–4
- ⁹⁸⁴ Liu, Samuel, et al. "Bayesian neural networks using magnetic tunnel junction-based probabilistic in-memory computing." *Frontiers in Nanotechnology* 4 (2022): 1021943.
- ⁹⁸⁵ R. Landauer, "Irreversibility and Heat Generation in the Computing Process," *IBM Journal of Research and Development*, vol. 5, no. 3, pp. 183–191, July 1961. doi:10.1147/rd.53.0183.
- ⁹⁸⁶ M. P. Frank, "Why reversible computing is the only way forward for general digital computing," invited talk, *IEEE International Nanodevices and Computing Conference*, Grenoble, France, Apr. 2019. <https://cfwebprod.sandia.gov/cfdocs/CompResearch/docs/INC19-talk-v5.pdf>.
- ⁹⁸⁷ C. H. Bennett, "Logical Reversibility of Computation," *IBM Journal of Research and Development*, vol. 17, no. 6, pp. 525–532, Nov. 1973. doi:10.1147/rd.176.0525.
- ⁹⁸⁸ M. P. Frank and M.J. Ammer, "Relativized separation of reversible and irreversible space-time complexity classes," preprint, arxiv:1708.08480 [cs.ET], 2017.
- ⁹⁸⁹ E. D. Demaine, J. Lynch, G. J. Mirano, and N. Tyagi, "Energy-efficient algorithms," in *Proc. 2016 ACM Conf. on Innovations in Theoretical Computer Science*, pp. 321–332, 2016.
- ⁹⁹⁰ A. H. Earley, "On the performance and programming of reversible molecular computers," Ph.D. thesis, University of Cambridge, 2021. doi:10.17863/CAM.79111.
- ⁹⁹¹ S. Pidaparathi and C. Lent, "Exponentially Adiabatic Switching in Quantum-Dot Cellular Automata," *Journal of Low Power Electronics and Applications* vol. 8, no. 3, 2018. doi:10.3390/jlpea8030030.
- ⁹⁹² S. S. Pidaparathi and C. S. Lent, "Energy dissipation during two-state switching for quantum-dot cellular automata," *Journal of Applied Physics* vol. 129, art. 024304, 2021. doi:10.1063/5.0033633.
- ⁹⁹³ M. P. Frank and T. F. Knight Jr., "Ultimate theoretical models of nanocomputers." *Nanotechnology* 9, no. 3, 1998. doi:10.1088/0957-4484/9/3/005.
- ⁹⁹⁴ S. G. Younis and T. F. Knight, Jr., "Asymptotically zero energy split-level charge recovery logic," *International Workshop on Low Power Design*, pp. 177–182, Apr. 1994.
- ⁹⁹⁵ S. G. Younis, "Asymptotically zero energy computing using split-level charge recovery logic," Ph.D. dissertation, MIT Dept. of EECS, 1994. <https://dspace.mit.edu/handle/1721.1/11620>
- ⁹⁹⁶ C. J. Vieri, "Pendulum—A reversible computer architecture," M.S. thesis, MIT Dept. of EECS, 1995
- ⁹⁹⁷ M. P. Frank, "Reversibility for Efficient Computing," Ph.D. dissertation, MIT Dept. of EECS, 1999. <https://dspace.mit.edu/handle/1721.1/9464>.
- ⁹⁹⁸ C. J. Vieri, "Reversible computer engineering and architecture," Ph.D. dissertation, MIT Dept. of EECS, 1999. <https://dspace.mit.edu/handle/1721.1/80144>.
- ⁹⁹⁹ M. P. Frank, "Reversible Computing and Truly Adiabatic Circuits: The Next Great Challenge for Digital Engineering," *5th IEEE Dallas Circuits and Systems Workshop on Design, Applications, Integration and Software (DCAS-06)*, Oct. 2006, Dallas, TX. doi:10.1109/DCAS.2006.321027.
- ¹⁰⁰⁰ A. Vetuli, S. Di Pascoli, and L. M. Reyneri, "Positive feedback in adiabatic logic," *Electronics Letters* vo. 32, no. 20, 1996.
- ¹⁰⁰¹ P. Teichman, "Adiabatic Logic: Future Trend and System Level Perspective," *Springer Science & Business Media*, vol. 24, New York: Springer, 2012.

- ¹⁰⁰² Y. Moon and D.-K. Jeong, "Efficient charge recovery logic," Digest of Technical Papers, Symposium on VLSI Circuits, IEEE, 1995.
- ¹⁰⁰³ S. Maheshwari, V. A. Bartlett and I. Kale, "Adiabatic flip-flops and sequential circuit design using novel resettable adiabatic buffers," *2017 European Conference on Circuit Theory and Design (ECCTD)*, Catania, pp. 1–4, 2017.
- ¹⁰⁰⁴ M. Avital, H. Dagan, I. Levi, O. Keren and A. Fish, "DPA-Secured Quasi-Adiabatic Logic (SQAL) for Low-Power Passive RFID Tags Employing S-Boxes," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 62, no. 1, pp. 149–156, Jan. 2015.
- ¹⁰⁰⁵ S. D. Kumar, H. Thapliyal and A. Mohammad, "EE-SPFAL: A Novel Energy-Efficient Secure Positive Feedback Adiabatic Logic for DPA Resistant RFID and Smart Card," *IEEE Transactions on Emerging Topics in Computing*, vol. 7, no. 2, pp. 281–293, Apr.-Jun. 2019.
- ¹⁰⁰⁶ I. K. Hanninen, C. O. Campos-Aguillon, R. Celis-Cordova, and G. L. Snider, "Design and Fabrication of a Microprocessor using Adiabatic CMOS and Bennett Clocking" *Lecture Notes in Computer Science, Reversible Computation: 7th International Conference*, Springer, Cham, 2015.
- ¹⁰⁰⁷ R. Celis-Cordova, A. O. Orlov, T. Lu, J.M. Kulick, and G. L. Snider, "The design of 16-bit Adiabatic Microprocessor". *2019 IEEE International Conference on Rebooting Computing (ICRC)*, San Mateo, CA, 2019.
- ¹⁰⁰⁸ M. P. Frank, R. W. Brocato, B. D. Tierney, N. A. Missert, and A. H. Hsia, "Reversible Computing with Fast, Fully Static, Fully Adiabatic CMOS," arxiv:2009.00448v2 [cs.AR], Sep. 2020.
- ¹⁰⁰⁹ E. P. DeBenedictis, "Inversion for S2LAL," Zettaflops LLC tech. report ZF004, Sep. 2020, <https://zettaflops.org/zf004/>.
- ¹⁰¹⁰ M. P. Frank and A. J. Edwards, "Limits of CMOS and Prospects for Adiabatic/Reversible CMOS," invited talk presented at the Texas Symposium on Computing with Emerging Technologies (ComET '23), University of Texas at Dallas, Oct. 2023. https://www.sandia.gov/app/uploads/sites/210/2023/11/Comet23-slides_SAND.pdf; video at <https://www.youtube.com/watch?v=vALCJJs9Dtw>.
- ¹⁰¹¹ K. Loe, E. Goto, "Analysis of flux input and output Josephson pair device," *IEEE Trans. Magn.*, vol. 21, pp. 884–887, 1985.
- ¹⁰¹² V. K. Semenov, G. V. Danilov, D. V. Averin, "Negative-inductance SQUID as the basic element of reversible Josephson-junction circuits," *IEEE Trans. Appl. Supercond.*, vol. 13, pp. 938–943, 2003.
- ¹⁰¹³ N. Takeuchi, Y. Yamanashi, N. Yoshikawa, "Energy efficiency of adiabatic superconductor logic," *Supercond. Sci. Technol.*, vol. 28, art. 015003, 2015.
- ¹⁰¹⁴ N. Takeuchi, Y. Yamanashi, N. Yoshikawa, "Thermodynamic Study of Energy Dissipation in Adiabatic Superconductor Logic," *Phys. Rev. Applied*, vol. 4, art. 034007, 2015.
- ¹⁰¹⁵ N. Takeuchi, Y. Yamanashi and N. Yoshikawa, "Measurement of 10 zJ energy dissipation of adiabatic quantum-flux-parametron logic using a superconducting resonator," *Appl. Phys. Lett.*, vol. 102, art. 052602, 2013.
- ¹⁰¹⁶ N. Takeuchi, T. Yamae, C. L. Ayala, H. Suzuki, N. Yoshikawa, "An adiabatic superconductor 8-bit adder with $24 k_B T$ energy dissipation per junction," *Appl. Phys. Lett.*, vol. 114, art. 042602, 2019.
- ¹⁰¹⁷ N. Takeuchi, T. Yamae, and N. Yoshikawa, "A reversible 8-bit adder using adiabatic quantum-flux-parametron logic," *IEEE Applied Superconductivity Conference*, 2020.
- ¹⁰¹⁸ T. Yamae, N. Takeuchi, and N. Yoshikawa, "Design and demonstration of an 8 word by 1 bit reversible quantum flux parametron register file," *IEEE Applied Superconductivity Conference*, 2020.
- ¹⁰¹⁹ T. Yamae, N. Takeuchi, and N. Yoshikawa, "Minimum energy dissipation required for information processing using adiabatic quantum-flux-parametron circuits," *Journal of Applied Physics*, vol. 135, art. no. 063902, 2024.
- ¹⁰²⁰ K. E. Drexler, "Molecular engineering: An approach to the development of general capabilities for molecular manipulation." *Proceedings of the National Academy of Sciences*, vol. 78, no. 9, pp. 5275–5278, 1981.
- ¹⁰²¹ K. E. Drexler, "Rod logic and thermal noise in the mechanical nanocomputer," in *Molecular Electronic Devices*, F. L. Carter, R. Siatkowski and H. Wohltjen, eds. Amsterdam: North-Holland, 1988.
- ¹⁰²² K. E. Drexler, *Nanosystems: molecular machinery, manufacturing, and computation*, New York: Wiley, 1992.
- ¹⁰²³ K. K. Likharev and A. N. Korotkov, "Single-electron parametron: reversible computation in a discrete-state system," *Science*, vol. 273, no. 5276, pp. 763–765, 1996.
- ¹⁰²⁴ E. Blair, "Electric-Field Inputs for Molecular Quantum-Dot Cellular Automata Circuits," *IEEE Transactions on Nanotechnology*, vol. 18, pp. 453–460, 2019.
- ¹⁰²⁵ K. Walus, M. Mazur, G. Schulhof, and G.A. Jullien, "Simple 4-bit processor based on quantum-dot cellular automata (QCA)." *2005 IEEE International Conference on Application-Specific Systems, Architecture Processors (ASAP '05)*, pp. 288–293, IEEE, July 2005.
- ¹⁰²⁶ E. Fazzion, O.L. Fonseca, J. A. M. Nacif, O. P. V. Neto, A. O. Fernandes, and D. S. Silva, "A quantum-dot cellular automata processor design," *Proceedings of the 27th Symposium on Integrated Circuits and Systems Design*, p. 29, ACM, Sep. 2014.
- ¹⁰²⁷ G. Pillonnet, H. Fanet, and S. Houri, "Adiabatic Capacitive Logic: a paradigm for low-power logic," in *IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE: Baltimore, MD, 2017.
- ¹⁰²⁸ R. Celis-Cordova, A. O. Orlov and G. L. Snider, "Adiabatic Capacitive Logic using Voltage-controlled Variable Capacitors," *2020 IEEE Silicon Nanoelectronics Workshop (SNW)*, pp. 115–116, 2020.
- ¹⁰²⁹ Y. Perrin, A. Galisultanov, L. Hutin, P. Basset, H. Fanet and G. Pillonnet, "Contact-Free MEMS Devices for Reliable and Low-Power Logic Operations," in *IEEE Transactions on Electron Devices* vol. 68, no. 6, pp. 2938–2943, June 2021.
- ¹⁰³⁰ S. V. Polonsky, V. K. Semenov, D. F. Schneider, "Transmission of single-flux-quantum pulses along superconducting microstrip lines," *IEEE Trans. Appl. Supercond.*, vol. 3, no. 1, pp. 2598–2600, Mar. 1993.

- ¹⁰³¹ R. D. Parmentier, K. Lonngren, A. C. Scott, "Fluxons in long Josephson junctions" in *Solitons in Action*, Cambridge, MA, USA:Acad. Press, pp. 173–196, 1978.
- ¹⁰³² M. P. Frank, E. P. DeBenedictis, "Ballistic Reversible Superconducting Memory Element," U.S. Patent Application, Pub. No. US 2022/0036943 A1. United States Patent and Trademark Office, Feb. 2022.
- ¹⁰³³ M. P. Frank, "Fundamental Physical Limits of Reversible Computing—An Introduction," presented at the CCC Workshop on Physics & Engineering Issues in Adiabatic/Reversible Classical Computing, Oct. 2020. Slides: <https://cra.org/ccc/wp-content/uploads/sites/2/2020/10/CCC-Frank-Day1-v2.pdf>, video: https://youtu.be/Oj_cRr828-c, <https://youtu.be/PEqiOSLzFRE>.
- ¹⁰³⁴ K. Shukla, "Foundations of the Lindbladian Approach to Adiabatic and Reversible Computing," presented at the CCC Workshop on Physics & Engineering Issues in Adiabatic/Reversible Classical Computing, Oct. 2020. Slides: <https://cra.org/ccc/wp-content/uploads/sites/2/2020/12/Shukla-Foundations-of-the-Lindbladian-Approach-to-Adiabatic-and-Reversible-Computing-2.pdf>, video: <https://youtu.be/mzoL6m-2rrA>.
- ¹⁰³⁵ D. Guéry-Odelin, A. Ruschhaupt, A. Kiely, E. Torrontegui, S. Martínez-Garaot, and J. G. Muga, "Shortcuts to adiabaticity: concepts, methods, and applications," *Reviews of Modern Physics*, vol. 91, no. 4, art. 045001, 2019.
- ¹⁰³⁶ W. M. Itano, D. J. Heinzen, J. J. Bollinger, and D. J. Wineland, "Quantum zeno effect," *Physical Review A*, vol. 41, no. 5, p. 2295, 1990.
- ¹⁰³⁷ P. Facchi, H. Nakazato, and S. Pascazio, "From the quantum Zeno to the inverse quantum Zeno effect," *Physical Review Letters*, vol. 86, no. 13, p. 2699, 2001.
- ¹⁰³⁸ F. Gao, G. Tziantzioulis, and D. Wentzlaff, "ComputeDRAM: In-Memory Compute Using Off-the-Shelf DRAMs," in *MICRO*, 2019.
- ¹⁰³⁹ N. Hajinazar, G. F. Oliveira, S. Gregorio, J. D. Ferreira, N. M. Ghiasi, M. Patel, M. Alser, S. Ghose, J. Gomez-Luna, and O. Mutlu, "SIMDRAM: A Framework for Bit-Serial SIMD Processing Using DRAM," in *ASPLOS*, 2021.
- ¹⁰⁴⁰ S. Li, D. Niu, K. T. Malladi, H. Zheng, B. Brennan, and Y. Xie, "DRISA: A DRAM-Based Reconfigurable In-Situ Accelerator," in *MICRO*, 2017.
- ¹⁰⁴¹ V. Seshadri, D. Lee, T. Mullins, H. Hassan, A. Boroumand, J. Kim, M. A. Kozuch, O. Mutlu, P. B. Gibbons, and T. C. Mowry, "Ambit: In-Memory Accelerator for Bulk Bitwise Operations Using Commodity DRAM Technology," in *MICRO*, 2017.
- ¹⁰⁴² S. Aga, S. Jeloka, A. Subramaniyan, S. Narayanasamy, D. Blaauw, and R. Das, "Compute Caches," in *HPCA*, 2017.
- ¹⁰⁴³ D. Fujiki, S. Mahlke, and R. Das, "Duality Cache for Data Parallel Acceleration," in *ISCA*, 2019.
- ¹⁰⁴⁴ S. Jeloka, N. B. Akesh, D. Sylvester, and D. Blaauw, "A 28 nm Configurable Memory (TCAM/BCAM/SRAM) Using Push-Rule 6T Bit Cell Enabling Logic-in-Memory," in *JSSC*, 2016.
- ¹⁰⁴⁵ M. Kang, E. P. Kim, M.-S. Keel, and N. R. Shanbhag, "Energy-Efficient and High Throughput Sparse Distributed Memory Architecture," in *ISCAS*, 2015.
- ¹⁰⁴⁶ J. Wang, X. Wang, C. Eckert, A. Subramaniyan, R. Das, D. T. Blaauw, and D. Sylvester, "A Compute SRAM with Bit-Serial Integer/Floating-Point Operations for Programmable In-Memory Vector Acceleration," in *ISSCC*, 2019.
- ¹⁰⁴⁷ V. Seshadri, D. Lee, T. Mullins, H. Hassan, A. Boroumand, J. Kim, M. A. Kozuch, O. Mutlu, P. B. Gibbons, and T. C. Mowry, "Ambit: In-Memory Accelerator for Bulk Bitwise Operations Using Commodity DRAM Technology," in *MICRO*, 2017.
- ¹⁰⁴⁸ V. Seshadri, D. Lee, T. Mullins, H. Hassan, A. Boroumand, J. Kim, M. A. Kozuch, O. Mutlu, P. B. Gibbons, and T. C. Mowry, "Ambit: In-Memory Accelerator for Bulk Bitwise Operations Using Commodity DRAM Technology," in *MICRO*, 2017.
- ¹⁰⁴⁹ S. Jeloka, N. B. Akesh, D. Sylvester, and D. Blaauw, "A 28 nm Configurable Memory (TCAM/BCAM/SRAM) Using Push-Rule 6T Bit Cell Enabling Logic-in-Memory," in *JSSC*, 2016.
- ¹⁰⁵⁰ S. Gupta, M. Imani, and T. Rosing, "FELIX: Fast and Energy-Efficient Logic in Memory," in *ICCAD*, 2018.
- ¹⁰⁵¹ S. Hamdioui, L. Xie, H. A. D. Nguyen, M. Taouil, K. Bertels, H. Corporaal, H. Jiao, F. Cathoor, D. Wouters, L. Eike, and J. van Lunteren, "Memristor-Based Computation-in-Memory Architecture for Data-Intensive Applications," in *DATE*, 2015.
- ¹⁰⁵² S. Kvantinsky, D. Belousov, S. Liman, G. Satat, N. Wald, E. G. Friedman, A. Kolodny, and U. C. Weiser, "MAGIC: Memristor-Aided Logic," *TCAS II*, Sep. 2014.
- ¹⁰⁵³ S. Li, C. Xu, Q. Zou, J. Zhao, Y. Lu, and Y. Xie, "Pinatubo: A Processing-in-Memory Architecture for Bulk Bitwise Operations in Emerging Non-Volatile Memories," in *DAC*, 2016.
- ¹⁰⁵⁴ S. Angizi, Z. He, A. Awad, and D. Fan, "MRIMA: An MRAM-Based In-Memory Accelerator," *TCAD*, May 2020.
- ¹⁰⁵⁵ J. Louis, B. Hoffer, and S. Kvatinsky, "Performing Memristor-Aided Logic (MAGIC) Using STT-MRAM," in *ICECS*, 2019.
- ¹⁰⁵⁶ S. Angizi, Z. He, and D. Fan, "PIMA-Logic: A Novel Processing-in-Memory Architecture for Highly Flexible and Energy-Efficient Logic Computation," in *DAC*, 2018.
- ¹⁰⁵⁷ M. S. Q. Truong, E. Chen, D. Su, A. Glass, L. Shen, L. R. Carley, J. A. Bain, and S. Ghose, "RACER: Bit-Pipelined Processing Using Resistive Memory," in *MICRO*, 2021.
- ¹⁰⁵⁸ S. Ghose, A. Boroumand, J. S. Kim, J. Go'mez-Luna, and O. Mutlu, "Processing-in-Memory: A Workload-Driven Perspective," *IBM JRD*, Nov. 2019.
- ¹⁰⁵⁹ M. He, C. Song, I. Kim, C. Jeong, S. Kim, I. Park, M. Thottethodi, and T. N. Vijaykumar, "Newton: A DRAM-Maker's Accelerator-in-Memory (AiM) Architecture for Machine Learning," in *MICRO*, 2020.
- ¹⁰⁶⁰ Y. C. Kwon, et al., "A 20nm 6GB Function-In-Memory DRAM, Based on HBM2 with a 1.2TFLOPS Programmable Computing Unit Using Bank-Level Parallelism, for Machine Learning Applications," in *ISSCC*, 2021.

- ¹⁰⁶¹ S. Lee, S.-h. Kang, J. Lee, H. Kim, E. Lee, S. Seo, H. Yoon, S. Lee, K. Lim, H. Shin, J. Kim, S. O, A. Iyer, D. Wang, K. Sohn, and N. S. Kim, "Hardware Architecture and Software Stack for PIM Based on Commercial DRAM Technology," in *ISCA*, 2021.
- ¹⁰⁶² H. Shin, D. Kim, S. P. Eunhyeok Park, Y. Park, and S. Yoo, "McDRAM: Low Latency and Energy-Efficient Matrix Computations in DRAM," *TCAD*, Nov. 2018.
- ¹⁰⁶³ A. Yazdanbakhsh, C. Song, J. Sacks, P. Lotfi-Kamran, H. Esmailzadeh, and N. S. Kim, "In-DRAM Near-Data Approximate Acceleration for GPUs," in *PACT*, 2018.
- ¹⁰⁶⁴ Y. Eckert, N. Jayasena, and G. H. Loh, "Thermal Feasibility of Die-Stacked Processing in Memory," in *WoNDP*, 2014.
- ¹⁰⁶⁵ S. H. Pugsley, J. Jestes, H. Zhang, R. Balasubramonian, V. Srinivasan, A. Buyuktosunoglu, A. Davis, and F. Li, "NDC: Analyzing the Impact of 3D-Stacked Memory+Logic Devices on MapReduce Workloads," in *ISPASS*, 2014.
- ¹⁰⁶⁶ D. Zhang, N. Jayasena, A. Lyashevsky, J. L. Greathouse, L. Xu, and M. Ignatowski, "TOP-PIM: Throughput-Oriented Programmable Processing in Memory," in *HPDC*, 2014.
- ¹⁰⁶⁷ JEDEC Solid State Technology Assn., JESD235C: High Bandwidth Memory (HBM) DRAM, Jan. 2020.
- ¹⁰⁶⁸ Hybrid Memory Cube Consortium, Hybrid Memory Cube Specification Rev. 2.0, 2013.
- ¹⁰⁶⁹ Q. Zhu, B. Akin, H. E. Sumbul, J. C. Hoe, L. Pileggi, and F. Franchetti, "A 3D-Stacked Logic-in-Memory Accelerator for Application-Specific Data Intensive Computing," in *3DIC*, 2013.
- ¹⁰⁷⁰ [Y. Zhu, B. Wang, D. Li, and J. Zhao, "Integrated Thermal Analysis for Processing in Die-Stacking Memory," in *MEMSYS*, 2016.
- ¹⁰⁷¹ S. Aga, S. Jeloka, A. Subramaniam, S. Narayanasamy, D. Blaauw, and R. Das, "Compute Caches," *HPCA*, 2017.
- ¹⁰⁷² F. Devaux, "The True Processing in Memory Accelerator," in *HotChips*, 2019.
- ¹⁰⁷³ UPMEM SAS, *Technology*, <https://www.upmem.com/technology/>.
- ¹⁰⁷⁴ Samsung Electronics Co., Ltd., HBM Processing in Memory, <https://www.samsung.com/semiconductor/solutions/technology/hbm-processing-in-memory/>.
- ¹⁰⁷⁵ Schneider, Michael, et al. "SuperMind: A survey of the potential of superconducting electronics for neuromorphic computing." *Superconductor Science and Technology* 35.5 (2022): 053001
- ¹⁰⁷⁶ P. Crotty, D. Schult and K. Segall, "Josephson junction simulation of neurons." *Phys. Rev. E*, vol. **82**, no. 8, doi:10.1103/PhysRevE.82.011914, 2010.
- ¹⁰⁷⁷ M. L. Schneider, et al., "Ultralow power artificial synapses using nanotextured magnetic Josephson junctions," *Science Advances* vol. 4, e1701329, doi:10.1126/sciadv.1701329, 2018.
- ¹⁰⁷⁸ Segall, K., et al. "A superconducting synapse exhibiting spike-timing dependent plasticity." *Applied Physics Letters* 122.24 (2023).
- ¹⁰⁷⁹ Ucpinar, Beyza Zeynep, et al. "An On-Chip Trainable Neuron Circuit for SFQ-Based Spiking Neural Networks." *IEEE Transactions on Applied Superconductivity* (2024).
- ¹⁰⁸⁰ Cheng, Ran, Uday S. Goteti, and Michael C. Hamilton. "High-speed and low-power superconducting neuromorphic circuits based on quantum phase-slip junctions." *IEEE Transactions on Applied Superconductivity* 31.5 (2021): 1-8.
- ¹⁰⁸¹ Rizzo, Nicholas D., et al. "A fully functional 64 Mb DDR3 ST-MRAM built on 90 nm CMOS technology." *IEEE Transactions on Magnetics* 49.7 (2013): 4441-4446
- ¹⁰⁸² D. Apalkov, B. Dieny, and J. M. Slaughter, "Magnetoresistive Random Access Memory," *Proceedings of the IEEE* vol. 104, pp. 1796–1830, doi:10.1109/jproc.2016.2590142, 2016.
- ¹⁰⁸³ S. K. Tolpygo, "Superconductor digital electronics: Scalability and energy efficiency issues (Review Article)." *Low Temperature Physics* vol. 42, 361–379, doi:10.1063/1.4948618, 2016.
- ¹⁰⁸⁴ Jué, Emilie, et al. "Perspectives on nanoclustered magnetic Josephson junctions as artificial synapses." *Applied Physics Letters* 121.24 (2022).
- ¹⁰⁸⁵ Schegolev, Andrey, et al. "Learning cell for superconducting neural networks." *Superconductor Science and Technology* 34.1 (2020): 015006
- ¹⁰⁸⁶ Hamerly, Ryan, et al. "Large-scale optical neural networks based on photoelectric multiplication." *Physical Review X* 9.2 (2019): 021032.
- ¹⁰⁸⁷ Tait, Alexander N., et al. "Neuromorphic photonic networks using silicon photonic weight banks." *Scientific reports* 7.1 (2017): 7430
- ¹⁰⁸⁸ Shen, Yichen, et al. "Deep learning with coherent nanophotonic circuits." *Nature Photonics* 11.7 (2017): 441.
- ¹⁰⁸⁹ Yang, Lin, et al. "On-chip CMOS-compatible optical signal processor." *Optics express* 20.12 (2012): 13560-13565.
- ¹⁰⁹⁰ Tait, Alexander N., et al. "Microring weight banks." *IEEE Journal of Selected Topics in Quantum Electronics* 22.6 (2016): 312-325.
- ¹⁰⁹¹ Jouppi, Norman P., et al. "In-datacenter performance analysis of a tensor processing unit." *2017 ACM/IEEE 44th Annual International Symposium on Computer Architecture (ISCA)*. IEEE, 2017
- ¹⁰⁹² Papamichael, Jeremy Fowers Kalin Ovtcharov Michael, et al. "A Configurable Cloud-Scale DNN Processor for Real-Time AI." (2018).
- ¹⁰⁹³ Nicol, Chris. "A coarse grain reconfigurable array (cgra) for statically scheduled data flow computing." *Wave Computing White Paper* (2017).
- ¹⁰⁹⁴ Schrauwen, Jonathan, Dries Van Thourhout, and Roel Baets. "Trimming of silicon ring resonator by electron beam induced compaction and strain." *Optics express* 16.6 (2008): 3738-3743.
- ¹⁰⁹⁵ Atabaki, Amir H., et al. "Accurate post-fabrication trimming of ultra-compact resonators on silicon." *Optics express* 21.12 (2013): 14139-14145.

- ¹⁰⁹⁶ Atabaki, Amir H., et al. "Accurate post-fabrication trimming of ultra-compact resonators on silicon." *Optics express* 21.12 (2013): 14139-14145.
- ¹⁰⁹⁷ Padmaraju, Kishore, and Keren Bergman. "Resolving the thermal challenges for silicon microring resonator devices." *Nanophotonics* 3.4-5 (2014): 269-281.
- ¹⁰⁹⁸ Krishnamoorthy, Ashok V., et al. "Exploiting CMOS manufacturing to reduce tuning requirements for resonant optical devices." *IEEE Photonics Journal* 3.3 (2011): 567-579.
- ¹⁰⁹⁹ Su, Zhan, et al. "Reduced wafer-scale frequency variation in adiabatic microring resonators." *OFC 2014*. IEEE, 2014.
- ¹¹⁰⁰ Mekis, Attila, et al. "A grating-coupler-enabled CMOS photonics platform." *IEEE Journal of Selected Topics in Quantum Electronics* 17.3 (2010): 597-608.
- ¹¹⁰¹ Bogaerts, Wim, et al. "Silicon-on-insulator spectral filters fabricated with CMOS technology." *IEEE journal of selected topics in quantum electronics* 16.1 (2010): 33-44.
- ¹¹⁰² Assefa, Solomon, Fengnian Xia, and Yurii A. Vlasov. "Reinventing germanium avalanche photodetector for nanophotonic on-chip optical interconnects." *Nature* 464.7285 (2010): 80.
- ¹¹⁰³ Nahmias, Mitchell A., et al. "Photonic multiply-accumulate operations for neural networks." *IEEE Journal of Selected Topics in Quantum Electronics* 26.1 (2019): 1-18.
- ¹¹⁰⁴ Glick, Madeleine, Lionel C. Kimmerling, and Robert C. Pfahl. "A roadmap for integrated photonics." *Optics and Photonics News* 29.3 (2018): 36-41.
- ¹¹⁰⁵ I. Ahmed, P.-W. Chiu, W. Moy, and C. H. Kim, "A Probabilistic Compute Fabric Based on Coupled Ring Oscillators for Solving Combinatorial Optimization Problems," *IEEE Journal of Solid-State Circuits*, vol. 56, no. 9, 2021.
- ¹¹⁰⁶ C. T. Van Degriфт, "Tunnel diode oscillator for 0.001 ppm measurements at low temperatures," *Review of Scientific Instruments*, vol. 46, 599, 1975.
- ¹¹⁰⁷ J. M. Manley and H. E. Rowe, "Some general properties of nonlinear elements – Part I. general energy relations," *Proceedings of the Institute of Radio Engineers*, vol. 44, no. 7, pp. 904-913, 1956.
- ¹¹⁰⁸ A. M. Niknejad, "Inductors and Transformers," *IEEE Solid-State Circuits Magazine*, 2014.
- ¹¹⁰⁹ L. Iotti, S. Krishnamurthy, G. LaCaille, A. M. Niknejad, "A Low-Power 70-100 GHz Mixer-First RX Leveraging Frequency-Translational Feedback," *IEEE Journal of Solid-State Circuits*, vol. 55, no. 8, pp. 2043-2054, 2020.
- ¹¹¹⁰ K. Yogendra, D. Fan, and K. Roy, "Coupled Spin Torque Nano Oscillators for Low Power Neural Computation," *IEEE Transactions on Magnetics*, vol. 51, no. 10, pp. 1–9, Oct. 2015.
- ¹¹¹¹ A. Sengupta, P. Panda, P. Wijesinghe, Y. Kim, and K. Roy, "Magnetic Tunnel Junction Mimics Stochastic Cortical Spiking Neurons," *Scientific Reports*, vol. 6, p. srep30039, Jul. 2016.
- ¹¹¹² S. Kaka, M. R. Pufall, W. H. Rippard, T. J. Silva, S. E. Russek, and J. A. Katine, "Mutual phase-locking of microwave spin torque nano-oscillators," *Nature*, vol. 437, no. 7057, pp. 389–392, Sep. 2005.
- ¹¹¹³ Y. Li, X. de Milly, F. Abreu Araujo, O. Klein, V. Cros, J. Grollier, and G. de Loubens, "Probing Phase Coupling Between Two Spin-Torque Nano-Oscillators with an External Source," *Physical Review Letters*, vol. 118, no. 24, p. 247202, Jun. 2017.
- ¹¹¹⁴ R. Lebrun, S. Tsunegi, P. Bortolotti, H. Kubota, A. S. Jenkins, M. Romera, K. Yakushiji, A. Fukushima, J. Grollier, S. Yuasa, and V. Cros, "Mutual synchronization of spin torque nano-oscillators through a long-range and tunable electrical coupling scheme," *Nature Communications*, vol. 8, p. ncomms15825, Jun. 2017.
- ¹¹¹⁵ M. R. Pufall, W. H. Rippard, G. Csaba, D. E. Nikonov, G. I. Bourianoff, and W. Porod, "Physical Implementation of Coherently Coupled Oscillator Networks," *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, vol. 1, pp. 76–84, Dec. 2015.
- ¹¹¹⁶ B. C. McGoldrick, J. Z. Sun, and L. Liu, "Ising Machine Based on Electrically Coupled Spin Hall Nano-Oscillators," *Physical Review Applied*, vol. 17, no. 1, pp. 014006, 2022.
- ¹¹¹⁷ W. A. Vitale, C. F. Moldovan, A. Paone, A. Schöler, and A. M. Ionescu, "Fabrication of CMOS-compatible abrupt electronic switches based on vanadium dioxide," *Microelectronic Engineering*, vol. 145, pp. 117–119, Sep. 2015.
- ¹¹¹⁸ H.-T. Kim, B.-G. Chae, D.-H. Youn, S.-L. Maeng, G. Kim, K.-Y. Kang, and Y.-S. Lim, "Mechanism and observation of Mott transition in VO₂-based two- and three-terminal devices," *New Journal of Physics*, vol. 6, no. 1, p. 52, 2004.
- ¹¹¹⁹ N. Shukla, A. Parihar, E. Freeman, H. Paik, G. Stone, V. Narayanan, H. Wen, Z. Cai, V. Gopalan, R. Engel-Herbert, D. G. Schlom, A. Raychowdhury, and S. Datta, "Synchronized charge oscillations in correlated electron systems," *Scientific Reports*, vol. 4, p. 4964, May 2014.
- ¹¹²⁰ S. Kumar, J. P. Strachan, and R. S. Williams, "Chaotic dynamics in nanoscale NbO₂ Mott memristors for analogue computing," *Nature*, vol. 548, p. 318, 2017.
- ¹¹²¹ M. D. Pickett, G. Medeiros-Ribeiro, and R. S. Williams, "A scalable neuristor built with Mott memristors," *Nature Materials*, vol. 12, pp. 114-7, Feb 2013.
- ¹¹²² P.-Y. Chen, J.-s. Seo, Y. Cao, and S. Yu, "Compact oscillation neuron exploiting metal-insulator-transition for neuromorphic computing," presented at the *Proceedings of the 35th International Conference on Computer-Aided Design*, Austin, Texas, 2016.
- ¹¹²³ S. Dutta, et al., "An Ising Hamiltonian solver based on coupled stochastic phase-transition nano-oscillators," *Nature Electronics*, vol. 4, pp. 502-512, 2021.

- ¹¹²⁴ A. Raychowdhury, et al., "Computing with Networks of Oscillatory Dynamical Systems," *Proceedings of the IEEE*, vol. 107, no. 1, pp. 73–89, 2019.
- ¹¹²⁵ S. Dutta, et al., "Programmable coupled oscillators for synchronized locomotion," *Nature Communications*, vol. 10, 3299, 2019.
- ¹¹²⁶ S. Dutta, A. Khanna, J. Gomez, K. Ni, Z. Toroczka, and S. Datta, "Experimental Demonstration of Phase Transition Nano-Oscillator Based Ising Machine," *IEEE International Electron Devices Meeting (IEDM)*, 2019.
- ¹¹²⁷ C. Li, E. Wasige, L. Wang, J. Wang, and B. Romeira, "28 GHz MMIC resonant tunnelling diode oscillator of around 1 mW output power," *Electronics Letters*, vol. 49, no. 13, pp. 816–818, Jun. 2013.
- ¹¹²⁸ M. Feiginov, C. Sydlo, O. Cojocari, and P. Meissner, "Resonant-tunnelling-diode oscillators operating at frequencies above 1.1 THz," *Applied Physics Letters*, vol. 99, no. 23, p. 233506, 2011.
- ¹¹²⁹ S. Suzuki, M. Asada, A. Teranishi, H. Sugiyama, and H. Yokoyama, "Fundamental oscillation of resonant tunneling diodes above 1 THz at room temperature," *Applied Physics Letters*, vol. 97, no. 24, p. 242102, 2010.
- ¹¹³⁰ H. Eisele, "State of the art and future of electronic sources at terahertz frequencies," *Electronics Letters*, vol. 46, no. 26, p. S8, 2010.
- ¹¹³¹ T. Silva and W. Rippard, "Developments in nano-oscillators based upon spin-transfer point-contact devices," *Journal of Magnetism and Magnetic Materials*, vol. 320, no. 7, pp. 1260–1271, Apr. 2008.
- ¹¹³² C. L. Tang, W. R. Bosenberg, T. Ukachi, R. J. Lane, and L. K. Cheng, "Optical parametric oscillators," *Proceedings of the IEEE*, vol. 80, no. 3, pp. 365–374, 1992.
- ¹¹³³ L. Maleki, "Sources: The optoelectronic oscillator," *Nature Photonics*, vol. 5, no. 12, pp. 728–730, 2011.
- ¹¹³⁴ I. Mahboob and H. Yamaguchi, "Piezoelectrically pumped parametric amplification and Q enhancement in an electromechanical oscillator," *Applied Physics Letters*, vol. 92, 173109, 2008.
- ¹¹³⁵ S. Ikeda, J. Hayakawa, Y. Ashizawa, Y. Lee, K. Miura, H. Hasegawa, M. Tsunoda, F. Matsukura, and H. Ohno, "Tunnel magnetoresistance of 604% at 300K by suppression of Ta diffusion in CoFeB/MgO/CoFeB pseudo-spin-valves annealed at high temperature," *Applied Physics Letters* 93, 082508 (2008).
- ¹¹³⁶ W. Butler, X.-G. Zhang, T. Schulthess, and J. MacLaren, "Spin-dependent tunneling conductance of Fe|MgO|Fe sandwiches," *Physical Review B* 63, 054416 (2001).
- ¹¹³⁷ Safranski, Christopher, et al. "Demonstration of nanosecond operation in stochastic magnetic tunnel junctions." *Nano Letters* 21.5 (2021): 2040-2045.
- ¹¹³⁸ Hayakawa, Keisuke, et al. "Nanosecond random telegraph noise in in-plane magnetic tunnel junctions." *Physical Review Letters* 126.11 (2021): 117202.
- ¹¹³⁹ Funatsu, T., Kanai, S., Ieda, J. I., Fukami, S., & Ohno, H. (2022). Local bifurcation with spin-transfer torque in superparamagnetic tunnel junctions. *Nature communications*, 13(1), 4079.
- ¹¹⁴⁰ L. Schnitzspan, M. Klau, and G. Jakob, Nanosecond true-random-number generation with superparamagnetic tunnel junctions: Identification of joule heating and spin-transfer-torque effects, *Phys. Rev. Appl.* 20, 024002 (2023).
- ¹¹⁴¹ Soumah, L., Desplat, L., Phan, N. T., Valli, A. S. E., Madhavan, A., Disdier, F., ... & Talatchian, P. (2024). v. arXiv preprint arXiv:2402.03452.
- ¹¹⁴² Daniel, J., Sun, Z., Zhang, X., Tan, Y., Dilley, N., Chen, Z., & Appenzeller, J. (2024). Experimental demonstration of an on-chip p-bit core based on stochastic magnetic tunnel junctions and 2D MoS2 transistors. *Nature Communications*, 15(1), 1-11.
- ¹¹⁴³ Kerem Y Camsari, Mustafa Mert Torunbalci, William A Borders, Hideo Ohno, and Shunsuke Fukami. Double-free-layer magnetic tunnel junctions for probabilistic bits. *Physical Review Applied*, 15(4):044049, 2021.
- ¹¹⁴⁴ Jonathan Z. Sun, Christopher Safranski, Philip Trouilloud, Christopher D'Emic, Pouya Hashemi, and Guohan Hu. Easy-plane dominant stochastic magnetic tunnel junction with synthetic antiferromagnetic layers. *Phys. Rev. B*, 108:064418, 2023.
- ¹¹⁴⁵ Selcuk, K., Kanai, S., Ota, R., Ohno, H., Fukami, S., & Camsari, K. Y. (2024). Double-free-layer stochastic magnetic tunnel junctions with synthetic antiferromagnets. *Physical Review Applied*, 21(5), 054002.
- ¹¹⁴⁶ Stanco, A. et al. Efficient random number generation techniques for CMOS single-photon avalanche diode array exploiting fast time tagging units. *Phys. Rev. Res.* 2, 023287 (2020).
- ¹¹⁴⁷ Whitehead, W., Nelson, Z., Camsari, K. Y. & Theogarajan, L. CMOS-compatible Ising and Potts annealing using single photon avalanche diodes. *Nat. Electronics* 6, 1009–1019 (2023).
- ¹¹⁴⁸ R. K. Henderson, E. A. G. Webster, and R. Walker, "A Gate Modulated Avalanche Bipolar Transistor in 130nm CMOS Technology," *Solid-State Device Research Conference (ESSDERC), 2012 Proceedings of the European*, pp. 226-229, IEEE (2012)
- ¹¹⁴⁹ E. A.G. Webster, J. A. Richardson, L. A. Grant, R. K. Henderson, "A single electron bipolar avalanche transistor implemented in 90 nm CMOS" *Solid-State Electronics*, 76, pp.116-118 (2012)
- ¹¹⁵⁰ J. Rajendran et al., "Nano meets security: Exploring nanoelectronic devices for security applications," *Proceedings of the IEEE*, vol. 103, no. 5, pp. 829–849, 2015
- ¹¹⁵¹ M. L. Gallo, T. Tuma, F. Zipoli, A. Sebastian, and E. Eleftheriou, "Inherent stochasticity in phase-change memory devices," *2016 46th European Solid-State Device Research Conference (ESSDERC)*, 2016, pp. 373–376.
- ¹¹⁵² H. Jiang, D. Belkin, S. E. Savel'ev, S. Lin, Z. Wang, Y. Li, et al., "A novel true random number generator based on a stochastic diffusive memristor," *Nature Communications*, vol. 8, 2017.
- ¹¹⁵³ Z. Wang, S. Joshi, S. E. Saveliev, H. Jiang, R. Midya, P. Lin, et al., "Memristors with diffusive dynamics as synaptic emulators for neuromorphic computing," *Nature Materials*, vol. 16, pp. 101–108, 09/26/online 2016.

- 1154 Y. H. Tseng, C.-E. Huang, C.-H. Kuo, Y.-D. Chih, and C. J. Lin, "High density and ultra-small cell size of contact ReRAM (CR-RAM) in 90 nm CMOS logic technology and circuits," *Electron Devices Meeting (IEDM), 2009 IEEE International*, 2009, pp. 1–4: IEEE
- 1155 C.-Y. Huang, W. C. Shen, Y.-H. Tseng, Y.-C. King, and C.-J. Lin, "A contact-resistive random-access-memory-based true random number generator," *IEEE Electron Device Letters*, vol. 33, no. 8, pp. 1108–1110, 2012
- 1156 M. Park, J. C. Rodgers, and D. P. Lathrop, "True random number generation using CMOS Boolean chaotic oscillator," *Microelectronics Journal* 46, 1364 (2015).
- 1157 Russek, S. E. et al. in 2016 IEEE International Conference on Rebooting Computing (ICRC). 1-5.
- 1158 Schneider, M. L. et al. Ultralow power artificial synapses using nanotextured magnetic Josephson junctions. *Science Advances* 4, e1701329, doi:10.1126/sciadv.1701329 (2018).
- 1159 Yamanashi, Y., Umeda, K. & Yoshikawa, N. Pseudo Sigmoid Function Generator for a Superconductive Neural Network. *IEEE Transactions on Applied Superconductivity* 23, 1701004-1701004, doi:10.1109/TASC.2012.2228531 (2013).
- 1160 Onomi, T., Kondo, T. & Nakajima, K. Implementation of high-speed single flux-quantum up/down counter for the neural computation using stochastic logic. *IEEE Transactions on Applied Superconductivity* 19, 626-629 (2009).
- 1161 Crotty, Patrick, Kenneth Segall, and Daniel Schult. "Biologically realistic behaviors from a superconducting neuron model." *IEEE Transactions on Applied Superconductivity* 33.4 (2023): 1-6.
- 1162 Segall, Ken, et al. "Synchronization dynamics on the picosecond time scale in coupled Josephson junction neurons." *Physical Review E* 95.3 (2017): 032220.
- 1163 Chalkiadakis, Dimitrios, and Johanne Hizanidis. "Dynamical properties of neuromorphic Josephson junctions." *Physical Review E* 106.4 (2022): 044206.
- 1164 S. Oh et al., "Bayesian Neural Network Implemented by Dynamically Programmable Noise in Vanadium Oxide," *IEEE International Electron Devices Meeting (IEDM)*, pp. 1-4, 2023.
- 1165 G. M. Kim et al., "Self-clocking fast and variation tolerant true random number generator based on a stochastic mott memristor," *Nature Comm.*, 12, 1, 2021.
- 1166 R. D. Chakraborty, W. M. Postiglione, S. Ghosh, K. A. Mkhoyan, C. Leighton, and V. E. Ferry, "Optical Properties of Electrochemically Gated $\text{La}_{1-x}\text{Sr}_x\text{CoO}_{3-\delta}$ as a Topotactic Phase-Change Material," *Advanced Optical Materials*, 11, 16, 2023.
- 1167 D. E. Nikonov and I. A. Young, "Overview of Beyond-CMOS Devices and a Uniform Methodology for Their Benchmarking," *Proceedings of the IEEE*, vol. 101, no. 12, pp. 2498–2533, 2013
- 1168 D. E. Nikonov and I. A. Young, "Benchmarking of Beyond-CMOS Exploratory Devices for Logic Integrated Circuits," *Exploratory Solid-State Computational Devices and Circuits, IEEE Journal on*, vol. 1, pp. 3–11, 2015
- 1169 C. Pan and A. Nacemi, "An Expanded Benchmarking of Beyond-CMOS Devices Based on Boolean and Neuromorphic Representative Circuits," *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, vol. 3, pp. 101–110, 2017
- 1170 A. Iyengar, K. Ramclan, and S. Ghosh, "DWM-PUF: A low-overhead, memory-based security primitive," *Hardware-Oriented Security and Trust (HOST), 2014 IEEE International Symposium on*, 2014, pp. 154–159: IEEE
- 1171 G. S. Rose, N. McDonald, Y. Lok-Kwong, and B. Wysocki, "A write-time based memristive PUF for hardware security applications," *Computer-Aided Design (ICCAD), 2013 IEEE/ACM International Conference on*, 2013, pp. 830–833
- 1172 S. T. C. Konigsmark, L. K. Hwang, C. Deming, and M. D. F. Wong, "CNPUF: A Carbon Nanotube-based Physically Unclonable Function for secure low-energy hardware design," *Design Automation Conference (ASP-DAC), 2014 19th Asia and South Pacific*, 2014, pp. 73–78
- 1173 J. Rajendran et al., "Nano meets security: Exploring nanoelectronic devices for security applications," *Proceedings of the IEEE*, vol. 103, no. 5, pp. 829–849, 2015
- 1174 G. E. Suh, C. W. O'Donnell, I. Sachdev, and S. Devadas, "Design and implementation of the AEGIS single-chip secure processor using physical random functions," *ACM SIGARCH Computer Architecture News*, 2005, vol. 33, no. 2, pp. 25–36: IEEE Computer Society.
- 1175 J. Guajardo, S. S. Kumar, G.-J. Schrijen, and P. Tuyls, "Physical unclonable functions and public-key crypto for FPGA IP protection," *Field Programmable Logic and Applications, 2007. FPL 2007. International Conference on*, 2007, pp. 189–195: IEEE.
- 1176 T. Marukame, T. Tanamoto, and Y. Mitani, "Extracting physically unclonable function from spin transfer switching characteristics in magnetic tunnel junctions," *IEEE Transactions on Magnetics*, vol. 50, no. 11, pp. 1–4, 2014.
- 1177 J. Das, K. Scott, S. Rajaram, D. Burgett, and S. Bhanja, "MRAM PUF: A novel geometry based magnetic PUF with integrated CMOS," *IEEE Transactions on Nanotechnology*, vol. 14, no. 3, pp. 436–443, 2015.
- 1178 L. Zhang, Z. H. Kong, C.-H. Chang, A. Cabrini, and G. Torelli, "Exploiting process variations and programming sensitivity of phase change memory for reconfigurable physical unclonable functions," *IEEE Transactions on Information Forensics and Security*, vol. 9, no. 6, pp. 921–932, 2014.
- 1179 Y. Pang, H. Wu, B. Gao, N. Deng, D. Wu, R. Liu, S. Yu, A. Chen, and H. Qian, "Optimization of RRAM-Based Physical Unclonable Function With a Novel Differential Read-Out Method," *IEEE Electron. Dev. Lett.* 38(2), 168-171 (2017).
- 1180 Y. Pang, H. Wu, B. Gao, D. Wu, A. Chen, and H. Qian, "A Novel PUF Against Machine Learning Attack: Implementation on a 16 Mb RRAM Chip," *IEDM* 290-293 (2017)
- 1181 U. Rührmair, C. Jaeger, C. Hilgers, M. Algasinger, G. Csaba, and M. Stutzmann, "Security applications of diodes with unique current-voltage characteristics," *International Conference on Financial Cryptography and Data Security*, 2010, pp. 328–335: Springer.
- 1182 D. P. C. Dimitrakopoulos, and J. Smith, "Authentication using graphene based devices as physical unclonable functions," 2004. Available: <http://www.google.com/patents/US20140159040>

- 1183 S. C. Konigsmark, L. K. Hwang, D. Chen, and M. D. Wong, "Cnpuf: A carbon nanotube-based physically unclonable function for secure low-energy hardware design," *Design Automation Conference (ASP-DAC)*, 2014 19th Asia and South Pacific, 2014, pp. 73–78: IEEE
- 1184 G. S. Rose, N. McDonald, L.-K. Yan, and B. Wysocki, "A write-time based memristive PUF for hardware security applications," *Proceedings of the International Conference on Computer-Aided Design*, 2013, pp. 830–833: IEEE Press
- 1185 P. Koeberl, Ü. Kocabaş, and A.-R. Sadeghi, "Memristor PUFs: a new generation of memory-based physically unclonable functions," *Proceedings of the Conference on Design, Automation and Test in Europe*, 2013, pp. 428–431: EDA Consortium
- 1186 Y. H. Tseng, C.-E. Huang, C.-H. Kuo, Y.-D. Chih, and C. J. Lin, "High density and ultra-small cell size of contact ReRAM (CR-RAM) in 90 nm CMOS logic technology and circuits," *Electron Devices Meeting (IEDM)*, 2009 IEEE International, 2009, pp. 1–4: IEEE
- 1187 C.-Y. Huang, W. C. Shen, Y.-H. Tseng, Y.-C. King, and C.-J. Lin, "A contact-resistive random-access-memory-based true random number generator," *IEEE Electron Device Letters*, vol. 33, no. 8, pp. 1108–1110, 2012
- 1188 A. Colli, S. Pisana, A. Fasoli, J. Robertson, and A. C. Ferrari, "Electronic transport in ambipolar silicon nanowires," *physica status solidi (b)*, vol. 244, no. 11, pp. 4161–4164, 2007
- 1189 R. Martel et al., "Ambipolar Electrical Transport in Semiconducting Single-Wall Carbon Nanotubes," *Physical Review Letters*, vol. 87, no. 25, p. 256805, 12/03/ 2001
- 1190 A. K. Geim and K. S. Novoselov, "The rise of graphene," *Nat Mater*, 10.1038/nmat1849 vol. 6, no. 3, pp. 183–191, 03//print 2007
- 1191 L. Yu-Ming, J. Appenzeller, J. Knoch, and P. Avouris, "High-performance carbon nanotube field-effect transistor with tunable polarities," *Nanotechnology*, *IEEE Transactions on*, vol. 4, no. 5, pp. 481–489, 2005
- 1192 N. Harada, K. Yagi, S. Sato, and N. Yokoyama, "A polarity-controllable graphene inverter," *Applied Physics Letters*, vol. 96, no. 1, p. 012102, 2010
- 1193 A. Heinzig, S. Slesazek, F. Kreupl, T. Mikolajick, and W. M. Weber, "Reconfigurable Silicon Nanowire Transistors," *Nano Letters*, vol. 12, no. 1, pp. 119–124, 2012/01/11 2012
- 1194 S. Das and J. Appenzeller, "WSe₂ field effect transistors with enhanced ambipolar characteristics," *Applied Physics Letters*, vol. 103, no. 10, p. 103501, 2013
- 1195 M. De Marchi et al., "Polarity control in double-gate, gate-all-around vertically stacked silicon nanowire FETs," *Electron Devices Meeting (IEDM)*, 2012 IEEE International, 2012, pp. 8.4.1–8.4.4
- 1196 T. Vasen, "Investigation of III-V Tunneling Field-Effect Transistors," A Dissertation submitted to the University of Notre Dame, 2014.
- 1197 Y. Bi et al., "Emerging Technology Based Design of Primitives for Hardware Security," *ACM Journal on Emerging Technologies in Computing Systems (JETC)*, vol. 13, no. 1, Article 3, 2016
- 1198 B. Yu, P. E. Gaillardon, X. S. Hu, M. Niemier, Y. Jiann-Shiun, and J. Yier, "Leveraging Emerging Technology for Hardware Security - Case Study on Silicon Nanowire FETs and Graphene SymFETs," *Test Symposium (ATS)*, 2014 IEEE 23rd Asian, 2014, pp. 342–347
- 1199 A. Chen, X. S. Hu, Y. Jin, M. Niemier, and X. Yin, "Using Emerging Technologies for Hardware Security Beyond PUFs," *Design, Automation, and Test in Europe (DATE)*, Dresden, Germany, 2016
- 1200 J. Rajendran et al., "Fault Analysis-Based Logic Encryption," *Computers*, *IEEE Transactions on*, vol. 64, no. 2, pp. 410–424, 2015
- 1201 Y. Bi et al., "Emerging Technology Based Design of Primitives for Hardware Security," *ACM Journal on Emerging Technologies in Computing Systems (JETC)*, vol. 13, no. 1, Article 3, 2016
- 1202 B. Yu, P. E. Gaillardon, X. S. Hu, M. Niemier, Y. Jiann-Shiun, and J. Yier, "Leveraging Emerging Technology for Hardware Security - Case Study on Silicon Nanowire FETs and Graphene SymFETs," *Test Symposium (ATS)*, 2014 IEEE 23rd Asian, 2014, pp. 342–347
- 1203 Y. Bi, K. Shamsi, J.-S. Yuan, F.-X. Standaert, and Y. Jin, "Leverage emerging technologies for dpa-resilient block cipher design," *Proceedings of the 2016 Conference on Design, Automation & Test in Europe*, 2016, pp. 1538–1543: EDA Consortium
- 1204 Y. Bi, K. Shamsi, J.-S. Yuan, Y. Jin, M. Niemier, and X. S. Hu, "Tunnel fet current mode logic for dpa-resilient circuit designs," *IEEE Transactions on Emerging Topics in Computing*, vol. 5, no. 3, pp. 340–352, 2017
- 1205 A. C. Seabaugh and Z. Qin, "Low-Voltage Tunnel Transistors for Beyond CMOS Logic," *Proceedings of the IEEE*, vol. 98, no. 12, pp. 2095–2110, 2010
- 1206 H. Lu and A. Seabaugh, "Tunnel field-effect transistors: State-of-the-art," *IEEE Journal of the Electron Devices Society*, vol. 2, no. 4, pp. 44–49, 2014
- 1207 L. Britnell et al., "Resonant tunnelling and negative differential conductance in graphene transistors," *Nat Commun*, Article vol. 4, p. 1794, 04/30/online 2013
- 1208 MishchenkoA et al., "Twist-controlled resonant tunnelling in graphene/boron nitride/graphene heterostructures," *Nat Nano*, Letter vol. 9, no. 10, pp. 808–813, 10//print 2014
- 1209 M. O. Li, D. Esseni, J. J. Nahas, D. Jena, and H. G. Xing, "Two-Dimensional Heterojunction Interlayer Tunneling Field Effect Transistors (Thin-TFETs)," *Electron Devices Society*, *IEEE Journal of the*, vol. 3, no. 3, pp. 200–207, 2015
- 1210 H. Mamiya, A. Miyaji, and H. Morimoto, "Efficient Countermeasures against RPA, DPA, and SPA," *Cryptographic Hardware and Embedded Systems – CHES 2004*, vol. 3156, M. Joye and J.-J. Quisquater, Eds. (Lecture Notes in Computer Science: Springer Berlin Heidelberg, 2004, pp. 343–356
- 1211 B. Sedighi, J. J. Nahas, M. Niemier, and X. S. Hu, "Boolean circuit design using emerging tunneling devices," *Computer Design (ICCD)*, 2014 32nd IEEE International Conference on, 2014, pp. 355–360
- 1212 P. Kocher, "Design and validation strategies for obtaining assurance in countermeasures to power analysis and related attacks," NIST Physical Security Workshop, September 26-29 (2005).
- 1213 P. Kocher, J. Jaffe, and B. Jun, "Differential power analysis," *Annual International Cryptology Conference*, 1999, pp. 388–397: Springer

- ¹²¹⁴ M.-L. Akkar and C. Giraud, "An implementation of DES and AES, secure against some attacks," *International Workshop on Cryptographic Hardware and Embedded Systems*, 2001, pp. 309–318: Springer
- ¹²¹⁵ K. Tiri, M. Akmal, and I. Verbauwhede, "A dynamic and differential CMOS logic with signal independent power consumption to withstand differential power analysis on smart cards," *Solid-State Circuits Conference, 2002. ESSCIRC 2002. Proceedings of the 28th European*, 2002, pp. 403–406: IEEE
- ¹²¹⁶ Y. Bi, K. Shamsi, J.-S. Yuan, Y. Jin, M. Niemier, and X. S. Hu, "Tunnel fet current mode logic for dpa-resilient circuit designs," *IEEE Transactions on Emerging Topics in Computing*, vol. 5, no. 3, pp. 340–352, 2017
- ¹²¹⁷ C. De Canniere, O. Dunkelman, and M. Knežević, "KATAN and KTANTAN—a family of small and efficient hardware-oriented block ciphers," *Cryptographic Hardware and Embedded Systems-CHES 2009*: Springer, 2009, pp. 272–288
- ¹²¹⁸ A. Cevrero, F. Regazzoni, M. Schwander, S. Badel, P. lenne, and Y. Leblebici, "Power-gated mos current mode logic (pg-mcml): A power aware dpa-resistant standard cell library," *Design Automation Conference (DAC), 2011 48th ACM/EDAC/IEEE*, 2011, pp. 1014–1019: IEEE
- ¹²¹⁹ D. Gall, "The search for the most conductive metal for narrow interconnect lines" *J. Appl. Phys.* 127, 050901 (2020): <https://doi.org/10.1063/1.5133671>
- ¹²²⁰ M.S. Fuhrer, M.T. Edmonds, D. Culcer, M. Nadeem, X. Wang, N. Medhekar, Y. Yin, J and .H Cole, "Proposal for a Negative Capacitance Topological Quantum Field-Effect Transistor", in 2021 IEEE International Electron Devices Meeting (IEDM), 2021, pp. 38.2.1-38.2.4.
- ¹²²¹ F. Zuo, P. Panda, M. Kotiuga, J. Li, M. Kang, C. Mazzoli, H. Zhou, A. Barbour, S. Wilkins, B. Narayanan, M. Cherukara, Z. Zhang, S. K. R. S. Sankaranarayanan, R. Comin, K. M. Rabe, K. Roy and S. Ramanathan, "Habituation based synaptic plasticity and organismic learning in a quantum perovskite", *Nature Communications* 8, 240 (2017): <https://doi.org/10.1038/s41467-017-00248-6>
- ¹²²² S.-J. Kim, K. Ohkoda, M. Aono, H. Shima, M. Takahashi, Y. Naitoh, and H. Akinaga, "Reinforcement Learning System Comprising Resistive Analog Neuromorphic Devices", *IEEE International Reliability Physics Symposium* (2019): 10.1109/IRPS.2019.8720428
- ¹²²³ Z. Wang, C. Li, W. Song, M. Rao, D. Belkin, Y. Li, P. Yan, H. Jiang, P. Lin, M. Hu, J. P. Strachan, N. Ge, M. Barnell, Q. Wu, A. G. Barto, Q. Qiu, R. S. Williams, Q. Xia 1 and J. J. Yang, "Reinforcement learning with analogue memristor arrays", *Nature Electronics* 2, 115 (2019): <https://doi.org/10.1038/s41928-019-0221-6>
- ¹²²⁴ T. Machida, Y. Sun, S. Pyon, S. Takeda, Y. Kohsaka, T. Hanaguri, T. Sasagawa, and T. Tamegai, "Zero-energy vortex bound state in the superconducting topological surface state of Fe (Se,Te)", *Nature Materials* 18, 811–815 (2019): <https://doi.org/10.1038/s41563-019-0397-1>
- ¹²²⁵ W. A. Borders, A. Z. Pervaiz, S. Fukami, K. Y. Camsari, H. Ohno, and S. Datta, "Integer factorization using stochastic magnetic tunnel junctions", *Nature* 573, 390-393 (2019): <https://doi.org/10.1038/s41586-019-1557-9>
- ¹²²⁶ Z. Wang, H. Wu, G. W. Burr, C. S. Hwang, K. L. Wang, Q. Xia, and J. J. Yang, "Resistive switching materials for information processing", *Nat. Rev. Mater.* 5, 173–195 (2020). <https://doi.org/10.1038/s41578-019-0159-3>
- ¹²²⁷ Y. Shi, X. Liang, B. Yuan, V. Chen, H. Li, F. Hui, Z. Yu, F. Yuan, E. Pop, H.-S. Philip Wong, and M. Lanza, "Electronic synapses made of layered two-dimensional materials", *Nat. Electron.* 1, 458–465 (2018): <https://doi.org/10.1038/s41928-018-0118-9>
- ¹²²⁸ S. Chen, M. R. Mahmoodi, Y. Shi, C. Mahata, B. Yuan, X. Liang, C. Wen, F. Hui, D. Akinwande, D. B. Strukov, and M. Lanza, "Wafer-scale integration of two-dimensional materials in high-density memristive crossbar arrays for artificial neural networks", *Nat. Electron.* 3, 638–645 (2020): <https://doi.org/10.1038/s41928-020-00473-w>
- ¹²²⁹ S. K. Bose, C. P. Lawrence, Z. Liu, K. S. Makarenko, R. M. J. van Damme, H. J. Broersma and W. G. van der Wiel, "Evolution of a designless nanoparticle network into reconfigurable Boolean logic", *Nature Nanotechnology*, Vol. 10, 1048-1053 (2015): <http://dx.doi.org/10.1038/nnano.2015.207>
- ¹²³⁰ Minkyung Lee, Hyojin Seung, Jong Ik Kwon, Moon Kee Choi, Dae-Hyeong Kim, and Changsoon Choi, "Nanomaterial-Based Synaptic Optoelectronic Devices for In-Sensor Preprocessing of Image Data", *ACS Omega* 8 (6), 5209-5224 (2023)
- ¹²³¹ Jinyoung Seo, Sungi Kim, Ha H. Park, Da Yeon Choi, and Jwa-Min Nam, "Nano-bio-computing lipid nanotablet", *Science Advances*, 5 eaau2124 (2019)
- ¹²³² Sungi Kim, Namjun Kim, Jinyoung Seo, Jeong-Eun Park, Eun Ho Song, So Young Choi, Ji Eun Kim, Seungsang Cha, Ha H. Park, and Jwa-Min Nam, "Nanoparticle-based computing architecture for nanoparticle neural networks", *Science Advances*, 6. eabb3348 (2020)
- ¹²³³ K. Sugawara, H. Shima, M. Takahashi, Y. Naitoh, H. Suga, and H. Akinaga, "Low-Frequency-Noise Spectroscopy of TaOx-based Resistive Switching Memory", *Adv. Electron. Mater.*, 2100758, 2021: <https://doi.org/10.1002/aelm.202100758>
- ¹²³⁴ S.-W. Hwang, J.-K. Song, X. Huang, H. Cheng, S.-K. Kang, B. H. Kim, J.-H. Kim, S. Yu, Y. Huang, and J. A. Rogers, "High-Performance Biodegradable/Transient Electronics on Biodegradable Polymers", *Adv. Mater.* 26, 3905–3911 (2014): <https://doi.org/10.1002/adma.201306050>
- ¹²³⁵ A. Dutta and H. Cheng, "Pathway of transient electronics towards connected biomedical applications", *Nanoscale*, (2023), Advance Article: <https://doi.org/10.1039/D2NR06068J>
- ¹²³⁶ K. K. Fu, Z. Wang, J. Dai, M. Carter, and L. Hu, "Transient Electronics: Materials and Devices" *Chem. Mater.* 28, 3527–3539 (2016): <https://doi.org/10.1021/acs.chemmater.5b04931>
- ¹²³⁷ Y. Gao, Y. Zhang, X. Wang, K. Sim, J. Liu, J. Chen, X. Feng, H. Xu, and C. Yu, "Moisture-triggered physically transient electronics", *Sci. Adv.* 3, e1701222 (2017): <https://doi.org/10.1126/sciadv.1701222>
- ¹²³⁸ C. Arima, A. Suzuki, A. Kassai, T. Tsuruoka, and T. Hasegawa, "Development of a molecular gap-type atomic switch and its stochastic operation", *J. Appl. Phys.* 124, 152114 (2018): <https://doi.org/10.1063/1.5037657>

- ¹²³⁹ H. Sato, H. Shima, T. Nokami, T. Itoh, Y. Honma, Y. Naitoh, H. Akinaga, and K. Kinoshita, "Memristors with controllable data volatility by loading metal ion-added ionic liquids", *Front. Nanotechnol.* 3, 660563 (2021): <https://doi.org/10.3389/fnano.2021.660563>
- ¹²⁴⁰ F. Oba and Y. Kumagai, "Design and exploration of semiconductors from first principles: A review of recent advances", *Appl. Phys. Express* 11, 060101 (2018): <https://doi.org/10.7567/APEX.11.060101>
- ¹²⁴¹ A. Ueda, Y. Zhang, N. Sano, H. Imamura and Y. Iwasa, "Ambipolar device simulation based on the drift-diffusion model in ion-gated transition metal dichalcogenide transistors", *npj Comput Mater* 6, 24 (2020): <https://doi.org/10.1038/s41524-020-0293-x>
- ¹²⁴² E. Lidrikis, et al., "Coupling Length Scales for Multiscale Atomistics-Continuum Simulations: Atomistically induced stress distributions in Si/Si₃N₄ nanopixels", *Phys. Rev. Lett.* 87, 086104 (2001): <https://doi.org/10.1103/PhysRevLett.87.086104>
- ¹²⁴³ M. Andersson, J. Yuan, and B. Sundén, "Review on modeling development for multiscale chemical reactions coupled transport phenomena in solid oxide fuel cells", *Applied Energy* 87, 1461-1476 (2010): <https://doi.org/10.1016/j.apenergy.2009.11.013>
- ¹²⁴⁴ H. Kobayashi et al., "Hopping and band mobilities of pentacene, rubrene, and 2,7-dioctyl[1]benzothieno[3,2-b][1]benzothiophene (C8-BTBT) from first principle calculations", *J. Chem. Phys.* 139, 014707 (2013): <https://doi.org/10.1063/1.4812389>
- ¹²⁴⁵ H. Kobayashi, and Y. Tokita, "Modeling of hole transport across grain boundaries in organic semiconductors for mesoscale simulations", *Appl. Phys. Exp.* 8, 051602 (2015): <https://doi.org/10.7567/APEX.8.051602>
- ¹²⁴⁶ H. Kobayashi et al., "Carrier mobility in mesoscale heterogeneous organic materials: Effects of crystallinity and anisotropy on efficient charge transport", *Appl. Phys. Lett.* 111, 033301 (2017): <https://doi.org/10.1063/1.4995243>
- ¹²⁴⁷ R. Liu et al., *Integrating Materials and Manufacturing Innovation* 4, 192 (2015): <https://doi.org/10.1186/s40192-015-0042-z>
- ¹²⁴⁸ (a) G V Kunte, S A Shivashankar, and A M Umarji, "Thermogravimetric evaluation of the suitability of precursors for MOCVD," *Measurement Science and Technology*, vol. 19, p 025704. (b) M. B. E. Griffiths, Z. S. Dubrawski, G. Bačić, A. Japahuge, J. D. Masuda, T. Zeng, and S. T. Barry, "Controlling the Thermal Stability and Volatility of Organogold(I) Compounds for Vapor Deposition with Complementary Ligand Design," *European Journal of Inorganic Chemistry*, pp. 4927-4938, 2019.
- ¹²⁴⁹ A. L. Johnson and J. D. Parish, "Recent developments in molecular precursors for atomic layer deposition," *Organometallic Chemistry*, vol. 45, pp. 1-53, 2019.
- ¹²⁵⁰ G. N. Parsons and R. D. Clark, "Area-Selective Deposition: Fundamentals, Applications, and Future Outlook," *Chemistry of Materials*, vol. 32, pp. 4920-4953, 2020.
- ¹²⁵¹ J. Huang, J. Mu, Y. Cho, C. Winter, V. Wang, Z. Zhang, K., C. Kim, A. Yadav, K. Wong, S. Nemani, E. Yieh, and A. Kummel, "Low- k SiO_x/AlO_x Nanolaminate Dielectric on Dielectric Achieved by Hybrid Pulsed Chemical Vapor Deposition," *ACS Applied Materials & Interfaces*, vol. 15, pp. 56556-56566, 2023.
- ¹²⁵² K. Iwanaga, T. Hayakawa, Y. Yamamoto, H. Oike, R. Ebihara, and K. Tada, "Development of Precursors for Area Selective Deposition", *ADMETA Plus 32nd Asian Session*, S5-2, 2023.
- ¹²⁵³ C. Zhang, E. Tois, M. Leskelä, and M. Ritala, "Substrate-Dependent Area-Selective Atomic Layer Deposition of Noble Metals from Metal β-Diketonate Precursors," *Chemistry of Materials*, vol. 34, pp. 8379-8388, 2022.
- ¹²⁵⁴ Keyes, R.W., "The evolution of digital electronics towards VLSI," *IEEE Transactions on Electron Devices*, Volume 26, Issue 4, Apr 1979 Page(s):271 – 279.
- ¹²⁵⁵ Doug Matzke, "Will Physical Scalability Sabotage Performance Gains?" *IEEE Computer*, Volume 30, Issue 9, September, 1997, Page: 37 – 39.
- ¹²⁵⁶ J. Welser and K. Bernstein, "Challenges for Post-CMOS Devices & Architectures," *IEEE Device Research Conference Technical Digest*, Santa Barbara, CA, Jun 2011, pp. 183-186.
- ¹²⁵⁷ K. Bernstein, R.K. Cavin, W. Porod, A. Seabaugh, and J. Welser, "Device and Architecture Outlook for Beyond CMOS Switches," *Proceedings of the IEEE Special Issue - Nanoelectronics Research: Beyond CMOS Information Processing*, Volume 98, Issue 12, Dec 2010, pp. 2169-2184.
- ¹²⁵⁸ D.E. Nikonov and I.A. Young, "Uniform Methodology for Benchmarking Beyond-CMOS Logic Devices," *IEDM Tech. Dig.*, pp. 573–576, Dec. 2012.
- ¹²⁵⁹ D.E. Nikonov and I.A. Young, "Benchmarking of beyond-CMOS exploratory devices for logic integrated circuits," *IEEE J. Expl. Sol-State Comp. Dev. & Circ.*, vol. 1, pp. 3–11 (2015).
- ¹²⁶⁰ T. N. Theis and P. M. Solomon, "In Quest of the 'Next Switch': Prospects for Greatly Reduced Power Dissipation in a Successor to the Silicon Field-Effect Transistor," *Proceedings of the IEEE Special Issue - Nanoelectronics Research: Beyond CMOS Information Processing*, Volume 98, Issue 12, Dec 2010, pp. 2005-2014.
- ¹²⁶¹ C. Pan and A. Naeemi, "An Expanded Benchmarking of Beyond-CMOS Devices Based on Boolean and Neuromorphic Representative Circuits," *IEEE J. Exploratory Solid-State Computational Devices and Circuits*, vol. 3, pp 101-110, December 2017.
- ¹²⁶² I. Sutherland et al., *Logical Effort: Design Fast CMOS Circuits*, 1st ed. San Mateo, CA: Morgan Kaufmann, Feb. 1999, ISBN: 10:1558605576